

SCHAUM'S OUTLINE SERIES
THEORY AND PROBLEMS OF

FOURIER ANALYSIS

with applications to

**BOUNDARY VALUE
PROBLEMS**

MURRAY R. SPIEGEL

INCLUDING 205 SOLVED PROBLEMS

SCHAUM'S OUTLINE SERIES

McGRAW-HILL BOOK COMPANY

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THEORY AND PROBLEMS

of

FOURIER ANALYSIS

**with Applications to
Boundary Value Problems**

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by

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Preface

In the early years of the 19th century the French mathematician J. B. J. Fourier in his researches on heat conduction was led to the remarkable discovery of certain trigonometric series which now bear his name. Since that time Fourier series, and generalizations to Fourier integrals and orthogonal series, have become an essential part of the background of scientists, engineers and mathematicians from both an applied and theoretical point of view.

The purpose of this book is to present the fundamental concepts and applications of Fourier series, Fourier integrals and orthogonal functions (Bessel, Legendre, Hermite, and Laguerre functions, as well as others).

The book is designed to be used either as a textbook for a formal course in Fourier Analysis or as a comprehensive supplement to all current standard texts. It should be of considerable value to those taking courses in engineering, science or mathematics in which these important methods are frequently used. It should also prove useful as a book of reference to research workers employing Fourier methods or to those interested in the field for self-study.

Each chapter begins with a clear statement of pertinent definitions, principles and theorems, together with illustrative and other descriptive material. The solved problems serve to illustrate and amplify the theory and to provide the repetition of basic principles so vital to effective learning. Numerous proofs of theorems and derivations of formulas are included among the solved problems. The large number of supplementary problems with answers serve as a complete review of the material of each chapter.

Considerably more material has been included here than can be covered in most first courses. This has been done to make the book more flexible, to provide a more useful book of reference, and to stimulate further interest in the topics.

I wish to take this opportunity to thank Henry Hayden and David Beckwith for their splendid cooperation.

M. R. SPIEGEL

January 1974

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Chapter 1

Boundary Value Problems

MATHEMATICAL FORMULATION AND SOLUTION OF PHYSICAL PROBLEMS

In solving problems of science and engineering the following steps are generally taken.

1. **Mathematical formulation.** To achieve such formulation we usually adopt *mathematical models* which serve to approximate the real objects under investigation.

Example 1.

To investigate the motion of the earth or other planet about the sun we can choose *points* as mathematical models of the sun and earth. On the other hand, if we wish to investigate the motion of the earth about its axis, the mathematical model cannot be a point but might be a sphere or even more accurately an ellipsoid.

In the mathematical formulation we use known *physical laws* to set up *equations* describing the problem. If the laws are unknown we may even be led to set up *experiments* in order to discover them.

Example 2.

In describing the motion of a planet about the sun we use *Newton's laws* to arrive at a *differential equation* involving the distance of the planet from the sun at any time.

2. **Mathematical solution.** Once a problem has been successfully formulated in terms of equations, we need to solve them for the unknowns involved, subject to the various conditions which are given or implied in the physical problem. One important consideration is whether such solutions actually *exist* and, if they do exist, whether they are *unique*.

In the attempt to find solutions, the need for new kinds of mathematical analysis — leading to new mathematical problems — may arise.

Example 3.

J.B.J. Fourier, in attempting to solve a problem in heat flow which he had formulated in terms of partial differential equations, was led to the mathematical problem of expansion of functions into series involving sines and cosines. Such series, now called *Fourier series*, are of interest from the point of view of mathematical theory and in physical applications, as we shall see in Chapter 2.

3. **Physical interpretation.** After a solution has been obtained, it is useful to interpret it physically. Such interpretations may be of value in suggesting other kinds of problems, which could lead to new knowledge of a mathematical or physical nature.

In this book we shall be mainly concerned with the mathematical formulation of physical problems in terms of *partial differential equations* and with the solution of such equations by methods commonly called *Fourier methods*.

DEFINITIONS PERTAINING TO PARTIAL DIFFERENTIAL EQUATIONS

A *partial differential equation* is an equation containing an unknown function of two or more variables and its partial derivatives with respect to these variables.

The *order* of a partial differential equation is the order of the highest derivative present.

Example 4.

$\frac{\partial^2 u}{\partial x \partial y} = 2x - y$ is a partial differential equation of order two, or a second-order partial differential equation. Here u is the *dependent variable* while x and y are *independent variables*.

A *solution* of a partial differential equation is any function which satisfies the equation identically.

The *general solution* is a solution which contains a number of arbitrary independent functions equal to the order of the equation.

A *particular solution* is one which can be obtained from the general solution by particular choice of the arbitrary functions.

Example 5.

As seen by substitution, $u = x^2y - \frac{1}{2}xy^2 + F(x) + G(y)$ is a *solution* of the partial differential equation of Example 4. Because it contains two arbitrary independent functions $F(x)$ and $G(y)$, it is the *general solution*. If in particular $F(x) = 2 \sin x$, $G(y) = 3y^4 - 5$, we obtain the *particular solution*

$$u = x^2y - \frac{1}{2}xy^2 + 2 \sin x + 3y^4 - 5$$

A *singular solution* is one which cannot be obtained from the general solution by particular choice of the arbitrary functions.

Example 6.

If $u = x \frac{\partial u}{\partial x} - \left(\frac{\partial u}{\partial x} \right)^2$, where u is a function of x and y , we see by substitution that both $u = xF(y) - [F(y)]^2$ and $u = x^2/4$ are solutions. The first is the general solution involving one arbitrary function $F(y)$. The second, which cannot be obtained from the general solution by any choice of $F(y)$, is a singular solution.

A *boundary value problem* involving a partial differential equation seeks all solutions of the equation which satisfy conditions called *boundary conditions*. Theorems relating to the existence and uniqueness of such solutions are called *existence* and *uniqueness theorems*.

LINEAR PARTIAL DIFFERENTIAL EQUATIONS

The general *linear partial differential equation* of order two in two independent variables has the form

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G \quad (1)$$

where $A, B, \dots, G$ may depend on x and y but not on u . A second-order equation with independent variables x and y which does not have the form (1) is called *nonlinear*.

If $G = 0$ identically the equation is called *homogeneous*, while if $G \neq 0$ it is called *non-homogeneous*. Generalizations to higher-order equations are easily made.

Because of the nature of the solutions of (1), the equation is often classified as *elliptic*, *hyperbolic*, or *parabolic* according as $B^2 - 4AC$ is less than, greater than, or equal to zero, respectively.

SOME IMPORTANT PARTIAL DIFFERENTIAL EQUATIONS

1. Vibrating string equation

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$$

This equation is applicable to the small transverse vibrations of a taut, flexible string, such as a violin string, initially located on the x -axis and set into motion (see Fig. 1-1). The function $y(x, t)$ is the displacement of any point x of the string at time t . The constant $a^2 = \tau/\mu$, where τ is the (constant) tension in the string and μ is the (constant) mass per unit length of the string. It is assumed that no external forces act on the string and that it vibrates only due to its elasticity.

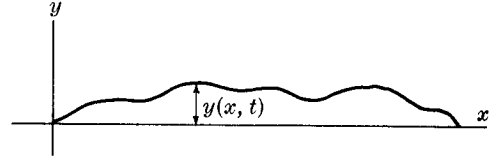


Fig. 1-1

The equation can easily be generalized to higher dimensions, as for example the vibrations of a membrane or drumhead in two dimensions. In two dimensions, the equation is

$$\frac{\partial^2 z}{\partial t^2} = a^2 \left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \right)$$

2. Heat conduction equation

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u$$

Here $u(x, y, z, t)$ is the temperature at position (x, y, z) in a solid at time t . The constant κ , called the *diffusivity*, is equal to $K/\sigma\mu$, where the *thermal conductivity* K , the *specific heat* σ and the density (mass per unit volume) μ are assumed constant. We call $\nabla^2 u$ the *Laplacian* of u ; it is given in three-dimensional rectangular coordinates (x, y, z) by

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

3. Laplace's equation

$$\nabla^2 v = 0$$

This equation occurs in many fields. In the theory of heat conduction, for example, v is the *steady-state temperature*, i.e. the temperature after a long time has elapsed, whose equation is obtained by putting $\partial u/\partial t = 0$ in the heat conduction equation above. In the theory of gravitation or electricity v represents the *gravitational* or *electric potential* respectively. For this reason the equation is often called the *potential equation*.

The problem of solving $\nabla^2 v = 0$ inside a region $\mathcal{R}$ when v is some given function on the boundary of $\mathcal{R}$ is often called a *Dirichlet problem*.

4. Longitudinal vibrations of a beam

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

This equation describes the motion of a beam (Fig. 1-2, page 4) which can vibrate longitudinally (i.e. in the x -direction) the vibrations being assumed small. The variable $u(x, t)$ is the longitudinal displacement from the equilibrium position of the cross section at x . The constant $c^2 = E/\mu$, where E is the modulus of elasticity (stress divided by strain) and depends on the properties of the beam, μ is the density (mass per unit volume).

Note that this equation is the same as that for a vibrating string.

5. Transverse vibrations of a beam

$$\frac{\partial^2 y}{\partial t^2} + b^2 \frac{\partial^4 y}{\partial x^4} = 0$$

This equation describes the motion of a beam (initially located on the x -axis, see Fig. 1-3) which is vibrating transversely (i.e. perpendicular to the x -direction) assuming small vibrations. In this case $y(x, t)$ is the transverse displacement or deflection at any time t of any point x . The constant $b^2 = EI/A\mu$, where E is the modulus of elasticity, I is the moment of inertia of any cross section about the x -axis, A is the area of cross section and μ is the mass per unit length. In case an external transverse force $F(x, t)$ is applied, the right-hand side of the equation is replaced by $b^2 F(x, t)/EI$.

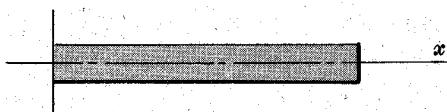


Fig. 1-2

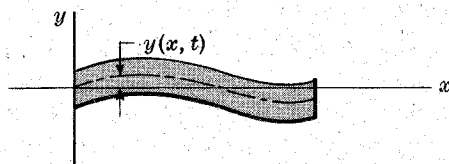


Fig. 1-3

THE LAPLACIAN IN DIFFERENT COORDINATE SYSTEMS

The Laplacian $\nabla^2 u$ often arises in partial differential equations of science and engineering. Depending on the type of problem involved, the choice of coordinate system may be important in obtaining solutions. For example, if the problem involves a cylinder, it will often be convenient to use *cylindrical coordinates*; while if it involves a sphere, it will be convenient to use *spherical coordinates*.

The Laplacian in cylindrical coordinates (ρ, ϕ, z) (see Fig. 1-4) is given by

$$\nabla^2 u = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} \quad (2)$$

The transformation equations between rectangular and cylindrical coordinates are

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z \quad (3)$$

where $\rho \geq 0$, $0 \leq \phi < 2\pi$, $-\infty < z < \infty$.

The Laplacian in spherical coordinates (r, θ, ϕ) (see Fig. 1-5) is given by

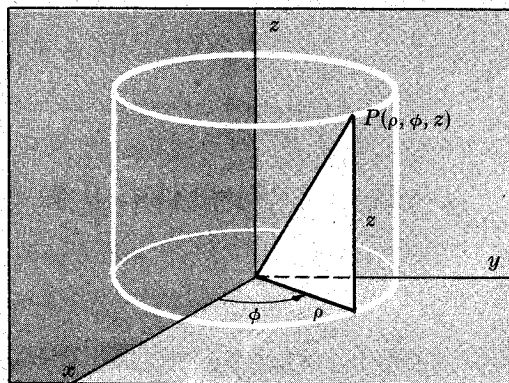


Fig. 1-4

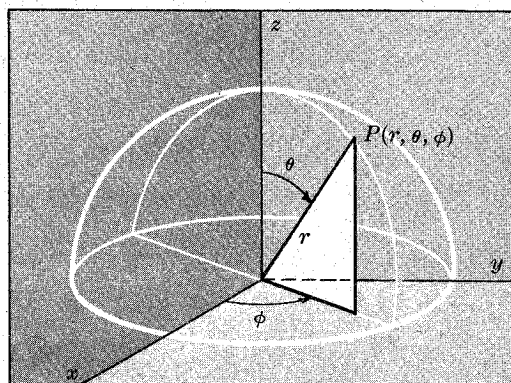


Fig. 1-5

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} \quad (4)$$

The transformation equations between rectangular and spherical coordinates are

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta \quad (5)$$

where $r \geq 0$, $0 \leq \theta \leq \pi$, $0 \leq \phi < 2\pi$.

METHODS OF SOLVING BOUNDARY VALUE PROBLEMS

There are many methods by which boundary value problems involving linear partial differential equations can be solved. In this book we shall be concerned with two methods which represent somewhat opposing points of view.

In the first method we seek to find the general solution of the partial differential equation and then particularize it to obtain the actual solution by using the boundary conditions. In the second method we first find particular solutions of the partial differential equation and then build up the actual solution by use of these particular solutions. Of the two methods the second will be found to be of far greater applicability than the first.

1. **General solutions.** In this method we first find the general solution and then that particular solution which satisfies the boundary conditions. The following theorems are of fundamental importance.

Theorem 1-1 (Superposition principle): If $u_1, u_2, \dots, u_n$ are solutions of a linear homogeneous partial differential equation, then $c_1 u_1 + c_2 u_2 + \dots + c_n u_n$, where $c_1, c_2, \dots, c_n$ are constants, is also a solution.

Theorem 1-2: The general solution of a linear nonhomogeneous partial differential equation is obtained by adding a particular solution of the nonhomogeneous equation to the general solution of the homogeneous equation.

We can sometimes find general solutions by using the methods of ordinary differential equations. See Problems 1.15 and 1.16.

If $A, B, \dots, F$ in (1) are constants, then the general solution of the homogeneous equation can be found by assuming that $u = e^{ax+by}$, where a and b are constants to be determined. See Problems 1.17–1.20.

2. **Particular solutions by separation of variables.** In this method, which is simple but powerful, it is assumed that a solution can be expressed as a product of unknown functions each of which depends on only one of the independent variables. The success of the method hinges on being able to write the resulting equation so that one side depends on only one variable while the other side depends on the remaining variables—from which it is concluded that each side must be a constant. By repetition of this, the unknown functions can be determined. Superposition of these solutions can then be used to find the actual solution. See Problems 1.21–1.25.

Solved Problems

MATHEMATICAL FORMULATION OF PHYSICAL PROBLEMS

1.1. Derive the vibrating string equation on page 3.

Referring to Fig. 1-6, assume that Δs represents an element of arc of the string. Since the tension is assumed constant, the net upward vertical force acting on Δs is given by

$$\tau \sin \theta_2 - \tau \sin \theta_1 \quad (1)$$

Since $\sin \theta = \tan \theta$, approximately, for small angles, this force is

$$\tau \frac{\partial y}{\partial x} \Big|_{x+\Delta x} - \tau \frac{\partial y}{\partial x} \Big|_x \quad (2)$$

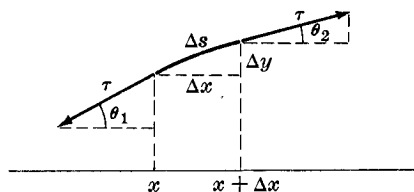


Fig. 1-6

using the fact that the slope is $\tan \theta = \frac{\partial y}{\partial x}$. We use here the notation $\frac{\partial y}{\partial x} \Big|_x$ and $\frac{\partial y}{\partial x} \Big|_{x+\Delta x}$ for the partial derivatives of y with respect to x evaluated at x and $x + \Delta x$, respectively. By Newton's law this net force is equal to the mass of the string ($\mu \Delta s$) times the acceleration of Δs , which is given by $\frac{\partial^2 y}{\partial t^2} + \epsilon$ where $\epsilon \rightarrow 0$ as $\Delta s \rightarrow 0$. Thus we have approximately

$$\tau \left[\frac{\partial y}{\partial x} \Big|_{x+\Delta x} - \frac{\partial y}{\partial x} \Big|_x \right] = (\mu \Delta s) \left(\frac{\partial^2 y}{\partial t^2} + \epsilon \right) \quad (3)$$

If the vibrations are small, then $\Delta s = \Delta x$ approximately, so that (3) becomes on division by $\mu \Delta x$:

$$\frac{\tau}{\mu} \frac{\frac{\partial y}{\partial x} \Big|_{x+\Delta x} - \frac{\partial y}{\partial x} \Big|_x}{\Delta x} = \frac{\partial^2 y}{\partial t^2} + \epsilon \quad (4)$$

Taking the limit as $\Delta x \rightarrow 0$ (in which case $\epsilon \rightarrow 0$ also), we have

$$\frac{\tau}{\mu} \frac{\partial}{\partial x} \left(\frac{\partial y}{\partial x} \right) = \frac{\partial^2 y}{\partial t^2} \quad \text{or} \quad \frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}, \quad \text{where } a^2 = \tau/\mu$$

1.2. Write the boundary conditions for a vibrating string of length L for which (a) the ends $x = 0$ and $x = L$ are fixed, (b) the initial shape is given by $f(x)$, (c) the initial velocity distribution is given by $g(x)$, (d) the displacement at any point x at time t is bounded.

(a) If the string is fixed at $x = 0$ and $x = L$, then the displacement $y(x, t)$ at $x = 0$ and $x = L$ must be zero for all times $t > 0$, i.e.

$$y(0, t) = 0, \quad y(L, t) = 0 \quad t > 0$$

(b) Since the string has an initial shape given by $f(x)$, we must have

$$y(x, 0) = f(x) \quad 0 < x < L$$

(c) Since the initial velocity of the string at any point x is $g(x)$, we must have

$$y_t(x, 0) = g(x) \quad 0 < x < L$$

Note that $y_t(x, 0)$ is the same as $\partial y / \partial t$ evaluated at $t = 0$.

(d) Since $y(x, t)$ is bounded, we can find a constant M independent of x and t such that

$$|y(x, t)| < M \quad 0 < x < L, \quad t > 0$$

1.3. Write boundary conditions for a vibrating string for which (a) the end $x = 0$ is moving so that its displacement is given in terms of time by $G(t)$, (b) the end $x = L$ is not fixed but is free to move.

(a) The displacement at $x = 0$ is given by $y(0, t)$. Thus we have

$$y(0, t) = G(t) \quad t > 0$$

- (b) If τ is the tension, the transverse force acting at any point x is

$$\tau \frac{\partial y}{\partial x} = \tau y_x(x, t)$$

Since the end $x = L$ is free to move so that there is no force acting on it, the boundary condition is given by

$$\tau y_x(L, t) = 0 \quad \text{or} \quad y_x(L, t) = 0 \quad t > 0$$

- 1.4. Suppose that in Problem 1.1 the tension in the string is variable, i.e. depends on the particular point taken. Denoting this tension by $\tau(x)$, show that the equation for the vibrating string is

$$\frac{\partial}{\partial x} \left[\tau(x) \frac{\partial y}{\partial x} \right] = \mu \frac{\partial^2 y}{\partial t^2}$$

In this case we write (2) of Problem 1.1 as

$$\tau(x) \frac{\partial y}{\partial x} \Big|_{x+\Delta x} - \tau(x) \frac{\partial y}{\partial x} \Big|_x$$

so that the corresponding equation (4) is

$$\frac{\tau(x) \frac{\partial y}{\partial x} \Big|_{x+\Delta x} - \tau(x) \frac{\partial y}{\partial x} \Big|_x}{\mu \Delta x} = \frac{\partial^2 y}{\partial t^2} + \epsilon$$

Thus, taking the limit as $\Delta x \rightarrow 0$ (in which case $\epsilon \rightarrow 0$), we obtain

$$\frac{\partial}{\partial x} \left[\tau(x) \frac{\partial y}{\partial x} \right] = \mu \frac{\partial^2 y}{\partial t^2}$$

after multiplying by μ .

- 1.5. Show that the heat flux across a plane in a conducting medium is given by $-K \frac{\partial u}{\partial n}$, where u is the temperature, n is a normal in a direction perpendicular to the plane and K is the thermal conductivity of the medium.

Suppose we have two parallel planes I and II a distance Δn apart (Fig. 1-7), having temperatures u and $u + \Delta u$, respectively. Then the heat flows from the plane of higher temperature to the plane of lower temperature. Also, the amount of heat per unit area per unit time, called the *heat flux*, is directly proportional to the difference in temperature Δu and inversely proportional to the distance Δn . Thus we have

$$\text{Heat flux from I to II} = -K \frac{\Delta u}{\Delta n} \quad (1)$$

where K is the constant of proportionality, called the *thermal conductivity*. The minus sign occurs in (1) since if $\Delta u > 0$ the heat flow actually takes place from II to I.

By taking the limit of (1) as Δn and thus Δu approaches zero, we have as required:

$$\text{Heat flux across plane I} = -K \frac{\partial u}{\partial n} \quad (2)$$

We sometimes call $\frac{\partial u}{\partial n}$ the *gradient* of u which in vector form is ∇u , so that (2) can be written

$$\text{Heat flux across plane I} = -K \nabla u \quad (3)$$

- 1.6. If the temperature at any point (x, y, z) of a solid at time t is $u(x, y, z, t)$ and if K, σ and μ are respectively the thermal conductivity, specific heat and density of the solid, all assumed constant, show that

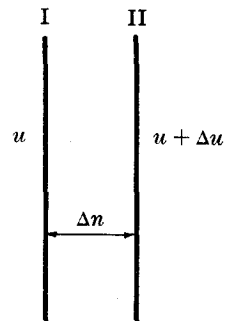


Fig. 1-7

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u \quad \text{where } \kappa = K/\sigma\mu$$

Consider a small volume element of the solid V , as indicated in Fig. 1-8 and greatly enlarged in Fig. 1-9. By Problem 1.5 the amount of heat per unit area per unit time entering the element through face $PQRS$ is $-K \frac{\partial u}{\partial x} \Big|_x$, where $\frac{\partial u}{\partial x} \Big|_x$ indicates the derivative of u with respect to x evaluated at the position x . Since the area of face $PQRS$ is $\Delta y \Delta z$, the total amount of heat entering the element through face $PQRS$ in time Δt is

$$-K \frac{\partial u}{\partial x} \Big|_x \Delta y \Delta z \Delta t \quad (1)$$

Similarly, the amount of heat leaving the element through face $NWZT$ is

$$-K \frac{\partial u}{\partial x} \Big|_{x+\Delta x} \Delta y \Delta z \Delta t \quad (2)$$

where $\frac{\partial u}{\partial x} \Big|_{x+\Delta x}$ indicates the derivative of u with respect to x evaluated at $x + \Delta x$.

The amount of heat which remains in the element is given by the amount entering minus the amount leaving, which is, from (1) and (2),

$$\left\{ K \frac{\partial u}{\partial x} \Big|_{x+\Delta x} - K \frac{\partial u}{\partial x} \Big|_x \right\} \Delta y \Delta z \Delta t \quad (3)$$

In a similar way we can show that the amounts of heat remaining in the element due to heat transfer taking place in the y - and z -directions are given by

$$\left\{ K \frac{\partial u}{\partial y} \Big|_{y+\Delta y} - K \frac{\partial u}{\partial y} \Big|_y \right\} \Delta x \Delta z \Delta t \quad (4)$$

and

$$\left\{ K \frac{\partial u}{\partial z} \Big|_{z+\Delta z} - K \frac{\partial u}{\partial z} \Big|_z \right\} \Delta x \Delta y \Delta t \quad (5)$$

respectively.

The total amount of heat gained by the element is given by the sum of (3), (4) and (5). This amount of heat serves to raise its temperature by the amount Δu . Now, we know that the heat needed to raise the temperature of a mass m by Δu is given by $m\sigma \Delta u$, where σ is the specific heat. If the density of the solid is μ , the mass is $m = \mu \Delta x \Delta y \Delta z$. Thus the quantity of heat given by the sum of (3), (4) and (5) is equal to

$$\sigma \mu \Delta x \Delta y \Delta z \Delta u \quad (6)$$

If we now equate the sum of (3), (4) and (5) to (6), and divide by $\Delta x \Delta y \Delta z \Delta t$, we find

$$\left\{ \frac{K \frac{\partial u}{\partial x} \Big|_{x+\Delta x} - K \frac{\partial u}{\partial x} \Big|_x}{\Delta x} \right\} + \left\{ \frac{K \frac{\partial u}{\partial y} \Big|_{y+\Delta y} - K \frac{\partial u}{\partial y} \Big|_y}{\Delta y} \right\} + \left\{ \frac{K \frac{\partial u}{\partial z} \Big|_{z+\Delta z} - K \frac{\partial u}{\partial z} \Big|_z}{\Delta z} \right\} = \sigma \mu \frac{\Delta u}{\Delta t}$$

In the limit as $\Delta x, \Delta y, \Delta z$ and Δt all approach zero the above equation becomes

$$\frac{\partial}{\partial x} \left(K \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(K \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(K \frac{\partial u}{\partial z} \right) = \sigma \mu \frac{\partial u}{\partial t} \quad (7)$$

or, as K is a constant,

$$K \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = \sigma \mu \frac{\partial u}{\partial t} \quad (8)$$

This can be rewritten as

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u \quad (9)$$

where $\kappa = \frac{K}{\sigma\mu}$ is called the *diffusivity*.

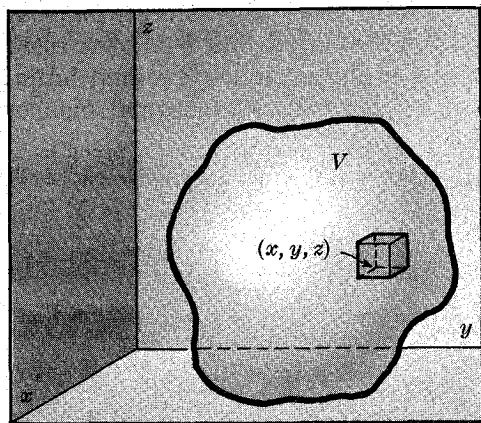


Fig. 1-8

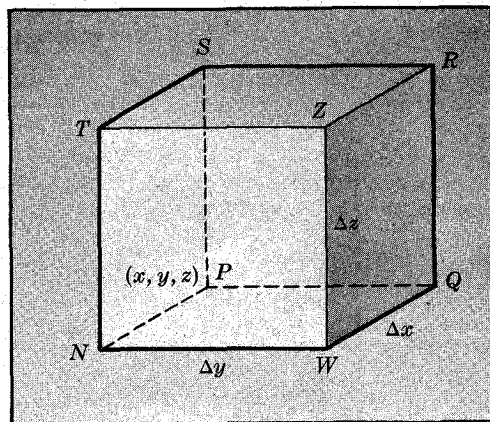


Fig. 1-9

1.7. Work Problem 1.6 by using vector methods.

Let V be an arbitrary volume lying within the solid, and let S denote its surface (see Fig. 1-8). The total flux of heat across S , or the quantity of heat leaving S per unit time, is

$$\iint_S (-K \nabla u) \cdot \mathbf{n} \, dS$$

where $\mathbf{n}$ is an outward-drawn unit normal to S . Thus the quantity of heat entering S per unit time is

$$\iint_S (K \nabla u) \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot (K \nabla u) \, dV \quad (1)$$

by the divergence theorem. The heat contained in a volume V is given by

$$\iiint_V \sigma_\mu u \, dV$$

Then the time rate of increase of heat is

$$\frac{\partial}{\partial t} \iiint_V \sigma_\mu u \, dV = \iiint_V \sigma_\mu \frac{\partial u}{\partial t} \, dV \quad (2)$$

Equating the right-hand sides of (1) and (2),

$$\iiint_V \left[\sigma_\mu \frac{\partial u}{\partial t} - \nabla \cdot (K \nabla u) \right] dV = 0$$

and since V is arbitrary, the integrand, assumed continuous, must be identically zero, so that

$$\sigma_\mu \frac{\partial u}{\partial t} = \nabla \cdot (K \nabla u)$$

or if K, σ, μ are constants,

$$\frac{\partial u}{\partial t} = \frac{K}{\sigma_\mu} \nabla \cdot \nabla u = \kappa \nabla^2 u \quad (3)$$

1.8. Show that for steady-state heat flow the heat conduction equation of Problem 1.6 or 1.7 reduces to Laplace's equation, $\nabla^2 u = 0$.

In the case of steady-state heat flow the temperature u does not depend on time t , so that $\frac{\partial u}{\partial t} = 0$. Thus the equation $\frac{\partial u}{\partial t} = \kappa \nabla^2 u$ becomes $\nabla^2 u = 0$.

1.9. A thin bar of diffusivity κ has its ends at $x=0$ and $x=L$ on the x -axis (see Fig. 1-10). Its lateral surface is insulated so that heat cannot enter or escape.

(a) If the initial temperature is $f(x)$ and the ends are kept at temperature zero, set up the boundary value problem. (b) Work part (a) if the end $x = L$ is insulated. (c) Work part (a) if the end $x = L$ radiates into the surrounding medium, which is assumed to be at temperature u_0 .

This is a problem in *one-dimensional* heat conduction since the temperature can only depend on the position x at any time t and can thus be denoted by $u(x, t)$. The heat conduction equation is thus given by

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \quad 0 < x < L, \quad t > 0 \quad (1)$$

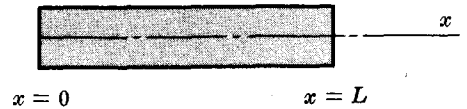


Fig. 1-10

(a) Since the ends are kept at temperature zero, we have

$$u(0, t) = 0, \quad u(L, t) = 0 \quad t > 0 \quad (2)$$

Since the initial temperature is $f(x)$, we have

$$u(x, 0) = f(x) \quad 0 < x < L \quad (3)$$

Also, from physical considerations the temperature must be bounded; hence

$$|u(x, t)| < M \quad 0 < x < L, \quad t > 0 \quad (4)$$

The problem of solving (1) subject to conditions (2), (3) and (4) is the required boundary value problem. A problem exactly equivalent to that considered above is that of an *infinite slab* of conducting material bounded by the planes $x = 0$ and $x = L$, where the planes are kept at temperature zero and where the temperature distribution initially is $f(x)$.

(b) If the end $x = L$ is insulated instead of being at temperature zero, then we must find a replacement for the condition $u(L, t) = 0$ in (2). To do this we note that if the end $x = L$ is insulated then the flux at $x = L$ is zero. Thus we have

$$-K \frac{\partial u}{\partial x} \Big|_{x=L} = 0 \quad \text{or equivalently} \quad u_x(L, t) = 0 \quad (5)$$

which is the required boundary condition.

(c) It is known from physical laws of heat transfer that the heat flux of radiation from one object at temperature U_1 to another object at temperature U_2 is given by $\alpha(U_1^4 - U_2^4)$, where α is a constant and the temperatures U_1 and U_2 are given in *absolute* or *Kelvin* temperature which is the number of Celsius (centigrade) degrees plus 273. This law is often called *Stefan's radiation law*. From this we obtain the boundary condition

$$-Ku_x(L, t) = \alpha(u_1^4 - u_0^4) \quad \text{where} \quad u_1 = u(L, t) \quad (6)$$

If u_1 and u_0 do not differ too greatly from each other, we can write

$$\begin{aligned} u_1^4 - u_0^4 &= (u_1 - u_0)(u_1^3 + u_1^2 u_0 + u_1 u_0^2 + u_0^3) \\ &= (u_1 - u_0)u_0^3 \left[\left(\frac{u_1}{u_0} \right)^3 + \left(\frac{u_1}{u_0} \right)^2 + \frac{u_1}{u_0} + 1 \right] \\ &\approx 4u_0^3(u_1 - u_0) \end{aligned}$$

since $(u_1/u_0)^3$, $(u_1/u_0)^2$, (u_1/u_0) are approximately equal to 1. Using this approximation, which is often referred to as *Newton's law of cooling*, we can write (6) as

$$-Ku_x(L, t) = \beta(u_1 - u_0) \quad (7)$$

where β is a constant.

CLASSIFICATION OF PARTIAL DIFFERENTIAL EQUATIONS

1.10. Determine whether each of the following partial differential equations is linear or nonlinear, state the order of each equation, and name the dependent and independent variables.

- (a) $\frac{\partial u}{\partial t} = 4 \frac{\partial^2 u}{\partial x^2}$ linear, order 2, dep. var. u , ind. var. x, t
- (b) $x^2 \frac{\partial^3 R}{\partial y^3} = y^3 \frac{\partial^2 R}{\partial x^2}$ linear, order 3, dep. var. R , ind. var. x, y
- (c) $W \frac{\partial^2 W}{\partial r^2} = rst$, nonlinear, order 2, dep. var. W , ind. var. r, s, t
- (d) $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$ linear, order 2, dep. var. ϕ , ind. var. x, y, z
- (e) $\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 = 1$ nonlinear, order 1, dep. var. z , ind. var. u, v

1.11. Classify each of the following equations as elliptic, hyperbolic or parabolic.

- (a) $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$
 $u = \phi$, $A = 1$, $B = 0$, $C = 1$; $B^2 - 4AC = -4 < 0$ and the equation is elliptic.
- (b) $\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}$
 $y = t$, $A = \kappa$, $B = 0$, $C = 0$; $B^2 - 4AC = 0$ and the equation is parabolic.
- (c) $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$
 $y = t$, $u = y$, $A = a^2$, $B = 0$, $C = -1$; $B^2 - 4AC = 4a^2 > 0$ and the equation is hyperbolic.
- (d) $\frac{\partial^2 u}{\partial x^2} + 3 \frac{\partial^2 u}{\partial x \partial y} + 4 \frac{\partial^2 u}{\partial y^2} + 5 \frac{\partial u}{\partial x} - 2 \frac{\partial u}{\partial y} + 4u = 2x - 3y$
 $A = 1$, $B = 3$, $C = 4$; $B^2 - 4AC = -7 < 0$ and the equation is elliptic.
- (e) $x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial y^2} + 3y^2 \frac{\partial u}{\partial x} = 0$
 $A = x$, $B = 0$, $C = y$; $B^2 - 4AC = -4xy$. Hence, in the region $xy > 0$ the equation is elliptic; in the region $xy < 0$ the equation is hyperbolic; if $xy = 0$, the equation is parabolic.

SOLUTIONS OF PARTIAL DIFFERENTIAL EQUATIONS

1.12. Show that $u(x, t) = e^{-8t} \sin 2x$ is a solution to the boundary value problem

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = \sin 2x$$

From $u(x, t) = e^{-8t} \sin 2x$ we have

$$u(0, t) = e^{-8t} \sin 0 = 0, \quad u(\pi, t) = e^{-8t} \sin 2\pi = 0, \quad u(x, 0) = e^{-0} \sin 2x = \sin 2x$$

and the boundary conditions are satisfied.

$$\text{Also} \quad \frac{\partial u}{\partial t} = -8e^{-8t} \sin 2x, \quad \frac{\partial u}{\partial x} = 2e^{-8t} \cos 2x, \quad \frac{\partial^2 u}{\partial x^2} = -4e^{-8t} \sin 2x$$

Then substituting into the differential equation, we have

$$-8e^{-8t} \sin 2x = 2(-4e^{-8t} \sin 2x)$$

which is an identity.

- 1.13. (a) Show that $v = F(y - 3x)$, where F is an arbitrary differentiable function, is a general solution of the equation

$$\frac{\partial v}{\partial x} + 3 \frac{\partial v}{\partial y} = 0$$

- (b) Find the particular solution which satisfies the condition $v(0, y) = 4 \sin y$.

- (a) Let $y - 3x = u$. Then $v = F(u)$ and

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial u} \frac{\partial u}{\partial x} = F'(u)(-3) = -3F'(u)$$

$$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial u} \frac{\partial u}{\partial y} = F'(u)(1) = F'(u)$$

Thus
$$\frac{\partial v}{\partial x} + 3 \frac{\partial v}{\partial y} = 0$$

Since the equation is of order one, the solution $v = F(u) = F(y - 3x)$, which involves one arbitrary function, is a general solution.

- (b) $v(x, y) = F(y - 3x)$. Then $v(0, y) = F(y) = 4 \sin y$. But if $F(y) = 4 \sin y$, then $v(x, y) = F(y - 3x) = 4 \sin(y - 3x)$ is the required solution.

- 1.14. (a) Show that $y(x, t) = F(2x + 5t) + G(2x - 5t)$ is a general solution of

$$4 \frac{\partial^2 y}{\partial t^2} = 25 \frac{\partial^2 y}{\partial x^2}$$

- (b) Find a particular solution satisfying the conditions

$$y(0, t) = y(\pi, t) = 0, \quad y(x, 0) = \sin 2x, \quad y_t(x, 0) = 0$$

- (a) Let $2x + 5t = u$, $2x - 5t = v$. Then $y = F(u) + G(v)$.

$$\frac{\partial y}{\partial t} = \frac{\partial F}{\partial u} \frac{\partial u}{\partial t} + \frac{\partial G}{\partial v} \frac{\partial v}{\partial t} = F'(u)(5) + G'(v)(-5) = 5F'(u) - 5G'(v) \quad (1)$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{\partial}{\partial t} (5F'(u) - 5G'(v)) = 5 \frac{\partial F'}{\partial u} \frac{\partial u}{\partial t} - 5 \frac{\partial G'}{\partial v} \frac{\partial v}{\partial t} = 25F''(u) + 25G''(v) \quad (2)$$

$$\frac{\partial y}{\partial x} = \frac{\partial F}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial G}{\partial v} \frac{\partial v}{\partial x} = F'(u)(2) + G'(v)(2) = 2F'(u) + 2G'(v) \quad (3)$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\partial}{\partial x} [2F'(u) + 2G'(v)] = 2 \frac{\partial F'}{\partial u} \frac{\partial u}{\partial x} + 2 \frac{\partial G'}{\partial v} \frac{\partial v}{\partial x} = 4F''(u) + 4G''(v) \quad (4)$$

From (2) and (4), $4 \frac{\partial^2 y}{\partial t^2} = 25 \frac{\partial^2 y}{\partial x^2}$ and the equation is satisfied. Since the equation is of order 2 and the solution involves two arbitrary functions, it is a general solution.

- (b) We have from $y(x, t) = F(2x + 5t) + G(2x - 5t)$,

$$y(x, 0) = F(2x) + G(2x) = \sin 2x \quad (5)$$

Also
$$y_t(x, t) = \frac{\partial y}{\partial t} = 5F'(2x + 5t) - 5G'(2x - 5t)$$

so that
$$y_t(x, 0) = 5F'(2x) - 5G'(2x) = 0 \quad (6)$$

Differentiating (5),
$$2F'(2x) + 2G'(2x) = 2 \cos 2x \quad (7)$$

From (6),
$$F'(2x) = G'(2x) \quad (8)$$

Then from (7), and (8),
$$F'(2x) = G'(2x) = \frac{1}{2} \cos 2x$$

from which $F(2x) = \frac{1}{2} \sin 2x + c_1$, $G(2x) = \frac{1}{2} \sin 2x + c_2$

i.e. $y(x, t) = \frac{1}{2} \sin (2x + 5t) + \frac{1}{2} \sin (2x - 5t) + c_1 + c_2$

Using $y(0, t) = 0$ or $y(\pi, t) = 0$, $c_1 + c_2 = 0$ so that

$$y(x, t) = \frac{1}{2} \sin (2x + 5t) + \frac{1}{2} \sin (2x - 5t) = \sin 2x \cos 5t$$

which can be checked as the required solution.

METHODS OF FINDING SOLUTIONS OF PARTIAL DIFFERENTIAL EQUATIONS

1.15. (a) Solve the equation $\frac{\partial^2 z}{\partial x \partial y} = x^2 y$.

(b) Find the particular solution for which $z(x, 0) = x^2$, $z(1, y) = \cos y$.

(a) Write the equation as $\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = x^2 y$. Then integrating with respect to x , we find

$$\frac{\partial z}{\partial y} = \frac{1}{3} x^3 y + F(y) \quad (1)$$

where $F(y)$ is arbitrary.

Integrating (1) with respect to y ,

$$z = \frac{1}{6} x^3 y^2 + \int F(y) dy + G(x) \quad (2)$$

where $G(x)$ is arbitrary. The result (2) can be written

$$z = z(x, y) = \frac{1}{6} x^3 y^2 + H(y) + G(x) \quad (3)$$

which has two arbitrary (independent) functions and is therefore a general solution.

(b) Since $z(x, 0) = x^2$, we have from (3)

$$x^2 = H(0) + G(x) \quad \text{or} \quad G(x) = x^2 - H(0) \quad (4)$$

Thus

$$z = \frac{1}{6} x^3 y^2 + H(y) + x^2 - H(0) \quad (5)$$

Since $z(1, y) = \cos y$, we have from (5)

$$\cos y = \frac{1}{6} y^2 + H(y) + 1 - H(0) \quad \text{or} \quad H(y) = \cos y - \frac{1}{6} y^2 - 1 + H(0) \quad (6)$$

Using (6) in (5), we find the required solution

$$z = \frac{1}{6} x^3 y^2 + \cos y - \frac{1}{6} y^2 + x^2 - 1$$

1.16. Solve $t \frac{\partial^2 u}{\partial x \partial t} + 2 \frac{\partial u}{\partial x} = x^2$.

Write the equation as $\frac{\partial}{\partial x} \left[t \frac{\partial u}{\partial t} + 2u \right] = x^2$. Integrating with respect to x ,

$$t \frac{\partial u}{\partial t} + 2u = x^2 t + F(t) \quad \text{or} \quad \frac{\partial u}{\partial t} + \frac{2}{t} u = x^2 + \frac{F(t)}{t}$$

This is a linear equation having integrating factor $e^{\int (2/t) dt} = e^{2 \ln t} = e^{\ln t^2} = t^2$. Then

$$\frac{\partial}{\partial t} (t^2 u) = x^2 t^2 + t F(t)$$

Integrating, $t^2 u = \frac{1}{3} x^2 t^3 + \int t F(t) dt + H(x) = \frac{1}{3} x^2 t^3 + G(t) + H(x)$
and this is the required general solution.

1.17. Find solutions of $\frac{\partial^2 u}{\partial x^2} + 3 \frac{\partial^2 u}{\partial x \partial y} + 2 \frac{\partial^2 u}{\partial y^2} = 0$.

Assume $u = e^{ax+by}$. Substituting in the given equation, we find

$$(a^2 + 3ab + 2b^2)e^{ax+by} = 0 \quad \text{or} \quad a^2 + 3ab + 2b^2 = 0$$

Then $(a+b)(a+2b) = 0$ and $a = -b$, $a = -2b$. If $a = -b$, $e^{-bx+by} = e^{b(y-x)}$ is a solution for any value of b . If $a = -2b$, $e^{-2bx+by} = e^{b(y-2x)}$ is a solution for any value of b .

Since the equation is linear and homogeneous, sums of these solutions are solutions (Theorem 1-1). For example, $3e^{2(y-x)} - 2e^{3(y-x)} + 5e^{\pi(y-x)}$ is a solution (among many others), and one is thus led to $F(y-x)$ where F is arbitrary, which can be verified as a solution. Similarly, $G(y-2x)$, where G is arbitrary, is a solution. The general solution found by addition is then given by

$$u = F(y-x) + G(y-2x)$$

1.18. Find a general solution of (a) $2\frac{\partial u}{\partial x} + 3\frac{\partial u}{\partial y} = 2u$, (b) $4\frac{\partial^2 u}{\partial x^2} - 4\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$.

(a) Let $u = e^{ax+by}$. Then $2a + 3b = 2$, $a = \frac{2-3b}{2}$, and $e^{[(2-3b)/2]x+by} = e^{x(b/2)(2y-3x)}$ is a solution.

Thus $u = e^{x(b/2)(2y-3x)}$ is a general solution.

(b) Let $u = e^{ax+by}$. Then $4a^2 - 4ab + b^2 = 0$ and $b = 2a$, $2a$. From this $u = e^{a(x+2y)}$ and so $F(x+2y)$ is a solution.

By analogy with repeated roots for ordinary differential equations we might be led to believe $xG(x+2y)$ or $yG(x+2y)$ to be another solution, and that this is in fact true is easy to verify. Thus a general solution is

$$u = F(x+2y) + xG(x+2y) \quad \text{or} \quad u = F(x+2y) + yG(x+2y)$$

1.19. Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 10e^{2x+y}$.

The homogeneous equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ has general solution $u = F(x+iy) + G(x-iy)$ by Problem 1.39(c).

To find a particular solution of the given equation assume $u = \alpha e^{2x+y}$ where α is an unknown constant. This is the *method of undetermined coefficients* as in ordinary differential equations. We find $\alpha = 2$, so that the required general solution is

$$u = F(x+iy) + G(x-iy) + 2e^{2x+y}$$

1.20. Solve $\frac{\partial^2 u}{\partial x^2} - 4\frac{\partial^2 u}{\partial y^2} = e^{2x+y}$.

The homogeneous equation has general solution

$$u = F(2x+y) + G(2x-y)$$

To find a particular solution, we would normally assume $u = \alpha e^{2x+y}$ as in Problem 1.19 but this assumed solution is already included in $F(2x+y)$. Hence we assume as in ordinary differential equations that $u = \alpha x e^{2x+y}$ (or $u = \alpha y e^{2x+y}$). Substituting, we find $\alpha = \frac{1}{4}$.

Then a general solution is

$$u = F(2x+y) + G(2x-y) + \frac{1}{4}x e^{2x+y}$$

SEPARATION OF VARIABLES

1.21. Solve the boundary value problem

$$\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial y}, \quad u(0, y) = 8e^{-3y}$$

by the method of separation of variables.

Let $u = XY$ in the given equation, where X depends only on x and Y depends only on y .

$$\text{Then} \quad X'Y = 4XY' \quad \text{or} \quad X'/4X = Y'/Y$$

where $X' = dX/dx$ and $Y' = dY/dy$.

Since X depends only on x and Y depends only on y and since x and y are independent variables, each side must be a constant, say c .

Then $X' - 4cX = 0$, $Y' - cY = 0$, whose solutions are $X = Ae^{4cx}$, $Y = Be^{cy}$.

A solution is thus given by

$$u(x, y) = XY = ABe^{c(4x+y)} = Ke^{c(4x+y)}$$

From the boundary condition,

$$u(0, y) = Ke^{cy} = 8e^{-3y}$$

which is possible if and only if $K = 8$ and $c = -3$. Then $u(x, y) = 8e^{-3(4x+y)} = 8e^{-12x-3y}$ is the required solution.

1.22. Solve Problem 1.21 if $u(0, y) = 8e^{-3y} + 4e^{-5y}$.

As before a solution is $Ke^{c(4x+y)}$. Then $K_1e^{c_1(4x+y)}$ and $K_2e^{c_2(4x+y)}$ are solutions and by the principle of superposition so also is their sum; i.e. a solution is

$$u(x, y) = K_1e^{c_1(4x+y)} + K_2e^{c_2(4x+y)}$$

From the boundary condition,

$$u(0, y) = K_1e^{c_1y} + K_2e^{c_2y} = 8e^{-3y} + 4e^{-5y}$$

which is possible if and only if $K_1 = 8$, $K_2 = 4$, $c_1 = -3$, $c_2 = -5$.

Then $u(x, y) = 8e^{-3(4x+y)} + 4e^{-5(4x+y)} = 8e^{-12x-3y} + 4e^{-20x-5y}$ is the required solution.

1.23. Solve $\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2}$, $0 < x < 3$, $t > 0$, given that $u(0, t) = u(3, t) = 0$,

$$u(x, 0) = 5 \sin 4\pi x - 3 \sin 8\pi x + 2 \sin 10\pi x, \quad |u(x, t)| < M$$

where the last condition states that u is bounded for $0 < x < 3$, $t > 0$.

Let $u = XT$. Then $XT' = X''T$ and $X''/X = T'/2T$. Each side must be a constant, which we call $-\lambda^2$. (If we use $+\lambda^2$, the resulting solution obtained does not satisfy the boundedness condition for real values of λ .) Then

$$X'' + \lambda^2 X = 0, \quad T' + 2\lambda^2 T = 0$$

with solutions $X = A_1 \cos \lambda x + B_1 \sin \lambda x$, $T = c_1 e^{-2\lambda^2 t}$

A solution of the partial differential equation is thus given by

$$u(x, t) = XT = c_1 e^{-2\lambda^2 t} (A_1 \cos \lambda x + B_1 \sin \lambda x) = e^{-2\lambda^2 t} (A \cos \lambda x + B \sin \lambda x)$$

Since $u(0, t) = 0$, $e^{-2\lambda^2 t} (A) = 0$ or $A = 0$. Then

$$u(x, t) = Be^{-2\lambda^2 t} \sin \lambda x$$

Since $u(3, t) = 0$, $Be^{-2\lambda^2 t} \sin 3\lambda = 0$. If $B = 0$, the solution is identically zero, so we must have $\sin 3\lambda = 0$ or $3\lambda = m\pi$, $\lambda = m\pi/3$, where $m = 0, \pm 1, \pm 2, \dots$. Thus a solution is

$$u(x, t) = Be^{-2m^2\pi^2 t/9} \sin \frac{m\pi x}{3}$$

Also, by the principle of superposition,

$$u(x, t) = B_1 e^{-2m_1^2\pi^2 t/9} \sin \frac{m_1\pi x}{3} + B_2 e^{-2m_2^2\pi^2 t/9} \sin \frac{m_2\pi x}{3} + B_3 e^{-2m_3^2\pi^2 t/9} \sin \frac{m_3\pi x}{3} \quad (1)$$

is a solution. By the last boundary condition,

$$\begin{aligned}
 u(x, 0) &= B_1 \sin \frac{m_1 \pi x}{3} + B_2 \sin \frac{m_2 \pi x}{3} + B_3 \sin \frac{m_3 \pi x}{3} \\
 &= 5 \sin 4\pi x - 3 \sin 8\pi x + 2 \sin 10\pi x
 \end{aligned}$$

which is possible if and only if $B_1 = 5$, $m_1 = 12$, $B_2 = -3$, $m_2 = 24$, $B_3 = 2$, $m_3 = 30$.

Substituting these in (1), the required solution is

$$u(x, t) = 5e^{-32\pi^2 t} \sin 4\pi x - 3e^{-128\pi^2 t} \sin 8\pi x + 2e^{-200\pi^2 t} \sin 10\pi x \quad (2)$$

This boundary value problem has the following interpretation as a heat flow problem. A bar whose surface is insulated (Fig. 1-11) has a length of 3 units and a diffusivity of 2 units. If its ends are kept at temperature zero units and its initial temperature $u(x, 0) = 5 \sin 4\pi x - 3 \sin 8\pi x + 2 \sin 10\pi x$, find the temperature at position x at time t , i.e. find $u(x, t)$.

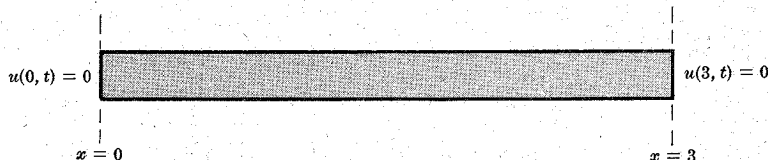


Fig. 1-11

- 1.24. Solve $\frac{\partial^2 y}{\partial t^2} = 16 \frac{\partial^2 y}{\partial x^2}$, $0 < x < 2$, $t > 0$, subject to the conditions $y(0, t) = 0$, $y(2, t) = 0$, $y(x, 0) = 6 \sin \pi x - 3 \sin 4\pi x$, $y_t(x, 0) = 0$, $|y(x, t)| < M$.

Let $y = XT$, where X depends only on x , T depends only on t . Then substitution in the differential equation yields

$$XT'' = 16X''T \quad \text{or} \quad X''/X = T''/16T$$

on separating the variables. Since each side must be a constant, say $-\lambda^2$, we have

$$X'' + \lambda^2 X = 0, \quad T'' + 16\lambda^2 T = 0$$

Solving these we find

$$X = a_1 \cos \lambda x + b_1 \sin \lambda x, \quad T = a_2 \cos 4\lambda t + b_2 \sin 4\lambda t$$

Thus a solution is

$$y(x, t) = (a_1 \cos \lambda x + b_1 \sin \lambda x)(a_2 \cos 4\lambda t + b_2 \sin 4\lambda t) \quad (1)$$

To find the constants it is simpler to proceed by using first those boundary conditions involving two zeros, such as $y(0, t) = 0$, $y_t(x, 0) = 0$. From $y(0, t) = 0$ we see from (1) that

$$a_1(a_2 \cos 4\lambda t + b_2 \sin 4\lambda t) = 0$$

so that to obtain a non zero solution (1) we must have $a_1 = 0$. Thus (1) becomes

$$y(x, t) = (b_1 \sin \lambda x)(a_2 \cos 4\lambda t + b_2 \sin 4\lambda t) \quad (2)$$

Differentiation of (2) with respect to t yields

$$y_t(x, t) = (b_1 \sin \lambda x)(-4\lambda a_2 \sin 4\lambda t + 4\lambda b_2 \cos 4\lambda t)$$

so that we have on putting $t = 0$ and using the condition $y_t(x, 0) = 0$

$$y_t(x, 0) = (b_1 \sin \lambda x)(4\lambda b_2) = 0 \quad (3)$$

In order to obtain a solution (2) which is not zero we see from (3) that we must have $b_2 = 0$. Thus (2) becomes

$$y(x, t) = B \sin \lambda x \cos 4\lambda t$$

on putting $b_2 = 0$ and writing $B = b_1 a_2$.

From $y(2, t) = 0$ we now find

$$B \sin 2\lambda \cos 4\lambda t = 0$$

and we see that we must have $\sin 2\lambda = 0$, i.e. $2\lambda = m\pi$ or $\lambda = m\pi/2$ where $m = 0, \pm 1, \pm 2, \dots$

Thus
$$y(x, t) = B \sin \frac{m\pi x}{2} \cos 2m\pi t \quad (4)$$

is a solution. Since this solution is bounded, the condition $|y(x, t)| < M$ is automatically satisfied.

In order to satisfy the last condition, $y(x, 0) = 5 \sin \pi x - 3 \sin 4\pi x$, we first use the principle of superposition to obtain the solution

$$y(x, t) = B_1 \sin \frac{m_1 \pi x}{2} \cos 2m_1 \pi t + B_2 \sin \frac{m_2 \pi x}{2} \cos 2m_2 \pi t \quad (5)$$

Then putting $t = 0$ we arrive at

$$\begin{aligned} y(x, 0) &= B_1 \sin \frac{m_1 \pi x}{2} + B_2 \sin \frac{m_2 \pi x}{2} \\ &= 6 \sin \pi x - 3 \sin 4\pi x \end{aligned}$$

This is possible if and only if $B_1 = 6$, $m_1 = 2$, $B_2 = -3$, $m_2 = 8$. Thus the required solution (5) is

$$y(x, t) = 6 \sin \pi x \cos 4\pi t - 3 \sin 4\pi x \cos 16\pi t \quad (6)$$

This boundary value problem can be interpreted physically in terms of the vibrations of a string. The string has its ends fixed at $x = 0$ and $x = 2$ and is given an initial shape $f(x) = 6 \sin \pi x - 3 \sin 4\pi x$. It is then released so that its initial velocity is zero. Then (6) gives the displacement of any point x of the string at any later time t .

- 1.25. Solve $\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}$, $0 < x < 3$, $t > 0$, given that $u(0, t) = u(3, t) = 0$, $u(x, 0) = f(x)$, $|u(x, t)| < M$.

This problem differs from Problem 1.23 only in the condition $u(x, 0) = f(x)$. In seeking to satisfy this last condition we see that taking a finite number of terms, as in (I) of Problem 1.23, will be insufficient for arbitrary $f(x)$. Thus we are led to assume that infinitely many terms are taken, i.e.

$$u(x, t) = \sum_{m=1}^{\infty} B_m e^{-2m^2 \pi^2 t/9} \sin \frac{m\pi x}{3}$$

The condition $u(x, 0) = f(x)$ then leads to

$$f(x) = \sum_{m=1}^{\infty} B_m \sin \frac{m\pi x}{3}$$

or the problem of expansion of a function into a sine series. Such trigonometric expansions, or *Fourier series*, will be considered in detail in the next chapter.

Supplementary Problems

MATHEMATICAL FORMULATION OF PHYSICAL PROBLEMS

- 1.26. If a taut, horizontal string with fixed ends vibrates in a vertical plane under the influence of gravity, show that its equation is

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} - g$$

where g is the acceleration due to gravity.

- 1.27. A thin bar located on the x -axis has its ends at $x = 0$ and $x = L$. The initial temperature of the bar is $f(x)$, $0 < x < L$, and the ends $x = 0$, $x = L$ are maintained at constant temperatures T_1 , T_2 respectively. Assuming the surrounding medium is at temperature u_0 and that Newton's law of cooling applies, show that the partial differential equation for the temperature of the bar at any point at any time is given by

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} - \beta(u - u_0)$$

and write the corresponding boundary conditions.

- 1.28. Write the boundary conditions in Problem 1.27 if (a) the ends $x = 0$ and $x = L$ are insulated, (b) the ends $x = 0$ and $x = L$ radiate into the surrounding medium according to Newton's law of cooling.
- 1.29. The gravitational potential v at any point (x, y, z) outside of a mass m located at the point (X, Y, Z) is defined as the mass m divided by the distance of the point (x, y, z) from (X, Y, Z) . Show that v satisfies Laplace's equation $\nabla^2 v = 0$.
- 1.30. Extend the result of Problem 1.29 to a solid body.
- 1.31. A string has its ends fixed at $x = 0$ and $x = L$. It is displaced a distance h at its midpoint and then released. Formulate a boundary value problem for the displacement $y(x, t)$ of any point x of the string at time t .

CLASSIFICATION OF PARTIAL DIFFERENTIAL EQUATIONS

- 1.32. Determine whether each of the following partial differential equations is linear or nonlinear, state the order of each equation, and name the dependent and independent variables.

$$\begin{array}{lll} (a) \quad \frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0 & (c) \quad \phi \frac{\partial \phi}{\partial x} = \frac{\partial^3 \phi}{\partial y^3} & (e) \quad \frac{\partial z}{\partial r} + \frac{\partial z}{\partial s} = \frac{1}{z^2} \\ (b) \quad (x^2 + y^2) \frac{\partial^4 T}{\partial z^4} = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} & (d) \quad \frac{\partial^2 y}{\partial t^2} - 4 \frac{\partial^2 y}{\partial x^2} = x^2 & \end{array}$$

- 1.33. Classify each of the following equations as elliptic, hyperbolic or parabolic.

$$\begin{array}{ll} (a) \quad \frac{\partial^2 \phi}{\partial x^2} - \frac{\partial^2 \phi}{\partial y^2} = 0 & (e) \quad (x^2 - 1) \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + (y^2 - 1) \frac{\partial^2 u}{\partial y^2} \\ & = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \\ (b) \quad \frac{\partial u}{\partial x} + \frac{\partial^2 u}{\partial x \partial y} = 4 & \\ (c) \quad \frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = x + 3y & (f) \quad (M^2 - 1) \frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0, \quad M > 0 \\ (d) \quad x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0 & \end{array}$$

SOLUTIONS OF PARTIAL DIFFERENTIAL EQUATIONS

- 1.34. Show that $z(x, y) = 4e^{-3x} \cos 3y$ is a solution to the boundary value problem

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0, \quad z(x, \pi/2) = 0, \quad z(x, 0) = 4e^{-3x}$$

- 1.35. (a) Show that $v(x, y) = xF(2x + y)$ is a general solution of $x \frac{\partial v}{\partial x} - 2x \frac{\partial v}{\partial y} = v$.
 (b) Find a particular solution satisfying $v(1, y) = y^2$.

- 1.36. Find a partial differential equation having general solution $u = F(x - 3y) + G(2x + y)$.

- 1.37. Find a partial differential equation having general solution

$$(a) \quad z = e^x f(2y - 3x), \quad (b) \quad z = f(2x + y) + g(x - 2y)$$

GENERAL SOLUTIONS OF PARTIAL DIFFERENTIAL EQUATIONS

- 1.38. (a) Solve $x \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial z}{\partial y} = 0$.
 (b) Find the particular solution for which $z(x, 0) = x^5 + x - \frac{68}{x}$, $z(2, y) = 3y^4$.

1.39. Find general solutions of each of the following.

$$\begin{aligned} (a) \quad \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 u}{\partial y^2} & (b) \quad \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} &= 3u & (c) \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0 \\ (d) \quad \frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} - 3 \frac{\partial^2 z}{\partial y^2} &= 0 & (e) \quad \frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} &= 0 \end{aligned}$$

1.40. Find general solutions of each of the following.

$$\begin{aligned} (a) \quad \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} &= x & (c) \quad \frac{\partial^4 u}{\partial x^4} + 2 \frac{\partial^4 u}{\partial x^3 \partial y} &= 4 \\ (b) \quad \frac{\partial^2 y}{\partial x^2} &= \frac{\partial^2 y}{\partial t^2} + 12t^2 & (d) \quad \frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} &= x \sin y \end{aligned}$$

1.41. Solve $\frac{\partial^4 u}{\partial x^4} + 2 \frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^4 u}{\partial y^4} = 16.$

1.42. Show that a general solution of $\frac{\partial^2 v}{\partial r^2} + \frac{2}{r} \frac{\partial v}{\partial r} = \frac{1}{c^2} \frac{\partial^2 v}{\partial t^2}$ is $v = \frac{F(r-ct) + G(r+ct)}{r}.$

SEPARATION OF VARIABLES

1.43. Solve each of the following boundary value problems by the method of separation of variables.

$$\begin{aligned} (a) \quad 3 \frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} &= 0, \quad u(x, 0) = 4e^{-x} \\ (b) \quad \frac{\partial u}{\partial x} &= 2 \frac{\partial u}{\partial y} + u, \quad u(x, 0) = 3e^{-5x} + 2e^{-3x} \\ (c) \quad \frac{\partial u}{\partial t} &= 4 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(\pi, t) = 0, \quad u(x, 0) = 2 \sin 3x - 4 \sin 5x \\ (d) \quad \frac{\partial u}{\partial t} &= \frac{\partial^2 u}{\partial x^2}, \quad u_x(0, t) = 0, \quad u(2, t) = 0, \quad u(x, 0) = 8 \cos \frac{3\pi x}{4} - 6 \cos \frac{9\pi x}{4} \\ (e) \quad \frac{\partial u}{\partial t} &= 3 \frac{\partial u}{\partial x}, \quad u(x, 0) = 8e^{-2x} \\ (f) \quad \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x} - 2u, \quad u(x, 0) = 10e^{-x} - 6e^{-4x} \\ (g) \quad \frac{\partial u}{\partial t} &= \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 0, \quad u(4, t) = 0, \quad u(x, 0) = 6 \sin \frac{\pi x}{2} + 3 \sin \pi x \end{aligned}$$

1.44. Solve and give a physical interpretation to the boundary value problem

$$\frac{\partial^2 y}{\partial t^2} = 4 \frac{\partial^2 y}{\partial x^2}, \quad y(0, t) = y(5, t) = 0, \quad y(x, 0) = 0, \quad y_t(x, 0) = f(x) \quad (0 < x < 5, t > 0)$$

if (a) $f(x) = 5 \sin \pi x$, (b) $f(x) = 3 \sin 2\pi x - 2 \sin 5\pi x$.

1.45. Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - 2u$ if $u(0, t) = 0$, $u(3, t) = 0$, $u(x, 0) = 2 \sin \pi x - \sin 4\pi x$.

1.46. Suppose that in Problem 1.24 we have $y(x, 0) = f(x)$, where $0 < x < 2$. Show how the problem can be solved if we know how to expand $f(x)$ in a series of sines.

1.47. Suppose that in Problem 1.25 the boundary conditions are $u_x(0, t) = 0$, $u(3, t) = 0$, $u(x, 0) = f(x)$. Show how the problem can be solved if we know how to expand $f(x)$ in a series of cosines. Give a physical interpretation of this problem.

Chapter 2

Fourier Series and Applications

THE NEED FOR FOURIER SERIES

In Problem 1.25, page 17, we saw that to obtain a solution to a particular boundary value problem we should need to know how to expand a function into a trigonometric series. In this chapter we shall investigate the theory of such series and shall use the theory to solve many boundary value problems.

Since each term of the trigonometric series considered in Problem 1.25 is periodic, it is clear that if we are to expand functions in such series, the functions should also be periodic. We therefore turn now to the consideration of periodic functions.

PERIODIC FUNCTIONS

A function $f(x)$ is said to have a *period* P or to be *periodic* with period P if for all x , $f(x+P) = f(x)$, where P is a positive constant. The least value of $P > 0$ is called the *least period* or simply *the period* of $f(x)$.

Example 1.

The function $\sin x$ has periods $2\pi, 4\pi, 6\pi, \dots$, since $\sin(x+2\pi), \sin(x+4\pi), \sin(x+6\pi), \dots$ all equal $\sin x$. However, 2π is the *least period* or *the period* of $\sin x$.

Example 2.

The period of $\sin nx$ or $\cos nx$, where n is a positive integer, is $2\pi/n$.

Example 3.

The period of $\tan x$ is π .

Example 4.

A constant has any positive number as a period.

Other examples of periodic functions are shown in the graphs of Fig. 2-1.

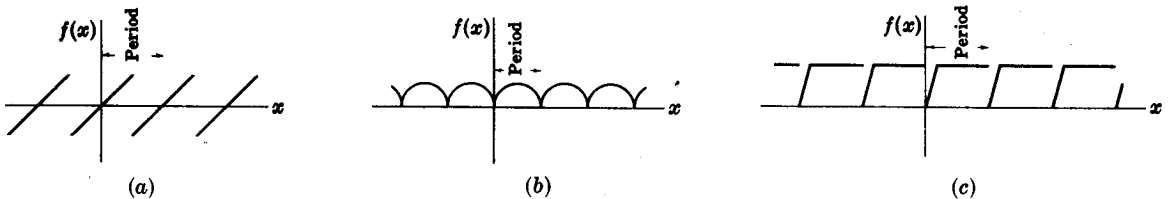


Fig. 2-1

PIECEWISE CONTINUOUS FUNCTIONS

A function $f(x)$ is said to be *piecewise continuous* in an interval if (i) the interval can be divided into a finite number of subintervals in each of which $f(x)$ is continuous and (ii) the limits of $f(x)$ as x approaches the endpoints of each subinterval are finite. Another way of stating this is to say that a piecewise continuous function is one that has at most a finite number of finite discontinuities. An example of a piecewise continuous function is shown in Fig. 2-2. The functions of Fig. 2-1(a) and (c) are piecewise continuous. The function of Fig. 2-1(b) is continuous.

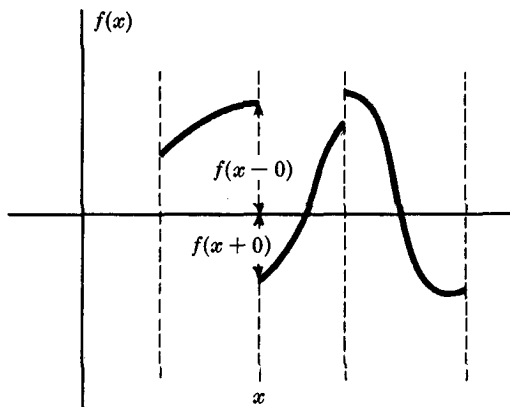


Fig. 2-2

The *limit of $f(x)$ from the right* or the *right-hand limit of $f(x)$* is often denoted by $\lim_{\epsilon \rightarrow 0} f(x + \epsilon) = f(x + 0)$, where $\epsilon > 0$. Similarly, the *limit of $f(x)$ from the left* or the *left-hand limit of $f(x)$* is denoted by $\lim_{\epsilon \rightarrow 0} f(x - \epsilon) = f(x - 0)$, where $\epsilon > 0$. The values $f(x + 0)$ and $f(x - 0)$ at the point x in Fig. 2-2 are as indicated. The fact that $\epsilon \rightarrow 0$ and $\epsilon > 0$ is sometimes indicated briefly by $\epsilon \rightarrow 0+$. Thus, for example, $\lim_{\epsilon \rightarrow 0+} f(x + \epsilon) = f(x + 0)$, $\lim_{\epsilon \rightarrow 0+} f(x - \epsilon) = f(x - 0)$.

DEFINITION OF FOURIER SERIES

Let $f(x)$ be defined in the interval $(-L, L)$ and determined outside of this interval by $f(x + 2L) = f(x)$, i.e. assume that $f(x)$ has the period $2L$. The *Fourier series* or *Fourier expansion* corresponding to $f(x)$ is defined to be

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \quad (1)$$

where the *Fourier coefficients* a_n and b_n are

$$\begin{cases} a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \\ b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \end{cases} \quad n = 0, 1, 2, \dots \quad (2)$$

Motivation for this definition is supplied in Problem 2.4.

If $f(x)$ has the period $2L$, the coefficients a_n and b_n can be determined equivalently from

$$\begin{cases} a_n = \frac{1}{L} \int_c^{c+2L} f(x) \cos \frac{n\pi x}{L} dx \\ b_n = \frac{1}{L} \int_c^{c+2L} f(x) \sin \frac{n\pi x}{L} dx \end{cases} \quad n = 0, 1, 2, \dots \quad (3)$$

where c is any real number. In the special case $c = -L$, (3) becomes (2). Note that the constant term in (1) is equal to $\frac{a_0}{2} = \frac{1}{2L} \int_{-L}^L f(x) dx$, which is the *mean* of $f(x)$ over a period.

If $L = \pi$, the series (1) and the coefficients (2) or (3) are particularly simple. The function in this case has the period 2π .

It should be emphasized that the series (1) is only the series which *corresponds* to $f(x)$. We do not know whether this series converges or even, if it does converge, whether it con-

verges to $f(x)$. This problem of convergence was examined by *Dirichlet*, who developed conditions for convergence of Fourier series which we now consider.

DIRICHLET CONDITIONS

Theorem 2-1: Suppose that

- (i) $f(x)$ is defined and single-valued except possibly at a finite number of points in $(-L, L)$
- (ii) $f(x)$ is periodic with period $2L$
- (iii) $f(x)$ and $f'(x)$ are piecewise continuous in $(-L, L)$

Then the series (1) with coefficients (2) or (3) converges to

- (a) $f(x)$ if x is a point of continuity
- (b) $\frac{f(x+0) + f(x-0)}{2}$ if x is a point of discontinuity

For a proof see Problems 2.18–2.23.

According to this result we can write

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \quad (4)$$

at any point of continuity x . However, if x is a point of discontinuity, then the left side is replaced by $\frac{1}{2}[f(x+0) + f(x-0)]$, so that the series converges to the mean value of $f(x+0)$ and $f(x-0)$.

The conditions (i), (ii) and (iii) imposed on $f(x)$ are *sufficient* but not *necessary*, i.e. if the conditions are satisfied the convergence is guaranteed. However, if they are not satisfied the series may or may not converge. The conditions above are generally satisfied in cases which arise in science or engineering.

There are at present no known necessary and sufficient conditions for convergence of Fourier series. It is of interest that continuity of $f(x)$ does not alone insure convergence of a Fourier series.

ODD AND EVEN FUNCTIONS

A function $f(x)$ is called *odd* if $f(-x) = -f(x)$. Thus x^3 , $x^5 - 3x^3 + 2x$, $\sin x$, $\tan 3x$ are odd functions.

A function $f(x)$ is called *even* if $f(-x) = f(x)$. Thus x^4 , $2x^6 - 4x^2 + 5$, $\cos x$, $e^x + e^{-x}$ are even functions.

The functions portrayed graphically in Fig. 2-1(a) and 2-1(b) are odd and even respectively, but that of Fig. 2-1(c) is neither odd nor even.

In the Fourier series corresponding to an odd function, only sine terms can be present. In the Fourier series corresponding to an even function, only cosine terms (and possibly a constant, which we shall consider to be a cosine term) can be present.

HALF-RANGE FOURIER SINE OR COSINE SERIES

A half-range Fourier sine or cosine series is a series in which only sine terms or only cosine terms are present, respectively. When a half-range series corresponding to a given

function is desired, the function is generally defined in the interval $(0, L)$ [which is half of the interval $(-L, L)$, thus accounting for the name *half-range*] and then the function is specified as odd or even, so that it is clearly defined in the other half of the interval, namely $(-L, 0)$. In such case, we have

$$\begin{cases} a_n = 0, & b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx & \text{for half-range sine series} \\ b_n = 0, & a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx & \text{for half-range cosine series} \end{cases} \quad (5)$$

PARSEVAL'S IDENTITY states that

$$\frac{1}{L} \int_{-L}^L \{f(x)\}^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \quad (6)$$

if a_n and b_n are the Fourier coefficients corresponding to $f(x)$ and if $f(x)$ satisfies the Dirichlet conditions.

UNIFORM CONVERGENCE

Suppose that we have an infinite series $\sum_{n=1}^{\infty} u_n(x)$. We define the R th partial sum of the series to be the sum of the first R terms of the series, i.e.

$$S_R(x) = \sum_{n=1}^R u_n(x) \quad (7)$$

Now by definition the infinite series is said to *converge* to $f(x)$ in some interval if given any positive number ϵ , there exists for each x in the interval a positive number N such that

$$|S_R(x) - f(x)| < \epsilon \quad \text{whenever } R > N \quad (8)$$

The number N depends in general not only on ϵ but also on x . We call $f(x)$ the *sum* of the series.

An important case occurs when N depends on ϵ but *not* on the value of x in the interval. In such case we say that the series *converges uniformly* or is *uniformly convergent* to $f(x)$.

Two very important properties of uniformly convergent series are summarized in the following two theorems.

Theorem 2-2: If each term of an infinite series is continuous in an interval (a, b) and the series is uniformly convergent to the sum $f(x)$ in this interval, then

1. $f(x)$ is also continuous in the interval
2. the series can be integrated term by term, i.e.

$$\int_a^b \left\{ \sum_{n=1}^{\infty} u_n(x) \right\} dx = \sum_{n=1}^{\infty} \int_a^b u_n(x) dx \quad (9)$$

Theorem 2-3: If each term of an infinite series has a derivative and the series of derivatives is uniformly convergent, then the series can be differentiated term by term, i.e.

$$\frac{d}{dx} \sum_{n=1}^{\infty} u_n(x) = \sum_{n=1}^{\infty} \frac{d}{dx} u_n(x) \quad (10)$$

There are various ways of proving the uniform convergence of a series. The most obvious way is to actually find the sum $S_R(x)$ in closed form and then apply the definition directly. A second and most powerful way is to use a theorem called the *Weierstrass M test*.

Theorem 2-4 (Weierstrass M test): If there exists a set of constants M_n , $n = 1, 2, \dots$, such that for all x in an interval $|u_n(x)| \leq M_n$, and if furthermore $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} u_n(x)$ converges uniformly in the interval. Incidentally, the series is also *absolutely convergent*, i.e. $\sum_{n=1}^{\infty} |u_n(x)|$ converges, under these conditions.

Example 5.

The series $\sum_{n=1}^{\infty} \frac{\sin nx}{n^2}$ converges uniformly in the interval $(-\pi, \pi)$ [or, in fact, in any interval], since a set of constants $M_n = 1/n^2$ can be found such that

$$\left| \frac{\sin nx}{n^2} \right| \leq \frac{1}{n^2} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges}$$

INTEGRATION AND DIFFERENTIATION OF FOURIER SERIES

Integration and differentiation of Fourier series can be justified by using Theorems 2-2 and 2-3, which hold for series in general. It must be emphasized, however, that those theorems provide sufficient conditions and are not necessary. The following theorem for integration is especially useful.

Theorem 2-5: The Fourier series corresponding to $f(x)$ may be integrated term by term from a to x , and the resulting series will converge uniformly to $\int_a^x f(u) du$, provided that $f(x)$ is piecewise continuous in $-L \leq x \leq L$ and both a and x are in this interval.

COMPLEX NOTATION FOR FOURIER SERIES

Using Euler's identities,

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad e^{-i\theta} = \cos \theta - i \sin \theta \quad (11)$$

where i is the imaginary unit such that $i^2 = -1$, the Fourier series for $f(x)$ can be written in complex form as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L} \quad (12)$$

where

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx \quad (13)$$

In writing the equality (12), we are supposing that the Dirichlet conditions are satisfied and further that $f(x)$ is continuous at x . If $f(x)$ is discontinuous at x , the left side of (12) should be replaced by $\frac{f(x+0) + f(x-0)}{2}$.

DOUBLE FOURIER SERIES

The idea of a Fourier series expansion for a function of a single variable x can be extended to the case of functions of two variables x and y , i.e. $f(x, y)$. For example, we can expand $f(x, y)$ into a *double Fourier sine series*

$$f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin \frac{m\pi x}{L_1} \sin \frac{n\pi y}{L_2} \quad (14)$$

where

$$B_{mn} = \frac{4}{L_1 L_2} \int_0^{L_1} \int_0^{L_2} f(x, y) \sin \frac{m\pi x}{L_1} \sin \frac{n\pi y}{L_2} dx dy \quad (15)$$

Similar results can be obtained for cosine series or for series having both sines and cosines. These ideas can be generalized to *triple Fourier series*, etc.

APPLICATIONS OF FOURIER SERIES

There are numerous applications of Fourier series to solutions of boundary value problems. For example:

1. Heat flow. See Problems 2.25–2.29.
2. Laplace's equation. See Problems 2.30, 2.31.
3. Vibrating systems. See Problems 2.32, 2.33.

Solved Problems

FOURIER SERIES

2.1. Graph each of the following functions.

$$(a) \quad f(x) = \begin{cases} 3 & 0 < x < 5 \\ -3 & -5 < x < 0 \end{cases} \quad \text{Period} = 10$$

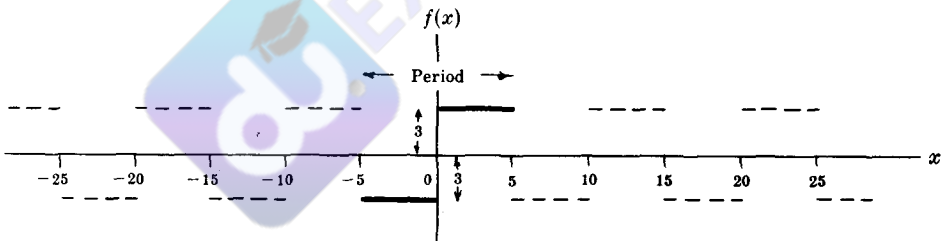


Fig. 2-3

Since the period is 10, that portion of the graph in $-5 < x < 5$ (indicated heavy in Fig. 2-3 above) is extended periodically outside this range (indicated dashed). Note that $f(x)$ is not defined at $x = 0, 5, -5, 10, -10, 15, -15$, etc. These values are the *discontinuities* of $f(x)$.

$$(b) \quad f(x) = \begin{cases} \sin x & 0 \leq x \leq \pi \\ 0 & \pi < x < 2\pi \end{cases} \quad \text{Period} = 2\pi$$

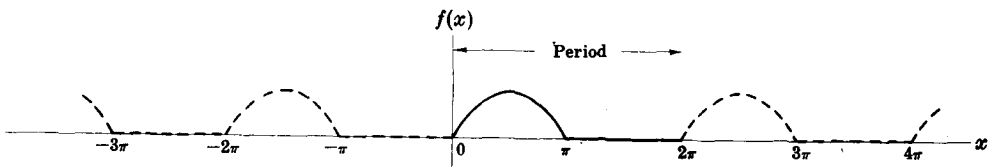


Fig. 2-4

Refer to Fig. 2-4 above. Note that $f(x)$ is defined for all x and is continuous everywhere.

$$(c) \quad f(x) = \begin{cases} 0 & 0 \leq x < 2 \\ 1 & 2 \leq x < 4 \\ 0 & 4 \leq x < 6 \end{cases} \quad \text{Period} = 6$$

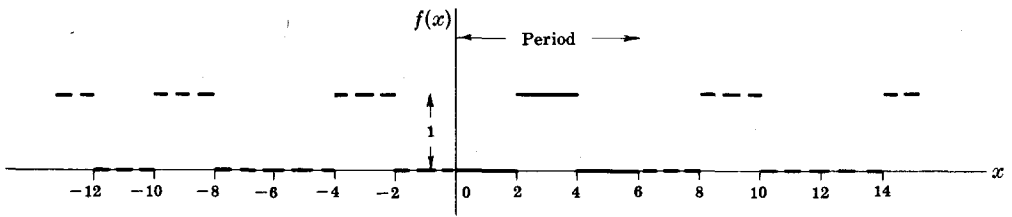


Fig. 2-5

Refer to Fig. 2-5 above. Note that $f(x)$ is defined for all x and is discontinuous at $x = \pm 2, \pm 4, \pm 8, \pm 10, \pm 14, \dots$

2.2. Prove $\int_{-L}^L \sin \frac{k\pi x}{L} dx = \int_{-L}^L \cos \frac{k\pi x}{L} dx = 0$ if $k = 1, 2, 3, \dots$

$$\int_{-L}^L \sin \frac{k\pi x}{L} dx = -\frac{L}{k\pi} \cos \frac{k\pi x}{L} \Big|_{-L}^L = -\frac{L}{k\pi} \cos k\pi + \frac{L}{k\pi} \cos(-k\pi) = 0$$

$$\int_{-L}^L \cos \frac{k\pi x}{L} dx = \frac{L}{k\pi} \sin \frac{k\pi x}{L} \Big|_{-L}^L = \frac{L}{k\pi} \sin k\pi - \frac{L}{k\pi} \sin(-k\pi) = 0$$

2.3. Prove (a) $\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = \begin{cases} 0 & m \neq n \\ L & m = n \end{cases}$

(b) $\int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0$

where m and n can assume any of the values $1, 2, 3, \dots$

(a) From trigonometry:

$$\cos A \cos B = \frac{1}{2} \{ \cos(A - B) + \cos(A + B) \}, \quad \sin A \sin B = \frac{1}{2} \{ \cos(A - B) - \cos(A + B) \}$$

Then, if $m \neq n$, we have by Problem 2.2,

$$\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \left\{ \cos \frac{(m-n)\pi x}{L} + \cos \frac{(m+n)\pi x}{L} \right\} dx = 0$$

Similarly, if $m \neq n$,

$$\int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \left\{ \cos \frac{(m-n)\pi x}{L} - \cos \frac{(m+n)\pi x}{L} \right\} dx = 0$$

If $m = n$, we have

$$\int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \left(1 + \cos \frac{2n\pi x}{L} \right) dx = L$$

$$\int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \left(1 - \cos \frac{2n\pi x}{L} \right) dx = L$$

Note that if $m = n = 0$ these integrals are equal to $2L$ and 0 respectively.

(b) We have $\sin A \cos B = \frac{1}{2} \{ \sin(A - B) + \sin(A + B) \}$. Then by Problem 2.2, if $m \neq n$,

$$\int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \left\{ \sin \frac{(m-n)\pi x}{L} + \sin \frac{(m+n)\pi x}{L} \right\} dx = 0$$

If $m = n$,

$$\int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \sin \frac{2n\pi x}{L} dx = 0$$

The results of parts (a) and (b) remain valid when the limits of integration $-L, L$ are replaced by $c, c + 2L$ respectively.

2.4. If the series $A + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$ converges uniformly to $f(x)$ in $(-L, L)$, show that for $n = 1, 2, 3, \dots$,

$$(a) \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad (b) \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx, \quad (c) \quad A = \frac{a_0}{2}.$$

$$(a) \text{ Multiplying} \quad f(x) = A + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \quad (1)$$

by $\cos \frac{m\pi x}{L}$ and integrating from $-L$ to L , using Problem 2.3, we have

$$\begin{aligned} \int_{-L}^L f(x) \cos \frac{m\pi x}{L} dx &= A \int_{-L}^L \cos \frac{m\pi x}{L} dx \\ &\quad + \sum_{n=1}^{\infty} \left\{ a_n \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx + b_n \int_{-L}^L \cos \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx \right\} \\ &= a_m L \quad \text{if } m \neq 0 \end{aligned} \quad (2)$$

$$\text{Thus} \quad a_m = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{m\pi x}{L} dx \quad \text{if } m = 1, 2, 3, \dots$$

(b) Multiplying (1) by $\sin \frac{m\pi x}{L}$ and integrating from $-L$ to L , using Problem 2.3, we have

$$\begin{aligned} \int_{-L}^L f(x) \sin \frac{m\pi x}{L} dx &= A \int_{-L}^L \sin \frac{m\pi x}{L} dx \\ &\quad + \sum_{n=1}^{\infty} \left\{ a_n \int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx + b_n \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx \right\} \\ &= b_m L \end{aligned}$$

$$\text{Thus} \quad b_m = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{m\pi x}{L} dx \quad \text{if } m = 1, 2, 3, \dots$$

(c) Integration of (1) from $-L$ to L , using Problem 2.2, gives

$$\int_{-L}^L f(x) dx = 2AL \quad \text{or} \quad A = \frac{1}{2L} \int_{-L}^L f(x) dx$$

Putting $m = 0$ in the result of part (a), we find $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$ and so $A = \frac{a_0}{2}$.

The above results also hold when the integration limits $-L, L$ are replaced by $c, c + 2L$.

Note that in all parts above, interchange of summation and integration is valid because the series is assumed to converge uniformly to $f(x)$ in $(-L, L)$. Even when this assumption is not warranted, the coefficients a_m and b_m as obtained above are called *Fourier coefficients* corresponding to $f(x)$, and the corresponding series with these values of a_m and b_m is called the *Fourier series* corresponding to $f(x)$. An important problem in this case is to investigate conditions under which this series actually converges to $f(x)$. Sufficient conditions for this convergence are the *Dirichlet conditions* established below in Problems 2.18–2.23.

2.5. (a) Find the Fourier coefficients corresponding to the function

$$f(x) = \begin{cases} 0 & -5 < x < 0 \\ 3 & 0 < x < 5 \end{cases} \quad \text{Period} = 10$$

(b) Write the corresponding Fourier series.

(c) How should $f(x)$ be defined at $x = -5$, $x = 0$ and $x = 5$ in order that the Fourier series will converge to $f(x)$ for $-5 \leq x \leq 5$?

The graph of $f(x)$ is shown in Fig. 2-6 below.

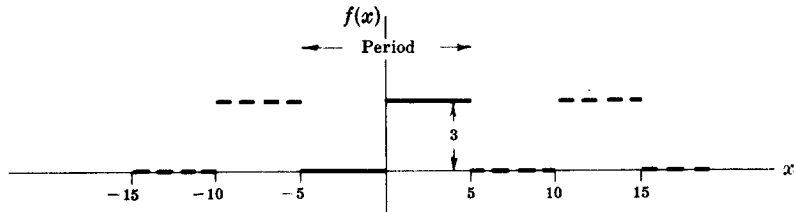


Fig. 2-6

(a) Period $= 2L = 10$ and $L = 5$. Choose the interval c to $c + 2L$ as -5 to 5 , so that $c = -5$. Then

$$\begin{aligned} a_n &= \frac{1}{L} \int_c^{c+2L} f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{5} \int_{-5}^5 f(x) \cos \frac{n\pi x}{5} dx \\ &= \frac{1}{5} \left\{ \int_{-5}^0 (0) \cos \frac{n\pi x}{5} dx + \int_0^5 (3) \cos \frac{n\pi x}{5} dx \right\} = \frac{3}{5} \int_0^5 \cos \frac{n\pi x}{5} dx \\ &= \frac{3}{5} \left(\frac{5}{n\pi} \sin \frac{n\pi x}{5} \right) \Big|_0^5 = 0 \quad \text{if } n \neq 0 \end{aligned}$$

$$\text{If } n = 0, \quad a_n = a_0 = \frac{3}{5} \int_0^5 \cos \frac{0\pi x}{5} dx = \frac{3}{5} \int_0^5 dx = 3.$$

$$\begin{aligned} b_n &= \frac{1}{L} \int_c^{c+2L} f(x) \sin \frac{n\pi x}{L} dx = \frac{1}{5} \int_{-5}^5 f(x) \sin \frac{n\pi x}{5} dx \\ &= \frac{1}{5} \left\{ \int_{-5}^0 (0) \sin \frac{n\pi x}{5} dx + \int_0^5 (3) \sin \frac{n\pi x}{5} dx \right\} = \frac{3}{5} \int_0^5 \sin \frac{n\pi x}{5} dx \\ &= \frac{3}{5} \left(-\frac{5}{n\pi} \cos \frac{n\pi x}{5} \right) \Big|_0^5 = \frac{3(1 - \cos n\pi)}{n\pi} \end{aligned}$$

(b) The corresponding Fourier series is

$$\begin{aligned} \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) &= \frac{3}{2} + \sum_{n=1}^{\infty} \frac{3(1 - \cos n\pi)}{n\pi} \sin \frac{n\pi x}{5} \\ &= \frac{3}{2} + \frac{6}{\pi} \left(\sin \frac{\pi x}{5} + \frac{1}{3} \sin \frac{3\pi x}{5} + \frac{1}{5} \sin \frac{5\pi x}{5} + \dots \right) \end{aligned}$$

(c) Since $f(x)$ satisfies the Dirichlet conditions, we can say that the series converges to $f(x)$ at all points of continuity and to $\frac{f(x+0) + f(x-0)}{2}$ at points of discontinuity. At $x = -5, 0$ and 5 , which are points of discontinuity, the series converges to $(3+0)/2 = 3/2$, as seen from the graph. The series will converge to $f(x)$ for $-5 \leq x \leq 5$ if we redefine $f(x)$ as follows:

$$f(x) = \begin{cases} 3/2 & x = -5 \\ 0 & -5 < x < 0 \\ 3/2 & x = 0 \\ 3 & 0 < x < 5 \\ 3/2 & x = 5 \end{cases} \quad \text{Period} = 10$$

2.6. Expand $f(x) = x^2$, $0 < x < 2\pi$, in a Fourier series if the period is 2π .

The graph of $f(x)$ with period 2π is shown in Fig. 2-7.

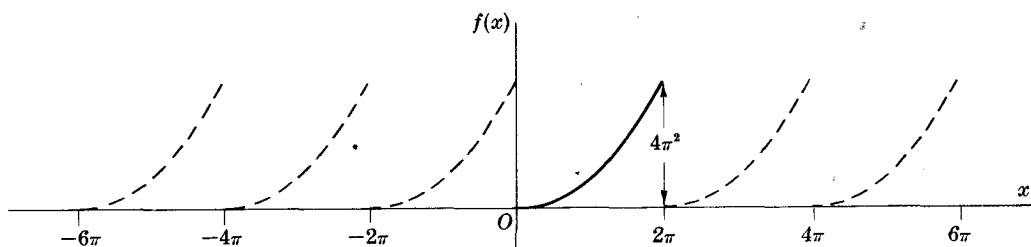


Fig. 2-7

Period = $2L = 2\pi$ and $L = \pi$. Choosing $c = 0$, we have

$$\begin{aligned} a_n &= \frac{1}{L} \int_c^{c+2L} f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx \, dx \\ &= \frac{1}{\pi} \left\{ (x^2) \left(\frac{\sin nx}{n} \right) - (2x) \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) \right\} \Big|_0^{2\pi} = \frac{4}{n^2}, \quad n \neq 0 \end{aligned}$$

If $n = 0$, $a_0 = \frac{1}{\pi} \int_0^{2\pi} x^2 dx = \frac{8\pi^2}{3}$.

$$\begin{aligned} b_n &= \frac{1}{L} \int_c^{c+2L} f(x) \sin \frac{n\pi x}{L} dx = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx \, dx \\ &= \frac{1}{\pi} \left\{ (x^2) \left(-\frac{\cos nx}{n} \right) - (2x) \left(-\frac{\sin nx}{n^2} \right) + (2) \left(\frac{\cos nx}{n^3} \right) \right\} \Big|_0^{2\pi} = \frac{-4\pi}{n} \end{aligned}$$

Then $f(x) = x^2 = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} \cos nx - \frac{4\pi}{n} \sin nx \right)$ for $0 < x < 2\pi$.

2.7. Using the results of Problem 2.6, prove that $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$.

At $x = 0$ the Fourier series of Problem 2.6 reduces to $\frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2}$.

But by the Dirichlet conditions, the series converges at $x = 0$ to $\frac{1}{2}(0 + 4\pi^2) = 2\pi^2$.

Hence the desired result.

ODD AND EVEN FUNCTIONS. HALF-RANGE FOURIER SERIES

2.8. Classify each of the following functions according as they are even, odd, or neither even nor odd.

(a) $f(x) = \begin{cases} 2 & 0 < x < 3 \\ -2 & -3 < x < 0 \end{cases}$ Period = 6

From Fig. 2-8 below it is seen that $f(-x) = -f(x)$, so that the function is odd.

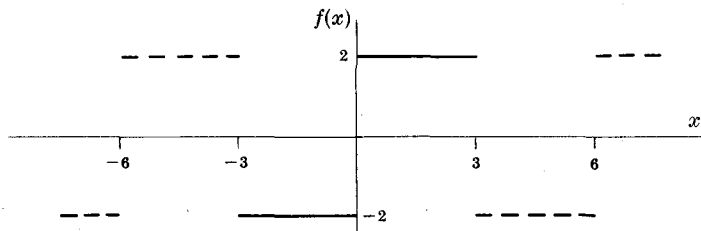


Fig. 2-8

$$(b) \quad f(x) = \begin{cases} \cos x & 0 < x < \pi \\ 0 & \pi < x < 2\pi \end{cases} \quad \text{Period} = 2\pi$$

From Fig. 2-9 below it is seen that the function is neither even nor odd.

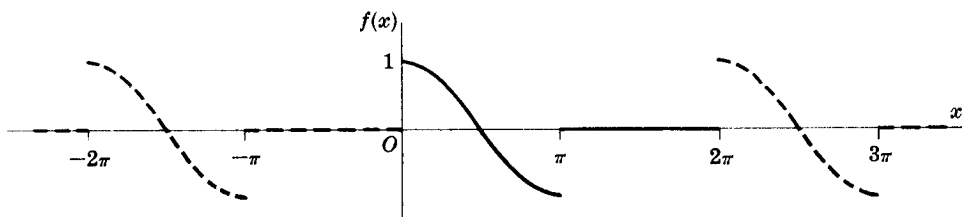


Fig. 2-9

$$(c) \quad f(x) = x(10 - x), \quad 0 < x < 10, \quad \text{Period} = 10.$$

From Fig. 2-10 below the function is seen to be even.

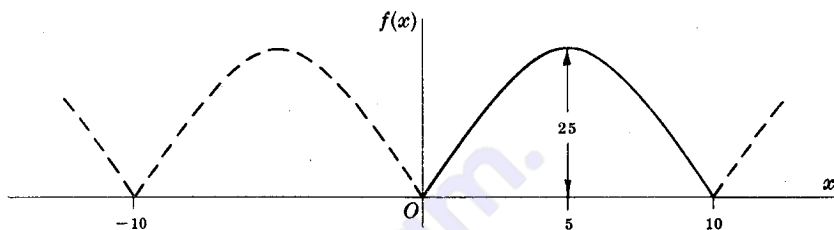


Fig. 2-10

2.9. Show that an even function can have no sine terms in its Fourier expansion.

Method 1.

No sine terms appear if $b_n = 0$, $n = 1, 2, 3, \dots$. To show this, let us write

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx = \frac{1}{L} \int_{-L}^0 f(x) \sin \frac{n\pi x}{L} dx + \frac{1}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx \quad (1)$$

If we make the transformation $x = -u$ in the first integral on the right of (1), we obtain

$$\begin{aligned} \frac{1}{L} \int_{-L}^0 f(x) \sin \frac{n\pi x}{L} dx &= \frac{1}{L} \int_0^L f(-u) \sin \left(-\frac{n\pi u}{L} \right) du = -\frac{1}{L} \int_0^L f(-u) \sin \frac{n\pi u}{L} du \\ &= -\frac{1}{L} \int_0^L f(u) \sin \frac{n\pi u}{L} du = -\frac{1}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx \end{aligned} \quad (2)$$

where we have used the fact that for an even function $f(-u) = f(u)$ and in the last step that the dummy variable of integration u can be replaced by any other symbol, in particular x . Thus from (1), using (2), we have

$$b_n = -\frac{1}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx + \frac{1}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx = 0$$

Method 2.

Assuming convergence

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

Then

$$f(-x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} - b_n \sin \frac{n\pi x}{L} \right)$$

If $f(x)$ is even, $f(-x) = f(x)$. Hence

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} - b_n \sin \frac{n\pi x}{L} \right)$$

and so
$$\sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} = 0, \quad \text{i.e.} \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}$$

and no sine terms appear. This method is weaker than Method 1 since convergence is assumed.

In a similar manner we can show that an odd function has no cosine terms (or constant term) in its Fourier expansion.

2.10. If $f(x)$ is even, show that (a) $a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$, (b) $b_n = 0$.

$$(a) \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{L} \int_{-L}^0 f(x) \cos \frac{n\pi x}{L} dx + \frac{1}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

Letting $x = -u$,

$$\frac{1}{L} \int_{-L}^0 f(x) \cos \frac{n\pi x}{L} dx = \frac{1}{L} \int_0^L f(-u) \cos \left(\frac{-n\pi u}{L} \right) du = \frac{1}{L} \int_0^L f(u) \cos \frac{n\pi u}{L} du$$

since by definition of an even function $f(-u) = f(u)$. Then

$$a_n = \frac{1}{L} \int_0^L f(u) \cos \frac{n\pi u}{L} du + \frac{1}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx$$

(b) This follows by Method 1 of Problem 2.9.

2.11. Expand $f(x) = \sin x$, $0 < x < \pi$, in a Fourier cosine series.

A Fourier series consisting of cosine terms alone is obtained only for an even function. Hence we extend the definition of $f(x)$ so that it becomes even (dashed part of Fig. 2-11). With this extension, $f(x)$ is defined in an interval of length 2π . Taking the period as 2π , we have $2L = 2\pi$, so that $L = \pi$.

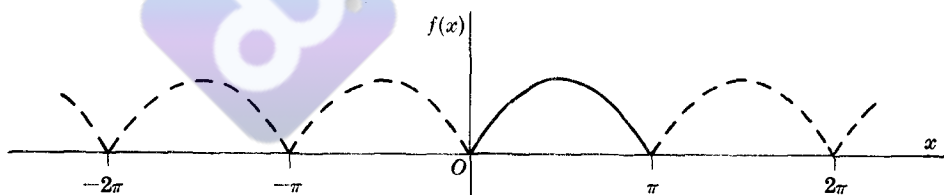


Fig. 2-11

By Problem 2.10, $b_n = 0$ and

$$\begin{aligned} a_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx \, dx \\ &= \frac{1}{\pi} \int_0^{\pi} \{\sin(x+nx) + \sin(x-nx)\} \, dx = \frac{1}{\pi} \left\{ -\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right\} \Big|_0^{\pi} \\ &= \frac{1}{\pi} \left\{ \frac{1 - \cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi - 1}{n-1} \right\} = \frac{1}{\pi} \left\{ -\frac{1 + \cos n\pi}{n+1} - \frac{1 + \cos n\pi}{n-1} \right\} \\ &= \frac{-2(1 + \cos n\pi)}{\pi(n^2 - 1)} \quad \text{if } n \neq 1 \end{aligned}$$

$$\text{For } n = 1, \quad a_1 = \frac{2}{\pi} \int_0^{\pi} \sin x \cos x \, dx = \frac{2}{\pi} \left. \frac{\sin^2 x}{2} \right|_0^{\pi} = 0.$$

Then

$$\begin{aligned} f(x) &= \frac{2}{\pi} - \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{(1 + \cos n\pi)}{n^2 - 1} \cos nx \\ &= \frac{2}{\pi} - \frac{4}{\pi} \left(\frac{\cos 2x}{2^2 - 1} + \frac{\cos 4x}{4^2 - 1} + \frac{\cos 6x}{6^2 - 1} + \dots \right) \end{aligned}$$

2.12. Expand $f(x) = x$, $0 < x < 2$, in a half-range (a) sine series, (b) cosine series.

- (a) Extend the definition of the given function to that of the odd function of period 4 shown in Fig. 2-12 below. This is sometimes called the *odd extension* of $f(x)$. Then $2L = 4$, $L = 2$.

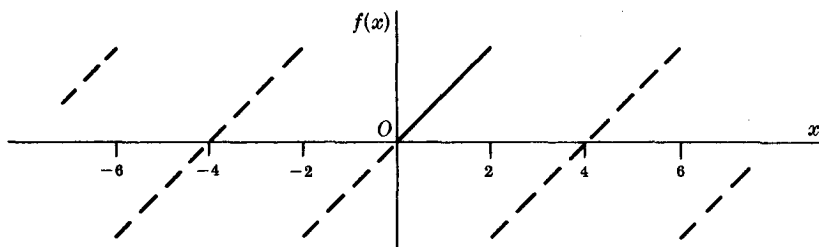


Fig. 2-12

Thus $a_n = 0$ and

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx = \frac{2}{2} \int_0^2 x \sin \frac{n\pi x}{2} dx \\ &= \left\{ (x) \left(\frac{-2}{n\pi} \cos \frac{n\pi x}{2} \right) - (1) \left(\frac{-4}{n^2 \pi^2} \sin \frac{n\pi x}{2} \right) \right\} \bigg|_0^2 = \frac{-4}{n\pi} \cos n\pi \end{aligned}$$

Then

$$\begin{aligned} f(x) &= \sum_{n=1}^{\infty} \frac{-4}{n\pi} \cos n\pi \sin \frac{n\pi x}{2} \\ &= \frac{4}{\pi} \left(\sin \frac{\pi x}{2} - \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} - \dots \right) \end{aligned}$$

- (b) Extend the definition of $f(x)$ to that of the even function of period 4 shown in Fig. 2-13 below. This is the *even extension* of $f(x)$. Then $2L = 4$, $L = 2$.

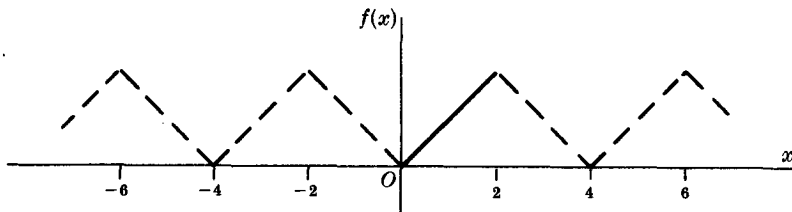


Fig. 2-13

Thus $b_n = 0$,

$$\begin{aligned} a_n &= \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx = \frac{2}{2} \int_0^2 x \cos \frac{n\pi x}{2} dx \\ &= \left\{ (x) \left(\frac{2}{n\pi} \sin \frac{n\pi x}{2} \right) - (1) \left(\frac{-4}{n^2 \pi^2} \cos \frac{n\pi x}{2} \right) \right\} \bigg|_0^2 \\ &= \frac{-4}{n^2 \pi^2} (\cos n\pi - 1) \quad \text{if } n \neq 0 \end{aligned}$$

If $n = 0$, $a_0 = \int_0^2 x dx = 2$.

$$\begin{aligned} \text{Then} \quad f(x) &= 1 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} (\cos n\pi - 1) \cos \frac{n\pi x}{2} \\ &= 1 - \frac{8}{\pi^2} \left(\cos \frac{\pi x}{2} + \frac{1}{3^2} \cos \frac{3\pi x}{2} + \frac{1}{5^2} \cos \frac{5\pi x}{2} + \dots \right) \end{aligned}$$

It should be noted that although both series of (a) and (b) represent $f(x)$ in the interval $0 < x < 2$, the second series converges more rapidly.

PARSEVAL'S IDENTITY

2.13. Assuming that the Fourier series corresponding to $f(x)$ converges uniformly to $f(x)$ in $(-L, L)$, prove Parseval's identity

$$\frac{1}{L} \int_{-L}^L \{f(x)\}^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$$

where the integral is assumed to exist.

If $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$, then multiplying by $f(x)$ and integrating term by term from $-L$ to L (which is justified since the series is uniformly convergent), we obtain

$$\begin{aligned} \int_{-L}^L \{f(x)\}^2 dx &= \frac{a_0}{2} \int_{-L}^L f(x) dx + \sum_{n=1}^{\infty} \left\{ a_n \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx + b_n \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \right\} \\ &= \frac{a_0^2}{2} L + L \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \end{aligned} \quad (1)$$

where we have used the results

$$\int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx = La_n, \quad \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx = Lb_n, \quad \int_{-L}^L f(x) dx = La_0 \quad (2)$$

obtained from the Fourier coefficients.

The required result follows on dividing both sides of (1) by L . Parseval's identity is valid under less restrictive conditions than imposed here. In Chapter 3 we shall discuss the significance of Parseval's identity in connection with generalizations of Fourier series known as *orthonormal series*.

2.14. (a) Write Parseval's identity corresponding to the Fourier series of Problem 2.12(b).

(b) Determine from (a) the sum S of the series $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots + \frac{1}{n^4} + \dots$.

(a) Here $L = 2$; $a_0 = 2$; $a_n = \frac{4}{n^2 \pi^2} (\cos n\pi - 1)$, $n \neq 0$; $b_n = 0$.

Then Parseval's identity becomes

$$\begin{aligned} \frac{1}{2} \int_{-2}^2 \{f(x)\}^2 dx &= \frac{1}{2} \int_{-2}^2 x^2 dx = \frac{(2)^2}{2} + \sum_{n=1}^{\infty} \frac{16}{n^4 \pi^4} (\cos n\pi - 1)^2 \\ \text{or } \frac{8}{3} &= 2 + \frac{64}{\pi^4} \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right), \quad \text{i.e. } \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}. \end{aligned}$$

$$\begin{aligned} (b) \quad S &= \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots = \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right) + \left(\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots \right) \\ &= \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right) + \frac{1}{2^4} \left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \right) \\ &= \frac{\pi^4}{96} + \frac{S}{16}, \quad \text{from which } S = \frac{\pi^4}{90} \end{aligned}$$

2.15. Prove that for all positive integers M ,

$$\frac{a_0^2}{2} + \sum_{n=1}^M (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^L \{f(x)\}^2 dx$$

where a_n and b_n are the Fourier coefficients corresponding to $f(x)$, and $f(x)$ is assumed piecewise continuous in $(-L, L)$.

$$\text{Let} \quad S_M(x) = \frac{a_0}{2} + \sum_{n=1}^M \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \quad (1)$$

For $M = 1, 2, 3, \dots$ this is the sequence of partial sums of the Fourier series corresponding to $f(x)$.

$$\text{We have} \quad \int_{-L}^L \{f(x) - S_M(x)\}^2 dx \geq 0 \quad (2)$$

since the integrand is non-negative. Expanding the integrand, we obtain

$$2 \int_{-L}^L f(x) S_M(x) dx - \int_{-L}^L S_M^2(x) dx \leq \int_{-L}^L \{f(x)\}^2 dx \quad (3)$$

Multiplying both sides of (1) by $2f(x)$ and integrating from $-L$ to L , using equations (2) of Problem 2.13, gives

$$2 \int_{-L}^L f(x) S_M(x) dx = 2L \left\{ \frac{a_0^2}{2} + \sum_{n=1}^M (a_n^2 + b_n^2) \right\} \quad (4)$$

Also, squaring (1) and integrating from $-L$ to L , using Problem 2.3, we find

$$\int_{-L}^L S_M^2(x) dx = L \left\{ \frac{a_0^2}{2} + \sum_{n=1}^M (a_n^2 + b_n^2) \right\} \quad (5)$$

Substitution of (4) and (5) into (3) and dividing by L yields the required result.

Taking the limit as $M \rightarrow \infty$, we obtain *Bessel's inequality*

$$\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \leq \frac{1}{L} \int_{-L}^L \{f(x)\}^2 dx \quad (6)$$

If the equality holds, we have Parseval's identity (Problem 2.13).

We can think of $S_M(x)$ as representing an *approximation* to $f(x)$, while the left hand side of (2), divided by $2L$, represents the *mean square error* of the approximation. Parseval's identity indicates that as $M \rightarrow \infty$ the mean square error approaches zero, while Bessel's inequality indicates the possibility that this mean square error does not approach zero.

The results are connected with the idea of *completeness*. If, for example, we were to leave out one or more terms in a Fourier series ($\cos 4\pi x/L$, say), we could never get the mean square error to approach zero, no matter how many terms we took. We shall return to these ideas from a generalized viewpoint in Chapter 3.

INTEGRATION AND DIFFERENTIATION OF FOURIER SERIES

2.16. (a) Find a Fourier series for $f(x) = x^2$, $0 < x < 2$, by integrating the series of Problem 2.12(a). (b) Use (a) to evaluate the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$.

(a) From Problem 2.12(a),

$$x = \frac{4}{\pi} \left(\sin \frac{\pi x}{2} - \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} - \dots \right) \quad (1)$$

Integrating both sides from 0 to x (applying Theorem 2-5, page 24) and multiplying by 2, we find

$$x^2 = C - \frac{16}{\pi^2} \left(\cos \frac{\pi x}{2} - \frac{1}{2^2} \cos \frac{2\pi x}{2} + \frac{1}{3^2} \cos \frac{3\pi x}{2} - \dots \right) \quad (2)$$

$$\text{where} \quad C = \frac{16}{\pi^2} \left(1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right).$$

- (b) To determine C in another way, note that (2) represents the Fourier cosine series for x^2 in $0 < x < 2$. Then since $L = 2$ in this case,

$$C = \frac{a_0}{2} = \frac{1}{L} \int_0^L f(x) dx = \frac{1}{2} \int_0^2 x^2 dx = \frac{4}{3}$$

Then from the value of C in (a), we have

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{16} \cdot \frac{4}{3} = \frac{\pi^2}{12}$$

2.17. Show that term by term differentiation of the series in Problem 2.12(a) is not valid.

Term by term differentiation yields $2 \left(\cos \frac{\pi x}{2} - \cos \frac{2\pi x}{2} + \cos \frac{3\pi x}{2} - \cdots \right)$. Since the n th term of this series does not approach 0, the series does not converge for any value of x .

CONVERGENCE OF FOURIER SERIES

2.18. Prove that (a) $\frac{1}{2} + \cos t + \cos 2t + \cdots + \cos Mt = \frac{\sin(M + \frac{1}{2})t}{2 \sin \frac{1}{2}t}$

$$(b) \frac{1}{\pi} \int_0^{\pi} \frac{\sin(M + \frac{1}{2})t}{2 \sin \frac{1}{2}t} dt = \frac{1}{2}, \quad \frac{1}{\pi} \int_{-\pi}^0 \frac{\sin(M + \frac{1}{2})t}{2 \sin \frac{1}{2}t} dt = \frac{1}{2}.$$

(a) We have $\cos nt \sin \frac{1}{2}t = \frac{1}{2} \{ \sin(n + \frac{1}{2})t - \sin(n - \frac{1}{2})t \}$. Then summing from $n = 1$ to M ,

$$\begin{aligned} \sin \frac{1}{2}t \{ \cos t + \cos 2t + \cdots + \cos Mt \} &= (\sin \frac{3}{2}t - \sin \frac{1}{2}t) + (\sin \frac{5}{2}t - \sin \frac{3}{2}t) \\ &\quad + \cdots + [\sin(M + \frac{1}{2})t - \sin(M - \frac{1}{2})t] \\ &= \frac{1}{2} \{ \sin(M + \frac{1}{2})t - \sin \frac{1}{2}t \} \end{aligned}$$

On dividing by $\sin \frac{1}{2}t$ and adding $\frac{1}{2}$, the required result follows.

(b) Integrate the result in (a) from 0 to π and $-\pi$ to 0 respectively. This gives the required results, since the integrals of all the cosine terms are zero.

2.19. Prove that $\lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} f(x) \sin nx dx = \lim_{n \rightarrow \infty} \int_{-\pi}^{\pi} f(x) \cos nx dx = 0$ if $f(x)$ is piecewise continuous.

This follows at once from Problem 2.15, since if the series $\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$ is convergent, $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = 0$.

The result is sometimes called *Riemann's theorem*.

2.20. Prove that $\lim_{M \rightarrow \infty} \int_{-\pi}^{\pi} f(x) \sin(M + \frac{1}{2})x dx = 0$ if $f(x)$ is piecewise continuous. We have

$$\int_{-\pi}^{\pi} f(x) \sin(M + \frac{1}{2})x dx = \int_{-\pi}^{\pi} \{f(x) \sin \frac{1}{2}x\} \cos Mx dx + \int_{-\pi}^{\pi} \{f(x) \cos \frac{1}{2}x\} \sin Mx dx$$

Then the required result follows at once by using the result of Problem 2.19, with $f(x)$ replaced by $f(x) \sin \frac{1}{2}x$ and $f(x) \cos \frac{1}{2}x$, respectively, which are piecewise continuous if $f(x)$ is.

The result can also be proved when the integration limits are a and b instead of $-\pi$ and π .

2.21. Assuming that $L = \pi$, i.e. that the Fourier series corresponding to $f(x)$ has period $2L = 2\pi$, show that

$$S_M(x) = \frac{a_0}{2} + \sum_{n=1}^M (a_n \cos nx + b_n \sin nx) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t+x) \frac{\sin(M+\frac{1}{2})t}{2 \sin \frac{1}{2}t} dt$$

Using the formulas for the Fourier coefficients with $L = \pi$, we have

$$\begin{aligned} a_n \cos nx + b_n \sin nx &= \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(u) \cos nu \, du \right) \cos nx + \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(u) \sin nu \, du \right) \sin nx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(u) (\cos nu \cos nx + \sin nu \sin nx) \, du \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(u) \cos n(u-x) \, du \end{aligned}$$

$$\text{Also,} \quad \frac{a_0}{2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(u) \, du$$

$$\begin{aligned} \text{Then} \quad S_M(x) &= \frac{a_0}{2} + \sum_{n=1}^M (a_n \cos nx + b_n \sin nx) \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(u) \, du + \frac{1}{\pi} \sum_{n=1}^M \int_{-\pi}^{\pi} f(u) \cos n(u-x) \, du \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(u) \left\{ \frac{1}{2} + \sum_{n=1}^M \cos n(u-x) \right\} \, du \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(u) \frac{\sin(M+\frac{1}{2})(u-x)}{2 \sin \frac{1}{2}(u-x)} \, du \end{aligned}$$

using Problem 2.18. Letting $u-x = t$, we have

$$S_M(x) = \frac{1}{\pi} \int_{-\pi-x}^{\pi-x} f(t+x) \frac{\sin(M+\frac{1}{2})t}{2 \sin \frac{1}{2}t} dt$$

Since the integrand has period 2π , we can replace the interval $-\pi-x, \pi-x$ by any other interval of length 2π , in particular $-\pi, \pi$. Thus we obtain the required result.

2.22. Prove that

$$\begin{aligned} S_M(x) - \left(\frac{f(x+0) + f(x-0)}{2} \right) &= \frac{1}{\pi} \int_{-\pi}^0 \frac{f(t+x) - f(x-0)}{2 \sin \frac{1}{2}t} \sin(M+\frac{1}{2})t \, dt \\ &\quad + \frac{1}{\pi} \int_0^{\pi} \frac{f(t+x) - f(x+0)}{2 \sin \frac{1}{2}t} \sin(M+\frac{1}{2})t \, dt \end{aligned}$$

From Problem 2.21,

$$S_M(x) = \frac{1}{\pi} \int_{-\pi}^0 f(t+x) \frac{\sin(M+\frac{1}{2})t}{2 \sin \frac{1}{2}t} dt + \frac{1}{\pi} \int_0^{\pi} f(t+x) \frac{\sin(M+\frac{1}{2})t}{2 \sin \frac{1}{2}t} dt \quad (1)$$

Multiplying the integrals of Problem 2.18(b) by $f(x-0)$ and $f(x+0)$ respectively,

$$\frac{f(x+0) + f(x-0)}{2} = \frac{1}{\pi} \int_{-\pi}^0 f(x-0) \frac{\sin(M+\frac{1}{2})t}{2 \sin \frac{1}{2}t} dt + \frac{1}{\pi} \int_0^{\pi} f(x+0) \frac{\sin(M+\frac{1}{2})t}{2 \sin \frac{1}{2}t} dt \quad (2)$$

Subtracting (2) from (1) yields the required result.

2.23. If $f(x)$ and $f'(x)$ are piecewise continuous in $(-\pi, \pi)$, prove that

$$\lim_{M \rightarrow \infty} S_M(x) = \frac{f(x+0) + f(x-0)}{2}$$

The function $\frac{f(t+x) - f(x+0)}{2 \sin \frac{1}{2}t}$ is piecewise continuous in $0 < t \leq \pi$ because $f(x)$ is piecewise continuous.

Also,

$$\lim_{t \rightarrow 0+} \frac{f(t+x) - f(x+0)}{2 \sin \frac{1}{2}t} = \lim_{t \rightarrow 0+} \frac{f(x+t) - f(x+0)}{t} \cdot \frac{\frac{1}{2}t}{\sin \frac{1}{2}t} = \lim_{t \rightarrow 0+} \frac{f(x+t) - f(x+0)}{t}$$

exists, since by hypothesis $f'(x)$ is piecewise continuous, so that the right-hand derivative of $f(x)$ at each x exists.

Thus $\frac{f(t+x) - f(x+0)}{2 \sin \frac{1}{2}t}$ is piecewise continuous in $0 \leq t \leq \pi$.

Similarly, $\frac{f(t+x) - f(x-0)}{2 \sin \frac{1}{2}t}$ is piecewise continuous in $-\pi \leq t \leq 0$.

Then from Problems 2.20 and 2.22, we have

$$\lim_{M \rightarrow \infty} \left\{ S_M(x) - \frac{f(x+0) + f(x-0)}{2} \right\} = 0 \quad \text{or} \quad \lim_{M \rightarrow \infty} S_M(x) = \frac{f(x+0) + f(x-0)}{2}$$

DOUBLE FOURIER SERIES

2.24. Obtain formally the Fourier coefficients (15), page 24, for the double Fourier sine series (14).

$$\text{Suppose that} \quad f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin \frac{m\pi x}{L_1} \sin \frac{n\pi y}{L_2} \quad (1)$$

We can write this as

$$f(x, y) = \sum_{m=1}^{\infty} C_m \sin \frac{m\pi x}{L_1} \quad (2)$$

where

$$C_m = \sum_{n=1}^{\infty} B_{mn} \sin \frac{n\pi y}{L_2} \quad (3)$$

Now we can consider (2) as a Fourier series in which y is kept constant so that the Fourier coefficients C_m are given by

$$C_m = \frac{2}{L_1} \int_0^{L_1} f(x, y) \sin \frac{m\pi x}{L_1} dx \quad (4)$$

On noting that C_m is a function of y , we see that (3) can be considered as a Fourier series for which the coefficients B_{mn} are given by

$$B_{mn} = \frac{2}{L_2} \int_0^{L_2} C_m \sin \frac{n\pi y}{L_2} dy \quad (5)$$

If we now use (4) in (5), we see that

$$B_{mn} = \frac{4}{L_1 L_2} \int_0^{L_1} \int_0^{L_2} f(x, y) \sin \frac{m\pi x}{L_1} \sin \frac{n\pi y}{L_2} dx dy \quad (6)$$

APPLICATIONS TO HEAT CONDUCTION

2.25. Find the temperature of the bar in Problem 1.23, page 15, if the initial temperature is 25°C .

This problem is identical with Problem 1.23, except that to satisfy the initial condition $u(x, 0) = 25$ it is necessary to superimpose an *infinite number of solutions*, i.e. we must replace equation (1) of that problem by

$$u(x, t) = \sum_{m=1}^{\infty} B_m e^{-2m^2\pi^2 t/9} \sin \frac{m\pi x}{3}$$

which for $t = 0$ yields

$$25 = \sum_{m=1}^{\infty} B_m \sin \frac{m\pi x}{3} \quad 0 < x < 3$$

This amounts to expanding 25 in a *Fourier sine series*. By the methods of this chapter we then find

$$B_m = \frac{2}{L} \int_0^L f(x) \sin \frac{m\pi x}{L} dx = \frac{2}{3} \int_0^3 25 \sin \frac{m\pi x}{3} dx = \frac{50(1 - \cos m\pi)}{m\pi}$$

The result can be written

$$\begin{aligned} u(x, t) &= \sum_{m=1}^{\infty} \frac{50(1 - \cos m\pi)}{m\pi} e^{-2m^2\pi^2 t/9} \sin \frac{m\pi x}{3} \\ &= \frac{100}{\pi} \left\{ e^{-2\pi^2 t/9} \sin \frac{\pi x}{3} + \frac{1}{3} e^{-2\pi^2 t} \sin \pi x + \dots \right\} \end{aligned}$$

which can be verified as the required solution.

This problem illustrates the importance of Fourier series in solving boundary value problems.

2.26. Solve the boundary value problem

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = 10, \quad u(3, t) = 40, \quad u(x, 0) = 25, \quad |u(x, t)| < M$$

This is the same as Problem 1.23, page 15, except that the ends of the bar are at temperatures 10°C and 40°C instead of 0°C . As far as the solution goes, this makes quite a difference since we can no longer conclude that $A = 0$ and $\lambda = m\pi/3$ as in that problem.

To solve the present problem assume that $u(x, t) = v(x, t) + \psi(x)$ where $\psi(x)$ is to be suitably determined. In terms of $v(x, t)$ the boundary value problem becomes

$$\frac{\partial v}{\partial t} = 2 \frac{\partial^2 v}{\partial x^2} + 2\psi''(x), \quad v(0, t) + \psi(0) = 10, \quad v(3, t) + \psi(3) = 40, \quad v(x, 0) + \psi(x) = 25, \quad |v(x, t)| < M$$

This can be simplified by choosing

$$\psi''(x) = 0, \quad \psi(0) = 10, \quad \psi(3) = 40$$

from which we find $\psi(x) = 10x + 10$, so that the resulting boundary value problem is

$$\frac{\partial v}{\partial t} = 2 \frac{\partial^2 v}{\partial x^2}, \quad v(0, t) = 0, \quad v(3, t) = 0, \quad v(x, 0) = 15 - 10x$$

As in Problem 1.23 we find from the first three of these,

$$v(x, t) = \sum_{m=1}^{\infty} B_m e^{-2m^2\pi^2 t/9} \sin \frac{m\pi x}{3}$$

The last condition yields

$$15 - 10x = \sum_{m=1}^{\infty} B_m \sin \frac{m\pi x}{3}$$

from which

$$B_m = \frac{2}{3} \int_0^3 (15 - 10x) \sin \frac{m\pi x}{3} dx = \frac{30}{m\pi} (\cos m\pi - 1)$$

Since $u(x, t) = v(x, t) + \psi(x)$, we have finally

$$u(x, t) = 10x + 10 + \sum_{m=1}^{\infty} \frac{30}{m\pi} (\cos m\pi - 1) e^{-2m^2\pi^2 t/9} \sin \frac{m\pi x}{3}$$

as the required solution.

The term $10x + 10$ is the *steady-state temperature*, i.e. the temperature after a long time has elapsed.

2.27. A bar of length L whose entire surface is insulated including its ends at $x = 0$ and $x = L$ has initial temperature $f(x)$. Determine the subsequent temperature of the bar.

In this case, the boundary value problem is

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \tag{1}$$

$$|u(x, t)| < M, \quad u_x(0, t) = 0, \quad u_x(L, t) = 0, \quad u(x, 0) = f(x) \tag{2}$$

Letting $u = XT$ in (1) and separating the variables, we find

$$XT' = \kappa X''T \quad \text{or} \quad \frac{T'}{\kappa T} = \frac{X''}{X}$$

Setting each side equal to the constant $-\lambda^2$, we find

$$T' + \kappa\lambda^2 T = 0, \quad X'' + \lambda^2 X = 0$$

so that

$$X = a \cos \lambda x + b \sin \lambda x, \quad T = ce^{-\kappa\lambda^2 t}$$

A solution is thus given by

$$u(x, t) = e^{-\kappa\lambda^2 t} (A \cos \lambda x + B \sin \lambda x)$$

where $A = ac$, $B = bc$.

From $u_x(0, t) = 0$ we have $B = 0$ so that

$$u(x, t) = Ae^{-\kappa\lambda^2 t} \cos \lambda x$$

Then from $u_x(L, t) = 0$ we have

$$\sin \lambda L = 0 \quad \text{or} \quad \lambda L = m\pi, \quad m = 0, 1, 2, 3, \dots$$

Thus

$$u(x, t) = Ae^{-\kappa m^2 \pi^2 t / L^2} \cos \frac{m\pi x}{L} \quad m = 0, 1, 2, \dots$$

To satisfy the last condition, $u(x, 0) = f(x)$, we use the superposition principle to obtain

$$u(x, t) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m e^{-\kappa m^2 \pi^2 t / L^2} \cos \frac{m\pi x}{L}$$

Then from $u(x, 0) = f(x)$ we see that

$$f(x) = \frac{A_0}{2} + \sum_{m=1}^{\infty} A_m e^{-\kappa m^2 \pi^2 t / L^2} \cos \frac{m\pi x}{L}$$

Thus, from Fourier series we find

$$A_m = \frac{2}{L} \int_0^L f(x) \cos \frac{m\pi x}{L} dx$$

$$\text{and} \quad u(x, t) = \frac{1}{L} \int_0^L f(x) dx + \frac{2}{L} \sum_{m=1}^{\infty} \left(e^{-\kappa m^2 \pi^2 t / L^2} \cos \frac{m\pi x}{L} \right) \int_0^L f(x) \cos \frac{m\pi x}{L} dx$$

- 2.28.** A circular plate of unit radius, whose faces are insulated, has half of its boundary kept at constant temperature u_1 and the other half at constant temperature u_2 (see Fig. 2-14). Find the steady-state temperature of the plate.

In polar coordinates (ρ, ϕ) the partial differential equation for steady-state heat flow is

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} = 0 \quad (1)$$

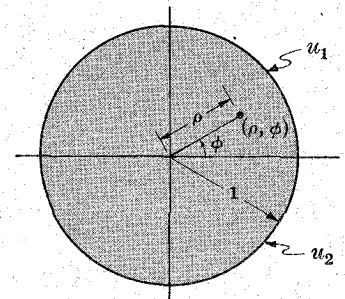


Fig. 2-14

The boundary conditions are

$$u(1, \phi) = \begin{cases} u_1 & 0 < \phi < \pi \\ u_2 & \pi < \phi < 2\pi \end{cases} \quad (2)$$

$$|u(\rho, \phi)| < M, \text{ i.e. } u \text{ is bounded in the region} \quad (3)$$

Let $u(\rho, \phi) = P\Phi$ where P is a function of ρ and Φ is a function of ϕ . Then (1) becomes

$$P''\Phi + \frac{1}{\rho}P'\Phi + \frac{1}{\rho^2}P\Phi'' = 0$$

Dividing by $P\Phi$, multiplying by ρ^2 and rearranging terms,

$$\frac{\rho^2 P''}{P} + \frac{\rho P'}{P} = -\frac{\Phi''}{\Phi}$$

Setting each side equal to λ^2 ,

$$\Phi'' + \lambda^2 \Phi = 0 \quad \rho^2 P'' + \rho P' - \lambda^2 P = 0 \quad (4)$$

The first equation in (4) has general solution

$$\Phi = A_1 \cos \lambda \phi + B_1 \sin \lambda \phi$$

By letting $P = \rho^k$ in the second equation of (4), which is a *Cauchy* or *Euler differential equation*, we find $k = \pm \lambda$; so that ρ^λ and $\rho^{-\lambda}$ are solutions. Thus we obtain the general solution

$$P = A_2 \rho^\lambda + B_2 \rho^{-\lambda} \quad (5)$$

Since $u(\rho, \phi)$ must have period 2π in ϕ , we must have $\lambda = m = 0, 1, 2, 3, \dots$

Also, since u must be bounded at $\rho = 0$, we must have $B_2 = 0$. Thus

$$u = P\Phi = A_2 \rho^m (A_1 \cos m\phi + B_1 \sin m\phi) = \rho^m (A \cos m\phi + B \sin m\phi)$$

By superposition, a solution is

$$u(\rho, \phi) = \frac{A_0}{2} + \sum_{m=1}^{\infty} \rho^m (A_m \cos m\phi + B_m \sin m\phi)$$

from which

$$u(1, \phi) = \frac{A_0}{2} + \sum_{m=1}^{\infty} (A_m \cos m\phi + B_m \sin m\phi)$$

Then from the theory of Fourier series,

$$\begin{aligned} A_m &= \frac{1}{\pi} \int_0^{2\pi} u(1, \phi) \cos m\phi \, d\phi \\ &= \frac{1}{\pi} \int_0^\pi u_1 \cos m\phi \, d\phi + \frac{1}{\pi} \int_\pi^{2\pi} u_2 \cos m\phi \, d\phi = \begin{cases} 0 & \text{if } m > 0 \\ u_1 + u_2 & \text{if } m = 0 \end{cases} \end{aligned}$$

$$\begin{aligned} B_m &= \frac{1}{\pi} \int_0^{2\pi} u(1, \phi) \sin m\phi \, d\phi \\ &= \frac{1}{\pi} \int_0^\pi u_1 \sin m\phi \, d\phi + \frac{1}{\pi} \int_\pi^{2\pi} u_2 \sin m\phi \, d\phi = \frac{(u_1 - u_2)}{m\pi} (1 - \cos m\pi) \end{aligned}$$

$$\begin{aligned} \text{Then: } u(\rho, \phi) &= \frac{u_1 + u_2}{2} + \sum_{m=1}^{\infty} \frac{(u_1 - u_2)(1 - \cos m\pi)}{m\pi} \rho^m \sin m\phi \\ &= \frac{u_1 + u_2}{2} + \frac{2(u_1 - u_2)}{\pi} (\rho \sin \phi + \frac{1}{3}\rho^3 \sin 3\phi + \frac{1}{5}\rho^5 \sin 5\phi + \dots) \\ &= \frac{u_1 + u_2}{2} + \frac{u_1 - u_2}{\pi} \tan^{-1} \left(\frac{2\rho \sin \phi}{1 - \rho^2} \right) \end{aligned}$$

on making use of Problem 2.54.

2.29. A square plate with sides of unit length has its faces insulated and its sides kept at 0°C . If the initial temperature is specified, determine the subsequent temperature at any point of the plate.

Choose a coordinate system as shown in Fig. 2-15. Then the equation for the temperature $u(x, y, t)$ at any point (x, y) at time t is

$$\frac{\partial u}{\partial t} = \kappa \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (1)$$

The boundary conditions are given by

$$|u(x, y, t)| < M$$

$$u(0, y, t) = u(1, y, t) = u(x, 0, t) = u(x, 1, t) = 0$$

$$u(x, y, 0) = f(x, y)$$

where $0 < x < 1$, $0 < y < 1$, $t > 0$.

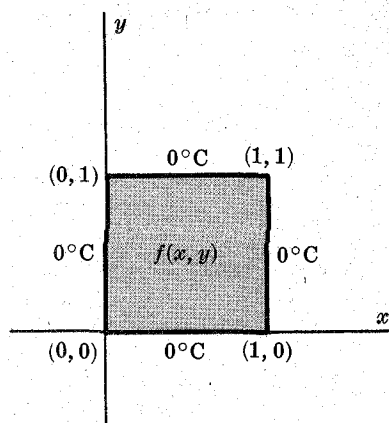


Fig. 2-15

To solve the boundary value problem let $u = XYT$, where X, Y, T are functions of x, y, t respectively. Then (1) becomes

$$XYT' = \kappa(X''YT + XY''T)$$

Dividing by κXYT yields

$$\frac{T'}{\kappa T} = \frac{X''}{X} + \frac{Y''}{Y}$$

Since the left side is a function of t alone, while the right side is a function of x and y , we see that each side must be a constant, say $-\lambda^2$ (which is needed for boundedness). Thus

$$T' + \kappa\lambda^2 T = 0 \quad \frac{X''}{X} + \frac{Y''}{Y} = -\lambda^2 \quad (2)$$

The second equation can be written as

$$\frac{X''}{X} = -\frac{Y''}{Y} - \lambda^2$$

and since the left side depends only on x while the right side depends only on y each side must be a constant, say $-\mu^2$. Thus

$$X'' + \mu^2 X = 0 \quad Y'' + (\lambda^2 - \mu^2)Y = 0 \quad (3)$$

Solutions to the two equations in (3) and the first equation in (2) are given by

$$X = a_1 \cos \mu x + b_1 \sin \mu x, \quad Y = a_2 \cos \sqrt{\lambda^2 - \mu^2} y + b_2 \sin \sqrt{\lambda^2 - \mu^2} y, \quad T = a_3 e^{-\kappa\lambda^2 t}$$

It follows that a solution to (1) is given by

$$u(x, y, t) = (a_1 \cos \mu x + b_1 \sin \mu x)(a_2 \cos \sqrt{\lambda^2 - \mu^2} y + b_2 \sin \sqrt{\lambda^2 - \mu^2} y)(a_3 e^{-\kappa\lambda^2 t})$$

From the boundary condition $u(0, y, t) = 0$ we see that $a_1 = 0$. From $u(x, 0, t) = 0$ we see that $a_2 = 0$. Thus the solution satisfying these two conditions is

$$u(x, y, t) = B e^{-\kappa\lambda^2 t} \sin \mu x \sin \sqrt{\lambda^2 - \mu^2} y$$

where we have written $B = b_1 b_2 a_3$.

From the boundary condition $u(1, y, t) = 0$ we see that $\mu = m\pi$, $m = 1, 2, 3, \dots$. From $u(x, 1, t) = 0$ we see that $\sqrt{\lambda^2 - \mu^2} = n\pi$, $n = 1, 2, 3, \dots$, or $\lambda = \sqrt{m^2 + n^2}\pi$.

It follows that a solution satisfying all the conditions except $u(x, y, 0) = f(x, y)$ is given by

$$u(x, y, t) = B e^{-\kappa(m^2 + n^2)\pi^2 t} \sin m\pi x \sin n\pi y$$

Now, by the superposition theorem we can arrive at the possible solution

$$u(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} e^{-\kappa(m^2+n^2)\pi^2 t} \sin m\pi x \sin n\pi y \quad (4)$$

Letting $t = 0$ and using the condition $u(x, y, 0) = f(x, y)$, we arrive at

$$f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin m\pi x \sin n\pi y$$

As in Problem 2.24 we then find that

$$B_{mn} = 4 \int_0^1 \int_0^1 f(x, y) \sin m\pi x \sin n\pi y \, dx \, dy \quad (5)$$

Thus the formal solution to our problem is given by (4), where the B_{mn} are determined from (5).

LAPLACE'S EQUATION

2.30. Suppose that the square plate of Problem 2.29 has three sides kept at temperature zero, while the fourth side is kept at temperature u_1 . Determine the steady-state temperature everywhere in the plate.

Choose the side having temperature u_1 to be the one where $y = 1$, as shown in Fig. 2-16. Since we wish the steady-state temperature u , which does not depend on time t , the equation is obtained from (1) of Problem 2.29 by setting $\partial u / \partial t = 0$; i.e. Laplace's equation in two dimensions:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad (1)$$

The boundary conditions are

$$u(0, y) = u(1, y) = u(x, 0) = 0, \quad u(x, 1) = u_1$$

and $|u(x, y)| < M$.

To solve this boundary value problem let $u = XY$ in (1) to obtain

$$X''Y + XY'' = 0 \quad \text{or} \quad \frac{X''}{X} = -\frac{Y''}{Y}$$

Setting each side equal to $-\lambda^2$ yields

$$X'' + \lambda^2 X = 0 \quad Y'' - \lambda^2 Y = 0$$

from which

$$X = a_1 \cos \lambda x + b_1 \sin \lambda x \quad Y = a_2 \cosh \lambda y + b_2 \sinh \lambda y$$

Then a possible solution is

$$u(x, y) = (a_1 \cos \lambda x + b_1 \sin \lambda x)(a_2 \cosh \lambda y + b_2 \sinh \lambda y)$$

From $u(0, y) = 0$ we find $a_1 = 0$. From $u(x, 0) = 0$ we find $a_2 = 0$. From $u(1, y) = 0$ we find $\lambda = m\pi$, $m = 1, 2, 3, \dots$. Thus a solution satisfying all these conditions is

$$u(x, y) = B \sin m\pi x \sinh m\pi y$$

To satisfy the last condition, $u(x, 1) = u_1$, we must first use the principle of superposition to obtain the solution

$$u(x, y) = \sum_{m=1}^{\infty} B_m \sin m\pi x \sinh m\pi y \quad (2)$$

Then from $u(x, 1) = u_1$ we must have

$$u_1 = \sum_{m=1}^{\infty} (B_m \sinh m\pi) \sin m\pi x$$

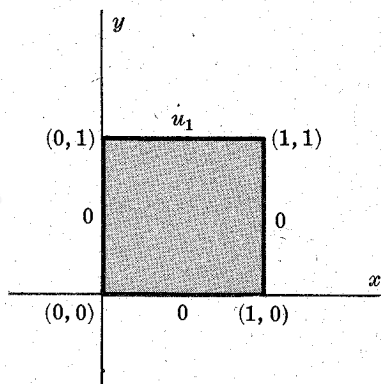


Fig. 2-16

Thus, using the theory of Fourier series,

$$B_m \sinh m\pi = 2 \int_0^1 u_1 \sin m\pi x = \frac{2u_1(1 - \cos m\pi)}{m\pi}$$

from which

$$B_m = \frac{2u_1(1 - \cos m\pi)}{m\pi \sinh m\pi} \quad (3)$$

From (2) and (3) we obtain

$$u(x, y) = \frac{2u_1}{\pi} \sum_{m=1}^{\infty} \frac{1 - \cos m\pi}{m \sinh m\pi} \sin m\pi x \sinh m\pi y$$

Note that this is a *Dirichlet problem*, since we are solving Laplace's equation $\nabla^2 u = 0$ for u inside a region $\mathcal{R}$ when u is specified on the boundary of $\mathcal{R}$.

- 2.31.** If the square plate of Problem 2.29 has its sides kept at constant temperatures u_1, u_2, u_3, u_4 , respectively, show how to determine the steady-state temperature.

The temperatures at which the sides are kept are indicated in Fig. 2-17. The fact that most of these temperatures are nonzero makes for the same type of difficulty considered in Problem 2.26. To overcome this difficulty we break the problem up into four problems of the type of Problem 2.30, where three of the four sides have temperature zero. We can then show that the solution to the given problem is the sum of solutions to the problems indicated by Figs. 2-18 to 2-21 below.

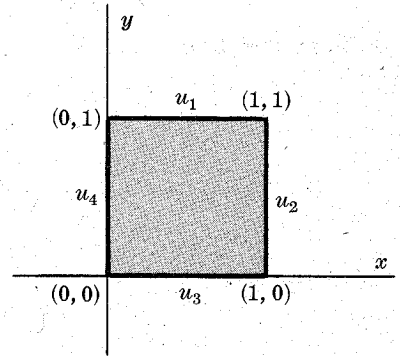


Fig. 2-17

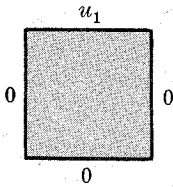


Fig. 2-18

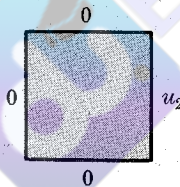


Fig. 2-19

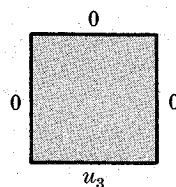


Fig. 2-20

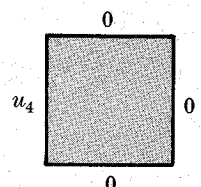


Fig. 2-21

The details are left to Problem 2.57 which provides a generalization to the case where the side temperatures may vary.

APPLICATIONS TO VIBRATING STRINGS AND MEMBRANES

- 2.32.** A string of length L is stretched between points $(0,0)$ and $(L,0)$ on the x -axis. At time $t=0$ it has a shape given by $f(x)$, $0 < x < L$, and it is released from rest. Find the displacement of the string at any later time.

The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \quad 0 < x < L, \quad t > 0$$

where $y(x, t)$ = displacement from x -axis at time t (Fig. 2-22).

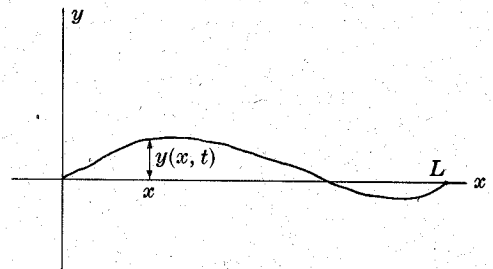


Fig. 2-22

Since the ends of the string are fixed at $x = 0$ and $x = L$,

$$y(0, t) = y(L, t) = 0 \quad t > 0$$

Since the initial shape of the string is given by $f(x)$,

$$y(x, 0) = f(x) \quad 0 < x < L$$

Since the initial velocity of the string is zero,

$$y_t(x, 0) = 0 \quad 0 < x < L$$

To solve this boundary value problem, let $y = XT$ as usual.

$$\text{Then} \quad XT'' = a^2 X''T \quad \text{or} \quad T''/a^2 T = X''/X$$

Calling the separation constant $-\lambda^2$, we have

$$T'' + \lambda^2 a^2 T = 0 \quad X'' + \lambda^2 X = 0$$

$$\text{and} \quad T = A_1 \sin \lambda a t + B_1 \cos \lambda a t \quad X = A_2 \sin \lambda x + B_2 \cos \lambda x$$

A solution is thus given by

$$y(x, t) = XT = (A_2 \sin \lambda x + B_2 \cos \lambda x)(A_1 \sin \lambda a t + B_1 \cos \lambda a t)$$

From $y(0, t) = 0$, $A_2 = 0$. Then

$$y(x, t) = B_2 \sin \lambda x (A_1 \sin \lambda a t + B_1 \cos \lambda a t) = \sin \lambda x (A \sin \lambda a t + B \cos \lambda a t)$$

From $y(L, t) = 0$, we have $\sin \lambda L (A \sin \lambda a t + B \cos \lambda a t) = 0$, so that $\sin \lambda L = 0$, $\lambda L = m\pi$ or $\lambda = m\pi/L$, since the second factor must not be equal to zero. Now,

$$y_t(x, t) = \sin \lambda x (A \lambda a \cos \lambda a t - B \lambda a \sin \lambda a t)$$

and $y_t(x, 0) = (\sin \lambda x)(A \lambda a) = 0$, from which $A = 0$. Thus

$$y(x, t) = B \sin \frac{m\pi x}{L} \cos \frac{m\pi a t}{L}$$

To satisfy the condition $y(x, 0) = f(x)$, it will be necessary to superpose solutions. This yields

$$y(x, t) = \sum_{m=1}^{\infty} B_m \sin \frac{m\pi x}{L} \cos \frac{m\pi a t}{L}$$

Then

$$y(x, 0) = f(x) = \sum_{m=1}^{\infty} B_m \sin \frac{m\pi x}{L}$$

and from the theory of Fourier series,

$$B_m = \frac{2}{L} \int_0^L f(x) \sin \frac{m\pi x}{L} dx$$

The final result is

$$y(x, t) = \sum_{m=1}^{\infty} \left(\frac{2}{L} \int_0^L f(x) \sin \frac{m\pi x}{L} dx \right) \sin \frac{m\pi x}{L} \cos \frac{m\pi a t}{L}$$

which can be verified as the solution.

The terms in this series represent the *natural* or *normal modes of vibration*. The frequency of the m th normal mode f_m is obtained from the term involving $\cos \frac{m\pi a t}{L}$ and is given by

$$2\pi f_m = \frac{m\pi a}{L} \quad \text{or} \quad f_m = \frac{ma}{2L} = \frac{m}{2L} \sqrt{\frac{\tau}{\mu}}$$

Since all the frequencies are integer multiples of the lowest frequency f_1 , the vibrations of the string will yield a musical tone, as in the case of a violin or piano string. The first three normal modes are illustrated in Fig. 2-23. As time increases the shapes of these modes vary from curves shown solid to curves shown dashed and then back again, the time for a complete cycle being the

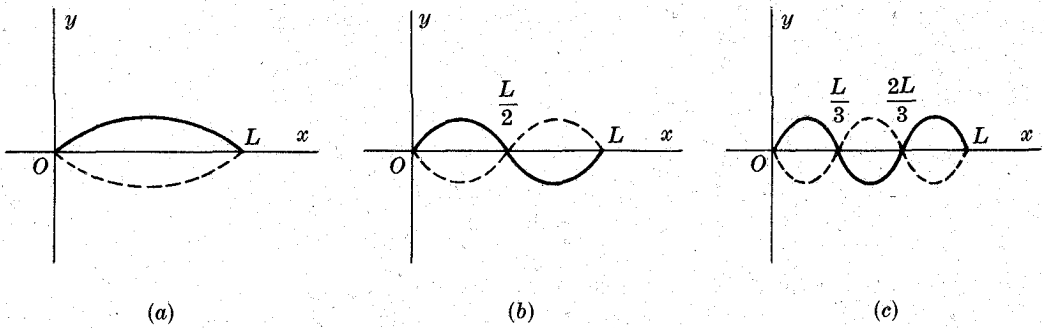


Fig. 2-23

period and the reciprocal of this period being the frequency. We call the mode (a) the *fundamental mode* or *first harmonic*, while (b) and (c) are called the *second* and *third harmonic* (or *first* and *second overtone*), respectively.

- 2.33. A square drumhead or membrane has edges which are fixed and of unit length. If the drumhead is given an initial transverse displacement and then released, determine the subsequent motion.

Assume a coordinate system as in Fig. 2-24 and suppose that the transverse displacement from the equilibrium position (i.e. the perpendicular distance from the xy -plane) of any point (x, y) at time t is given by $z(x, y, t)$.

Then the equation for the transverse motion is

$$\frac{\partial^2 z}{\partial t^2} = a^2 \left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \right) \quad (1)$$

where $a^2 = \tau/\mu$, the quantity τ being the tension per unit length along any line drawn in the drumhead, and μ is the mass per unit area.

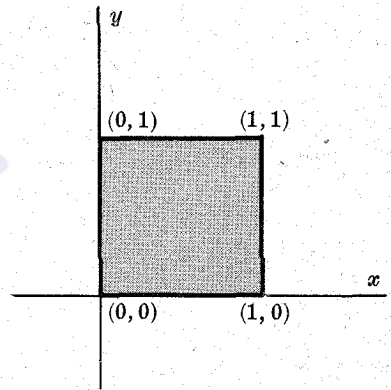


Fig. 2-24

Assuming the initial transverse displacement to be $f(x, y)$ and the initial velocity to be zero, we have the conditions

$$\begin{aligned} |z(x, y, t)| &< M, \quad z(0, y, t) = z(1, y, t) = z(x, 0, t) = z(x, 1, t) = 0, \\ z(x, y, 0) &= f(x, y), \quad z_t(x, y, 0) = 0 \end{aligned}$$

where we have in addition expressed the condition for boundedness and the conditions that the edges do not move.

To solve the boundary value problem we let $z = XYT$ in (1), where X, Y, T are functions of x, y , and t respectively. Then, proceeding as in Problem 2.29, we find

$$\frac{T''}{a^2 T} = \frac{X''}{X} + \frac{Y''}{Y}$$

and we are led exactly as in Problem 2.29 to the equation

$$T'' + \lambda^2 a^2 T = 0, \quad X'' + \mu^2 X = 0, \quad Y'' + (\lambda^2 - \mu^2) Y = 0$$

Solutions of these equations are

$$\begin{aligned} X &= a_1 \cos \mu x + b_1 \sin \mu x, & Y &= a_2 \cos \sqrt{\lambda^2 - \mu^2} y + b_2 \sin \sqrt{\lambda^2 - \mu^2} y \\ T &= a_3 \cos \lambda at + b_3 \sin \lambda at \end{aligned}$$

A solution of (1) is thus given by

$$z(x, y, t) = (a_1 \cos \mu x + b_1 \sin \mu x)(a_2 \cos \sqrt{\lambda^2 - \mu^2} y + b_2 \sin \sqrt{\lambda^2 - \mu^2} y)(a_3 \cos \lambda at + b_3 \sin \lambda at)$$

From $z(0, y, t) = 0$ we find $a_1 = 0$. From $z(x, 0, t) = 0$ we find $a_2 = 0$. From $z_t(x, y, 0) = 0$ we find $b_3 = 0$. Thus the solution satisfying these conditions (and the boundedness condition) is

$$z(x, y, t) = B \sin \mu x \sin \sqrt{\lambda^2 - \mu^2} y \cos \lambda at$$

From $z(1, y, t) = 0$ we see that $\mu = m\pi$, $m = 1, 2, 3, \dots$. From $z(x, 1, t) = 0$ we see that $\sqrt{\lambda^2 - \mu^2} = n\pi$, $n = 1, 2, 3, \dots$, i.e. $\lambda = \sqrt{m^2 + n^2} \pi$.

Thus a solution satisfying all conditions but $z(x, y, 0) = f(x, y)$ is given by

$$z(x, y, t) = B \sin m\pi x \sin n\pi y \cos \sqrt{m^2 + n^2} \pi at$$

By the superposition theorem we can arrive at the possible solution

$$z(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin m\pi x \sin n\pi y \cos \sqrt{m^2 + n^2} \pi at \quad (2)$$

Then, letting $t = 0$ and using $z(x, y, 0) = f(x, y)$, we arrive at

$$f(x, y) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin m\pi x \sin n\pi y$$

from which we are led as in Problem 2.24 to

$$B_{mn} = 4 \int_0^1 \int_0^1 f(x, y) \sin m\pi x \sin n\pi y \, dx \, dy \quad (3)$$

Thus the formal solution to our problem is given by (2), where the coefficients B_{mn} are determined from (3).

In this problem the natural modes have frequencies f_{mn} given by $2\pi f_{mn} = \sqrt{m^2 + n^2} \pi a$, i.e.

$$f_{mn} = \frac{1}{2} \sqrt{m^2 + n^2} \sqrt{\frac{\tau}{\mu}} \quad (4)$$

The lowest mode, $m = 0$, $n = 1$ or $m = 1$, $n = 0$, has frequency $\frac{1}{2} \sqrt{\tau/\mu}$. The next higher one has $m = 1$, $n = 1$ with frequency $\frac{1}{2} \sqrt{2\tau/\mu}$, which is not an integer multiple of the lowest (i.e. fundamental) frequency. Similarly, higher modes do not in general have frequencies which are integer multiples of the fundamental frequency. In such case we do not get *music*.

Supplementary Problems

FOURIER SERIES

2.34. Graph each of the following functions and find its corresponding Fourier series, using properties of even and odd functions wherever applicable.

$$\begin{aligned} (a) \quad f(x) &= \begin{cases} 8 & 0 < x < 2 \\ -8 & 2 < x < 4 \end{cases} \quad \text{Period 4} & (b) \quad f(x) &= \begin{cases} -x & -4 \leq x \leq 0 \\ x & 0 \leq x \leq 4 \end{cases} \quad \text{Period 8} \\ (c) \quad f(x) &= 4x, \quad 0 < x < 10, \quad \text{Period 10} & (d) \quad f(x) &= \begin{cases} 2x & 0 \leq x \leq 3 \\ 0 & -3 < x < 0 \end{cases} \quad \text{Period 6} \end{aligned}$$

2.35. In each part of Problem 2.34, tell where the discontinuities of $f(x)$ are located and to what value the series converges at these discontinuities.

2.36. Expand $f(x) = \begin{cases} 2 - x & 0 < x < 4 \\ x - 6 & 4 < x < 8 \end{cases}$ in a Fourier series of period 8.

2.37. (a) Expand $f(x) = \cos x$, $0 < x < \pi$, in a Fourier sine series.

(b) How should $f(x)$ be defined at $x = 0$ and $x = \pi$ so that the series will converge to $f(x)$ for $0 \leq x \leq \pi$?

2.38. (a) Expand in a Fourier series $f(x) = \cos x$, $0 < x < \pi$, if the period is π ; and (b) compare with the result of Problem 2.37, explaining the similarities and differences if any.

2.39. Expand $f(x) = \begin{cases} x & 0 < x < 4 \\ 8 - x & 4 < x < 8 \end{cases}$ in a series of (a) sines, (b) cosines.

2.40. Prove that for $0 \leq x \leq \pi$,

$$(a) \quad x(\pi - x) = \frac{\pi^2}{6} - \left(\frac{\cos 2x}{1^2} + \frac{\cos 4x}{2^2} + \frac{\cos 6x}{3^2} + \dots \right)$$

$$(b) \quad x(\pi - x) = \frac{8}{\pi} \left(\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \dots \right)$$

2.41. Use Problem 2.40 to show that

$$(a) \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad (b) \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}, \quad (c) \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)^3} = \frac{\pi^3}{32}.$$

2.42. Show that $\frac{1}{1^3} + \frac{1}{3^3} - \frac{1}{5^3} - \frac{1}{7^3} + \frac{1}{9^3} + \frac{1}{11^3} - \dots = \frac{3\pi^3\sqrt{2}}{128}.$

INTEGRATION AND DIFFERENTIATION OF FOURIER SERIES

2.43. (a) Show that for $-\pi < x < \pi$,

$$x = 2 \left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots \right)$$

(b) By integrating the result of (a), show that for $-\pi \leq x \leq \pi$,

$$x^2 = \frac{\pi^2}{3} - 4 \left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \dots \right)$$

(c) By integrating the result of (b), show that for $-\pi \leq x \leq \pi$,

$$x(\pi - x)(\pi + x) = 12 \left(\frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \dots \right)$$

(d) Show that the series on the right in parts (b) and (c) converge uniformly to the functions on the left.

2.44. (a) Show that for $-\pi < x < \pi$,

$$x \cos x = -\frac{1}{2} \sin x + 2 \left(\frac{2}{1 \cdot 3} \sin 2x - \frac{3}{2 \cdot 4} \sin 3x + \frac{4}{3 \cdot 5} \sin 4x - \dots \right)$$

(b) Use (a) to show that for $-\pi \leq x \leq \pi$,

$$x \sin x = 1 - \frac{1}{2} \cos x - 2 \left(\frac{\cos 2x}{1 \cdot 3} - \frac{\cos 3x}{2 \cdot 4} + \frac{\cos 4x}{3 \cdot 5} - \dots \right)$$

2.45. By differentiating the result of Problem 2.44(b), prove that for $0 \leq x \leq \pi$,

$$x = \frac{\pi}{2} - \frac{4}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right)$$

PARSEVAL'S IDENTITY

2.46. By using Problem 2.40 and Parseval's identity, show that

$$(a) \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90} \quad (b) \quad \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$$

- 2.47. Show that $\frac{1}{1^2 \cdot 3^2} + \frac{1}{3^2 \cdot 5^2} + \frac{1}{5^2 \cdot 7^2} + \cdots = \frac{\pi^2 - 8}{16}$. [Hint. Use Problem 2.11.]
- 2.48. Show that (a) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$, (b) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^6} = \frac{\pi^6}{960}$.
- 2.49. Show that $\frac{1}{1^2 \cdot 2^2 \cdot 3^2} + \frac{1}{2^2 \cdot 3^2 \cdot 4^2} + \frac{1}{3^2 \cdot 4^2 \cdot 5^2} + \cdots = \frac{4\pi^2 - 39}{16}$.

SOLUTIONS USING FOURIER SERIES

- 2.50. (a) Solve the boundary value problem

$$\frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2} \quad u(0, t) = u(4, t) = 0 \quad u(x, 0) = 25x$$

where $0 < x < 4$, $t > 0$.

- (b) Interpret physically the boundary value problem in (a).

- 2.51. (a) Show that the solution of the boundary value problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad u_x(0, t) = u_x(\pi, t) = 0 \quad u(x, 0) = f(x)$$

where $0 < x < \pi$, $t > 0$, is given by

$$u(x, t) = \frac{1}{\pi} \int_0^\pi f(x) dx + \frac{2}{\pi} \sum_{m=1}^{\infty} e^{-m^2 t} \cos mx \int_0^\pi f(x) \cos mx dx$$

- (b) Interpret physically the boundary value problem in (a).

- 2.52. Find the steady-state temperature in a bar whose ends are located at $x = 0$ and $x = 10$, if these ends are kept at 150°C and 100°C respectively.

- 2.53. A circular plate of unit radius (see Fig. 2-14, page 39) whose faces are insulated has its boundary kept at temperature $120 + 60 \cos 2\phi$. Find the steady-state temperature of the plate.

- 2.54. Show that $\rho \sin \phi + \frac{1}{3}\rho^3 \sin 3\phi + \frac{1}{5}\rho^5 \sin 5\phi + \cdots = \frac{1}{2} \tan^{-1} \left(\frac{2\rho \sin \phi}{1 - \rho^2} \right)$ and thus complete Problem 2.28.

- 2.55. A string 2 ft long is stretched between two fixed points $x = 0$ and $x = 2$. If the displacement of the string from the x -axis at $t = 0$ is given by $f(x) = 0.03 x(2 - x)$ and if the initial velocity is zero, find the displacement at any later time.

- 2.56. A square plate of side a has one side maintained at temperature $f(x)$ and the others at zero, as indicated in Fig. 2-25. Show that the steady-state temperature at any point of the plate is given by

$$u(x, y) = \sum_{k=1}^{\infty} \left[\frac{2}{a \sinh(k\pi)} \int_0^a f(x) \sin \frac{k\pi x}{a} dx \right] \sin \frac{k\pi x}{a} \sinh \frac{k\pi y}{a}$$

- 2.57. Work Problem 2.56 if the sides are maintained at temperatures $f_1(x)$, $g_1(y)$, $f_2(x)$, $g_2(y)$, respectively. [Hint. Use the principle of superposition and the result of Problem 2.56.]

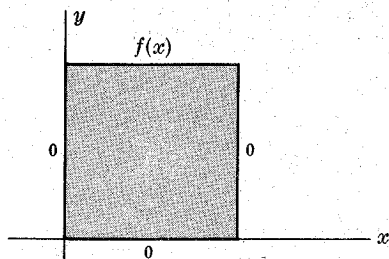


Fig. 2-25

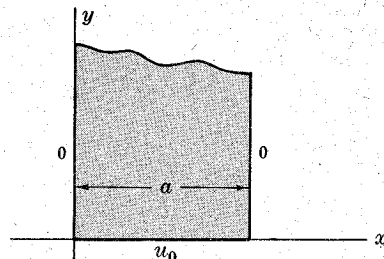


Fig. 2-26

- 2.58. An infinitely long plate of width a (indicated by the shaded region of Fig. 2-26) has its two parallel sides maintained at temperature 0 and its other side at constant temperature u_0 . (a) Show that the steady-state temperature is given by

$$u(x, y) = \frac{4u_0}{\pi} \left(e^{-y} \sin \frac{\pi x}{a} + \frac{1}{3} e^{-3y} \sin \frac{3\pi x}{a} + \frac{1}{5} e^{-5y} \sin \frac{5\pi x}{a} + \dots \right)$$

- (b) Use Problem 2.54 to show that

$$u(x, y) = \frac{2u_0}{\pi} \tan^{-1} \left[\frac{\sin(\pi x/a)}{\sinh y} \right]$$

- 2.59. Solve Problem 1.26 if the string has its ends fixed at $x = 0$ and $x = L$ and if its initial displacement and velocity are given by $f(x)$ and $g(x)$ respectively.

- 2.60. A square plate (Fig. 2-27) having sides of unit length has its edges fixed in the xy -plane and is set into transverse vibration.

- (a) Show that the transverse displacement $z(x, y, t)$ of any point (x, y) at time t is given by

$$\frac{\partial^2 z}{\partial t^2} = a^2 \left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} \right)$$

where a^2 is a constant.

- (b) Show that if the plate is given an initial shape $f(x, y)$ and released with velocity $g(x, y)$, then the displacement is given by

$$z(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos \lambda_{mn} t + B_{mn} \sin \lambda_{mn} t] \sin m\pi x \sin n\pi y$$

where

$$A_{mn} = 4 \int_0^1 \int_0^1 f(x, y) \sin m\pi x \sin n\pi y \, dx \, dy$$

$$B_{mn} = \frac{4}{a\lambda_{mn}} \int_0^1 \int_0^1 g(x, y) \sin m\pi x \sin n\pi y \, dx \, dy$$

and $\lambda_{mn} = \pi\sqrt{m^2 + n^2}$.

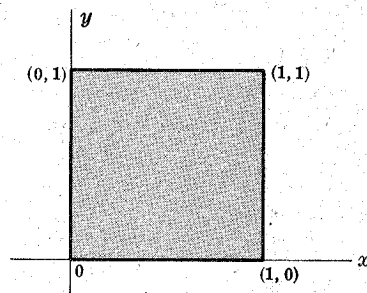


Fig. 2-27

- 2.61. Work Problem 2.60 for a rectangular plate of sides b and c .
- 2.62. Prove that the result for $u(x, t)$ obtained in Problem 2.25 actually satisfies the partial differential equation and the boundary conditions.
- 2.63. Solve the boundary value problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} - \alpha^2 u \quad 0 < x < L, \quad t > 0$$

$$u(0, t) = u_1, \quad u(L, t) = u_2, \quad u(x, 0) = 0$$

where α and L are constants, and interpret physically.

2.64. Work Problem 2.63 if $u(x, 0) = f(x)$.

2.65. Solve and interpret physically the boundary value problem

$$\frac{\partial^2 y}{\partial t^2} + b^2 \frac{\partial^4 y}{\partial x^4} = 0$$

where $y(0, t) = 0$, $y(L, t) = 0$, $y(x, 0) = f(x)$, $y_t(x, 0) = 0$, $y_{xx}(0, t) = 0$, $y_{xx}(L, t) = 0$, $|y(x, t)| < M$.

2.66. Work Problem 2.65 if $y_t(x, 0) = g(x)$.

2.67. A plate is bounded by two concentric circles of radius a and b , as shown in Fig. 2-28. The faces are insulated and the boundaries are kept at temperatures $f(\theta)$ and $g(\theta)$ respectively. Show that the steady-state temperature at any point (r, θ) is given by

$$u(r, \theta) = A_0 + B_0 \ln r + \sum_{n=1}^{\infty} \left\{ \left(A_n r^n + \frac{B_n}{r^n} \right) \cos n\theta + \left(C_n r^n + \frac{D_n}{r^n} \right) \sin n\theta \right\}$$

where A_0 and B_0 are determined from

$$A_0 + B_0 \ln a = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) d\theta$$

$$A_0 + B_0 \ln b = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) d\theta$$

A_n, B_n are determined from

$$A_n a^n + B_n a^{-n} = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta d\theta, \quad A_n b^n + B_n b^{-n} = \frac{1}{\pi} \int_0^{2\pi} g(\theta) \cos n\theta d\theta$$

and C_n, D_n are determined from

$$C_n a^n + D_n a^{-n} = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta d\theta, \quad C_n b^n + D_n b^{-n} = \frac{1}{\pi} \int_0^{2\pi} g(\theta) \sin n\theta d\theta$$

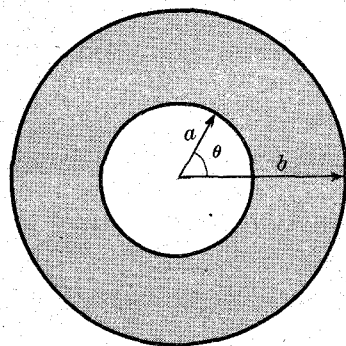


Fig. 2-28

2.68. Investigate the limiting cases of Problem 2.67 as (a) $a \rightarrow 0$, (b) $b \rightarrow \infty$, and give physical interpretations.

2.69. (a) Solve the boundary value problem

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} + \beta e^{-\gamma x}$$

where $u(0, t) = 0$, $u(L, t) = 0$, $u(x, 0) = f(x)$, $|u(x, t)| < M$, and (b) give a physical interpretation.

2.70. Work Problem 2.69 if $\beta e^{-\gamma x}$ is replaced by $u_0 \sin \alpha x$, where u_0 and α are constants.

2.71. Solve $\frac{\partial^2 y}{\partial t^2} = \alpha^2 \frac{\partial^2 y}{\partial x^2} - g$ where $y(0, t) = 0$, $y(L, t) = 0$, $y(x, 0) = f(x)$, $y_t(x, 0) = 0$, $|y(x, t)| < M$, and give a physical interpretation.

2.72. Find the steady-state temperature in a solid cube of unit side (Fig. 2-29) if the face in the xy -plane is kept at the prescribed temperature $F(x, y)$, while all other faces are kept at temperature zero.

- 2.73. How would you solve Problem 2.72 if temperatures were prescribed on the other faces also?
- 2.74. How would you solve Problem 2.72 if the initial temperature inside the cube was given and you wished to find the temperature inside the cube at any later time?
- 2.75. Generalize the result of Problem 2.72 to any rectangular parallelepiped.
- 2.76. A plate in the form of a sector of a circle of radius a has central angle β , as shown in Fig. 2-30. If the circular part is maintained at a temperature $f(\theta)$, $0 < \theta < \beta$, while the bounding radii are maintained at temperature zero, find the steady-state temperature everywhere in the sector.

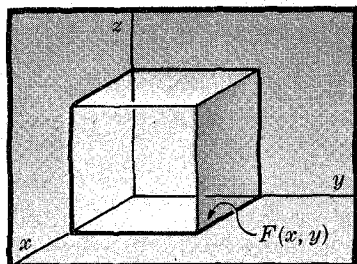


Fig. 2-29

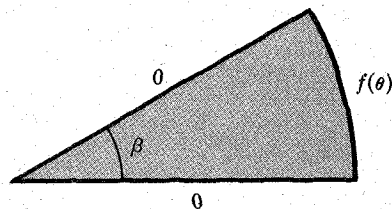


Fig. 2-30

Chapter 3

Orthogonal Functions

DEFINITIONS INVOLVING ORTHOGONAL FUNCTIONS. ORTHONORMAL SETS

Many properties of Fourier series considered in Chapter 2 depended on such results as

$$\int_0^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = 0, \quad \int_0^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0 \quad (m \neq n) \quad (1)$$

In this chapter we shall seek to generalize some ideas of Chapter 2. To do this we first recall some elementary properties of *vectors*.

Two vectors **A** and **B** are called *orthogonal* (perpendicular) if $\mathbf{A} \cdot \mathbf{B} = 0$ or $A_1B_1 + A_2B_2 + A_3B_3 = 0$, where $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$. Although not geometrically or physically obvious, these ideas can be generalized to include vectors with more than three components. In particular we can think of a function, say $A(x)$, as being a vector with an *infinity of components* (i.e. an *infinite-dimensional vector*), the value of each component being specified by substituting a particular value of x taken from some interval (a, b) . It is natural in such case to define two functions, $A(x)$ and $B(x)$, as *orthogonal* in (a, b) if

$$\int_a^b A(x) B(x) dx = 0 \quad (2)$$

The left side of (2) is often called the *scalar product* of $A(x)$ and $B(x)$.

A vector **A** is called a *unit vector* or *normalized vector* if its magnitude is unity, i.e. if $\mathbf{A} \cdot \mathbf{A} = A^2 = 1$. Extending the concept, we say that the function $A(x)$ is *normal* or *normalized* in (a, b) if

$$\int_a^b \{A(x)\}^2 dx = 1 \quad (3)$$

From the above it is clear that we can consider a set of functions $\{\phi_k(x)\}$, $k = 1, 2, 3, \dots$, having the properties

$$\int_a^b \phi_m(x) \phi_n(x) dx = 0 \quad m \neq n \quad (4)$$

$$\int_a^b \{\phi_m(x)\}^2 dx = 1 \quad m = 1, 2, 3, \dots \quad (5)$$

Each member of the set is orthogonal to every other member of the set and is also normalized. We call such a set of functions an *orthonormal set* in (a, b) .

The equations (4) and (5) can be summarized by writing

$$\int_a^b \phi_m(x) \phi_n(x) dx = \delta_{mn} \quad (6)$$

where δ_{mn} , called *Kronecker's symbol*, is defined as 0 if $m \neq n$ and 1 if $m = n$.

Example 1.

The set of functions

$$\phi_m(x) = \sqrt{\frac{2}{\pi}} \sin mx \quad m = 1, 2, 3, \dots$$

is an orthonormal set in the interval $0 \leq x \leq \pi$.

ORTHOGONALITY WITH RESPECT TO A WEIGHT FUNCTION

If
$$\int_a^b \psi_m(x) \psi_n(x) w(x) dx = \delta_{mn} \quad (7)$$

where $w(x) \geq 0$, we often say that the set $\{\psi_k(x)\}$ is orthonormal with respect to the *density function* or *weight function* $w(x)$. In such case the set $\phi_m(x) = \sqrt{w(x)} \psi_m(x)$, $m = 1, 2, 3, \dots$, is an orthonormal set in (a, b) .

EXPANSION OF FUNCTIONS IN ORTHONORMAL SERIES

Just as any vector $\mathbf{r}$ in 3 dimensions can be expanded in a set of mutually orthogonal unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ in the form $\mathbf{r} = c_1\mathbf{i} + c_2\mathbf{j} + c_3\mathbf{k}$, so we consider the possibility of expanding a function $f(x)$ in a set of orthonormal functions, i.e.

$$f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x) \quad a \leq x \leq b \quad (8)$$

Such series, called *orthonormal series*, are generalizations of Fourier series and are of great interest and utility both from theoretical and applied viewpoints.

Assuming that the series on the right of (8) converges to $f(x)$, we can formally multiply both sides by $\phi_m(x)$ and integrate both sides from a to b to obtain

$$c_m = \int_a^b f(x) \phi_m(x) dx \quad (9)$$

which are called the *generalized Fourier coefficients*. As in the case of Fourier series, an investigation should be made to determine whether the series on the right of (8) with coefficients (9) actually converges to $f(x)$. In practice, if $f(x)$ and $f'(x)$ are piecewise continuous in (a, b) , then the series on the right of (8) with coefficients given by (9) converges to $\frac{1}{2}[f(x+0) + f(x-0)]$ as in the case of Fourier series.

APPROXIMATIONS IN THE LEAST-SQUARES SENSE

Let $f(x)$ and $f'(x)$ be piecewise continuous in (a, b) . Let $\phi_m(x)$, $m = 1, 2, \dots$, be orthonormal in (a, b) . Suppose now that we consider the finite sum

$$S_M(x) = \sum_{n=1}^M \alpha_n \phi_n(x) \quad (10)$$

as an approximation to $f(x)$, where α_n , $n = 1, 2, 3, \dots$, are constants presently unknown. Then the *mean square error* of this approximation is given by

$$\text{Mean square error} = \frac{\int_a^b [f(x) - S_M(x)]^2 dx}{b-a} \quad (11)$$

and the *root mean square error* E_{rms} is given by the square root of (11), i.e.

$$E_{\text{rms}} = \sqrt{\frac{1}{b-a} \int_a^b [f(x) - S_M(x)]^2 dx} \quad (12)$$

We now seek to determine the constants α_n which will produce the *least* root mean square error. The result is supplied in the following theorem which is proved in Problem 3.5.

Theorem 3-1: The root mean square error (12) is least (i.e. a minimum) when the coefficients are equal to the generalized Fourier coefficients (9), i.e. when

$$\alpha_n = c_n = \int_a^b f(x) \phi_n(x) dx \quad (13)$$

We often say that $S_M(x)$ with coefficients c_n is an *approximation to $f(x)$ in the least-squares sense* or a *least-squares approximation to $f(x)$* .

It is of interest to note that once we have worked out an approximation to $f(x)$ in the least-squares sense by using the coefficients c_n , we do not have to recompute these coefficients if we wish to have a better approximation. This is sometimes referred to as the *principle of finality*.

PARSEVAL'S IDENTITY FOR ORTHONORMAL SERIES. COMPLETENESS

For the case where $\alpha_n = c_n$ we can show (see Problem 3.5) that the root mean square error is given by

$$E_{\text{rms}} = \frac{1}{\sqrt{b-a}} \left[\int_a^b [f(x)]^2 dx - \sum_{n=1}^M c_n^2 \right]^{1/2} \quad (14)$$

It is seen that E_{rms} depends on M . As $M \rightarrow \infty$ we would expect that $E_{\text{rms}} \rightarrow 0$, in which case we would have

$$\int_a^b [f(x)]^2 dx = \sum_{n=1}^{\infty} c_n^2 \quad (15)$$

Now, (15) could certainly not be true if, for example, we left out certain functions $\phi_n(x)$ in the series approximation, i.e. if the set of functions were incomplete. We are therefore led to define a set of functions $\phi_n(x)$ to be *complete* if and only if $E_{\text{rms}} \rightarrow 0$ as $M \rightarrow \infty$, so that (15) is valid. We refer to (15) as *Parseval's identity for orthonormal series of functions*. In (6) of Chapter 2, page 23, we have obtained Parseval's identity for the special case of Fourier series.

In the case where $E_{\text{rms}} \rightarrow 0$ as $M \rightarrow \infty$, i.e.

$$\lim_{M \rightarrow \infty} \int_a^b [f(x) - S_M(x)]^2 dx = 0 \quad (16)$$

we sometimes write

$$\text{l.i.m.}_{M \rightarrow \infty} S_M(x) = f(x) \quad (17)$$

This is read *the limit in mean of $S_M(x)$ as $M \rightarrow \infty$ equals $f(x)$* or *$S_M(x)$ converges in the mean to $f(x)$ as $M \rightarrow \infty$* and is equivalent to (16).

STURM-LIOUVILLE SYSTEMS. EIGENVALUES AND EIGENFUNCTIONS

A boundary value problem having the form

$$\left. \begin{aligned} \frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)] y &= 0 & a \leq x \leq b \\ \alpha_1 y(a) + \alpha_2 y'(a) &= 0, & \beta_1 y(b) + \beta_2 y'(b) &= 0 \end{aligned} \right\} \quad (18)$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are given constants; $p(x), q(x), r(x)$ are given functions which we shall assume to be differentiable and λ is an unspecified parameter independent of x , is called a *Sturm-Liouville boundary value problem* or *Sturm-Liouville system*. Such systems arise in practice on using the separation of variables method in solution of partial differential equations. In such case λ is the "separation constant." See Problem 3.14.

A nontrivial solution of this system, i.e. one which is not identically zero, exists in general only for a particular set of values of the parameter λ . These values are called the *characteristic values*, or more often *eigenvalues*, of the system. The corresponding solutions are called *characteristic functions* or *eigenfunctions* of the system. In general to each eigenvalue there is one eigenfunction, although exceptions can occur.

If $p(x)$ and $q(x)$ are real, then the eigenvalues are real. Also, the eigenfunctions form an orthogonal set with respect to the weight function $r(x)$, which is generally taken as non-negative, i.e. $r(x) \geq 0$. It follows that by suitable normalization the set of functions can be made an orthonormal set with respect to $r(x)$ in $a \leq x \leq b$. See Problems 3.8–3.11.

THE GRAM-SCHMIDT ORTHONORMALIZATION PROCESS

Given a finite or infinite set of linearly independent functions $\psi_1(x), \psi_2(x), \psi_3(x), \dots$ defined in an interval (a, b) it is possible to generate from these functions a set of orthonormal functions in (a, b) . To do this we first consider a new set of functions obtained from the $\psi_k(x)$ and given by

$$c_{11}\psi_1(x), \quad c_{21}\psi_1(x) + c_{22}\psi_2(x), \quad c_{31}\psi_1(x) + c_{32}\psi_2(x) + c_{33}\psi_3(x), \quad \dots \quad (19)$$

where the c 's are constants to be determined. We shall designate the functions in (19) by $\phi_1(x), \phi_2(x), \phi_3(x), \dots$

We now choose the constants $c_{11}, c_{21}, c_{22}, \dots$ so that the functions $\phi_1(x), \phi_2(x), \phi_3(x), \dots$ are mutually orthogonal and also normalized in (a, b) . The process, known as the *Gram-Schmidt orthonormalization process*, is illustrated in Problem 3.12.

An extension to the case where orthonormalization is with respect to a given weight function is easily made.

APPLICATIONS TO BOUNDARY VALUE PROBLEMS

In the course of solving boundary value problems using separation of variables we often arrive at Sturm-Liouville differential equations (see Problem 3.15, for example). The parameter λ in these equations is the separation constant, and the values of λ which are obtained represent the real eigenvalues. The solution of the boundary value problem is then obtained in terms of the corresponding mutually orthogonal eigenfunctions.

For an illustration which does not involve Fourier series, see Problem 3.13. Other illustrations involving this general procedure will be given in later chapters.

Solved Problems

ORTHOGONAL FUNCTIONS AND ORTHONORMAL SERIES

3.1. (a) Show that the set of functions

$$1, \sin \frac{\pi x}{L}, \cos \frac{\pi x}{L}, \sin \frac{2\pi x}{L}, \cos \frac{2\pi x}{L}, \sin \frac{3\pi x}{L}, \cos \frac{3\pi x}{L}, \dots$$

form an orthogonal set in the interval $(-L, L)$.

(b) Determine the corresponding normalizing constants for the set in (a) so that the set is orthonormal in $(-L, L)$.

(a) This follows at once from the results of Problems 2.2 and 2.3, page 26.

(b) By Problem 2.3,

$$\int_{-L}^L \sin^2 \frac{m\pi x}{L} dx = L, \quad \int_{-L}^L \cos^2 \frac{m\pi x}{L} dx = L$$

$$\text{Then} \quad \int_{-L}^L \left(\sqrt{\frac{1}{L}} \sin \frac{m\pi x}{L} \right)^2 dx = 1, \quad \int_{-L}^L \left(\sqrt{\frac{1}{L}} \cos \frac{m\pi x}{L} \right)^2 dx = 1$$

$$\text{Also,} \quad \int_{-L}^L (1)^2 dx = 2L \quad \text{or} \quad \int_{-L}^L \left(\frac{1}{\sqrt{2L}} \right)^2 dx = 1$$

Thus the required orthonormal set is given by

$$\frac{1}{\sqrt{2L}}, \quad \frac{1}{\sqrt{L}} \sin \frac{\pi x}{L}, \quad \frac{1}{\sqrt{L}} \cos \frac{\pi x}{L}, \quad \frac{1}{\sqrt{L}} \sin \frac{2\pi x}{L}, \quad \frac{1}{\sqrt{L}} \cos \frac{2\pi x}{L}, \quad \dots$$

3.2. Let $\{\phi_n(x)\}$ be a set of functions which are mutually orthonormal in (a, b) . Prove that if $\sum_{n=1}^{\infty} c_n \phi_n(x)$ converges uniformly to $f(x)$ in (a, b) , then

$$c_n = \int_a^b f(x) \phi_n(x) dx$$

Multiplying both sides of

$$f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x) \quad (1)$$

by $\phi_m(x)$ and integrating from a to b , we have

$$\int_a^b f(x) \phi_m(x) dx = \sum_{n=1}^{\infty} c_n \int_a^b \phi_m(x) \phi_n(x) dx \quad (2)$$

where the interchange of integration and summation is justified by the fact that the series converges uniformly to $f(x)$. Now since the functions $\{\phi_n(x)\}$ are mutually orthonormal in (a, b) , we have

$$\int_a^b \phi_m(x) \phi_n(x) dx = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

so that (2) becomes

$$\int_a^b f(x) \phi_m(x) dx = c_m \quad (3)$$

as required.

We call the coefficients c_m given by (3) the *generalized Fourier coefficients* corresponding to $f(x)$ even though nothing may be known about the convergence of the series in (1). As in the case of Fourier series, convergence of $\sum_{n=1}^{\infty} c_n \phi_n(x)$ is then investigated using the coefficients (3). The conditions of convergence depend of course on the types of orthonormal functions used. In the remainder of this book we shall be concerned with many examples of orthonormal functions and series.

LEAST-SQUARES APPROXIMATIONS. PARSEVAL'S IDENTITY AND COMPLETENESS

3.3. If $S_M(x) = \sum_{n=1}^M \alpha_n \phi_n(x)$, where $\phi_n(x)$, $n = 1, 2, \dots$, is orthonormal in (a, b) , prove that

$$\int_a^b [f(x) - S_M(x)]^2 dx = \int_a^b [f(x)]^2 dx - 2 \sum_{n=1}^M \alpha_n c_n + \sum_{n=1}^M \alpha_n^2$$

where $c_n = \int_a^b f(x) \phi_n(x) dx$ are the generalized Fourier coefficients corresponding to $f(x)$.

We have

$$f(x) - S_M(x) = f(x) - \sum_{n=1}^M \alpha_n \phi_n(x)$$

By squaring we obtain

$$[f(x) - S_M(x)]^2 = [f(x)]^2 - 2 \sum_{n=1}^M \alpha_n f(x) \phi_n(x) + \sum_{m=1}^M \sum_{n=1}^M \alpha_m \alpha_n \phi_m(x) \phi_n(x)$$

Integrating both sides from a to b using

$$c_n = \int_a^b f(x) \phi_n(x) dx, \quad \int_a^b \phi_m(x) \phi_n(x) dx = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

we obtain

$$\int_a^b [f(x) - S_M(x)]^2 dx = \int_a^b [f(x)]^2 dx - 2 \sum_{n=1}^M \alpha_n c_n + \sum_{n=1}^M \alpha_n^2$$

We have assumed that $f(x)$ is such that all the above integrals exist.

3.4. Show that

$$\int_a^b [f(x) - S_M(x)]^2 dx = \int_a^b [f(x)]^2 dx + \sum_{n=1}^M (\alpha_n - c_n)^2 - \sum_{n=1}^M c_n^2$$

This follows from Problem 3.3 by noting that

$$\begin{aligned} \int_a^b [f(x)]^2 dx - 2 \sum_{n=1}^M \alpha_n c_n + \sum_{n=1}^M \alpha_n^2 &= \int_a^b [f(x)]^2 dx + \sum_{n=1}^M (\alpha_n^2 - 2\alpha_n c_n) \\ &= \int_a^b [f(x)]^2 dx + \sum_{n=1}^M [(\alpha_n - c_n)^2 - c_n^2] \\ &= \int_a^b [f(x)]^2 dx + \sum_{n=1}^M (\alpha_n - c_n)^2 - \sum_{n=1}^M c_n^2 \end{aligned}$$

3.5. (a) Prove Theorem 3.1, page 54: The root mean square error is a minimum when the coefficients α_n equal the Fourier coefficients c_n .

(b) What is the value of the root mean square error in this case?

(a) From Problem 3.4 we have

$$\int_a^b [f(x) - S_M(x)]^2 dx = \int_a^b [f(x)]^2 dx + \sum_{n=1}^M (\alpha_n - c_n)^2 - \sum_{n=1}^M c_n^2$$

Now the root mean square error will be a minimum when the above is a minimum. However, it is clear that the right-hand side is a minimum when $\sum_{n=1}^M (\alpha_n - c_n)^2 = 0$, i.e. when $\alpha_n = c_n$ for all n .

(b) From part (a) we see that the minimum value of the root mean square error is given by

$$\begin{aligned} E_{\text{rms}} &= \left[\frac{1}{b-a} \int_a^b [f(x) - S_M(x)]^2 dx \right]^{1/2} \\ &= \frac{1}{\sqrt{b-a}} \left[\int_a^b [f(x)]^2 dx - \sum_{n=1}^{\infty} c_n^2 \right]^{1/2} \end{aligned}$$

3.6. Prove that if c_n , $n = 1, 2, 3, \dots$, denote the generalized Fourier coefficients corresponding to $f(x)$, then

$$\sum_{n=1}^{\infty} c_n^2 \leq \int_a^b [f(x)]^2 dx$$

From Problem 3.5 we see that, since the root mean square error must be nonnegative,

$$\sum_{n=1}^M c_n^2 \leq \int_a^b [f(x)]^2 dx \quad (1)$$

Then, taking the limit as $M \rightarrow \infty$ and noting that the right side does not depend on M , it follows that

$$\sum_{n=1}^{\infty} c_n^2 \leq \int_a^b [f(x)]^2 dx \quad (2)$$

This inequality is often called *Bessel's inequality*.

As a consequence of (2) we see that if the right side of (2) exists, then the series on the left must converge. In the special case where the equality holds in (2) we obtain *Parseval's identity*.

3.7. Show that $\lim_{n \rightarrow \infty} \int_a^b f(x) \phi_n(x) dx = 0$.

By definition we have $c_n = \int_a^b f(x) \phi_n(x) dx$. But since $\sum_{n=1}^{\infty} c_n^2$ converges by Problem 3.6, the n th term c_n^2 , and with it c_n , must approach zero as $n \rightarrow \infty$, which is the required result. Note that this result for the special case of Fourier series is *Riemann's theorem* (see Problem 2.19, page 35).

STURM-LIOUVILLE SYSTEMS. EIGENVALUES AND EIGENFUNCTIONS

3.8. (a) Verify that the system $y'' + \lambda y = 0$, $y(0) = 0$, $y(1) = 0$ is a Sturm-Liouville system. (b) Find the eigenvalues and eigenfunctions of the system. (c) Prove that the eigenfunctions are orthogonal in $(0, 1)$. (d) Find the corresponding set of normalized eigenfunctions. (e) Expand $f(x) = 1$ in a series of these orthonormal functions.

(a) The system is a special case of (18), page 54, with $p(x) = 1$, $q(x) = 0$, $r(x) = 1$, $a = 0$, $b = 1$, $\alpha_1 = 1$, $\alpha_2 = 0$, $\beta_1 = 1$, $\beta_2 = 0$ and thus is a Sturm-Liouville system.

(b) The general solution of $y'' + \lambda y = 0$ is $y = A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x$. From the boundary condition $y(0) = 0$ we have $A = 0$, i.e. $y = B \sin \sqrt{\lambda} x$. From the boundary condition $y(1) = 0$ we have $B \sin \sqrt{\lambda} = 0$; since B cannot be zero (otherwise the solution will be identically zero, i.e. trivial), we must have $\sin \sqrt{\lambda} = 0$. Then $\sqrt{\lambda} = m\pi$, $\lambda = m^2\pi^2$, where $m = 1, 2, 3, \dots$ are the required eigenvalues.

The eigenfunctions belonging to the eigenvalues $\lambda = m^2\pi^2$ can be designated by $B_m \sin m\pi x$, $m = 1, 2, 3, \dots$. Note that we exclude the value $m = 0$ or $\lambda = 0$ as an eigenvalue, since the corresponding eigenfunction is zero.

(c) The eigenfunctions are orthogonal since

$$\begin{aligned} \int_0^1 (B_m \sin m\pi x)(B_n \sin n\pi x) dx &= B_m B_n \int_0^1 \sin m\pi x \sin n\pi x dx \\ &= \frac{B_m B_n}{2} \int_0^1 [\cos(m-n)\pi x - \cos(m+n)\pi x] dx \\ &= \frac{B_m B_n}{2} \left[\frac{\sin(m-n)\pi x}{(m-n)\pi} - \frac{\sin(m+n)\pi x}{(m+n)\pi} \right] \Big|_0^1 = 0, \quad m \neq n \end{aligned}$$

(d) The eigenfunctions will be orthonormal if

$$\int_0^1 (B_m \sin m\pi x)^2 dx = 1$$

i.e. if $B_m^2 \int_0^1 \sin^2 m\pi x dx = \frac{B_m^2}{2} \int_0^1 (1 - \cos 2m\pi x) dx = \frac{B_m^2}{2} = 1$, or $B_m = \sqrt{2}$, taking the positive square root. Thus the set $\sqrt{2} \sin m\pi x$, $m = 1, 2, \dots$, is an orthonormal set.

(e) We must find constants $c_1, c_2, \dots$ such that

$$f(x) = \sum_{m=1}^{\infty} c_m \phi_m(x)$$

where $f(x) = 1$, $\phi_m(x) = \sqrt{2} \sin m\pi x$. By the methods of Fourier series,

$$c_m = \int_0^1 f(x) \phi_m(x) dx = \sqrt{2} \int_0^1 \sin m\pi x dx = \frac{\sqrt{2}(1 - \cos m\pi)}{m\pi}$$

Then the required series [Fourier series] is, assuming $0 < x < 1$,

$$1 = \sum_{m=1}^{\infty} \frac{2(1 - \cos m\pi)}{m\pi} \sin m\pi x$$

3.9. Show that the eigenvalues of a Sturm-Liouville system are real.

We have

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)] y = 0 \quad (1)$$

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \quad \beta_1 y(b) + \beta_2 y'(b) = 0 \quad (2)$$

Then assuming $p(x), q(x), r(x), \alpha_1, \alpha_2, \beta_1, \beta_2$ are real, while λ and y may be complex, we have on taking the complex conjugate (represented by using a bar, as in $\bar{y}, \bar{\lambda}$):

$$\frac{d}{dx} \left[p(x) \frac{d\bar{y}}{dx} \right] + [q(x) + \bar{\lambda} r(x)] \bar{y} = 0 \quad (3)$$

$$\alpha_1 \bar{y}(a) + \alpha_2 \bar{y}'(a) = 0, \quad \beta_1 \bar{y}(b) + \beta_2 \bar{y}'(b) = 0 \quad (4)$$

Multiplying equation (1) by $\bar{y}$, (3) by y and subtracting, we find after simplifying,

$$\frac{d}{dx} [p(x)(y\bar{y}' - \bar{y}y')] = (\lambda - \bar{\lambda})r(x)y\bar{y}$$

Then integrating from a to b , we have

$$(\lambda - \bar{\lambda}) \int_a^b r(x)|y|^2 dx = p(x)(y\bar{y}' - \bar{y}y') \Big|_a^b = 0 \quad (5)$$

on using the conditions (2) and (4). Since $r(x) \geq 0$ and is not identically zero in (a, b) , the integral on the left of (5) is positive and so $\lambda - \bar{\lambda} = 0$ or $\lambda = \bar{\lambda}$, so that λ is real.

3.10. Show that the eigenfunctions belonging to two different eigenvalues are orthogonal with respect to $r(x)$ in (a, b) .

If y_1 and y_2 are eigenfunctions belonging to the eigenvalues λ_1 and λ_2 respectively,

$$\frac{d}{dx} \left[p(x) \frac{dy_1}{dx} \right] + [q(x) + \lambda_1 r(x)] y_1 = 0 \quad (1)$$

$$\alpha_1 y_1(a) + \alpha_2 y_1'(a) = 0, \quad \beta_1 y_1(b) + \beta_2 y_1'(b) = 0 \quad (2)$$

$$\frac{d}{dx} \left[p(x) \frac{dy_2}{dx} \right] + [q(x) + \lambda_2 r(x)] y_2 = 0 \quad (3)$$

$$\alpha_1 y_2(a) + \alpha_2 y_2'(a) = 0, \quad \beta_1 y_2(b) + \beta_2 y_2'(b) = 0 \quad (4)$$

Then multiplying (1) by y_2 , (3) by y_1 and subtracting, we find as in Problem 3.9,

$$\frac{d}{dx} [p(x)(y_1 y_2' - y_2 y_1')] = (\lambda_1 - \lambda_2) r(x) y_1 y_2$$

Integrating from a to b , we have on using (2) and (4),

$$(\lambda_1 - \lambda_2) \int_a^b r(x) y_1 y_2 dx = p(x)(y_1 y_2' - y_2 y_1') \Big|_a^b = 0$$

and since $\lambda_1 \neq \lambda_2$ we have the required result

$$\int_a^b r(x) y_1 y_2 dx = 0$$

3.11. Given the Sturm-Liouville system $y'' + \lambda y = 0$, $y(0) = 0$, $y'(L) + \beta y(L) = 0$, where β and L are given constants. Find (a) the eigenvalues and (b) the normalized eigenfunctions of the system. (c) Expand $f(x)$, $0 < x < L$, in a series of these normalized eigenfunctions.

(a) The general solution of $y'' + \lambda y = 0$ is

$$y = A \cos \sqrt{\lambda} x + B \sin \sqrt{\lambda} x$$

Then from the condition $y(0) = 0$ we find $A = 0$, so that

$$y = B \sin \sqrt{\lambda} x$$

The condition $y'(L) + \beta y(L) = 0$ gives

$$B\sqrt{\lambda} \cos \sqrt{\lambda} L + \beta B \sin \sqrt{\lambda} L = 0 \quad \text{or} \quad \tan \sqrt{\lambda} L = -\frac{\sqrt{\lambda}}{\beta} \quad (1)$$

which is the equation for determining the eigenvalues λ . This equation cannot be solved exactly; however we can obtain approximate values graphically. To do this we let $v = \sqrt{\lambda} L$ so that the equation becomes

$$\tan v = -\frac{v}{\beta L} \quad (2)$$

The values of v , and from these the values of λ , can be obtained from the intersection points $v_1, v_2, v_3, \dots$ of the graphs of $w = \tan v$ and $w = -v/\beta L$, as indicated in Fig. 3-1. In construction of these we have assumed that β and L are positive. We also note that we need only find the positive roots of (1).

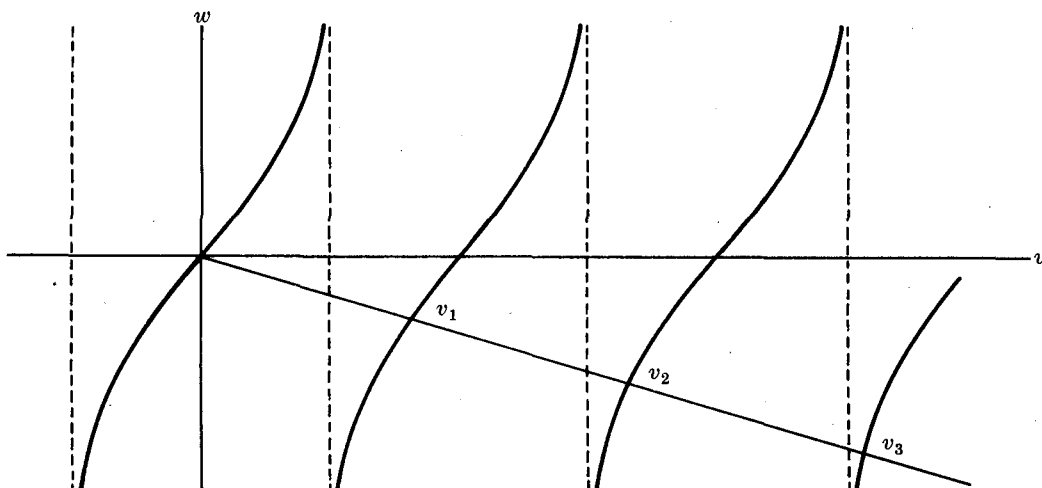


Fig. 3-1

(b) The eigenfunctions are given by

$$\phi_n(x) = B_n \sin \sqrt{\lambda_n} x \quad (3)$$

where λ_n , $n = 1, 2, 3, \dots$, represent the eigenvalues obtained in part (a). To normalize these we require

$$\int_0^L B_n^2 \sin^2 \sqrt{\lambda_n} x \, dx = 1$$

i.e.

$$\frac{B_n^2}{2} \int_0^L (1 - \cos 2\sqrt{\lambda_n} x) \, dx = 1$$

or

$$B_n^2 = \frac{4\sqrt{\lambda_n}}{2\sqrt{\lambda_n}L - \sin 2\sqrt{\lambda_n}L} \quad (4)$$

Thus a set of normalized eigenfunctions is given by

$$\phi_n(x) = \sqrt{\frac{4\sqrt{\lambda_n}}{2\sqrt{\lambda_n}L - \sin 2\sqrt{\lambda_n}L}} \sin \sqrt{\lambda_n} x \quad n = 1, 2, \dots \quad (5)$$

(c) If $f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x)$, then

$$c_n = \int_0^L f(x) \phi_n(x) \, dx = \sqrt{\frac{4\sqrt{\lambda_n}}{2\sqrt{\lambda_n}L - \sin 2\sqrt{\lambda_n}L}} \int_0^L f(x) \sin \sqrt{\lambda_n} x \, dx \quad (6)$$

Thus the required expansion is that with coefficients given by (6). The expansion for $f(x)$ can equivalently be written as

$$f(x) = \sum_{n=1}^{\infty} \frac{4\sqrt{\lambda_n}}{2\sqrt{\lambda_n}L - \sin 2\sqrt{\lambda_n}L} \left\{ \int_0^L f(x) \sin \sqrt{\lambda_n} x \, dx \right\} \sin \sqrt{\lambda_n} x \quad (7)$$

GRAM-SCHMIDT ORTHONORMALIZATION PROCESS

3.12. Generate a set of polynomials orthonormal in the interval $(-1, 1)$ from the sequence $1, x, x^2, x^3, \dots$.

According to the Gram-Schmidt process we consider the functions

$$\phi_1(x) = c_{11}, \quad \phi_2(x) = c_{21} + c_{22}x, \quad \phi_3(x) = c_{31} + c_{32}x + c_{33}x^2, \quad \dots$$

Since $\phi_2(x)$ must be orthogonal to $\phi_1(x)$ in $(-1, 1)$, we have

$$\int_{-1}^1 (c_{11})(c_{21} + c_{22}x) \, dx = 0 \quad \text{i.e.} \quad c_{11}(2c_{21}) = 0$$

from which $c_{21} = 0$, because $c_{11} \neq 0$. Thus we have

$$\phi_1(x) = c_{11} \quad \phi_2(x) = c_{22}x$$

In order that $\phi_1(x)$ and $\phi_2(x)$ be normalized in $(-1, 1)$ we must have

$$\int_{-1}^1 (c_{11})^2 \, dx = 1 \quad \int_{-1}^1 (c_{22}x)^2 \, dx = 1$$

from which

$$c_{11} = \pm \sqrt{\frac{1}{2}} \quad c_{22} = \pm \sqrt{\frac{3}{2}}$$

Since $\phi_3(x)$ must be orthogonal to $\phi_1(x)$ and $\phi_2(x)$ in $(-1, 1)$, we have

$$\int_{-1}^1 (c_{11})(c_{31} + c_{32}x + c_{33}x^2) \, dx = 0, \quad \int_{-1}^1 (c_{22}x)(c_{31} + c_{32}x + c_{33}x^2) \, dx = 0$$

from which

$$2c_{31} + \frac{2}{3}c_{33} = 0 \quad \text{or} \quad c_{33} = -3c_{31}, \quad c_{32} = 0$$

Thus

$$\phi_3(x) = c_{31}(1 - 3x^2)$$

In order that $\phi_3(x)$ be normalized in $(-1, 1)$ we must have

$$\int_{-1}^1 [c_{31}(1-3x^2)]^2 dx = 1 \quad \text{whence} \quad c_{31} = \pm \frac{1}{2} \sqrt{\frac{5}{2}}$$

The orthonormal functions thus far are given by

$$\phi_1(x) = \pm \sqrt{\frac{1}{2}}, \quad \phi_2(x) = \pm \sqrt{\frac{3}{2}}x, \quad \phi_3(x) = \pm \sqrt{\frac{5}{2}} \left(\frac{3x^2 - 1}{2} \right)$$

By continuing the process (see Problem 3.29) we find

$$\phi_4(x) = \pm \sqrt{\frac{7}{2}} \left(\frac{5x^3 - 3x}{2} \right), \quad \phi_5(x) = \pm \sqrt{\frac{9}{2}} \left(\frac{35x^4 - 30x^2 + 3}{8} \right), \quad \dots$$

From these we obtain the *Legendre polynomials*

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{3x^2 - 1}{2}, \quad P_3(x) = \frac{5x^3 - 3x}{2},$$

$$P_4(x) = \frac{35x^4 - 30x^2 + 3}{8}, \quad \dots$$

The polynomials are such that $P_n(1) = 1$, $n = 0, 1, 2, 3, \dots$. We shall investigate Legendre polynomials and applications in Chapter 7.

APPLICATIONS TO BOUNDARY VALUE PROBLEMS

3.13. A thin conducting bar whose ends are at $x = 0$ and $x = L$ has the end $x = 0$ at temperature zero, while at the end $x = L$ radiation takes place into a medium of temperature zero. Assuming that the surface is insulated and that the initial temperature is $f(x)$; $0 < x < L$, find the temperature at any point x of the bar at any time t .

The heat conduction equation for the temperature in a bar whose surface is insulated is

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \quad (1)$$

Assuming Newton's law of cooling applies at the end $x = L$, we obtain the condition

$$-Ku_x(L, t) = h[u(L, t) - 0]$$

or

$$u_x(L, t) = -\beta u(L, t) \quad (2)$$

where $\beta = K/h$, K being the thermal conductivity and h a constant of proportionality. The remaining boundary conditions are given by

$$u(0, t) = 0, \quad u(x, 0) = f(x), \quad |u(x, t)| < M$$

To solve this boundary value problem we let $u = XT$ in (1) to obtain the solution

$$u = e^{-\kappa\lambda^2 t} (A \cos \lambda x + \beta \sin \lambda x)$$

From $u(0, t) = 0$ we find $A = 0$, so that

$$u(x, t) = B e^{-\kappa\lambda^2 t} \sin \lambda x$$

The boundary condition (2) yields

$$\tan \lambda L = -\frac{\lambda}{\beta} \quad (3)$$

This equation is exactly the same as (1) on page 60 with λ replaced by λ^2 . Denoting the n th positive root of (3) by λ_n , $n = 1, 2, 3, \dots$, we see that solutions are

$$u(x, t) = B_n e^{-\kappa\lambda_n^2 t} \sin \lambda_n x$$

Using the principle of superposition we then arrive at a solution

$$u(x, t) = \sum_{n=1}^{\infty} B_n e^{-\kappa\lambda_n^2 t} \sin \lambda_n x$$

The last boundary condition, $u(x, 0) = f(x)$, now leads to

$$f(x) = \sum_{n=1}^{\infty} B_n \sin \lambda_n x$$

We can find B_m by multiplying both sides by $\sin \lambda_m x$ and then integrating, using the fact that

$$\int_0^L \sin \lambda_m x \sin \lambda_n x dx = 0 \quad m \neq n$$

However the result is already available to us from (6) of Problem 3.11 if we replace λ_n by λ_n^2 . Thus the solution is

$$u(x, t) = \sum_{n=1}^{\infty} \frac{4\lambda_n e^{-\kappa\lambda_n^2 t} \sin \lambda_n x}{2\lambda_n L - \sin 2\lambda_n L} \left\{ \int_0^L f(x) \sin \lambda_n x dx \right\}$$

3.14. (a) Show that separation of variables in the boundary value problem

$$g(x) \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left[K(x) \frac{\partial u}{\partial x} \right] + h(x)u \quad 0 < x < L, \quad t > 0$$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad u(x, 0) = f(x), \quad |u(x, t)| < M$$

leads to a Sturm-Liouville system. (b) Give a physical interpretation of the equation in (a). (c) How would you proceed to solve the boundary value problem in (a)?

(a) Letting $u = XT$ in the given equation, we find

$$g(x)XT' = T \frac{d}{dx} \left[K(x) \frac{dX}{dx} \right] + h(x)XT$$

Then dividing by $g(x)XT$ yields

$$\frac{T'}{T} = \frac{1}{g(x)X} \frac{d}{dx} \left[K(x) \frac{dX}{dx} \right] + h(x)$$

Setting each side equal to $-\lambda$, we find

$$T' + \lambda T = 0 \quad (1)$$

$$\frac{d}{dx} \left[K(x) \frac{dX}{dx} \right] + [h(x) + \lambda g(x)]X = 0 \quad (2)$$

Also, from the conditions $u(0, t) = 0$ and $u(L, t) = 0$ we are led to the conditions

$$X(0) = 0 \quad X(L) = 0 \quad (3)$$

The required Sturm-Liouville system is given by (2) and (3). Note that the Sturm-Liouville differential equation (2) corresponds to that of (18), page 54, if we choose $y = X$, $p(x) = K(x)$, $q(x) = h(x)$, $r(x) = g(x)$.

- (b) By comparison with the derivation of the heat conduction equation on page 9 we see that $u(x, t)$ can be interpreted as the temperature at any point x at time t . In such case $K(x)$ is the (nonconstant) thermal conductivity and $g(x)$ is the specific heat multiplied by the density. The term $h(x)u$ can represent the fact that a Newton's law of cooling type radiation into a medium of temperature zero is taking place at the surface of the bar, with a proportionality factor that depends on position.
- (c) From equation (2) subject to boundary conditions (3) we can find eigenvalues λ_n and normalized eigenfunctions $X_n(x)$, where $n = 1, 2, 3, \dots$. Equation (1) gives $T = ce^{-\lambda t}$. Thus a solution obtained by superposition is

$$u(x, t) = \sum_{n=1}^{\infty} c_n e^{-\lambda_n t} X_n(x)$$

From the boundary condition $u(x, 0) = f(x)$ we have

$$f(x) = \sum_{n=1}^{\infty} c_n X_n(x)$$

which leads to

$$c_n = \int_0^L f(x) X_n(x) dx$$

Thus we obtain the solution

$$u(x, t) = \sum_{n=1}^{\infty} \left\{ \int_0^L f(x) X_n(x) dx \right\} e^{-\lambda_n t} X_n(x)$$

Supplementary Problems

ORTHOGONAL FUNCTIONS AND ORTHONORMAL SERIES

- 3.15. Given the functions $a_0, a_1 + a_2x, a_3 + a_4x + a_5x^2$ where $a_0, \dots, a_5$ are constants. Determine the constants so that these functions are mutually orthonormal in the interval $(0, 1)$.
- 3.16. Generalize Problem 3.15 to arbitrary finite intervals.
- 3.17. (a) Show that the functions $1, 1-x, 2-4x+x^2$ are mutually orthogonal in $(0, \infty)$ with respect to the density function e^{-x} . (b) Obtain a mutually orthonormal set.
- 3.18. Give a vector interpretation to functions which are orthonormal with respect to a density or weight function.
- 3.19. (a) Show that the functions $\cos(n \cos^{-1} x)$, $n = 0, 1, 2, 3, \dots$, are mutually orthogonal in $(-1, 1)$ with respect to the weight function $(1-x^2)^{-1/2}$. (b) Obtain a mutually orthonormal set of these functions.
- 3.20. Show how to expand $f(x)$ into a series $\sum_{n=1}^{\infty} c_n \phi_n(x)$, where $\phi_n(x)$ are mutually orthonormal in (a, b) with respect to the weight function $w(x)$.
- 3.21. (a) Expand $f(x)$ into a series having the form $\sum_{n=0}^{\infty} c_n \phi_n(x)$, where $\phi_n(x)$ are the mutually orthonormal functions of Problem 3.19. (b) Discuss the relationship of the series in (a) to Fourier series.

APPROXIMATIONS IN THE LEAST-SQUARES SENSE. PARSEVAL'S IDENTITY AND COMPLETENESS

- 3.22. Let $\mathbf{r}$ be any three-dimensional vector. Show that

$$(a) \quad (\mathbf{r} \cdot \mathbf{i})^2 + (\mathbf{r} \cdot \mathbf{j})^2 \leq r^2 \quad (b) \quad (\mathbf{r} \cdot \mathbf{i})^2 + (\mathbf{r} \cdot \mathbf{j})^2 + (\mathbf{r} \cdot \mathbf{k})^2 = r^2$$

where $r^2 = \mathbf{r} \cdot \mathbf{r}$ and discuss these with reference to Bessel's inequality and Parseval's identity. Compare with Problem 3.6.

- 3.23. Suppose that one term in any orthonormal series (such as a Fourier series) is omitted. (a) Can we expand an arbitrary function $f(x)$ in the series? (b) Can Parseval's identity be satisfied? (c) Can Bessel's inequality be satisfied? Justify your answers.
- 3.24. (a) Find c_1, c_2, c_3 such that $\int_{-\pi}^{\pi} [x - (c_1 \sin x + c_2 \sin 2x + c_3 \sin 3x)]^2 dx$ is a minimum.
- (b) What is the mean square error and root mean square error in approximating x by $c_1 \sin x + c_2 \sin 2x + c_3 \sin 3x$, where c_1, c_2, c_3 are the values obtained in (a)?

- (c) Suppose that it is desired to approximate x by $\alpha_1 \sin x + \alpha_2 \sin 2x + \alpha_3 \sin 3x + \alpha_4 \sin 4x$ in the least-squares sense in the interval $(-\pi, \pi)$. Are the values $\alpha_1, \alpha_2, \alpha_3$ the same as c_1, c_2, c_3 of part (a)? Explain and discuss the significance of this.

3.25. Verify that Bessel's inequality holds in Problem 3.24.

3.26. Discuss the relationship of Problem 3.24 with the expansion of $f(x) = x$ in a Fourier series in the interval $(-\pi, \pi)$.

3.27. Prove that the set of orthonormal functions $\phi_n(x)$, $n = 1, 2, 3, \dots$, cannot be complete in (a, b) if there exists some function $f(x)$ different from zero which is orthogonal to all members of the set, i.e. if

$$\int_a^b f(x) \phi_n(x) dx = 0 \quad n = 1, 2, 3, \dots$$

3.28. Is the converse of Problem 3.27 true? Explain.

GRAM-SCHMIDT ORTHONORMALIZATION PROCESS

3.29. Verify that a continuation of the process in Problem 3.12 produces the indicated results for $\phi_4(x)$ and $\phi_5(x)$.

3.30. Given the set of functions $1, x, x^2, x^3, \dots$, obtain from these a set of functions which are mutually orthonormal in $(-1, 1)$ with respect to the weight function x .

3.31. Work Problem 3.30 if the interval is $(0, \infty)$ and the weight function is e^{-x} . The polynomials thus obtained are *Laguerre polynomials*.

3.32. Is it possible to use the Gram-Schmidt process to obtain from $x, 1-x, 3+2x$ a set of functions orthonormal in $(0, 1)$? Explain.

STURM-LIOUVILLE SYSTEMS. EIGENVALUES AND EIGENFUNCTIONS

3.33. (a) Verify that the system $y'' + \lambda y = 0$, $y'(0) = 0$, $y(1) = 0$ is a Sturm-Liouville system.

(b) Find the eigenvalues and eigenfunctions of the system.

(c) Prove that the eigenfunctions are orthogonal and determine the corresponding orthonormal functions.

3.34. Work Problem 3.33, if the boundary conditions are (a) $y(0) = 0$, $y'(1) = 0$; (b) $y'(0) = 0$, $y'(1) = 0$.

3.35. Show that in Problem 3.11 we have

$$B_n^2 = \frac{2(\lambda_n + \beta^2)}{L\lambda_n + L\beta^2 + \beta}$$

3.36. Show that any equation having the form $a_0(x)y'' + a_1(x)y' + [a_2(x) + \lambda a_3(x)]y = 0$ can be written in Sturm-Liouville form as

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda r(x)]y = 0$$

with
$$p(x) = e^{\int (a_1/a_0) dx}; \quad q(x) = \frac{a_2}{a_0} p(x), \quad r(x) = \frac{a_3}{a_0} p(x)$$

3.37. Discuss Problem 3.13 if the boundary conditions are replaced by $u_x(0, t) = h_1 u(0, t)$, $u_x(L, t) = h_2 u(L, t)$.

- 3.38. (a) Show that separation of variables in the boundary value problem

$$g(x) \frac{\partial^2 y}{\partial t^2} = \frac{\partial}{\partial x} \left[\tau(x) \frac{\partial y}{\partial x} \right] + h(x)y$$

$$y(0, t) = 0, \quad y(L, t) = 0, \quad y(x, 0) = f(x), \quad y_t(x, 0) = 0, \quad |y(x, t)| < M$$

leads to a Sturm-Liouville system. (b) Give a physical interpretation of the equations in (a). (c) How would you solve the boundary value problem?

- 3.39. Discuss Problem 3.38 if the boundary conditions $y(0, t) = 0$, $y(L, t) = 0$ are replaced by $y_x(0, t) = h_1 y(0, t)$, $y_x(L, t) = h_2 y(L, t)$, respectively.

APPLICATIONS TO BOUNDARY VALUE PROBLEMS

- 3.40. (a) Solve the boundary value problem

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \quad 0 < x < L, \quad t > 0$$

$$u(0, t) = 0, \quad u_x(L, t) = 0, \quad u(x, 0) = f(x), \quad |u(x, t)| < M$$

and (b) interpret physically.

- 3.41. (a) Solve the boundary value problem

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$$

$$y(0, t) = 0, \quad y_x(L, t) = 0, \quad y(x, 0) = f(x), \quad y_t(x, 0) = 0, \quad |y(x, t)| < M$$

and (b) interpret physically.

- 3.42. (a) Solve the boundary value problem

$$\frac{\partial^2 y}{\partial t^2} + b^2 \frac{\partial^4 y}{\partial x^4} = 0 \quad 0 < x < L, \quad t > 0$$

$$y(0, t) = 0, \quad y_x(0, t) = 0, \quad y(L, t) = 0, \quad y_x(L, t) = 0, \quad y(x, 0) = f(x), \quad |y(x, t)| < M$$

and (b) interpret physically.

- 3.43. Show that the solution of the boundary value problem

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \quad 0 < x < l, \quad t > 0$$

$$u_x(0, t) = hu(0, t), \quad u_x(l, t) = -hu(l, t), \quad u(x, 0) = f(x)$$

where κ , h and l are constants, is

$$u(x, t) = \sum_{n=1}^{\infty} e^{-\kappa \lambda_n^2 t} \frac{\lambda_n \cos \lambda_n x + h \sin \lambda_n x}{(\lambda_n^2 + h^2)l + 2h} \int_0^l f(x)(\lambda_n \cos \lambda_n x + h \sin \lambda_n x) dx$$

where λ_n are solutions of the equation $\tan \lambda l = \frac{2h\lambda}{\lambda^2 - h^2}$. Give a physical interpretation.

Chapter 4

Gamma, Beta and Other Special Functions

SPECIAL FUNCTIONS

In the process of obtaining solutions to boundary value problems we often arrive at *special functions*. In this chapter we shall survey some special functions that will be employed in later chapters. If desired, the student may skip this chapter, returning to it should the need arise.

THE GAMMA FUNCTION

The *gamma function*, denoted by $\Gamma(n)$, is defined by

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx \quad (1)$$

which is convergent for $n > 0$.

A recurrence formula for the gamma function is

$$\Gamma(n+1) = n\Gamma(n) \quad (2)$$

where $\Gamma(1) = 1$ (see Problem 4.1). From (2), $\Gamma(n)$ can be determined for all $n > 0$ when the values for $1 \leq n < 2$ (or any other interval of unit length) are known (see table on page 68). In particular if n is a positive integer, then

$$\Gamma(n+1) = n! \quad n = 1, 2, 3, \dots \quad (3)$$

For this reason $\Gamma(n)$ is sometimes called the *factorial function*.

Examples. $\Gamma(2) = 1! = 1, \quad \Gamma(6) = 5! = 120, \quad \frac{\Gamma(5)}{\Gamma(3)} = \frac{4!}{2!} = 12$

It can be shown (Problem 4.4) that

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad (4)$$

The recurrence relation (2) is a difference equation which has (1) as a solution. By taking (1) as the definition of $\Gamma(n)$ for $n > 0$, we can generalize the gamma function to $n < 0$ by use of (2) in the form

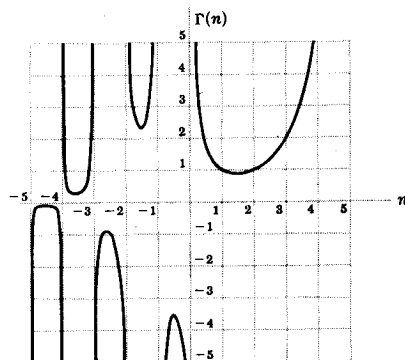
$$\Gamma(n) = \frac{\Gamma(n+1)}{n} \quad (5)$$

See Problem 4.7, for example. The process is called *analytic continuation*.

TABLE OF VALUES AND GRAPH OF THE GAMMA FUNCTION

n	$\Gamma(n)$
1.00	1.0000
1.10	0.9514
1.20	0.9182
1.30	0.8975
1.40	0.8873
1.50	0.8862
1.60	0.8935
1.70	0.9086
1.80	0.9314
1.90	0.9618
2.00	1.0000

Fig. 4-1

ASYMPTOTIC FORMULA FOR $\Gamma(n)$

If n is large, the computational difficulties inherent in a direct calculation of $\Gamma(n)$ are apparent. A useful result in such case is supplied by the relation

$$\Gamma(n+1) = \sqrt{2\pi n} n^n e^{-n} e^{\theta/12(n+1)} \quad 0 < \theta < 1 \quad (6)$$

For most practical purposes the last factor, which is very close to 1 for large n , can be omitted. If n is an integer, we can write

$$n! \sim \sqrt{2\pi n} n^n e^{-n} \quad (7)$$

where $\sim$ means "is approximately equal to for large n ". This is sometimes called *Stirling's factorial approximation* (or *asymptotic formula*) for $n!$.

MISCELLANEOUS RESULTS INVOLVING THE GAMMA FUNCTION

$$1. \quad \Gamma(x) \Gamma(1-x) = \frac{\pi}{\sin x\pi}$$

In particular if $x = \frac{1}{2}$, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ as in (4).

$$2. \quad 2^{2x-1} \Gamma(x) \Gamma(x+\frac{1}{2}) = \sqrt{\pi} \Gamma(2x)$$

This is called the *duplication formula* for the gamma function.

$$3. \quad \Gamma(x) \Gamma\left(x + \frac{1}{m}\right) \Gamma\left(x + \frac{2}{m}\right) \cdots \Gamma\left(x + \frac{m-1}{m}\right) = m^{(1/2)-mx} (2\pi)^{(m-1)/2} \Gamma(mx)$$

The duplication formula is a special case of this with $m = 2$.

$$4. \quad \Gamma(x+1) \sim \sqrt{2\pi x} x^x e^{-x} \left\{ 1 + \frac{1}{12x} + \frac{1}{288x^2} - \frac{139}{51,840x^3} + \cdots \right\}$$

This is called *Stirling's asymptotic series* for the gamma function. The series in braces is an asymptotic series as defined on page 70.

$$5. \quad \Gamma'(1) = \int_0^\infty e^{-x} \ln x \, dx = -\gamma$$

where γ is *Euler's constant* and is defined as

$$\lim_{M \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{M} - \ln M \right) = 0.5772156 \dots$$

$$6. \quad \frac{\Gamma'(p+1)}{\Gamma(p+1)} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p} - \gamma$$

THE BETA FUNCTION

The *beta function*, denoted by $B(m, n)$, is defined by

$$B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx \quad (8)$$

which is convergent for $m > 0, n > 0$.

The beta function is connected with the gamma function according to the relation

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \quad (9)$$

See Problem 4.12. Using (4) we can define $B(m, n)$ for $m < 0, n < 0$.

Many integrals can be evaluated in terms of beta or gamma functions. Two useful results are

$$\int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \frac{1}{2} B(m, n) = \frac{\Gamma(m) \Gamma(n)}{2 \Gamma(m+n)} \quad (10)$$

valid for $m > 0$ and $n > 0$ (see Problems 4.11 and 4.14) and

$$\int_0^\infty \frac{x^{p-1}}{1+x} dx = \Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin p\pi} \quad 0 < p < 1 \quad (11)$$

See Problem 4.18.

OTHER SPECIAL FUNCTIONS

Many other special functions are of importance in science and engineering. Some of these are given in the following list. Others will be considered in later chapters.

1. Error function. $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du = 1 - \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-u^2} du$
2. Complementary error function. $\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-u^2} du = 1 - \operatorname{erf}(x)$
3. Exponential integral. $Ei(x) = \int_x^\infty \frac{e^{-u}}{u} du$
4. Sine integral. $Si(x) = \int_0^x \frac{\sin u}{u} du = \frac{\pi}{2} - \int_x^\infty \frac{\sin u}{u} du$
5. Cosine integral. $Ci(x) = \int_x^\infty \frac{\cos u}{u} du$
6. Fresnel sine integral. $S(x) = \sqrt{\frac{2}{\pi}} \int_0^x \sin u^2 du = 1 - \sqrt{\frac{2}{\pi}} \int_x^\infty \sin u^2 du$
7. Fresnel cosine integral. $C(x) = \sqrt{\frac{2}{\pi}} \int_0^x \cos u^2 du = 1 - \sqrt{\frac{2}{\pi}} \int_x^\infty \cos u^2 du$

ASYMPTOTIC SERIES OR EXPANSIONS

Consider the series

$$S(x) = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} + \cdots \quad (12)$$

$$\text{and suppose that} \quad S_n(x) = a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \quad (13)$$

are the partial sums of the series.

If $R_n(x) = f(x) - S_n(x)$, where $f(x)$ is given, is such that for every n

$$\lim_{x \rightarrow \infty} x^n |R_n(x)| = 0 \quad (14)$$

then $S(x)$ is called an *asymptotic series* or *expansion* of $f(x)$ and we denote this by writing $f(x) \sim S(x)$.

In practice the series (12) diverges. However, by taking the sum of successive terms of the series, stopping just before the terms begin to increase, we may obtain a useful approximation for $f(x)$. The approximation becomes better the larger the value of x .

Various operations with asymptotic series are permissible. For example, asymptotic series may be multiplied together or integrated term by term to yield another asymptotic series.

Solved Problems

THE GAMMA FUNCTION

4.1. Prove: (a) $\Gamma(n+1) = n\Gamma(n)$, $n > 0$; (b) $\Gamma(n+1) = n!$, $n = 1, 2, 3, \dots$

$$\begin{aligned} (a) \quad \Gamma(n+1) &= \int_0^\infty x^n e^{-x} dx = \lim_{M \rightarrow \infty} \int_0^M x^n e^{-x} dx \\ &= \lim_{M \rightarrow \infty} \left\{ (x^n)(-e^{-x}) \Big|_0^M - \int_0^M (-e^{-x})(nx^{n-1}) dx \right\} \\ &= \lim_{M \rightarrow \infty} \left\{ -M^n e^{-M} + n \int_0^M x^{n-1} e^{-x} dx \right\} = n\Gamma(n) \quad \text{if } n > 0 \end{aligned}$$

$$(b) \quad \Gamma(1) = \int_0^\infty e^{-x} dx = \lim_{M \rightarrow \infty} \int_0^M e^{-x} dx = \lim_{M \rightarrow \infty} (1 - e^{-M}) = 1$$

Put $n = 1, 2, 3, \dots$ in $\Gamma(n+1) = n\Gamma(n)$. Then

$$\Gamma(2) = 1\Gamma(1) = 1, \quad \Gamma(3) = 2\Gamma(2) = 2 \cdot 1 = 2!, \quad \Gamma(4) = 3\Gamma(3) = 3 \cdot 2! = 3!$$

In general, $\Gamma(n+1) = n!$ if n is a positive integer.

4.2. Evaluate (a) $\frac{\Gamma(6)}{2\Gamma(3)}$, (b) $\frac{\Gamma(\frac{5}{2})}{\Gamma(\frac{1}{2})}$, (c) $\frac{\Gamma(3)\Gamma(2.5)}{\Gamma(5.5)}$, (d) $\frac{6\Gamma(\frac{3}{2})}{.5\Gamma(\frac{3}{2})}$.

$$(a) \quad \frac{\Gamma(6)}{2\Gamma(3)} = \frac{5!}{2 \cdot 2!} = \frac{5 \cdot 4 \cdot 3 \cdot 2}{2 \cdot 2} = 30$$

$$(b) \quad \frac{\Gamma(\frac{5}{2})}{\Gamma(\frac{1}{2})} = \frac{\frac{3}{2}\Gamma(\frac{3}{2})}{\Gamma(\frac{1}{2})} = \frac{\frac{3}{2} \cdot \frac{1}{2}\Gamma(\frac{1}{2})}{\Gamma(\frac{1}{2})} = \frac{3}{4}$$

$$(c) \quad \frac{\Gamma(3)\Gamma(2.5)}{\Gamma(5.5)} = \frac{2!(1.5)(0.5)\Gamma(0.5)}{(4.5)(3.5)(2.5)(1.5)(0.5)\Gamma(0.5)} = \frac{16}{315}$$

$$(d) \quad \frac{6 \Gamma(\frac{8}{3})}{5 \Gamma(\frac{2}{3})} = \frac{6(\frac{5}{3})(\frac{2}{3}) \Gamma(\frac{2}{3})}{5 \Gamma(\frac{2}{3})} = \frac{4}{3}$$

4.3. Evaluate (a) $\int_0^\infty x^3 e^{-x} dx$, (b) $\int_0^\infty x^6 e^{-2x} dx$.

$$(a) \quad \int_0^\infty x^3 e^{-x} dx = \Gamma(4) = 3! = 6$$

(b) Let $2x = y$. Then the integral becomes

$$\int_0^\infty \left(\frac{y}{2}\right)^6 e^{-y} \frac{dy}{2} = \frac{1}{2^7} \int_0^\infty y^6 e^{-y} dy = \frac{\Gamma(7)}{2^7} = \frac{6!}{2^7} = \frac{45}{8}$$

4.4. Prove that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

We have $\Gamma(\frac{1}{2}) = \int_0^\infty x^{-1/2} e^{-x} dx = 2 \int_0^\infty e^{-u^2} du$, on letting $x = u^2$. It follows that

$$\{\Gamma(\frac{1}{2})\}^2 = \left\{2 \int_0^\infty e^{-u^2} du\right\} \left\{2 \int_0^\infty e^{-v^2} dv\right\} = 4 \int_0^\infty \int_0^\infty e^{-(u^2+v^2)} du dv$$

Changing to polar coordinates (ρ, ϕ) , where $u = \rho \cos \phi$, $v = \rho \sin \phi$, the last integral becomes

$$4 \int_{\phi=0}^{\pi/2} \int_{\rho=0}^\infty e^{-\rho^2} \rho d\rho d\phi = 4 \int_{\phi=0}^{\pi/2} \left[-\frac{1}{2} e^{-\rho^2}\right]_{\rho=0}^\infty d\phi = \pi$$

and so $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

4.5. Evaluate (a) $\int_0^\infty \sqrt{y} e^{-y^3} dy$, (b) $\int_0^\infty 3^{-4z^2} dz$, (c) $\int_0^1 \frac{dx}{\sqrt{-\ln x}}$.

(a) Letting $y^3 = x$, the integral becomes

$$\int_0^\infty \sqrt{x^{1/3}} e^{-x} \cdot \frac{1}{3} x^{-2/3} dx = \frac{1}{3} \int_0^\infty x^{-1/2} e^{-x} dx = \frac{1}{3} \Gamma(\frac{1}{2}) = \frac{\sqrt{\pi}}{3}$$

(b) $\int_0^\infty 3^{-4z^2} dz = \int_0^\infty (e^{\ln 3})^{(-4z^2)} dz = \int_0^\infty e^{-(4 \ln 3)z^2} dz$. Let $(4 \ln 3)z^2 = x$ and the integral becomes

$$\int_0^\infty e^{-x} d\left(\frac{x^{1/2}}{\sqrt{4 \ln 3}}\right) = \frac{1}{2\sqrt{4 \ln 3}} \int_0^\infty x^{-1/2} e^{-x} dx = \frac{\Gamma(\frac{1}{2})}{2\sqrt{4 \ln 3}} = \frac{\sqrt{\pi}}{4\sqrt{\ln 3}}$$

(c) Let $-\ln x = u$. Then $x = e^{-u}$. When $x = 1$, $u = 0$; when $x = 0$, $u = \infty$. The integral becomes

$$\int_0^\infty \frac{e^{-u}}{\sqrt{u}} du = \int_0^\infty u^{-1/2} e^{-u} du = \Gamma(\frac{1}{2}) = \sqrt{\pi}$$

4.6. Evaluate $\int_0^\infty x^m e^{-ax^n} dx$, where m, n, a are positive constants.

Letting $ax^n = y$, the integral becomes

$$\int_0^\infty \left\{\left(\frac{y}{a}\right)^{1/n}\right\}^m e^{-y} d\left\{\left(\frac{y}{a}\right)^{1/n}\right\} = \frac{1}{na^{(m+1)/n}} \int_0^\infty y^{(m+1)/n-1} e^{-y} dy = \frac{1}{na^{(m+1)/n}} \Gamma\left(\frac{m+1}{n}\right)$$

4.7. Evaluate (a) $\Gamma(-1/2)$, (b) $\Gamma(-5/2)$.

We use the generalization to negative values defined by $\Gamma(n) = \frac{\Gamma(n+1)}{n}$.

(a) Letting $n = -\frac{1}{2}$, $\Gamma(-1/2) = \frac{\Gamma(1/2)}{-1/2} = -2\sqrt{\pi}$.

(b) Letting $n = -3/2$, $\Gamma(-3/2) = \frac{\Gamma(-1/2)}{-3/2} = \frac{-2\sqrt{\pi}}{-3/2} = \frac{4\sqrt{\pi}}{3}$, using (a).

Then $\Gamma(-5/2) = \frac{\Gamma(-3/2)}{-5/2} = -\frac{8}{15}\sqrt{\pi}$.

4.8. Prove that $\int_0^1 x^m (\ln x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$, where n is a positive integer and $m > -1$.

Letting $x = e^{-y}$, the integral becomes $(-1)^n \int_0^\infty y^n e^{-(m+1)y} dy$. If $(m+1)y = u$, this last integral becomes

$$(-1)^n \int_0^\infty \frac{u^n}{(m+1)^n} e^{-u} \frac{du}{m+1} = \frac{(-1)^n}{(m+1)^{n+1}} \int_0^\infty u^n e^{-u} du = \frac{(-1)^n}{(m+1)^{n+1}} \Gamma(n+1) = \frac{(-1)^n n!}{(m+1)^{n+1}}$$

4.9. Prove that $\int_0^\infty e^{-\alpha\lambda^2} \cos \beta\lambda d\lambda = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} e^{-\beta^2/4\alpha}$.

Let $I = I(\alpha, \beta) = \int_0^\infty e^{-\alpha\lambda^2} \cos \beta\lambda d\lambda$. Then

$$\begin{aligned} \frac{\partial I}{\partial \beta} &= \int_0^\infty (-\lambda e^{-\alpha\lambda^2}) \sin \beta\lambda d\lambda \\ &= \frac{e^{-\alpha\lambda^2}}{2\alpha} \sin \beta\lambda \Big|_0^\infty - \frac{\beta}{2\alpha} \int_0^\infty e^{-\alpha\lambda^2} \cos \beta\lambda d\lambda = -\frac{\beta}{2\alpha} I \end{aligned}$$

Thus $\frac{1}{I} \frac{\partial I}{\partial \beta} = -\frac{\beta}{2\alpha}$ or $\frac{\partial}{\partial \beta} \ln I = -\frac{\beta}{2\alpha}$ (1)

Integration with respect to β yields

$$\ln I = -\frac{\beta^2}{4\alpha} + c_1$$

or $I = I(\alpha, \beta) = C e^{-\beta^2/4\alpha}$ (2)

But $C = I(\alpha, 0) = \int_0^\infty e^{-\alpha\lambda^2} d\lambda = \frac{1}{2\sqrt{\alpha}} \int_0^\infty x^{-1/2} e^{-x} dx = \frac{\Gamma(\frac{1}{2})}{2\sqrt{\alpha}} = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}}$, on letting $x = \alpha\lambda^2$.

Thus, as required,

$$I = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} e^{-\beta^2/4\alpha}$$

4.10. A particle is attracted toward a fixed point O with a force inversely proportional to its instantaneous distance from O . If the particle is released from rest, find the time for it to reach O .

At time $t = 0$ let the particle be located on the x -axis at $x = a > 0$ and let O be the origin. Then by Newton's law

$$m \frac{d^2 x}{dt^2} = -\frac{k}{x} \quad (1)$$

where m is the mass of the particle and $k > 0$ is a constant of proportionality.

Let $\frac{dx}{dt} = v$, the velocity of the particle. Then $\frac{d^2 x}{dt^2} = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$ and (1) becomes

$$mv \frac{dv}{dx} = -\frac{k}{x} \quad \text{or} \quad \frac{mv^2}{2} = -k \ln x + c \quad (2)$$

upon integrating. Since $v = 0$ at $x = a$, we find $c = k \ln a$. Then

$$\frac{mv^2}{2} = k \ln \frac{a}{x} \quad \text{or} \quad v = \frac{dx}{dt} = -\sqrt{\frac{2k}{m}} \sqrt{\ln \frac{a}{x}} \quad (3)$$

where the negative sign is chosen since x is decreasing as t increases. We thus find that the time T taken for the particle to go from $x = a$ to $x = 0$ is given by

$$T = \sqrt{\frac{m}{2k}} \int_0^a \frac{dx}{\sqrt{\ln a/x}} \quad (4)$$

Letting $\ln a/x = u$ or $x = ae^{-u}$, this becomes

$$T = a \sqrt{\frac{m}{2k}} \int_0^\infty u^{-1/2} e^{-u} du = a \sqrt{\frac{m}{2k}} \Gamma\left(\frac{1}{2}\right) = a \sqrt{\frac{\pi m}{2k}}$$

THE BETA FUNCTION

4.11. Prove that (a) $B(m, n) = B(n, m)$, (b) $B(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$.

(a) Using the transformation $x = 1 - y$, we have

$$\begin{aligned} B(m, n) &= \int_0^1 x^{m-1} (1-x)^{n-1} dx = \int_0^1 (1-y)^{m-1} y^{n-1} dy \\ &= \int_0^1 y^{n-1} (1-y)^{m-1} dy = B(n, m) \end{aligned}$$

(b) Using the transformation $x = \sin^2 \theta$, we have

$$\begin{aligned} B(m, n) &= \int_0^1 x^{m-1} (1-x)^{n-1} dx = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (\cos^2 \theta)^{n-1} 2 \sin \theta \cos \theta d\theta \\ &= 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta \end{aligned}$$

4.12. Prove that $B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$ $m, n > 0$.

Letting $z = x^2$, we have $\Gamma(m) = \int_0^\infty z^{m-1} e^{-z} dz = 2 \int_0^\infty x^{2m-1} e^{-x^2} dx$.

Similarly, $\Gamma(n) = 2 \int_0^\infty y^{2n-1} e^{-y^2} dy$. Then

$$\begin{aligned} \Gamma(m) \Gamma(n) &= 4 \left(\int_0^\infty x^{2m-1} e^{-x^2} dx \right) \left(\int_0^\infty y^{2n-1} e^{-y^2} dy \right) \\ &= 4 \int_0^\infty \int_0^\infty x^{2m-1} y^{2n-1} e^{-(x^2+y^2)} dx dy \end{aligned}$$

Transforming to polar coordinates, $x = \rho \cos \phi$, $y = \rho \sin \phi$,

$$\begin{aligned} \Gamma(m) \Gamma(n) &= 4 \int_{\phi=0}^{\pi/2} \int_{\rho=0}^\infty \rho^{2(m+n)-1} e^{-\rho^2} \cos^{2m-1} \phi \sin^{2n-1} \phi d\rho d\phi \\ &= 4 \left(\int_{\rho=0}^\infty \rho^{2(m+n)-1} e^{-\rho^2} d\rho \right) \left(\int_{\phi=0}^{\pi/2} \cos^{2m-1} \phi \sin^{2n-1} \phi d\phi \right) \\ &= 2 \Gamma(m+n) \int_0^{\pi/2} \cos^{2m-1} \phi \sin^{2n-1} \phi d\phi = \Gamma(m+n) B(n, m) \\ &= \Gamma(m+n) B(m, n) \end{aligned}$$

using the results of Problem 4.11. Hence the required result follows.

The above argument can be made rigorous by using a limiting procedure.

4.13. Evaluate (a) $\int_0^1 x^4(1-x)^3 dx$, (b) $\int_0^2 \frac{x^2 dx}{\sqrt{2-x}}$, (c) $\int_0^a y^4 \sqrt{a^2 - y^2} dy$.

$$(a) \int_0^1 x^4(1-x)^3 dx = B(5, 4) = \frac{\Gamma(5)\Gamma(4)}{\Gamma(9)} = \frac{4!3!}{8!} = \frac{1}{280}$$

(b) Letting $x = 2v$, the integral becomes

$$4\sqrt{2} \int_0^1 \frac{v^2}{\sqrt{1-v}} dv = 4\sqrt{2} \int_0^1 v^2(1-v)^{-1/2} dv = 4\sqrt{2} B(3, \frac{1}{2}) = \frac{4\sqrt{2} \Gamma(3) \Gamma(1/2)}{\Gamma(7/2)} = \frac{64\sqrt{2}}{15}$$

(c) Letting $y^2 = a^2x$ or $y = a\sqrt{x}$, the integral becomes

$$\frac{a^6}{2} \int_0^1 x^{3/2}(1-x)^{1/2} dx = \frac{a^6}{2} B(5/2, 3/2) = \frac{a^6 \Gamma(5/2) \Gamma(3/2)}{2 \Gamma(4)} = \frac{\pi a^6}{32}$$

4.14. Show that $\int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \frac{\Gamma(m) \Gamma(n)}{2 \Gamma(m+n)} \quad m, n > 0$.

This follows at once from Problems 4.11 and 4.12.

4.15. Evaluate (a) $\int_0^{\pi/2} \sin^6 \theta d\theta$, (b) $\int_0^{\pi/2} \sin^4 \theta \cos^5 \theta d\theta$, (c) $\int_0^{\pi} \cos^4 \theta d\theta$.

(a) Let $2m-1 = 6$, $2n-1 = 0$, i.e. $m = 7/2$, $n = 1/2$, in Problem 4.14.

Then the required integral has the value $\frac{\Gamma(7/2) \Gamma(1/2)}{2 \Gamma(4)} = \frac{5\pi}{32}$.

(b) Letting $2m-1 = 4$, $2n-1 = 5$, the required integral has the value $\frac{\Gamma(5/2) \Gamma(3)}{2 \Gamma(11/2)} = \frac{8}{315}$.

(c) $\int_0^{\pi} \cos^4 \theta d\theta = 2 \int_0^{\pi/2} \cos^4 \theta d\theta$. Thus, letting $2m-1 = 0$, $2n-1 = 4$ in Problem 4.14, the value is $\frac{2 \Gamma(1/2) \Gamma(5/2)}{2 \Gamma(3)} = \frac{3\pi}{8}$.

4.16. Prove $\int_0^{\pi/2} \sin^p \theta d\theta = \int_0^{\pi/2} \cos^p \theta d\theta = (a) \frac{1 \cdot 3 \cdot 5 \cdots (p-1)}{2 \cdot 4 \cdot 6 \cdots p} \frac{\pi}{2}$ if p is an even positive integer, (b) $\frac{2 \cdot 4 \cdot 6 \cdots (p-1)}{1 \cdot 3 \cdot 5 \cdots p}$ if p is an odd positive integer.

From Problem 4.14 with $2m-1 = p$, $2n-1 = 0$, we have

$$\int_0^{\pi/2} \sin^p \theta d\theta = \frac{\Gamma[\frac{1}{2}(p+1)] \Gamma(\frac{1}{2})}{2 \Gamma[\frac{1}{2}(p+2)]}$$

(a) If $p = 2r$, the integral equals

$$\begin{aligned} \frac{\Gamma(r+\frac{1}{2}) \Gamma(\frac{1}{2})}{2 \Gamma(r+1)} &= \frac{(r-\frac{1}{2})(r-\frac{3}{2}) \cdots \frac{1}{2} \Gamma(\frac{1}{2}) \cdot \Gamma(\frac{1}{2})}{2r(r-1) \cdots 1} \\ &= \frac{(2r-1)(2r-3) \cdots 1}{2r(2r-2) \cdots 2} \frac{\pi}{2} = \frac{1 \cdot 3 \cdot 5 \cdots (2r-1)}{2 \cdot 4 \cdot 6 \cdots 2r} \frac{\pi}{2} \end{aligned}$$

(b) If $p = 2r+1$, the integral equals

$$\frac{\Gamma(r+1) \Gamma(\frac{1}{2})}{2 \Gamma(r+\frac{3}{2})} = \frac{r(r-1) \cdots 1 \cdot \sqrt{\pi}}{2(r+\frac{1}{2})(r-\frac{1}{2}) \cdots \frac{1}{2} \sqrt{\pi}} = \frac{2 \cdot 4 \cdot 6 \cdots 2r}{1 \cdot 3 \cdot 5 \cdots (2r+1)}$$

In both cases $\int_0^{\pi/2} \sin^p \theta d\theta = \int_0^{\pi/2} \cos^p \theta d\theta$, as seen by letting $\theta = \frac{\pi}{2} - \phi$.

4.17. Evaluate (a) $\int_0^{\pi/2} \cos^6 \theta \, d\theta$, (b) $\int_0^{\pi/2} \sin^3 \theta \cos^2 \theta \, d\theta$, (c) $\int_0^{2\pi} \sin^8 \theta \, d\theta$.

(a) From Problem 4.16 the integral equals $\frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{\pi}{2} = \frac{5\pi}{32}$ [compare Problem 4.15(a)].

(b) The integral equals

$$\int_0^{\pi/2} \sin^3 \theta (1 - \sin^2 \theta) \, d\theta = \int_0^{\pi/2} \sin^3 \theta \, d\theta - \int_0^{\pi/2} \sin^5 \theta \, d\theta = \frac{2}{1 \cdot 3} - \frac{2 \cdot 4}{1 \cdot 3 \cdot 5} = \frac{2}{15}$$

The method of Problem 4.15(b) can also be used.

(c) The given integral equals $4 \int_0^{\pi/2} \sin^8 \theta \, d\theta = 4 \left(\frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6 \cdot 8} \frac{\pi}{2} \right) = \frac{35\pi}{64}$.

4.18. Given $\int_0^\infty \frac{x^{p-1}}{1+x} dx = \frac{\pi}{\sin p\pi}$, show that $\Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin p\pi}$ where $0 < p < 1$.

Letting $\frac{x}{1+x} = y$ or $x = \frac{y}{1-y}$, the given integral becomes

$$\int_0^1 y^{p-1} (1-y)^{-p} dy = B(p, 1-p) = \Gamma(p) \Gamma(1-p)$$

and the result follows.

4.19. Evaluate $\int_0^\infty \frac{dy}{1+y^4}$.

Let $y^4 = x$. Then the integral becomes $\frac{1}{4} \int_0^\infty \frac{x^{-3/4}}{1+x} dx = \frac{\pi}{4 \sin(\pi/4)} = \frac{\pi\sqrt{2}}{4}$ by Problem 4.18 with $p = \frac{1}{4}$.

The result can also be obtained by letting $y^2 = \tan \theta$.

4.20. Show that $\int_0^2 x \sqrt[3]{8-x^3} \, dx = \frac{16\pi}{9\sqrt{3}}$.

Letting $x^3 = 8y$ or $x = 2y^{1/3}$, the integral becomes

$$\begin{aligned} \int_0^1 2y^{1/3} \cdot \sqrt[3]{8(1-y)} \cdot \frac{2}{3} y^{-2/3} dy &= \frac{8}{3} \int_0^1 y^{-1/3} (1-y)^{1/3} dy = \frac{8}{3} B\left(\frac{2}{3}, \frac{4}{3}\right) \\ &= \frac{8}{3} \frac{\Gamma(\frac{2}{3}) \Gamma(\frac{4}{3})}{\Gamma(2)} = \frac{8}{9} \frac{\Gamma(\frac{1}{3}) \Gamma(\frac{2}{3})}{\Gamma(\frac{1}{3}) \Gamma(\frac{2}{3})} = \frac{8}{9} \cdot \frac{\pi}{\sin \pi/3} = \frac{16\pi}{9\sqrt{3}} \end{aligned}$$

4.21. Prove the duplication formula: $2^{2p-1} \Gamma(p) \Gamma(p + \frac{1}{2}) = \sqrt{\pi} \Gamma(2p)$.

Let $I = \int_0^{\pi/2} \sin^{2p} x \, dx$, $J = \int_0^{\pi/2} \sin^{2p} 2x \, dx$.

Then $I = \frac{1}{2} B(p + \frac{1}{2}, \frac{1}{2}) = \frac{\Gamma(p + \frac{1}{2}) \sqrt{\pi}}{2 \Gamma(p + 1)}$

Letting $2x = u$, we find

$$J = \frac{1}{2} \int_0^\pi \sin^{2p} u \, du = \int_0^{\pi/2} \sin^{2p} u \, du = I$$

But

$$\begin{aligned} J &= \int_0^{\pi/2} (2 \sin x \cos x)^{2p} dx = 2^{2p} \int_0^{\pi/2} \sin^{2p} x \cos^{2p} x \, dx \\ &= 2^{2p-1} B(p + \frac{1}{2}, p + \frac{1}{2}) = \frac{2^{2p-1} \{\Gamma(p + \frac{1}{2})\}^2}{\Gamma(2p + 1)} \end{aligned}$$

Then since $I = J$,

$$\frac{\Gamma(p + \frac{1}{2})\sqrt{\pi}}{2p\Gamma(p)} = \frac{2^{2p-1}\{\Gamma(p + \frac{1}{2})\}^2}{2p\Gamma(2p)}$$

and the required result follows.

4.22. Prove that
$$\int_0^\infty \frac{\cos x}{x^p} dx = \frac{\pi}{2\Gamma(p)\cos(p\pi/2)}, \quad 0 < p < 1.$$

We have $\frac{1}{x^p} = \frac{1}{\Gamma(p)} \int_0^\infty u^{p-1} e^{-xu} du$. Then

$$\int_0^\infty \frac{\cos x}{x^p} dx = \frac{1}{\Gamma(p)} \int_0^\infty \int_0^\infty u^{p-1} e^{-xu} \cos x du dx = \frac{1}{\Gamma(p)} \int_0^\infty \frac{u^p}{1+u^2} du \quad (1)$$

where we have reversed the order of integration and used the fact that

$$\int_0^\infty e^{-xu} \cos x dx = \frac{u}{u^2 + 1} \quad (2)$$

Letting $u^2 = v$ in the last integral in (1), we have by Problem 4.18

$$\int_0^\infty \frac{u^p}{1+u^2} du = \frac{1}{2} \int_0^\infty \frac{v^{(p-1)/2}}{1+v} dv = \frac{\pi}{2 \sin(p+1)\pi/2} = \frac{\pi}{2 \cos p\pi/2} \quad (3)$$

Substitution of (3) in (1) yields the required result.

STIRLING'S FORMULA

4.23. Show that for large n , $n! = \sqrt{2\pi n} n^n e^{-n}$ approximately.

We have

$$\Gamma(n+1) = \int_0^\infty x^n e^{-x} dx = \int_0^\infty e^{n \ln x - x} dx \quad (1)$$

The function $n \ln x - x$ has a relative maximum for $x = n$, as is easily shown by elementary calculus. This leads us to the substitution $x = n + y$. Then (1) becomes

$$\begin{aligned} \Gamma(n+1) &= e^{-n} \int_{-n}^\infty e^{n \ln(n+y) - y} dy = e^{-n} \int_{-n}^\infty e^{n \ln n + n \ln(1+y/n) - y} dy \\ &= n^n e^{-n} \int_{-n}^\infty e^{n \ln(1+y/n) - y} dy \end{aligned} \quad (2)$$

Up to now the analysis is rigorous. The formal procedures which follow can be made rigorous by suitable limiting processes, but the proofs become involved and we shall omit them.

In (2) use the result

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \quad (3)$$

with $x = y/n$. Then on letting $y = \sqrt{n}v$, we find

$$\Gamma(n+1) = n^n e^{-n} \int_{-n}^\infty e^{-y^2/2n + y^3/3n^2 - \dots} dy = n^n e^{-n} \sqrt{n} \int_{-\sqrt{n}}^\infty e^{-v^2/2 + v^3/3\sqrt{n} - \dots} dv \quad (4)$$

When n is large a close approximation is

$$\Gamma(n+1) = n^n e^{-n} \sqrt{n} \int_{-\infty}^\infty e^{-v^2/2} dv = \sqrt{2\pi n} n^n e^{-n} \quad (5)$$

It is of interest that from (4) we can obtain the entire asymptotic series for the gamma function (result 4. on page 68). See Problem 4.36.

SPECIAL FUNCTIONS AND ASYMPTOTIC EXPANSIONS

4.24. (a) Prove that if $x > 0$, $p > 0$, then

$$I_p = \int_x^\infty \frac{e^{-u}}{u^p} du = S_n(x) + R_n(x)$$

where

$$S_n(x) = e^{-x} \left\{ \frac{1}{x^p} - \frac{p}{x^{p+1}} + \frac{p(p+1)}{x^{p+2}} - \cdots + (-1)^n \frac{p(p+1) \cdots (p+n)}{x^{p+n+1}} \right\}$$

$$R_n(x) = (-1)^{n+1} p(p+1) \cdots (p+n) \int_x^\infty \frac{e^{-u}}{u^{p+n+1}} du$$

(b) Prove that $\lim_{x \rightarrow \infty} x^n \left| \int_x^\infty \frac{e^{-u}}{u^p} du - S_n(x) \right| = \lim_{x \rightarrow \infty} x^n |R_n(x)| = 0$.

(c) Explain the significance of the results in (b).

(a) Integrating by parts, we have

$$I_p = \int_x^\infty \frac{e^{-u}}{u^p} du = \frac{e^{-x}}{x^p} - p \int_x^\infty \frac{e^{-u}}{u^{p+1}} du = \frac{e^{-x}}{x^p} - p I_{p+1}$$

Similarly $I_{p+1} = \frac{e^{-x}}{x^{p+1}} - (p+1) I_{p+2}$ so that

$$I_p = \frac{e^{-x}}{x^p} - p \left\{ \frac{e^{-x}}{x^{p+1}} - (p+1) I_{p+2} \right\} = \frac{e^{-x}}{x^p} - \frac{pe^{-x}}{x^{p+1}} + p(p+1) I_{p+2}$$

By continuing in this manner the required result follows.

$$\begin{aligned} (b) |R_n(x)| &= p(p+1) \cdots (p+n) \int_x^\infty \frac{e^{-u}}{u^{p+n+1}} du \leq p(p+1) \cdots (p+n) \int_x^\infty \frac{e^{-u}}{x^{p+n+1}} du \\ &\leq \frac{p(p+1) \cdots (p+n)}{x^{p+n+1}} \end{aligned}$$

since $\int_x^\infty e^{-u} du \leq \int_0^\infty e^{-u} du = 1$. Thus

$$\lim_{x \rightarrow \infty} x^n |R_n(x)| \leq \lim_{x \rightarrow \infty} \frac{p(p+1) \cdots (p+n)}{x^{p+1}} = 0$$

(c) Because of the results in (b), we can say that

$$\int_x^\infty \frac{e^{-u}}{u^p} du \sim e^{-x} \left\{ \frac{1}{x^p} - \frac{p}{x^{p+1}} + \frac{p(p+1)}{x^{p+2}} - \cdots \right\} \quad (1)$$

i.e. the series on the right is the asymptotic expansion of the function on the left.

4.25. Show that

$$\operatorname{erf}(x) \sim 1 - \frac{e^{-x^2}}{\sqrt{\pi}} \left(\frac{1}{x} - \frac{1}{2x^3} + \frac{1 \cdot 3}{2^2 x^5} - \frac{1 \cdot 3 \cdot 5}{2^3 x^7} + \cdots \right)$$

$$\begin{aligned} \text{We have } \operatorname{erf}(x) &= \frac{2}{\sqrt{\pi}} \int_0^x e^{-v^2} dv = \frac{1}{\sqrt{\pi}} \int_0^{x^2} u^{-1/2} e^{-u} du \\ &= 1 - \frac{1}{\sqrt{\pi}} \int_{x^2}^\infty u^{-1/2} e^{-u} du \end{aligned}$$

Now from the result (1) of Problem 4.24 we have, on letting $p = 1/2$ and replacing x by x^2 ,

$$\int_{x^2}^\infty u^{-1/2} e^{-u} du \sim e^{-x^2} \left(\frac{1}{x} - \frac{1}{2x^3} + \frac{1 \cdot 3}{2^2 x^5} - \frac{1 \cdot 3 \cdot 5}{2^3 x^7} + \cdots \right)$$

which gives the required result.

Supplementary Problems

THE GAMMA FUNCTION

- 4.26. Evaluate (a) $\frac{\Gamma(7)}{2\Gamma(4)\Gamma(3)}$, (b) $\frac{\Gamma(3)\Gamma(3/2)}{\Gamma(9/2)}$, (c) $\Gamma(1/2)\Gamma(3/2)\Gamma(5/2)$.
- 4.27. Evaluate (a) $\int_0^\infty x^4 e^{-x} dx$, (b) $\int_0^\infty x^6 e^{-3x} dx$, (c) $\int_0^\infty x^2 e^{-2x^3} dx$.
- 4.28. Find (a) $\int_0^\infty e^{-x^3} dx$, (b) $\int_0^\infty \sqrt[4]{x} e^{-\sqrt{x}} dx$, (c) $\int_0^\infty y^3 e^{-2y^5} dy$.
- 4.29. Show that $\int_0^\infty \frac{e^{-st}}{\sqrt{t}} dt = \sqrt{\frac{\pi}{s}}$, $s > 0$.
- 4.30. Prove that (a) $\Gamma(n) = \int_0^1 \left(\ln \frac{1}{x}\right)^{n-1} dx$, $n > 0$
 (b) $\int_0^1 x^p \left(\ln \frac{1}{x}\right)^q dx = \frac{\Gamma(p+1)}{(p+1)^{q+1}}$, $p > -1$, $q > -1$
- 4.31. Evaluate (a) $\int_0^1 (\ln x)^4 dx$, (b) $\int_0^1 (x \ln x)^3 dx$, (c) $\int_0^1 \sqrt[3]{\ln(1/x)} dx$.
- 4.32. Evaluate (a) $\Gamma(-7/2)$, (b) $\Gamma(-1/3)$.
- 4.33. Prove that $\lim_{x \rightarrow -m} |\Gamma(x)| = \infty$ where $m = 0, 1, 2, 3, \dots$.
- 4.34. Prove that if m is a positive integer, $\Gamma(-m + \frac{1}{2}) = \frac{(-1)^m 2^m \sqrt{\pi}}{1 \cdot 3 \cdot 5 \cdots (2m-1)}$
- 4.35. Prove that $\Gamma'(1) = \int_0^\infty e^{-x} \ln x dx$ is a negative number. (It is equal to $-\gamma$, where $\gamma = 0.577215\dots$ is called *Euler's constant*.)
- 4.36. Obtain the miscellaneous result 4. on page 68 from the result (4) of Problem 4.23.
 [Hint: Expand $e^{(v^3/3\sqrt{n})} - \dots$ in a power series and replace the lower limit of the integral by $-\infty$.]

THE BETA FUNCTION

- 4.37. Evaluate (a) $B(3, 5)$, (b) $B(3/2, 2)$, (c) $B(1/3, 2/3)$.
- 4.38. Find (a) $\int_0^1 x^2(1-x)^3 dx$, (b) $\int_0^1 \sqrt{(1-x)/x} dx$, (c) $\int_0^2 (4-x^2)^{3/2} dx$.
- 4.39. Evaluate (a) $\int_0^4 u^{3/2}(4-u)^{5/2} du$, (b) $\int_0^3 \frac{dx}{\sqrt{3x-x^2}}$.
- 4.40. Prove that $\int_0^a \frac{dy}{\sqrt{a^4-y^4}} = \frac{\{\Gamma(1/4)\}^2}{4a\sqrt{2\pi}}$.
- 4.41. Evaluate (a) $\int_0^{\pi/2} \sin^4 \theta \cos^4 \theta d\theta$, (b) $\int_0^{2\pi} \cos^6 \theta d\theta$.
- 4.42. Evaluate (a) $\int_0^\pi \sin^5 \theta d\theta$, (b) $\int_0^{\pi/2} \cos^5 \theta \sin^2 \theta d\theta$.

4.43. Prove that (a) $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \pi/\sqrt{2}$; (b) $\int_0^{\pi/2} \tan^p \theta d\theta = \frac{\pi}{2} \sec \frac{p\pi}{2}$, $0 < p < 1$.

4.44. Prove that (a) $\int_0^\infty \frac{x dx}{1+x^6} = \frac{\pi}{3\sqrt{3}}$, (b) $\int_0^\infty \frac{y^2 dy}{1+y^4} = \frac{\pi}{2\sqrt{2}}$.

4.45. Prove that $\int_{-\infty}^\infty \frac{e^{2x}}{ae^{3x}+b} dx = \frac{2\pi}{3\sqrt{3} a^{2/3} b^{1/3}}$, where $a, b > 0$.

4.46. Prove that $\int_{-\infty}^\infty \frac{e^{2x}}{(e^{3x}+1)^2} dx = \frac{2\pi}{9\sqrt{3}}$. [Hint: Differentiate with respect to b in Problem 4.45.]

SPECIAL FUNCTIONS AND ASYMPTOTIC EXPANSIONS

4.47. Show that $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \left(x - \frac{x^3}{3 \cdot 1!} + \frac{x^5}{5 \cdot 2!} - \frac{x^7}{7 \cdot 3!} + \cdots \right)$.

4.48. Obtain the asymptotic expansion $Ei(x) \sim \frac{e^{-x}}{x} \left(1 - \frac{1!}{x} + \frac{2!}{x^2} - \frac{3!}{x^3} + \cdots \right)$.

4.49. Show that (a) $Si(-x) = -Si(x)$, (b) $Si(\infty) = \pi/2$.

4.50. Obtain the asymptotic expansions

$$Si(x) \sim \frac{\pi}{2} - \frac{\sin x}{x} \left(\frac{1}{x} - \frac{3!}{x^3} + \frac{5!}{x^5} - \cdots \right) - \frac{\cos x}{x} \left(1 - \frac{2!}{x^2} + \frac{4!}{x^4} - \cdots \right)$$

$$Ci(x) \sim \frac{\cos x}{x} \left(\frac{1}{x} - \frac{3!}{x^3} + \frac{5!}{x^5} - \cdots \right) - \frac{\sin x}{x} \left(1 - \frac{2!}{x^2} + \frac{4!}{x^4} - \cdots \right)$$

4.51. Show that $\int_0^\infty \frac{\sin x}{x^p} dx = \frac{\pi}{2 \Gamma(p) \sin(p\pi/2)}$, $0 < p < 1$.

4.52. Show that $\int_0^\infty \sin x^2 dx = \int_0^\infty \cos x^2 dx = \frac{1}{2} \sqrt{\frac{\pi}{2}}$.

4.53. Prove that $\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}} = e$

Fourier Integrals and Applications

THE NEED FOR FOURIER INTEGRALS

In Chapter 2 we considered the theory and applications involving the expansion of a function $f(x)$ of period $2L$ into a Fourier series. One question which arises quite naturally is: what happens in the case where $L \rightarrow \infty$? We shall find that in such case the Fourier series becomes a *Fourier integral*. We shall discuss Fourier integrals and their applications in this chapter.

THE FOURIER INTEGRAL

Let us assume the following conditions on $f(x)$:

1. $f(x)$ and $f'(x)$ are piecewise continuous in every finite interval.
2. $\int_{-\infty}^{\infty} |f(x)| dx$ converges, i.e. $f(x)$ is absolutely integrable in $(-\infty, \infty)$.

Then *Fourier's integral theorem* states that

$$f(x) = \int_0^{\infty} \{A(\alpha) \cos \alpha x + B(\alpha) \sin \alpha x\} d\alpha \quad (1)$$

where

$$\left. \begin{aligned} A(\alpha) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \cos \alpha x dx \\ B(\alpha) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(x) \sin \alpha x dx \end{aligned} \right\} \quad (2)$$

The result (1) holds if x is a point of continuity of $f(x)$. If x is a point of discontinuity, we must replace $f(x)$ by $\frac{f(x+0) + f(x-0)}{2}$ as in the case of Fourier series. Note that the above conditions are sufficient but not necessary.

The similarity of (1) and (2) with corresponding results for Fourier series is apparent. The right-hand side of (1) is sometimes called a *Fourier integral expansion* of $f(x)$.

EQUIVALENT FORMS OF FOURIER'S INTEGRAL THEOREM

Fourier's integral theorem can also be written in the forms

$$f(x) = \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{\infty} f(u) \cos \alpha(x-u) du d\alpha \quad (3)$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) e^{i\alpha(x-u)} du d\alpha$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha x} d\alpha \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du \quad (4)$$

where it is understood that if $f(x)$ is not continuous at x the left side must be replaced by $\frac{f(x+0) + f(x-0)}{2}$.

These results can be simplified somewhat if $f(x)$ is either an odd or an even function, and we have

$$f(x) = \frac{2}{\pi} \int_0^\infty \sin \alpha x \, d\alpha \int_0^\infty f(u) \sin \alpha u \, du \quad \text{if } f(x) \text{ is odd} \quad (5)$$

$$f(x) = \frac{2}{\pi} \int_0^\infty \cos \alpha x \, d\alpha \int_0^\infty f(u) \cos \alpha u \, du \quad \text{if } f(x) \text{ is even} \quad (6)$$

FOURIER TRANSFORMS

From (4) it follows that if

$$F(\alpha) = \int_{-\infty}^\infty f(u) e^{-i\alpha u} \, du \quad (7)$$

then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^\infty F(\alpha) e^{i\alpha x} \, d\alpha \quad (8)$$

The function $F(\alpha)$ is called the *Fourier transform* of $f(x)$ and is sometimes written $F(\alpha) = \mathcal{F}\{f(x)\}$. The function $f(x)$ is the *inverse Fourier transform* of $F(\alpha)$ and is written $f(x) = \mathcal{F}^{-1}\{F(\alpha)\}$.

Note: The constants 1 and $1/2\pi$ preceding the integral signs in (7) and (8) could be replaced by any two constants whose product is $1/2\pi$. In this book, however, we shall keep to the above choice.

FOURIER SINE AND COSINE TRANSFORMS

If $f(x)$ is an odd function, then Fourier's integral theorem reduces to (5). If we let

$$F_s(\alpha) = \int_0^\infty f(u) \sin \alpha u \, du \quad (9)$$

then it follows from (5) that

$$f(x) = \frac{2}{\pi} \int_0^\infty F_s(\alpha) \sin \alpha x \, d\alpha \quad (10)$$

We call $F_s(\alpha)$ the *Fourier sine transform* of $f(x)$, while $f(x)$ is the *inverse Fourier sine transform* of $F_s(\alpha)$.

Similarly, if $f(x)$ is an even function, Fourier's integral theorem reduces to (6). Thus if we let

$$F_c(\alpha) = \int_0^\infty f(u) \cos \alpha u \, du \quad (11)$$

then it follows from (6) that

$$f(x) = \frac{2}{\pi} \int_0^\infty F_c(\alpha) \cos \alpha x \, d\alpha \quad (12)$$

We call $F_c(\alpha)$ the *Fourier cosine transform* of $f(x)$, while $f(x)$ is the *inverse Fourier cosine transform* of $F_c(\alpha)$.

PARSEVAL'S IDENTITIES FOR FOURIER INTEGRALS

In Chapter 2, page 23, we arrived at Parseval's identity for Fourier series. An analogy exists for Fourier integrals.

If $F(\alpha)$ and $G(\alpha)$ are Fourier transforms of $f(x)$ and $g(x)$ respectively we can show that

$$\int_{-\infty}^{\infty} f(x) g(x) dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) \overline{G(\alpha)} d\alpha \quad (13)$$

where the bar signifies the complex conjugate obtained on replacing i by $-i$. In particular, if $f(x) = g(x)$ and hence $F(\alpha) = G(\alpha)$, then we have

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\alpha)|^2 d\alpha \quad (14)$$

We can refer to (14), or to the more general (13), as *Parseval's identity* for Fourier integrals.

Corresponding results can be written involving sine and cosine transforms. If $F_s(\alpha)$ and $G_s(\alpha)$ are the Fourier sine transforms of $f(x)$ and $g(x)$, respectively, then

$$\int_0^{\infty} f(x) g(x) dx = \frac{2}{\pi} \int_0^{\infty} F_s(\alpha) G_s(\alpha) d\alpha \quad (15)$$

Similarly, if $F_c(\alpha)$ and $G_c(\alpha)$ are the Fourier cosine transforms of $f(x)$ and $g(x)$, respectively, then

$$\int_0^{\infty} f(x) g(x) dx = \frac{2}{\pi} \int_0^{\infty} F_c(\alpha) G_c(\alpha) d\alpha \quad (16)$$

In the special case where $f(x) = g(x)$, (15) and (16) become respectively

$$\int_0^{\infty} \{f(x)\}^2 dx = \frac{2}{\pi} \int_0^{\infty} \{F_s(\alpha)\}^2 d\alpha \quad (17)$$

$$\int_0^{\infty} \{f(x)\}^2 dx = \frac{2}{\pi} \int_0^{\infty} \{F_c(\alpha)\}^2 d\alpha \quad (18)$$

THE CONVOLUTION THEOREM FOR FOURIER TRANSFORMS

The *convolution* of the functions $f(x)$ and $g(x)$ is defined by

$$f * g = \int_{-\infty}^{\infty} f(u) g(x-u) du \quad (19)$$

An important theorem, often referred to as the *convolution theorem*, states that the Fourier transform of the convolution of $f(x)$ and $g(x)$ is equal to the product of the Fourier transforms of $f(x)$ and $g(x)$. In symbols,

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\} \mathcal{F}\{g\} \quad (20)$$

The convolution has other important properties. For example, we have for functions f , g , and h :

$$f * g = g * f, \quad f * (g * h) = (f * g) * h, \quad f * (g + h) = f * g + f * h \quad (21)$$

i.e., the convolution obeys the commutative, associative and distributive laws of algebra.

APPLICATIONS OF FOURIER INTEGRALS AND TRANSFORMS

Fourier integrals and transforms can be used in solving a variety of boundary value problems arising in science and engineering. See Problems 5.20–5.22.

Solved Problems

THE FOURIER INTEGRAL AND FOURIER TRANSFORMS

5.1. Show that (1) and (3), page 80, are equivalent forms of Fourier's integral theorem.

Let us start with the form

$$f(x) = \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{\infty} f(u) \cos \alpha(x-u) du d\alpha \quad (1)$$

which is proved later (see Problems 5.10–5.14). The result (1) can be written as

$$f(x) = \frac{1}{\pi} \int_{\alpha=0}^{\infty} \int_{u=-\infty}^{\infty} f(u) [\cos \alpha x \cos \alpha u + \sin \alpha x \sin \alpha u] du d\alpha$$

or

$$f(x) = \int_{\alpha=0}^{\infty} \{A(\alpha) \cos \alpha x + B(\alpha) \sin \alpha x\} d\alpha \quad (2)$$

where we let

$$\left. \begin{aligned} A(\alpha) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \cos \alpha u du \\ B(\alpha) &= \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \sin \alpha u du \end{aligned} \right\} \quad (3)$$

Conversely, by substituting (3) into (2) we obtain (1). Thus the two forms are equivalent.

5.2. Show that (3) and (4), page 80, are equivalent.

We have from (3), page 80, and the fact that $\cos \alpha(x-u)$ is an even function of α :

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) \cos \alpha(x-u) du d\alpha \quad (1)$$

Then, using the fact that $\sin \alpha(x-u)$ is an odd function of α , we have

$$0 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) \sin \alpha(x-u) du d\alpha \quad (2)$$

Multiplying (2) by i and adding to (1) we then have

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) [\cos \alpha(x-u) + i \sin \alpha(x-u)] du d\alpha \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) e^{i\alpha(x-u)} du d\alpha \end{aligned}$$

Similarly we can deduce that (3), page 80, follows from (4).

5.3. (a) Find the Fourier transform of $f(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases}$.

(b) Graph $f(x)$ and its Fourier transform for $a = 3$.

(a) The Fourier transform of $f(x)$ is

$$\begin{aligned} F(\alpha) &= \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du = \int_{-a}^a (1) e^{-i\alpha u} du = \frac{e^{-i\alpha u}}{-i\alpha} \Big|_{-a}^a \\ &= \frac{e^{i\alpha a} - e^{-i\alpha a}}{i\alpha} = 2 \frac{\sin \alpha a}{\alpha}, \quad \alpha \neq 0 \end{aligned}$$

For $\alpha = 0$, we obtain $F(\alpha) = 2a$.

(b) The graphs of $f(x)$ and $F(\alpha)$ for $a = 3$ are shown in Figs. 5-1 and 5-2 respectively.

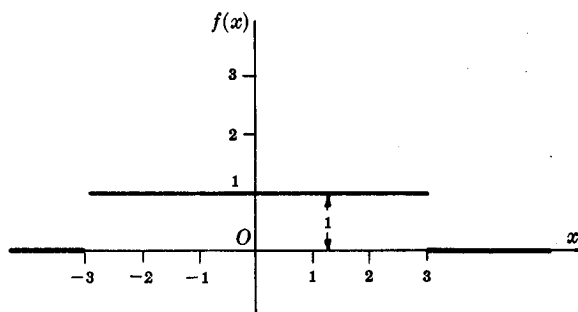


Fig. 5-1

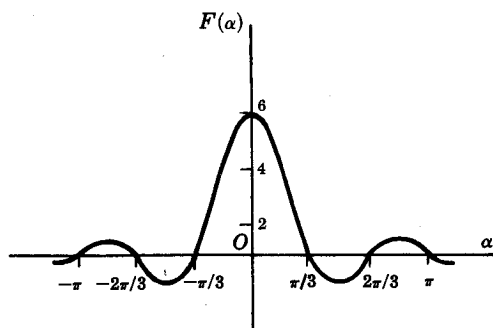


Fig. 5-2

5.4. (a) Use the result of Problem 5.3 to evaluate $\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha} d\alpha$.

(b) Deduce the value of $\int_0^{\infty} \frac{\sin u}{u} du$.

(a) From Fourier's integral theorem, if

$$F(\alpha) = \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du \quad \text{then} \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\alpha) e^{i\alpha x} d\alpha$$

Then from Problem 5.3,

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} 2 \frac{\sin \alpha a}{\alpha} e^{i\alpha x} d\alpha = \begin{cases} 1 & |x| < a \\ 1/2 & |x| = a \\ 0 & |x| > a \end{cases} \quad (1)$$

The left side of (1) is equal to

$$\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha} d\alpha + \frac{i}{\pi} \int_{-\infty}^{\infty} \frac{\sin \alpha a \sin \alpha x}{\alpha} d\alpha \quad (2)$$

The integrand in the second integral of (2) is odd and so the integral is zero. Then from (1) and (2), we have

$$\int_{-\infty}^{\infty} \frac{\sin \alpha a \cos \alpha x}{\alpha} d\alpha = \begin{cases} \pi & |x| < a \\ \pi/2 & |x| = a \\ 0 & |x| > a \end{cases} \quad (3)$$

(b) If $x = 0$ and $a = 1$ in the result of (a), we have

$$\int_{-\infty}^{\infty} \frac{\sin \alpha}{\alpha} d\alpha = \pi \quad \text{or} \quad \int_0^{\infty} \frac{\sin \alpha}{\alpha} d\alpha = \frac{\pi}{2}$$

since the integrand is even.

5.5. (a) Find the Fourier cosine transform of $f(x) = e^{-mx}$, $m > 0$.

(b) Use the result in (a) to show that

$$\int_0^{\infty} \frac{\cos pv}{v^2 + \beta^2} dv = \frac{\pi}{2\beta} e^{-p\beta} \quad (p > 0, \beta > 0)$$

(a) The Fourier cosine transform of $f(x) = e^{-mx}$ is by definition

$$\begin{aligned} F_C(\alpha) &= \int_0^{\infty} e^{-mu} \cos \alpha u \, du \\ &= \left. \frac{e^{-mu}(-m \cos \alpha u + \alpha \sin \alpha u)}{m^2 + \alpha^2} \right|_0^{\infty} \\ &= \frac{m}{m^2 + \alpha^2} \end{aligned}$$

(b) From (12), page 81, we have

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_C(\alpha) \cos \alpha x \, d\alpha$$

or

$$e^{-mx} = \frac{2}{\pi} \int_0^{\infty} \frac{m \cos \alpha x}{m^2 + \alpha^2} d\alpha$$

i.e.

$$\int_0^{\infty} \frac{\cos \alpha x}{m^2 + \alpha^2} d\alpha = \frac{\pi}{2m} e^{-mx}$$

Replacing α by v , x by p , and m by β , we have

$$\int_0^{\infty} \frac{\cos pv}{v^2 + \beta^2} dv = \frac{\pi}{2\beta} e^{-v\beta}, \quad p > 0, \beta > 0$$

5.6. Solve the integral equation

$$\int_0^{\infty} f(x) \sin \alpha x \, dx = \begin{cases} 1 - \alpha & 0 \leq \alpha \leq 1 \\ 0 & \alpha > 1 \end{cases}$$

If we write

$$F_S(\alpha) = \int_0^{\infty} f(x) \sin \alpha x \, dx = \begin{cases} 1 - \alpha & 0 \leq \alpha \leq 1 \\ 0 & \alpha > 1 \end{cases}$$

then, by (10), page 81,

$$\begin{aligned} f(x) &= \frac{2}{\pi} \int_0^{\infty} F_S(\alpha) \sin \alpha x \, d\alpha \\ &= \frac{2}{\pi} \int_0^1 (1 - \alpha) \sin \alpha x \, d\alpha \\ &= \frac{2(x - \sin x)}{\pi x^2} \end{aligned}$$

THE CONVOLUTION THEOREM

5.7. Prove the convolution theorem on page 82.

We have by definition of the Fourier transform

$$F(\alpha) = \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du, \quad G(\alpha) = \int_{-\infty}^{\infty} g(v) e^{-i\alpha v} dv \quad (1)$$

Then

$$F(\alpha) G(\alpha) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) g(v) e^{-i\alpha(u+v)} du dv \quad (2)$$

Let $u + v = x$ in the double integral (2) which we wish to transform from the variables (u, v) to the variables (u, x) . From advanced calculus we know that

$$du dv = \frac{\partial(u, v)}{\partial(u, x)} du dx \quad (3)$$

where the Jacobian of the transformation is given by

$$\frac{\partial(u, v)}{\partial(u, x)} = \begin{vmatrix} \frac{\partial u}{\partial u} & \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial u} & \frac{\partial v}{\partial x} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

Thus (2) becomes

$$\begin{aligned} F(\alpha) G(\alpha) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) g(x-u) e^{-i\alpha x} du dx \\ &= \int_{-\infty}^{\infty} e^{-i\alpha x} \left[\int_{-\infty}^{\infty} f(u) g(x-u) du \right] dx \\ &= \mathcal{F} \left\{ \int_{-\infty}^{\infty} f(u) g(x-u) du \right\} \\ &= \mathcal{F}\{f * g\} \end{aligned}$$

where $f * g = \int_{-\infty}^{\infty} f(u) g(x-u) du$ is the convolution of f and g .

From this we have equivalently

$$\begin{aligned} f * g &= \mathcal{F}^{-1}\{F(\alpha) G(\alpha)\} \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha x} F(\alpha) G(\alpha) d\alpha \end{aligned}$$

5.8. Show that $f * g = g * f$.

Let $x - u = v$. Then

$$\begin{aligned} f * g &= \int_{-\infty}^{\infty} f(u) g(x-u) du = \int_{-\infty}^{\infty} f(x-v) g(v) dv \\ &= \int_{-\infty}^{\infty} g(v) f(x-v) dv = g * f \end{aligned}$$

5.9. Solve the integral equation

$$y(x) = g(x) + \int_{-\infty}^{\infty} y(u) r(x-u) du$$

where $g(x)$ and $r(x)$ are given.

Suppose that the Fourier transforms of $y(x)$, $g(x)$ and $r(x)$ exist, and denote them by $Y(\alpha)$, $G(\alpha)$ and $R(\alpha)$ respectively. Then, taking the Fourier transform of both sides of the given integral equation, we have by the convolution theorem

$$Y(\alpha) = G(\alpha) + Y(\alpha) R(\alpha) \quad \text{or} \quad Y(\alpha) = \frac{G(\alpha)}{1 - R(\alpha)}$$

Then

$$y(x) = \mathcal{F}^{-1} \left\{ \frac{G(\alpha)}{1 - R(\alpha)} \right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \frac{G(\alpha)}{1 - R(\alpha)} \right\} e^{i\alpha x} d\alpha$$

assuming this integral exists.

5.10. Solve for $y(x)$ the integral equation

$$\int_{-\infty}^{\infty} \frac{y(u) du}{(x-u)^2 + a^2} = \frac{1}{x^2 + b^2} \quad 0 < a < b$$

We have

$$\mathcal{F}\left\{\frac{1}{x^2 + b^2}\right\} = \int_{-\infty}^{\infty} \frac{e^{-i\alpha x}}{x^2 + b^2} dx = 2 \int_0^{\infty} \frac{\cos \alpha x}{x^2 + b^2} dx = \frac{\pi}{b} e^{-b\alpha}$$

on making use of Problem 5.5(b). Then, taking the Fourier transform of both sides of the integral equation, we find

$$\mathcal{F}\{y\} \mathcal{F}\left\{\frac{1}{x^2 + a^2}\right\} = \mathcal{F}\left\{\frac{1}{x^2 + b^2}\right\}$$

$$\text{i.e.} \quad Y(\alpha) \frac{\pi}{a} e^{-a\alpha} = \frac{\pi}{b} e^{-b\alpha} \quad \text{or} \quad Y(\alpha) = \frac{a}{b} e^{-(b-a)\alpha}$$

$$\text{Thus} \quad y(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha x} Y(\alpha) d\alpha = \frac{a}{b\pi} \int_0^{\infty} e^{-(b-a)\alpha} \cos \alpha x d\alpha = \frac{(b-a)\alpha}{b\pi[x^2 + (b-a)^2]}$$

PROOF OF THE FOURIER INTEGRAL THEOREM

5.11. Present a heuristic demonstration of Fourier's integral theorem by use of a limiting form of Fourier series.

$$\text{Let} \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right) \quad (1)$$

$$\text{where} \quad a_n = \frac{1}{L} \int_{-L}^L f(u) \cos \frac{n\pi u}{L} du \quad \text{and} \quad b_n = \frac{1}{L} \int_{-L}^L f(u) \sin \frac{n\pi u}{L} du.$$

Then by substitution of these coefficients into (1) we find

$$f(x) = \frac{1}{2L} \int_{-L}^L f(u) du + \frac{1}{L} \sum_{n=1}^{\infty} \int_{-L}^L f(u) \cos \frac{n\pi}{L} (u-x) du \quad (2)$$

If we assume that $\int_{-\infty}^{\infty} |f(u)| du$ converges, the first term on the right of (2) approaches zero as $L \rightarrow \infty$, while the remaining part appears to approach

$$\lim_{L \rightarrow \infty} \frac{1}{L} \sum_{n=1}^{\infty} \int_{-\infty}^{\infty} f(u) \cos \frac{n\pi}{L} (u-x) du \quad (3)$$

This last step is not rigorous and makes the demonstration heuristic.

Calling $\Delta\alpha = \pi/L$, (3) can be written

$$f(x) = \lim_{\Delta\alpha \rightarrow 0} \sum_{n=1}^{\infty} \Delta\alpha F(n\Delta\alpha) \quad (4)$$

where we have written

$$F(\alpha) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(u) \cos \alpha(u-x) du \quad (5)$$

But the limit (4) is equal to

$$f(x) = \int_0^{\infty} F(\alpha) d\alpha = \frac{1}{\pi} \int_0^{\infty} d\alpha \int_{-\infty}^{\infty} f(u) \cos \alpha(u-x) du \quad (6)$$

which is Fourier's integral formula.

This demonstration merely provides a possible result. To be rigorous, we start with the double integral in (6) and examine the convergence. This method is considered in Problems 5.12-5.15.

5.12. Prove that: (a) $\lim_{\alpha \rightarrow \infty} \int_0^L \frac{\sin \alpha v}{v} dv = \frac{\pi}{2}$, (b) $\lim_{\alpha \rightarrow \infty} \int_{-L}^0 \frac{\sin \alpha v}{v} dv = \frac{\pi}{2}$.

(a) Let $\alpha v = y$. Then $\lim_{\alpha \rightarrow \infty} \int_0^L \frac{\sin \alpha v}{v} dv = \lim_{\alpha \rightarrow \infty} \int_0^{\alpha L} \frac{\sin y}{y} dy = \int_0^{\infty} \frac{\sin y}{y} dy = \frac{\pi}{2}$, as can be shown by using Problem 5.40.

(b) Let $\alpha v = -y$. Then $\lim_{\alpha \rightarrow \infty} \int_{-L}^0 \frac{\sin \alpha v}{v} dv = \lim_{\alpha \rightarrow \infty} \int_0^{\alpha L} \frac{\sin y}{y} dy = \frac{\pi}{2}$.

5.13. Riemann's theorem states that if $F(x)$ is piecewise continuous in (a, b) , then

$$\lim_{\alpha \rightarrow \infty} \int_a^b F(x) \sin \alpha x dx = 0$$

with a similar result for the cosine (see Problem 5.41). Use this to prove that

$$(a) \lim_{\alpha \rightarrow \infty} \int_0^L f(x+v) \frac{\sin \alpha v}{v} dv = \frac{\pi}{2} f(x+0)$$

$$(b) \lim_{\alpha \rightarrow \infty} \int_{-L}^0 f(x+v) \frac{\sin \alpha v}{v} dv = \frac{\pi}{2} f(x-0)$$

where $f(x)$ and $f'(x)$ are assumed piecewise continuous [see condition 1. on page 80].

(a) Using Problem 5.12(a), it is seen that a proof of the given result amounts to proving that

$$\lim_{\alpha \rightarrow \infty} \int_0^L \{f(x+v) - f(x+0)\} \frac{\sin \alpha v}{v} dv = 0$$

This follows at once from Riemann's theorem, because $F(v) = \frac{f(x+v) - f(x+0)}{v}$ is piecewise continuous in $(0, L)$ since $\lim_{v \rightarrow 0+} F(v)$ exists and $f(x)$ is piecewise continuous.

(b) A proof of this is analogous to that in part (a) if we make use of Problem 5.12(b).

5.14. If $f(x)$ satisfies the additional condition that $\int_{-\infty}^{\infty} |f(x)| dx$ converges, prove that

$$(a) \lim_{\alpha \rightarrow \infty} \int_0^{\infty} f(x+v) \frac{\sin \alpha v}{v} dv = \frac{\pi}{2} f(x+0), \quad (b) \lim_{\alpha \rightarrow \infty} \int_{-\infty}^0 f(x+v) \frac{\sin \alpha v}{v} dv = \frac{\pi}{2} f(x-0).$$

(a) We have

$$\int_0^{\infty} f(x+v) \frac{\sin \alpha v}{v} dv = \int_0^L f(x+v) \frac{\sin \alpha v}{v} dv + \int_L^{\infty} f(x+v) \frac{\sin \alpha v}{v} dv \quad (1)$$

$$\int_0^{\infty} f(x+0) \frac{\sin \alpha v}{v} dv = \int_0^L f(x+0) \frac{\sin \alpha v}{v} dv + \int_L^{\infty} f(x+0) \frac{\sin \alpha v}{v} dv \quad (2)$$

Subtracting,

$$\begin{aligned} & \int_0^{\infty} \{f(x+v) - f(x+0)\} \frac{\sin \alpha v}{v} dv \\ &= \int_0^L \{f(x+v) - f(x+0)\} \frac{\sin \alpha v}{v} dv + \int_L^{\infty} f(x+v) \frac{\sin \alpha v}{v} dv - \int_L^{\infty} f(x+0) \frac{\sin \alpha v}{v} dv \end{aligned} \quad (3)$$

Denoting the integrals in (3) by I, I_1, I_2 and I_3 respectively, we have $I = I_1 + I_2 + I_3$ so that

$$|I| \leq |I_1| + |I_2| + |I_3| \quad (4)$$

Now $|I_2| \leq \int_L^{\infty} \left| f(x+v) \frac{\sin \alpha v}{v} \right| dv \leq \frac{1}{L} \int_L^{\infty} |f(x+v)| dv$

Also $|I_3| \leq |f(x+0)| \left| \int_L^{\infty} \frac{\sin \alpha v}{v} dv \right|$

Since $\int_0^{\infty} |f(x)| dx$ and $\int_0^{\infty} \frac{\sin \alpha v}{v} dv$ both converge, we can choose L so large that $|I_2| \leq \epsilon/3$, $|I_3| \leq \epsilon/3$. Also, we can choose α so large that $|I_1| \leq \epsilon/3$. Then from (4) we have $|I| \leq \epsilon$ for α and L sufficiently large, so that the required result follows.

(b) This result follows by reasoning exactly analogous to that in part (a).

5.15. Prove Fourier's integral formula if $f(x)$ satisfies the conditions stated on page 80.

We must prove that $\lim_{L \rightarrow \infty} \frac{1}{\pi} \int_{\alpha=0}^L \int_{u=-\infty}^{\infty} f(u) \cos \alpha(x-u) du d\alpha = \frac{f(x+0) + f(x-0)}{2}$.

Since $\left| \int_{-\infty}^{\infty} f(u) \cos \alpha(x-u) du \right| \leq \int_{-\infty}^{\infty} |f(u)| du$, which converges, it follows by the Weierstrass M test for integrals that $\int_{-\infty}^{\infty} f(u) \cos \alpha(x-u) du$ converges absolutely and uniformly for all α . We can show from this that the order of integration can be reversed to obtain

$$\begin{aligned} \frac{1}{\pi} \int_{\alpha=0}^L d\alpha \int_{u=-\infty}^{\infty} f(u) \cos \alpha(x-u) du &= \frac{1}{\pi} \int_{u=-\infty}^{\infty} f(u) du \int_{\alpha=0}^L \cos \alpha(x-u) d\alpha \\ &= \frac{1}{\pi} \int_{u=-\infty}^{\infty} f(u) \frac{\sin L(u-x)}{u-x} du \\ &= \frac{1}{\pi} \int_{v=-\infty}^{\infty} f(x+v) \frac{\sin Lv}{v} dv \\ &= \frac{1}{\pi} \int_{-\infty}^0 f(x+v) \frac{\sin Lv}{v} dv + \frac{1}{\pi} \int_0^{\infty} f(x+v) \frac{\sin Lv}{v} dv \end{aligned}$$

where we have let $u = x + v$.

Letting $L \rightarrow \infty$, we see by Problem 5.14 that the given integral converges to $\frac{f(x+0) + f(x-0)}{2}$ as required.

SOLUTIONS USING FOURIER INTEGRALS

5.16. A semi-infinite thin bar ($x \geq 0$) whose surface is insulated has an initial temperature equal to $f(x)$. A temperature of zero is suddenly applied to the end $x = 0$ and maintained. (a) Set up the boundary value problem for the temperature $u(x, t)$ at any point x at time t . (b) Show that

$$u(x, t) = \frac{1}{\pi} \int_0^{\infty} \int_0^{\infty} f(v) e^{-\kappa \lambda^2 t} \sin \lambda v \sin \lambda x d\lambda dv$$

(a) The boundary value problem is

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2} \quad x > 0, t > 0 \quad (1)$$

$$u(x, 0) = f(x), \quad u(0, t) = 0, \quad |u(x, t)| < M \quad (2)$$

where the last condition is used since the temperature must be bounded for physical reasons.

(b) A solution of (1) obtained by separation of variables is

$$u(x, t) = e^{-\kappa \lambda^2 t} (A \cos \lambda x + B \sin \lambda x)$$

From the second of boundary conditions (2) we find $A = 0$ so that

$$u(x, t) = B e^{-\kappa \lambda^2 t} \sin \lambda x \quad (3)$$

Now since there is no restriction on λ we can replace B in (3) by a function $B(\lambda)$ and still have a solution. Furthermore we can integrate over λ from 0 to ∞ and still have a solution. This is the analog of the superposition theorem for discrete values of λ used in connection with Fourier series. We thus arrive at the possible solution

$$u(x, t) = \int_0^{\infty} B(\lambda) e^{-\kappa \lambda^2 t} \sin \lambda x d\lambda \quad (4)$$

From the first of boundary conditions (2) we find

$$f(x) = \int_0^\infty B(\lambda) \sin \lambda x \, d\lambda$$

which is an integral equation for the determination of $B(\lambda)$. From page 81, we see that since $f(x)$ must be an odd function, we have

$$B(\lambda) = \frac{2}{\pi} \int_0^\infty f(x) \sin \lambda x \, dx = \frac{2}{\pi} \int_0^\infty f(v) \sin \lambda v \, dv$$

Using this in (4) we find

$$u(x, t) = \frac{2}{\pi} \int_0^\infty \int_0^\infty f(v) e^{-\kappa \lambda^2 t} \sin \lambda v \sin \lambda x \, d\lambda \, dv$$

5.17. Show that the result of Problem 5.16 can be written

$$u(x, t) = \frac{1}{\sqrt{\pi}} \left[\int_{-x/2\sqrt{\kappa t}}^\infty e^{-w^2} f(2w\sqrt{\kappa t} + x) \, dw - \int_{x/2\sqrt{\kappa t}}^\infty e^{-w^2} f(2w\sqrt{\kappa t} - x) \, dw \right]$$

Since $\sin \lambda v \sin \lambda x = \frac{1}{2} [\cos \lambda(v-x) - \cos \lambda(v+x)]$, the result of Problem 5.16 can be written

$$\begin{aligned} u(x, t) &= \frac{1}{\pi} \int_0^\infty \int_0^\infty f(v) e^{-\kappa \lambda^2 t} [\cos \lambda(v-x) - \cos \lambda(v+x)] \, d\lambda \, dv \\ &= \frac{1}{\pi} \int_0^\infty f(v) \left[\int_0^\infty e^{-\kappa \lambda^2 t} \cos \lambda(v-x) \, d\lambda - \int_0^\infty e^{-\kappa \lambda^2 t} \cos \lambda(v+x) \, d\lambda \right] dv \end{aligned}$$

From the integral

$$\int_0^\infty e^{-\alpha \lambda^2} \cos \beta \lambda \, d\lambda = \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} e^{-\beta^2/4\alpha}$$

(see Problem 4.9, page 72) we find

$$u(x, t) = \frac{1}{2\sqrt{\pi \kappa t}} \left[\int_0^\infty f(v) e^{-(v-x)^2/4\kappa t} \, dv - \int_0^\infty f(v) e^{-(v+x)^2/4\kappa t} \, dv \right]$$

Letting $(v-x)/2\sqrt{\kappa t} = w$ in the first integral and $(v+x)/2\sqrt{\kappa t} = w$ in the second integral, we find that

$$u(x, t) = \frac{1}{\sqrt{\pi}} \left[\int_{-x/2\sqrt{\kappa t}}^\infty e^{-w^2} f(2w\sqrt{\kappa t} + x) \, dw - \int_{x/2\sqrt{\kappa t}}^\infty e^{-w^2} f(2w\sqrt{\kappa t} - x) \, dw \right]$$

5.18. In case the initial temperature $f(x)$ in Problem 5.16 is the constant u_0 , show that

$$u(x, t) = \frac{2u_0}{\sqrt{\pi}} \int_0^{x/2\sqrt{\kappa t}} e^{-w^2} \, dw = u_0 \operatorname{erf} (x/2\sqrt{\kappa t})$$

where $\operatorname{erf} (x/2\sqrt{\kappa t})$ is the error function (see page 69).

If $f(x, t) = u_0$, we obtain from Problem 5.17

$$\begin{aligned} u(x, t) &= \frac{u_0}{\sqrt{\pi}} \left[\int_{-x/2\sqrt{\kappa t}}^\infty e^{-w^2} \, dw - \int_{x/2\sqrt{\kappa t}}^\infty e^{-w^2} \, dw \right] \\ &= \frac{u_0}{\sqrt{\pi}} \int_{-x/2\sqrt{\kappa t}}^{x/2\sqrt{\kappa t}} e^{-w^2} \, dw = \frac{2u_0}{\sqrt{\pi}} \int_0^{x/2\sqrt{\kappa t}} e^{-w^2} \, dw = u_0 \operatorname{erf} (x/2\sqrt{\kappa t}) \end{aligned}$$

We can show that this actually is a solution of the corresponding boundary value problem (see Problem 5.48).

- 5.19. Find a bounded solution to Laplace's equation $\nabla^2 v = 0$ for the half plane $y > 0$ (Fig. 5-3) if v takes on the value $f(x)$ on the x -axis.

The boundary value problem for the determination of $v(x, y)$ is given by

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

$$v(x, 0) = f(x), \quad |v(x, y)| < M$$

To solve this, let $v = XY$ in the partial differential equation, where X depends only on x and Y depends only on y . Then, on separating the variables, we have

$$\frac{X''}{X} = -\frac{Y''}{Y}$$

Setting each side equal to $-\lambda^2$ we find

$$X'' + \lambda^2 X = 0, \quad Y'' - \lambda^2 Y = 0$$

so that

$$X = a_1 \cos \lambda x + b_1 \sin \lambda x, \quad Y = a_2 e^{\lambda y} + b_2 e^{-\lambda y}$$

Then the solution is

$$v(x, y) = (a_1 \cos \lambda x + b_1 \sin \lambda x)(a_2 e^{\lambda y} + b_2 e^{-\lambda y})$$

If $\lambda > 0$ the term in $e^{\lambda y}$ is unbounded as $y \rightarrow \infty$; so that to keep $v(x, y)$ bounded we must have $a_2 = 0$. This leads to the solution

$$v(x, y) = e^{-\lambda y} [A \cos \lambda x + B \sin \lambda x]$$

Since there is no restriction on λ , we can replace A by $A(\lambda)$, B by $B(\lambda)$ and integrate over λ to obtain

$$v(x, y) = \int_0^\infty e^{-\lambda y} [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] d\lambda \quad (1)$$

The boundary condition $v(x, 0) = f(x)$ yields

$$\int_0^\infty [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] d\lambda = f(x)$$

Thus, from Fourier's integral theorem we find

$$A(\lambda) = \frac{1}{\pi} \int_{-\infty}^\infty f(u) \cos \lambda u du, \quad B(\lambda) = \frac{1}{\pi} \int_{-\infty}^\infty f(u) \sin \lambda u du$$

Putting these in (1) we have finally:

$$v(x, y) = \frac{1}{\pi} \int_{\lambda=0}^\infty \int_{u=-\infty}^\infty e^{-\lambda y} f(u) \cos \lambda(u-x) du d\lambda \quad (2)$$

- 5.20. Show that the solution to Problem 5.19 can be written in the form

$$v(x, y) = \frac{1}{\pi} \int_{-\infty}^\infty \frac{y f(u)}{y^2 + (u-x)^2} du$$

Write the result (2) of Problem 5.19 as

$$v(x, y) = \frac{1}{\pi} \int_{-\infty}^\infty f(u) \left[\int_0^\infty e^{-\lambda y} \cos \lambda(u-x) d\lambda \right] du \quad (1)$$

Then by elementary integration we have

$$\int_0^\infty e^{-\lambda y} \cos \lambda(u-x) d\lambda = \frac{y}{y^2 + (u-x)^2} \quad (2)$$

so that (1) becomes

$$v(x, y) = \frac{1}{\pi} \int_{-\infty}^\infty \frac{y f(u)}{y^2 + (u-x)^2} du \quad (3)$$

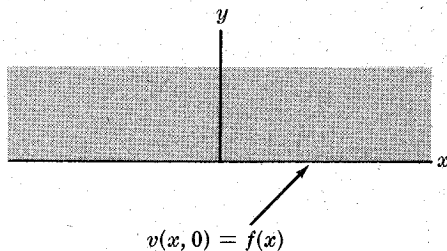


Fig. 5-3

SOLUTIONS BY USE OF FOURIER TRANSFORMS

5.21. By taking the Fourier transform with respect to the variable x , show that

$$(a) \mathcal{F}\left(\frac{\partial v}{\partial x}\right) = i\alpha \mathcal{F}(v), \quad (b) \mathcal{F}\left(\frac{\partial^2 v}{\partial x^2}\right) = -\alpha^2 \mathcal{F}(v), \quad (c) \mathcal{F}\left(\frac{\partial v}{\partial t}\right) = \frac{\partial}{\partial t} \mathcal{F}(v)$$

(a) By definition we have on using integration by parts:

$$\begin{aligned} \mathcal{F}\left(\frac{\partial v}{\partial x}\right) &= \int_{-\infty}^{\infty} \frac{\partial v}{\partial x} e^{-i\alpha x} dx \\ &= e^{-i\alpha x} v \Big|_{-\infty}^{\infty} + i\alpha \int_{-\infty}^{\infty} v e^{-i\alpha x} dx \\ &= i\alpha \int_{-\infty}^{\infty} v e^{-i\alpha x} dx \\ &= i\alpha \mathcal{F}(v) \end{aligned}$$

where we suppose that $v \rightarrow 0$ as $x \rightarrow \pm\infty$.

(b) Let $v = \partial w / \partial \alpha$ in part (a) then

$$\mathcal{F}\left(\frac{\partial^2 w}{\partial x^2}\right) = i\alpha \mathcal{F}\left(\frac{\partial w}{\partial x}\right) = (i\alpha)^2 \mathcal{F}(w)$$

Then if we formally replace w by v we have

$$\mathcal{F}\left(\frac{\partial^2 v}{\partial x^2}\right) = (i\alpha)^2 \mathcal{F}(v) = -\alpha^2 \mathcal{F}(v)$$

provided that v and $\frac{\partial v}{\partial x} \rightarrow 0$ as $x \rightarrow \pm\infty$.

In general we can show that

$$\mathcal{F}\left(\frac{\partial^n v}{\partial x^n}\right) = (i\alpha)^n \mathcal{F}(v)$$

if $v, \frac{\partial v}{\partial x}, \dots, \frac{\partial^{n-1} v}{\partial x^{n-1}} \rightarrow 0$ as $x \rightarrow \pm\infty$.

(c) By definition

$$\mathcal{F}\left(\frac{\partial v}{\partial t}\right) = \int_{-\infty}^{\infty} \frac{\partial v}{\partial t} e^{-i\alpha x} dx = \frac{\partial}{\partial t} \int_{-\infty}^{\infty} v e^{-i\alpha x} dx = \frac{\partial}{\partial t} \mathcal{F}(v)$$

5.22. (a) Use Fourier transforms to solve the boundary value problem

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = f(x), \quad |u(x, t)| < M$$

where $-\infty < x < \infty$, $t > 0$. (b) Give a physical interpretation.

(a) Taking the Fourier transform with respect to x of both sides of the given partial differential equation and using results (b) and (c) of Problem 5.21, we have

$$\frac{d}{dt} \mathcal{F}(u) = -\kappa \alpha^2 \mathcal{F}(u) \quad (1)$$

where we have written the total derivative since $\mathcal{F}(u)$ depends only on t and not on x . Solving the ordinary differential equation (1) for $\mathcal{F}(u)$, we obtain

$$\mathcal{F}(u) = C e^{-\kappa \alpha^2 t} \quad (2)$$

or more explicitly

$$\mathcal{F}\{u(x, t)\} = C e^{-\kappa \alpha^2 t} \quad (3)$$

Putting $t = 0$ in (3) we see that

$$\mathcal{F}\{u(x, 0)\} = \mathcal{F}\{f(x)\} = C \quad (4)$$

so that (2) becomes

$$\mathcal{F}\{u\} = \mathcal{F}\{f\}e^{-\kappa\alpha^2 t} \quad (5)$$

We can now apply the convolution theorem. By Problem 4.9, page 72,

$$e^{-\kappa\alpha^2 t} = \mathcal{F}\left\{\sqrt{\frac{1}{4\pi\kappa t}} e^{-(x^2/4\kappa t)}\right\} \quad (6)$$

$$\text{Hence } u(x, t) = f(x) * \sqrt{\frac{1}{4\pi\kappa t}} e^{-(x^2/4\kappa t)} = \int_{-\infty}^{\infty} f(w) \sqrt{\frac{1}{4\pi\kappa t}} e^{-[(x-w)^2/4\kappa t]} dw \quad (7)$$

If we now change variables from w to z according to the transformation $(x-w)^2/4\kappa t = z^2$ or $(x-w)/2\sqrt{\kappa t} = z$, (7) becomes

$$u(x, t) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} f(x - 2z\sqrt{\kappa t}) dz \quad (8)$$

- (b) The problem is that of determining the temperature in a thin infinite bar whose surface is insulated and whose initial temperature is $f(x)$.

5.23. An infinite string is given an initial displacement $y(x, 0) = f(x)$ and then released. Determine its displacement at any later time t .

The boundary value problem is

$$\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2} \quad (1)$$

$$y(x, 0) = f(x), \quad y_t(x, 0) = 0, \quad |y(x, t)| < M \quad (2)$$

where $-\infty < x < \infty$, $t > 0$.

Letting $y = XT$ in (1) we find in the usual manner that a solution satisfying the second boundary condition in (2) is given by

$$y(x, t) = (A \cos \lambda x + B \sin \lambda x) \cos \lambda at$$

By assuming that A and B are functions of λ and integrating from $\lambda = 0$ to ∞ we then arrive at the possible solution

$$y(x, t) = \int_0^{\infty} [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] \cos \lambda at d\lambda \quad (3)$$

Putting $t = 0$ in (3), we see from the first boundary condition in (2) that we must have

$$f(x) = \int_0^{\infty} [A(\lambda) \cos \lambda x + B(\lambda) \sin \lambda x] d\lambda$$

Then it follows from (1) and (2), page 80, that

$$A(\lambda) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos \lambda v dv, \quad B(\lambda) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \sin \lambda v dv \quad (4)$$

where we have changed the dummy variable from x to v .

Substitution of (4) into (3) yields

$$\begin{aligned} y(x, t) &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(v) [\cos \lambda x \cos \lambda v + \sin \lambda x \sin \lambda v] \cos \lambda at dv d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(v) \cos \lambda(x-v) \cos \lambda at dv d\lambda \\ &= \frac{1}{2\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(v) [\cos \lambda(x+at-v) + \cos \lambda(x-at-v)] dv d\lambda \end{aligned}$$

where in the last step we have used the trigonometric identity

$$\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$$

with $A = \lambda(x-v)$ and $B = \lambda at$.

By interchanging the order of integration, the result can be written

$$\begin{aligned} y(x, t) &= \frac{1}{2\pi} \int_0^\infty \int_{-\infty}^\infty f(v) \cos \lambda(x+at-v) dv d\lambda \\ &\quad + \frac{1}{2\pi} \int_0^\infty \int_{-\infty}^\infty f(v) \cos \lambda(x-at-v) dv d\lambda \end{aligned} \quad (5)$$

But we know from Fourier's integral theorem [equation (3), page 80] that

$$f(x) = \frac{1}{\pi} \int_0^\infty \int_{-\infty}^\infty f(v) \cos \lambda(x-v) dv d\lambda$$

Then, replacing x by $x+at$ and $x-at$ respectively, we see that (5) can be written

$$y(x, t) = \frac{1}{2}[f(x+at) + f(x-at)] \quad (6)$$

which is the required solution.

Supplementary Problems

THE FOURIER INTEGRAL AND FOURIER TRANSFORMS

- 5.24. (a) Find the Fourier transform of $f(x) = \begin{cases} 1/2\epsilon & |x| < 1 \\ 0 & |x| > 1 \end{cases}$.
 (b) Determine the limit of this transform as $\epsilon \rightarrow 0+$ and discuss the result.
- 5.25. (a) Find the Fourier transform of $f(x) = \begin{cases} 1-x^2 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$.
 (b) Evaluate $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$.
- 5.26. If $f(x) = \begin{cases} 1 & 0 \leq x < 1 \\ 0 & x \geq 1 \end{cases}$ find (a) the Fourier sine transform, (b) the Fourier cosine transform of $f(x)$. In each case obtain the graphs of $f(x)$ and its transform.
- 5.27. (a) Find the Fourier sine transform of e^{-x} , $x \geq 0$.
 (b) Show that $\int_0^\infty \frac{x \sin mx}{x^2 + 1} dx = \frac{\pi}{2} e^{-m}$, $m > 0$ by using the result in (a).
 (c) Explain from the viewpoint of Fourier's integral theorem why the result in (b) does not hold for $m = 0$.
- 5.28. Solve for $y(x)$ the integral equation
- $$\int_0^\infty y(x) \sin xt dx = \begin{cases} 1 & 0 \leq t < 1 \\ 2 & 1 \leq t < 2 \\ 0 & t \geq 2 \end{cases}$$
- and verify the solution by direct substitution.

- 5.29. If $F(x)$ is the Fourier transform of $f(x)$ show that it is possible to find a constant c so that $F(x) = f(x) = ce^{-x^2}$.

PARSEVAL'S IDENTITY

- 5.30. Evaluate (a) $\int_0^\infty \frac{dx}{(x^2+1)^2}$, (b) $\int_0^\infty \frac{x^2 dx}{(x^2+1)^2}$ by use of Parseval's identity.
[Hint. Use the Fourier sine and cosine transforms of e^{-x} , $x > 0$.]
- 5.31. Use Problem 5.25 to show that (a) $\int_0^\infty \left(\frac{1-\cos x}{x}\right)^2 dx = \frac{\pi}{2}$, (b) $\int_0^\infty \frac{\sin^4 x}{x^2} dx = \frac{\pi}{4}$.
- 5.32. Show that $\int_0^\infty \frac{(x \cos x - \sin x)^2}{x^6} dx = \frac{\pi}{15}$.
- 5.33. Prove the results given by (a) equation (13), page 82; (b) equation (14), page 82.
- 5.34. Establish the results of equations (15), (16), (17) and (18) on page 82.

CONVOLUTION THEOREM

- 5.35. Verify the convolution theorem for the functions $f(x) = g(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| > 1 \end{cases}$.
- 5.36. Verify the convolution theorem for the functions $f(x) = g(x) = e^{-x^2}$.
- 5.37. Solve the integral equation $\int_{-\infty}^\infty y(u) y(x-u) du = e^{-x^2}$.
- 5.38. Prove that $f * (g + h) = f * g + f * h$.
- 5.39. Prove that $f * (g * h) = (f * g) * h$.

PROOF OF FOURIER INTEGRAL THEOREM

- 5.40. By interchanging the order of integration in $\int_{y=0}^\infty \int_{x=0}^\infty e^{-xy} \sin y dx dy$, prove that

$$\int_0^\infty \frac{\sin y}{y} dy = \frac{\pi}{2}$$

and thus complete the proof in Problem 5.12.

- 5.41. Let n be any real number. Is Fourier's integral theorem valid for $f(x) = e^{-x^n}$? Explain.

SOLUTIONS USING FOURIER INTEGRALS

- 5.42. An infinite thin bar ($-\infty < x < \infty$) whose surface is insulated has an initial temperature given by

$$f(x) = \begin{cases} u_0 & |x| < a \\ 0 & |x| \geq a \end{cases}$$

Show that the temperature at any point x at any time t is

$$u(x, t) = \frac{u_0}{2} \left[\operatorname{erf} \left(\frac{x+a}{2\sqrt{\kappa t}} \right) - \operatorname{erf} \left(\frac{x-a}{2\sqrt{\kappa t}} \right) \right]$$

- 5.43. A semi-infinite solid ($x > 0$) has an initial temperature given by $f(x) = u_0 e^{-bx^2}$. If the plane face ($x = 0$) is insulated show that the temperature at any point x at any time t is

$$u(x, t) = \frac{u_0}{\sqrt{1 + 4\kappa b t}} e^{-bx^2/(1 + 4\kappa b t)}$$

- 5.44. Solve and physically interpret the following boundary value problem:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad y > 0$$

$$u(x, 0) = \begin{cases} -1 & x < 0 \\ 1 & x > 0 \end{cases} \quad |u(x, y)| < M$$

- 5.45. Show that if $u(x, 0) = \begin{cases} 0 & x < 0 \\ u_0 & x > 0 \end{cases}$ in Problem 5.44, then

$$u(x, y) = \frac{u_0}{2} + \frac{u_0}{\pi} \tan^{-1} \frac{x}{y}$$

- 5.46. Work Problem 5.44 if $u(x, 0) = \begin{cases} 0 & x < -1 \\ 1 & -1 < x < 1 \\ 0 & x > 1 \end{cases}$.

- 5.47. The region bounded by $x > 0$, $y > 0$ has one edge $x = 0$ kept at potential zero and the other edge $y = 0$ kept at potential $f(x)$. (a) Show that the potential at any point (x, y) is given by

$$v(x, y) = \frac{1}{\pi} \int_0^\infty y f(v) \left[\frac{1}{(v-x)^2 + y^2} - \frac{1}{(v+x)^2 + y^2} \right] dv$$

- (b) If $f(x) = 1$, show that $v(x, y) = \frac{2}{\pi} \tan^{-1} \frac{x}{y}$.

- 5.48. Verify that the result obtained in Problem 5.18 is actually a solution of the corresponding boundary value problem.

- 5.49. The lines $y = 0$ and $y = a$ in the xy -plane (see Fig. 5-4) are kept at potentials 0 and $f(x)$ respectively. Show that the potential at points (x, y) between these lines is given by

$$v(x, y) = \frac{1}{\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u) \frac{\sinh \lambda y}{\sinh \lambda a} \cos \lambda(u-x) du d\lambda$$

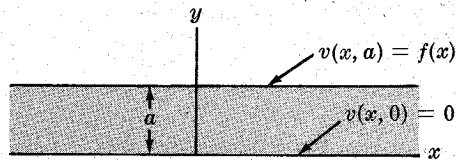


Fig. 5-4

- 5.50. An infinite string coinciding with the x -axis is given an initial shape $f(x)$ and an initial velocity $g(x)$. Assuming that gravity is neglected, show that the displacement of any point x of the string at time t is given by

$$y(x, t) = \frac{1}{2} [f(x+at) + f(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} g(u) du$$

- 5.51. Work Problem 5.50 if gravity is taken into account.

- 5.52. A semi-infinite cantilever beam ($x > 0$) clamped at $x = 0$ is given an initial shape $f(x)$ and released. Find the resulting displacement at any later time t .

Chapter 6

Bessel Functions and Applications

BESSEL'S DIFFERENTIAL EQUATION

Bessel functions arise as solutions of the differential equation

$$x^2 y'' + xy' + (x^2 - n^2)y = 0 \quad n \geq 0 \quad (1)$$

which is called *Bessel's differential equation*. The general solution of (1) is given by

$$y = c_1 J_n(x) + c_2 Y_n(x) \quad (2)$$

The solution $J_n(x)$, which has a finite limit as x approaches zero, is called a *Bessel function of the first kind of order n* . The solution $Y_n(x)$, which has no finite limit (i.e. is unbounded) as x approaches zero, is called a *Bessel function of the second kind of order n* or a *Neumann function*.

If the independent variable x in (1) is changed to λx , where λ is a constant, the resulting equation is

$$x^2 y'' + xy' + (\lambda^2 x^2 - n^2)y = 0 \quad (3)$$

with general solution

$$y = c_1 J_n(\lambda x) + c_2 Y_n(\lambda x) \quad (4)$$

The differential equation (1) or (3) is obtained, for example, from Laplace's equation $\nabla^2 u = 0$ expressed in cylindrical coordinates (ρ, ϕ, z) . See Problem 6.1.

THE METHOD OF FROBENIUS

An important method for obtaining solutions of differential equations such as Bessel's equation is known as the *method of Frobenius*. In this method we assume a solution of the form

$$y = \sum_{k=-\infty}^{\infty} c_k x^{k+\beta} \quad (5)$$

where $c_k = 0$ for $k < 0$, so that (5) actually begins with the term involving c_0 which is assumed different from zero.

By substituting (5) into a given differential equation we can obtain an equation for the constant β (called an *indicial equation*), as well as equations which can be used to determine the constants c_k . The process is illustrated in Problem 6.3.

BESSEL FUNCTIONS OF THE FIRST KIND

We define the Bessel function of the first kind of order n as

$$J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left\{ 1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \cdots \right\} \quad (6)$$

or

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{n+2r}}{r! \Gamma(n+r+1)} \quad (7)$$

where $\Gamma(n+1)$ is the gamma function (Chapter 4). If n is a positive integer, $\Gamma(n+1) = n!$, $\Gamma(1) = 1$. For $n = 0$, (6) becomes

$$J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 4^2} - \frac{x^6}{2^2 4^2 6^2} + \cdots \quad (8)$$

The series (6) or (7) converges for all x . Graphs of $J_0(x)$ and $J_1(x)$ are shown in Fig. 6-1.

If n is half an odd integer, $J_n(x)$ can be expressed in terms of sines and cosines. See Problems 6.6 and 6.9.

A function $J_{-n}(x)$, $n > 0$, can be defined by replacing n by $-n$ in (6) or (7). If n is an integer, then we can show that (see Problem 6.5)

$$J_{-n}(x) = (-1)^n J_n(x) \quad (9)$$

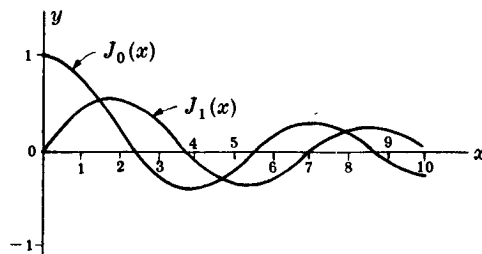


Fig. 6-1

If n is not an integer, $J_n(x)$ and $J_{-n}(x)$ are linearly independent, and for this case the general solution of (1) is

$$y = AJ_n(x) + BJ_{-n}(x) \quad n \neq 0, 1, 2, 3, \dots \quad (10)$$

BESSEL FUNCTIONS OF THE SECOND KIND

We shall define the Bessel function of the second kind of order n as

$$Y_n(x) = \begin{cases} \frac{J_n(x) \cos n\pi - J_{-n}(x)}{\sin n\pi} & n \neq 0, 1, 2, 3, \dots \\ \lim_{p \rightarrow n} \frac{J_p(x) \cos p\pi - J_{-p}(x)}{\sin p\pi} & n = 0, 1, 2, 3, \dots \end{cases} \quad (11)$$

For the case where $n = 0, 1, 2, 3, \dots$ we obtain the following series expansion for $Y_n(x)$:

$$Y_n(x) = \frac{2}{\pi} \{ \ln(x/2) + \gamma \} J_n(x) - \frac{1}{\pi} \sum_{k=0}^{n-1} \frac{(n-k-1)! (x/2)^{2k-n}}{k!} - \frac{1}{\pi} \sum_{k=0}^{\infty} (-1)^k \{ \Phi(k) + \Phi(n+k) \} \frac{(x/2)^{2k+n}}{k! (n+k)!} \quad (12)$$

where $\gamma = 0.5772156 \dots$ is Euler's constant (page 68) and

$$\Phi(p) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p}, \quad \Phi(0) = 0 \quad (13)$$

Graphs of the functions $Y_0(x)$ and $Y_1(x)$ are shown in Fig. 6-2. Note that these functions, as well as all the functions $Y_n(x)$ where $n > 0$, are unbounded at $x = 0$.

If n is half an odd integer $Y_n(x)$ can be expressed in terms of trigonometric functions.

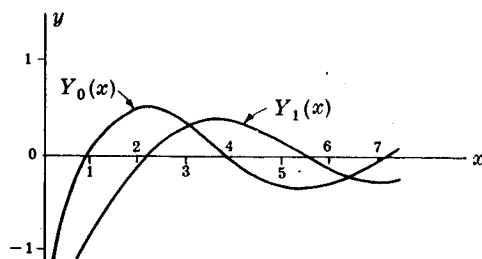


Fig. 6-2

GENERATING FUNCTION FOR $J_n(x)$

The function

$$e^{\frac{x}{2}(t - \frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x)t^n \quad (14)$$

is called the *generating function* for Bessel functions of the first kind of integral order. It is very useful in obtaining properties of these functions for integer values of n —properties which can then often be proved for all values of n .

RECURRENCE FORMULAS

The following results are valid for all values of n .

1. $J_{n+1}(x) = \frac{2n}{x}J_n(x) - J_{n-1}(x)$
2. $J'_n(x) = \frac{1}{2}[J_{n-1}(x) - J_{n+1}(x)]$
3. $xJ'_n(x) = nJ_n(x) - xJ_{n+1}(x)$
4. $xJ'_n(x) = xJ_{n-1}(x) - nJ_n(x)$
5. $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$
6. $\frac{d}{dx}[x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$

If n is an integer these can be proved by using the generating function. Note that results 3. and 4. are respectively equivalent to 5. and 6.

The functions $Y_n(x)$ satisfy exactly the same formulas, where $Y_n(x)$ replaces $J_n(x)$.

FUNCTIONS RELATED TO BESSEL FUNCTIONS

1. **Hankel functions of the first and second kinds** are defined respectively by

$$H_n^{(1)}(x) = J_n(x) + iY_n(x), \quad H_n^{(2)}(x) = J_n(x) - iY_n(x) \quad (15)$$

2. **Modified Bessel functions.** The *modified Bessel function of the first kind of order n* is defined as

$$I_n(x) = i^{-n}J_n(ix) = e^{-n\pi i/2}J_n(ix) \quad (16)$$

If n is an integer,

$$I_{-n}(x) = I_n(x) \quad (17)$$

but if n is not an integer, $I_n(x)$ and $I_{-n}(x)$ are linearly independent.

The *modified Bessel function of the second kind of order n* is defined as

$$K_n(x) = \begin{cases} \frac{\pi}{2} \left[\frac{I_{-n}(x) - I_n(x)}{\sin n\pi} \right] & n \neq 0, 1, 2, 3, \dots \\ \lim_{p \rightarrow n} \frac{\pi}{2} \left[\frac{I_{-p}(x) - I_p(x)}{\sin p\pi} \right] & n = 0, 1, 2, 3, \dots \end{cases} \quad (18)$$

These functions satisfy the differential equation

$$x^2 y'' + xy' - (x^2 + n^2)y = 0 \quad (19)$$

and the general solution of this equation is

$$y = c_1 I_n(x) + c_2 K_n(x) \quad (20)$$

or, if $n \neq 0, 1, 2, 3, \dots$,

$$y = A I_n(x) + B I_{-n}(x) \quad (21)$$

Graphs of the functions $I_0(x)$, $I_1(x)$, $K_0(x)$, $K_1(x)$ are shown in Figs. 6-3 and 6-4.

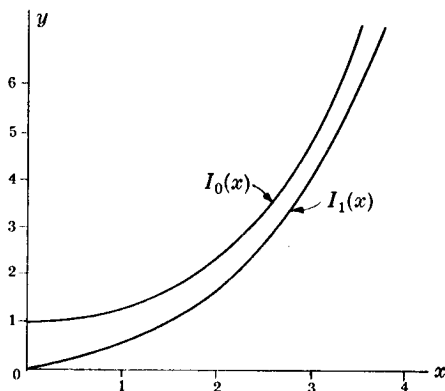


Fig. 6-3

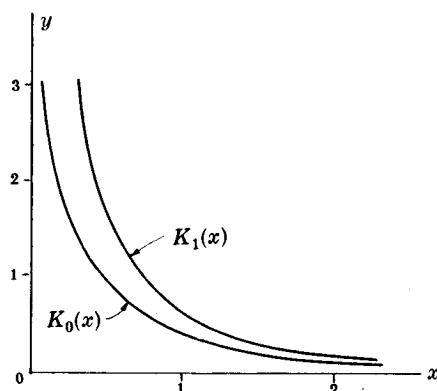


Fig. 6-4

3. **Ber, Bei, Ker, Kei functions.** The functions $\text{Ber}_n(x)$ and $\text{Bei}_n(x)$ are respectively the real and imaginary parts of $J_n(i^{3/2}x)$, where $i^{3/2} = e^{3\pi i/4} = (\sqrt{2}/2)(-1 + i)$, i.e.

$$J_n(i^{3/2}x) = \text{Ber}_n(x) + i \text{Bei}_n(x) \quad (22)$$

The functions $\text{Ker}_n(x)$ and $\text{Kei}_n(x)$ are respectively the real and imaginary parts of $e^{-n\pi i/2} K_n(i^{1/2}x)$, where $i^{1/2} = e^{\pi i/4} = (\sqrt{2}/2)(1 + i)$, i.e.

$$e^{-n\pi i/2} K_n(i^{1/2}x) = \text{Ker}_n(x) + i \text{Kei}_n(x) \quad (23)$$

The functions are useful in connection with the equation

$$x^2 y'' + xy' - (ix^2 + n^2)y = 0 \quad (24)$$

which arises in electrical engineering and other fields. The general solution of this equation is

$$y = c_1 J_n(i^{3/2}x) + c_2 K_n(i^{1/2}x) \quad (25)$$

If $n = 0$ we often denote $\text{Ber}_n(x)$, $\text{Bei}_n(x)$, $\text{Ker}_n(x)$, $\text{Kei}_n(x)$ by $\text{Ber}(x)$, $\text{Bei}(x)$, $\text{Ker}(x)$, $\text{Kei}(x)$, respectively. The graphs of these functions are shown in Figs. 6-5 and 6-6.

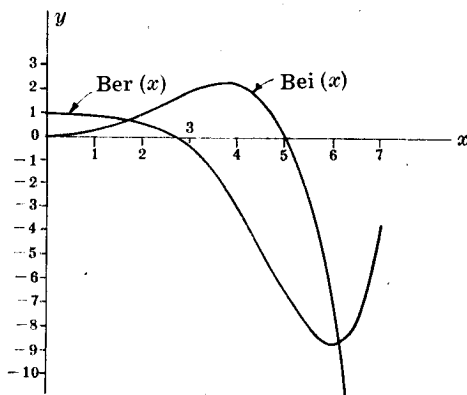


Fig. 6-5

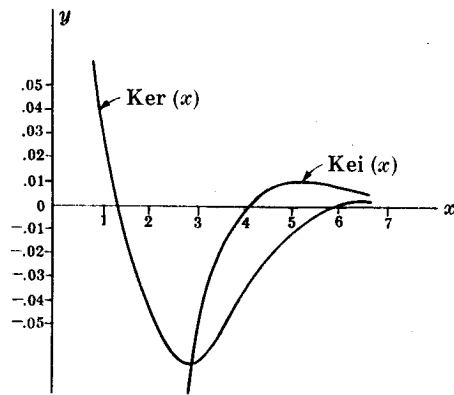


Fig. 6-6

EQUATIONS TRANSFORMABLE INTO BESSEL'S EQUATION

The equation

$$x^2 y'' + (2k+1)xy' + (\alpha^2 x^{2r} + \beta^2)y = 0 \quad (26)$$

where k, α, r, β are constants, has the general solution

$$y = x^{-k} [c_1 J_{\kappa/r}(\alpha x^r/r) + c_2 Y_{\kappa/r}(\alpha x^r/r)] \quad (27)$$

where $\kappa = \sqrt{k^2 - \beta^2}$. If $\alpha = 0$ the equation is an *Euler* or *Cauchy equation* (see Problem 6.79) and has solution

$$y = x^{-k}(c_3 x^\kappa + c_4 x^{-\kappa}) \quad (28)$$

ASYMPTOTIC FORMULAS FOR BESSEL FUNCTIONS

For large values of x we have the following asymptotic formulas:

$$J_n(x) \sim \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{\pi}{4} - \frac{n\pi}{2}\right), \quad Y_n(x) \sim \sqrt{\frac{2}{\pi x}} \sin\left(x - \frac{\pi}{4} - \frac{n\pi}{2}\right) \quad (29)$$

ZEROS OF BESSEL FUNCTIONS

We can show that if n is any real number, $J_n(x) = 0$ has an infinite number of roots which are all real. The difference between successive roots approaches π as the roots increase in value. This can be seen from (29). We can also show that the roots of $J_n(x) = 0$ [the *zeros* of $J_n(x)$] lie between those of $J_{n-1}(x) = 0$ and $J_{n+1}(x) = 0$. Similar remarks can be made for $Y_n(x)$. For a table giving zeros of Bessel functions see Appendix E, page 177.

ORTHOGONALITY OF BESSEL FUNCTIONS OF THE FIRST KIND

If λ and μ are two different constants, we can show (see Problem 6.23) that

$$\int_0^1 x J_n(\lambda x) J_n(\mu x) dx = \frac{\mu J_n(\lambda) J'_n(\mu) - \lambda J_n(\mu) J'_n(\lambda)}{\lambda^2 - \mu^2} \quad (30)$$

while (see Problem 6.24)

$$\int_0^1 x J_n^2(\lambda x) dx = \frac{1}{2} \left[J_n'^2(\lambda) + \left(1 - \frac{n^2}{\lambda^2}\right) J_n^2(\lambda) \right] \quad (31)$$

From (30) we can see that if λ and μ are any two different roots of the equation

$$R J_n(x) + S x J'_n(x) = 0 \quad (32)$$

where R and S are constants, then

$$\int_0^1 x J_n(\lambda x) J_n(\mu x) dx = 0 \quad (33)$$

which states that the functions $\sqrt{x} J_n(\lambda x)$ and $\sqrt{x} J_n(\mu x)$ are orthogonal in $(0, 1)$. Note that as special cases of (32) λ and μ can be any two different roots of $J_n(x) = 0$ or of $J'_n(x) = 0$. We can also say that the functions $J_n(\lambda x)$, $J_n(\mu x)$ are orthogonal with respect to the density or weight function x .

SERIES OF BESSEL FUNCTIONS OF THE FIRST KIND

As in the case of Fourier series, we can show that if $f(x)$ and $f'(x)$ are piecewise continuous then at every point of continuity of $f(x)$ in the interval of $0 < x < 1$ there will exist a Bessel series expansion having the form

$$f(x) = A_1 J_n(\lambda_1 x) + A_2 J_n(\lambda_2 x) + \cdots = \sum_{p=1}^{\infty} A_p J_n(\lambda_p x) \quad (34)$$

where $\lambda_1, \lambda_2, \lambda_3, \dots$ are the positive roots of (32) with $R/S \geq 0$, $S \neq 0$ and

$$A_p = \frac{2\lambda_p^2}{(\lambda_p^2 - n^2 + R^2/S^2)J_n^2(\lambda_p)} \int_0^1 x J_n(\lambda_p x) f(x) dx \quad (35)$$

At any point of discontinuity the series on the right in (34) converges to $\frac{1}{2}[f(x+0) + f(x-0)]$, which can be used in place of the left side of (34).

In case $S = 0$, so that $\lambda_1, \lambda_2, \dots$ are the roots of $J_n(x) = 0$,

$$A_p = \frac{2}{J_{n+1}^2(\lambda_p)} \int_0^1 x J_n(\lambda_p x) f(x) dx \quad (36)$$

If $R = 0$ and $n = 0$, then the series (34) starts out with the constant term

$$A_1 = 2 \int_0^1 x f(x) dx \quad (37)$$

In this case the positive roots are those of $J_n'(x) = 0$.

ORTHOGONALITY AND SERIES OF BESSEL FUNCTIONS OF THE SECOND KIND

The above results for Bessel functions of the first kind can be extended to Bessel functions of the second kind. See Problems 6.32 and 6.33.

SOLUTIONS TO BOUNDARY VALUE PROBLEMS USING BESSEL FUNCTIONS

The expansion of functions into Bessel series enables us to solve various boundary value problems arising in science and engineering. See Problems 6.28, 6.29, 6.31, 6.34, 6.35.

Solved Problems

BESSEL'S DIFFERENTIAL EQUATION

6.1. Show how Bessel's differential equation (3), page 97, is obtained from Laplace's equation $\nabla^2 u = 0$ expressed in cylindrical coordinates (ρ, ϕ, z) .

Laplace's equation in cylindrical coordinates is given by

$$\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad (1)$$

If we assume a solution of the form $u = P\Phi Z$, where P is a function of ρ , Φ is a function of ϕ and Z is a function of z , then (1) becomes

$$P''\Phi Z + \frac{1}{\rho} P'\Phi Z + \frac{1}{\rho^2} P\Phi''Z + P\Phi Z'' = 0 \quad (2)$$

where the primes denote derivatives with respect to the particular independent variable involved. Dividing (2) by $P\Phi Z$ yields

$$\frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{1}{\rho^2} \frac{\Phi''}{\Phi} + \frac{Z''}{Z} = 0 \quad (3)$$

Equation (3) can be written as

$$\frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{1}{\rho^2} \frac{\Phi''}{\Phi} = -\frac{Z''}{Z} \quad (4)$$

Since the right side depends only on z while the left side depends only on ρ and ϕ , it follows that each side must be a constant, say $-\lambda^2$. Thus we have

$$\frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{1}{\rho^2} \frac{\Phi''}{\Phi} = -\lambda^2 \quad (5)$$

and

$$Z'' - \lambda^2 Z = 0 \quad (6)$$

If we now multiply both sides of (5) by ρ^2 it becomes

$$\rho^2 \frac{P''}{P} + \rho \frac{P'}{P} + \frac{\Phi''}{\Phi} = -\lambda^2 \rho^2 \quad (7)$$

which can be written as

$$\rho^2 \frac{P''}{P} + \rho \frac{P'}{P} + \lambda^2 \rho^2 = -\frac{\Phi''}{\Phi} \quad (8)$$

Since the right side depends only on ϕ , while the left side depends only on ρ , it follows that each side must be a constant, say μ^2 . Thus we have

$$\rho^2 \frac{P''}{P} + \rho \frac{P'}{P} + \lambda^2 \rho^2 = \mu^2 \quad (9)$$

and

$$\Phi'' + \mu^2 \Phi = 0 \quad (10)$$

The equation (9) can be written as

$$\rho^2 P'' + \rho P' + (\lambda^2 \rho^2 - \mu^2) P = 0 \quad (11)$$

which is Bessel's differential equation (3) on page 97 with P instead of y , ρ instead of x and μ instead of n .

6.2. Show that if we let $\lambda\rho = x$ in equation (11) of Problem 6.1, then it becomes

$$x^2 y'' + xy' + (x^2 - \mu^2)y = 0$$

We have

$$\frac{dP}{d\rho} = \frac{dP}{dx} \frac{dx}{d\rho} = \frac{dP}{dx} \lambda = \lambda \frac{dy}{dx}$$

where $y(x)$, or briefly y , represents that function of x which $P(\rho)$ becomes when $\rho = x/\lambda$.

Similarly

$$\frac{d^2 P}{d\rho^2} = \frac{d}{d\rho} \left(\frac{dP}{d\rho} \right) = \frac{d}{dx} \left(\lambda \frac{dy}{dx} \right) \frac{dx}{d\rho} = \frac{d}{dx} \left(\lambda \frac{dy}{dx} \right) \lambda = \lambda^2 \frac{d^2 y}{dx^2}$$

Then equation (11) of Problem 6.1 which can be written

$$\rho^2 \frac{d^2 P}{d\rho^2} + \rho \frac{dP}{d\rho} + (\lambda^2 \rho^2 - \mu^2) P = 0$$

becomes

$$\left(\frac{x}{\lambda} \right)^2 \lambda^2 \frac{d^2 y}{dx^2} + \left(\frac{x}{\lambda} \right) \lambda \frac{dy}{dx} + (x^2 - \mu^2)y = 0$$

or

$$x^2 y'' + xy' + (x^2 - \mu^2)y = 0$$

as required.

6.3. Use the method of Frobenius to find series solutions of Bessel's differential equation $x^2y'' + xy' + (x^2 - n^2)y = 0$.

Assuming a solution of the form $y = \sum c_k x^{k+\beta}$ where k goes from $-\infty$ to ∞ and $c_k = 0$ for $k < 0$, we have

$$\begin{aligned}(x^2 - n^2)y &= \sum c_k x^{k+\beta+2} - \sum n^2 c_k x^{k+\beta} = \sum c_{k-2} x^{k+\beta} - \sum n^2 c_k x^{k+\beta} \\ xy' &= \sum (k+\beta) c_k x^{k+\beta} \\ x^2 y'' &= \sum (k+\beta)(k+\beta-1) c_k x^{k+\beta}\end{aligned}$$

Then by addition,

$$\sum [(k+\beta)(k+\beta-1)c_k + (k+\beta)c_k + c_{k-2} - n^2 c_k] x^{k+\beta} = 0$$

and since the coefficients of the $x^{k+\beta}$ must be zero, we find

$$[(k+\beta)^2 - n^2]c_k + c_{k-2} = 0 \quad (1)$$

Letting $k=0$ in (1) we obtain, since $c_{-2}=0$, the indicial equation $(\beta^2 - n^2)c_0 = 0$; or assuming $c_0 \neq 0$, $\beta^2 = n^2$. Then there are two cases, given by $\beta = -n$ and $\beta = n$. We shall consider first the case $\beta = n$ and obtain the second case by replacing n by $-n$.

Case 1: $\beta = n$.

In this case (1) becomes

$$k(2n+k)c_k + c_{k-2} = 0 \quad (2)$$

Putting $k=1, 2, 3, 4, \dots$ successively in (2), we have

$$c_1 = 0, \quad c_2 = \frac{-c_0}{2(2n+2)}, \quad c_3 = 0, \quad c_4 = \frac{-c_2}{4(2n+4)} = \frac{c_0}{2 \cdot 4(2n+2)(2n+4)}, \quad \dots$$

Thus the required series is

$$\begin{aligned}y &= c_0 x^n + c_2 x^{n+2} + c_4 x^{n+4} + \dots \\ &= c_0 x^n \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right] \quad (3)\end{aligned}$$

Case 2: $\beta = -n$.

On replacing n by $-n$ in Case 1, we find

$$y = c_0 x^{-n} \left[1 - \frac{x^2}{2(2-2n)} + \frac{x^4}{2 \cdot 4(2-2n)(4-2n)} - \dots \right] \quad (4)$$

Now if $n=0$, both of these series are identical. If $n=1, 2, \dots$ the second series fails to exist. However, if $n \neq 0, 1, 2, \dots$ the two series can be shown to be linearly independent, and so for this case the general solution is

$$\begin{aligned}y &= Cx^n \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right] \\ &\quad + Dx^{-n} \left[1 - \frac{x^2}{2(2-2n)} + \frac{x^4}{2 \cdot 4(2-2n)(4-2n)} - \dots \right] \quad (5)\end{aligned}$$

The cases where $n=0, 1, 2, 3, \dots$ are treated later (see Problems 6.17 and 6.18).

The first series in (5), with suitable choice of multiplicative constant, provides the definition of $J_n(x)$ given by (6), page 97.

BESSEL FUNCTIONS OF THE FIRST KIND

6.4. Using the definition (6) of $J_n(x)$ given on page 97, show that if $n \neq 0, 1, 2, \dots$, then the general solution of Bessel's equation is $y = AJ_n(x) + BJ_{-n}(x)$.

Note that the definition of $J_n(x)$ on page 97 agrees with the series of Case 1 in Problem 6.3, apart from a constant factor depending only on n . It follows that the result (5) can be written $y = AJ_n(x) + BJ_{-n}(x)$ for the cases $n \neq 0, 1, 2, \dots$.

6.5. (a) Prove that $J_{-n}(x) = (-1)^n J_n(x)$ for $n = 1, 2, 3, \dots$.

(b) Use (a) to explain why $AJ_n(x) + BJ_{-n}(x)$ is not the general solution of Bessel's equation for integer values of n .

(a) Replacing n by $-n$ in (6) or the equivalent (7) on page 98, we have

$$\begin{aligned} J_{-n}(x) &= \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{-n+2r}}{r! \Gamma(-n+r+1)} \\ &= \sum_{r=0}^{n-1} \frac{(-1)^r (x/2)^{-n+2r}}{r! \Gamma(-n+r+1)} + \sum_{r=n}^{\infty} \frac{(-1)^r (x/2)^{-n+2r}}{r! \Gamma(-n+r+1)} \end{aligned}$$

Now since $\Gamma(-n+r+1)$ is infinite for $r = 0, 1, \dots, n-1$, the first sum on the right is zero. Letting $r = n+k$ in the second sum, it becomes

$$\sum_{k=0}^{\infty} \frac{(-1)^{n+k} (x/2)^{n+2k}}{(n+k)! \Gamma(k+1)} = (-1)^n \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{n+2k}}{\Gamma(n+k+1)k!} = (-1)^n J_n(x)$$

(b) From (a) it follows that for integer values of n , $J_{-n}(x)$ and $J_n(x)$ are linearly dependent and so $AJ_n(x) + BJ_{-n}(x)$ cannot be a general solution of Bessel's equation. If n is not an integer, then we can show that $J_{-n}(x)$ and $J_n(x)$ are linearly independent, so that $AJ_n(x) + BJ_{-n}(x)$ is a general solution (see Problem 6.12).

6.6. Prove (a) $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$, (b) $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$.

$$\begin{aligned} (a) \quad J_{1/2}(x) &= \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{1/2+2r}}{r! \Gamma(r+3/2)} = \frac{(x/2)^{1/2}}{\Gamma(3/2)} - \frac{(x/2)^{5/2}}{1! \Gamma(5/2)} + \frac{(x/2)^{9/2}}{2! \Gamma(7/2)} - \dots \\ &= \frac{(x/2)^{1/2}}{(1/2)\sqrt{\pi}} - \frac{(x/2)^{5/2}}{1! (3/2)(1/2)\sqrt{\pi}} + \frac{(x/2)^{7/2}}{2! (5/2)(3/2)(1/2)\sqrt{\pi}} - \dots \\ &= \frac{(x/2)^{1/2}}{(1/2)\sqrt{\pi}} \left\{ 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \dots \right\} = \frac{(x/2)^{1/2}}{(1/2)\sqrt{\pi}} \frac{\sin x}{x} = \sqrt{\frac{2}{\pi x}} \sin x \end{aligned}$$

$$\begin{aligned} (b) \quad J_{-1/2}(x) &= \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{-1/2+2r}}{r! \Gamma(r+1/2)} = \frac{(x/2)^{-1/2}}{\Gamma(1/2)} - \frac{(x/2)^{3/2}}{1! \Gamma(3/2)} + \frac{(x/2)^{7/2}}{2! \Gamma(5/2)} - \dots \\ &= \frac{(x/2)^{-1/2}}{\sqrt{\pi}} \left\{ 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right\} = \sqrt{\frac{2}{\pi x}} \cos x \end{aligned}$$

6.7. Prove that for all n :

$$(a) \quad \frac{d}{dx} \{x^n J_n(x)\} = x^n J_{n-1}(x), \quad (b) \quad \frac{d}{dx} \{x^{-n} J_n(x)\} = -x^n J_{n+1}(x).$$

$$\begin{aligned} (a) \quad \frac{d}{dx} \{x^n J_n(x)\} &= \frac{d}{dx} \sum_{r=0}^{\infty} \frac{(-1)^r x^{2n+2r}}{2^{n+2r} r! \Gamma(n+r+1)} = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2n+2r-1}}{2^{n+2r-1} r! \Gamma(n+r)} \\ &= x^n \sum_{r=0}^{\infty} \frac{(-1)^r x^{n-1+2r}}{2^{(n-1)+2r} r! \Gamma[(n-1)+r+1]} = x^n J_{n-1}(x) \end{aligned}$$

$$\begin{aligned}
 (b) \quad \frac{d}{dx} \{x^{-n} J_n(x)\} &= \frac{d}{dx} \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{2^{n+2r} r! \Gamma(n+r+1)} \\
 &= x^{-n} \sum_{r=1}^{\infty} \frac{(-1)^r x^{n+2r-1}}{2^{n+2r-1} (r-1)! \Gamma(n+r+1)} \\
 &= x^{-n} \sum_{k=0}^{\infty} \frac{(-1)^{k+1} x^{n+2k+1}}{2^{n+2k+1} k! \Gamma(n+k+2)} = -x^{-n} J_{n+1}(x)
 \end{aligned}$$

6.8. Prove that for all n :

$$(a) \quad J'_n(x) = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)], \quad (b) \quad J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x).$$

From Problem 6.7(a), $x^n J'_n(x) + n x^{n-1} J_n(x) = x^n J_{n-1}(x)$

$$\text{or} \quad x J'_n(x) + n J_n(x) = x J_{n-1}(x) \quad (1)$$

From Problem 6.7(b), $x^{-n} J'_n(x) - n x^{-n-1} J_n(x) = -x^{-n} J_{n+1}(x)$

$$\text{or} \quad x J'_n(x) - n J_n(x) = -x J_{n+1}(x) \quad (2)$$

(a) Adding (1) and (2) and dividing by $2x$ gives

$$J'_n(x) = \frac{1}{2} [J_{n-1}(x) - J_{n+1}(x)]$$

(b) Subtracting (2) from (1) and dividing by x gives

$$J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x} J_n(x)$$

6.9. Show that (a) $J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x - x \cos x}{x} \right)$

(b) $J_{-3/2}(x) = -\sqrt{\frac{2}{\pi x}} \left(\frac{x \sin x + \cos x}{x} \right)$

(a) From Problems 6.8(b) and 6.6 we have on letting $n = 1/2$,

$$J_{3/2}(x) = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x - x \cos x}{x} \right)$$

(b) From Problems 6.8(b) and 6.6 we have on letting $n = -\frac{1}{2}$,

$$J_{-3/2}(x) = -\sqrt{\frac{2}{\pi x}} \left(\frac{x \sin x + \cos x}{x} \right)$$

6.10. Evaluate the integrals (a) $\int x^n J_{n-1}(x) dx$, (b) $\int \frac{J_{n+1}(x)}{x^n} dx$.

From Problem 6.7,

(a) $\frac{d}{dx} \{x^n J_n(x)\} = x^n J_{n-1}(x)$. Then $\int x^n J_{n-1}(x) dx = x^n J_n(x) + c$.

(b) $\frac{d}{dx} \{x^{-n} J_n(x)\} = -x^{-n} J_{n+1}(x)$. Then $\int \frac{J_{n+1}(x)}{x^n} dx = -x^{-n} J_n(x) + c$.

6.11. Evaluate (a) $\int x^4 J_1(x) dx$, (b) $\int x^3 J_3(x) dx$.

(a) **Method 1.** Integration by parts gives

$$\begin{aligned}\int x^4 J_1(x) dx &= \int (x^2)[x^2 J_1(x) dx] \\ &= x^2[x^2 J_2(x)] - \int [x^2 J_2(x)][2x dx] \\ &= x^4 J_2(x) - 2 \int x^3 J_2(x) dx \\ &= x^4 J_2(x) - 2x^3 J_3(x) + c\end{aligned}$$

Method 2. We have, using $J_1(x) = -J_0'(x)$ [Problem 6.7(b)],

$$\begin{aligned}\int x^4 J_1(x) dx &= -\int x^4 J_0'(x) dx = -\left\{x^4 J_0(x) - \int 4x^3 J_0(x) dx\right\} \\ \int x^3 J_0(x) dx &= \int x^2[x J_0(x) dx] = x^2[x J_1(x)] - \int [x J_1(x)][2x dx] \\ \int x^2 J_1(x) dx &= -\int x^2 J_0'(x) dx = -\left\{x^2 J_0(x) - \int 2x J_0(x) dx\right\} \\ &= -x^2 J_0(x) + 2x J_1(x)\end{aligned}$$

Then
$$\begin{aligned}\int x^4 J_1(x) dx &= -x^4 J_0(x) + 4[x^3 J_1(x) - 2\{-x^2 J_0(x) + 2x J_1(x)\}] + c \\ &= (8x^2 - x^4)J_0(x) + (4x^3 - 16x)J_1(x)\end{aligned}$$

(b)
$$\begin{aligned}\int x^3 J_3(x) dx &= \int x^5[x^{-2} J_3(x) dx] \\ &= x^5[-x^{-2} J_2(x)] - \int [-x^{-2} J_2(x)]5x^4 dx \\ &= -x^3 J_2(x) + 5 \int x^2 J_2(x) dx \\ \int x^2 J_2(x) dx &= \int x^3[x^{-1} J_2(x) dx] \\ &= x^3[-x^{-1} J_1(x)] - \int [-x^{-1} J_1(x)]3x^2 dx \\ &= -x^2 J_1(x) + 3 \int x J_1(x) dx\end{aligned}$$

$$\begin{aligned}\int x J_1(x) dx &= -\int x J_0'(x) dx = -\left[x J_0(x) - \int J_0(x) dx\right] \\ &= -x J_0(x) + \int J_0(x) dx\end{aligned}$$

Then
$$\begin{aligned}\int x^3 J_3(x) dx &= -x^3 J_2(x) + 5\left\{-x^2 J_1(x) + 3\left[-x J_0(x) + \int J_0(x) dx\right]\right\} \\ &= -x^3 J_2(x) - 5x^2 J_1(x) - 15x J_0(x) + 15 \int J_0(x) dx\end{aligned}$$

The integral $\int J_0(x) dx$ cannot be obtained in closed form. In general, $\int x^p J_q(x) dx$ can be obtained in closed form if $p+q \geq 0$ and $p+q$ is odd, where p and q are integers. If, however, $p+q$ is even, the result can be obtained in terms of $\int J_0(x) dx$.

6.12. (a) Prove that $J'_n(x)J_{-n}(x) - J'_{-n}(x)J_n(x) = \frac{2 \sin n\pi}{\pi x}$.

(b) Discuss the significance of the result of (a) with regard to the linear dependence of $J_n(x)$ and $J_{-n}(x)$.

(a) Since $J_n(x)$ and $J_{-n}(x)$, abbreviated J_n, J_{-n} respectively, satisfy Bessel's equation, we have

$$x^2 J''_n + x J'_n + (x^2 - n^2) J_n = 0, \quad x^2 J''_{-n} + x J'_{-n} + (x^2 - n^2) J_{-n} = 0$$

Multiply the first equation by J_{-n} , the second by J_n and subtract. Then

$$x^2 [J''_n J_{-n} - J''_{-n} J_n] + x [J'_n J_{-n} - J'_{-n} J_n] = 0$$

which can be written

$$x \frac{d}{dx} [J'_n J_{-n} - J'_{-n} J_n] + [J'_n J_{-n} - J'_{-n} J_n] = 0$$

or

$$\frac{d}{dx} \{x [J'_n J_{-n} - J'_{-n} J_n]\} = 0$$

Integrating, we find

$$J'_n J_{-n} - J'_{-n} J_n = c/x \quad (1)$$

To determine c use the series expansions for J_n and J_{-n} to obtain

$$J_n = \frac{x^n}{2^n \Gamma(n+1)} - \cdots, \quad J'_n = \frac{x^{n-1}}{2^n \Gamma(n)} - \cdots, \quad J_{-n} = \frac{x^{-n}}{2^{-n} \Gamma(-n+1)} - \cdots,$$

$$J'_{-n} = \frac{x^{-n-1}}{2^{-n} \Gamma(-n)} - \cdots$$

and then substitute in (1). We find

$$c = \frac{1}{\Gamma(n) \Gamma(1-n)} - \frac{1}{\Gamma(n+1) \Gamma(-n)} = \frac{2}{\Gamma(n) \Gamma(1-n)} = \frac{2 \sin n\pi}{\pi}$$

using the result 1, page 68. This gives the required result.

(b) The expression $J'_n J_{-n} - J'_{-n} J_n$ in (a) is the Wronskian of J_n and J_{-n} . If n is an integer, we see from (a) that the Wronskian is zero, so that J_n and J_{-n} are linearly dependent, as is also clear from Problem 6.5(a). On the other hand, if n is not an integer, they are linearly independent, since in such case the Wronskian differs from zero.

GENERATING FUNCTION AND MISCELLANEOUS RESULTS

6.13. Prove that $e^{\frac{x}{2}(t - \frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x) t^n$.

We have

$$e^{\frac{x}{2}(t - \frac{1}{t})} = e^{xt/2} e^{-x/2t} = \left\{ \sum_{r=0}^{\infty} \frac{(xt/2)^r}{r!} \right\} \left\{ \sum_{k=0}^{\infty} \frac{(-x/2t)^k}{k!} \right\} = \sum_{r=0}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{r+k} t^{r-k}}{r! k!}$$

Let $r-k = n$ so that n varies from $-\infty$ to ∞ . Then the sum becomes

$$\sum_{n=-\infty}^{\infty} \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{n+2k} t^n}{(n+k)! k!} = \sum_{n=-\infty}^{\infty} \left\{ \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{n+2k}}{k! (n+k)!} \right\} t^n = \sum_{n=-\infty}^{\infty} J_n(x) t^n$$

6.14. Prove (a) $\cos(x \sin \theta) = J_0(x) + 2J_2(x) \cos 2\theta + 2J_4(x) \cos 4\theta + \cdots$

(b) $\sin(x \sin \theta) = 2J_1(x) \sin \theta + 2J_3(x) \sin 3\theta + 2J_5(x) \sin 5\theta + \cdots$

Let $t = e^{i\theta}$ in Problem 6.13. Then

$$e^{\frac{1}{2}x(e^{i\theta} - e^{-i\theta})} = e^{ix \sin \theta} = \sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta} = \sum_{n=-\infty}^{\infty} J_n(x) [\cos n\theta + i \sin n\theta]$$

$$= \{J_0(x) + [J_{-1}(x) + J_1(x)] \cos \theta + [J_{-2}(x) + J_2(x)] \cos 2\theta + \cdots\}$$

$$+ i\{[J_1(x) - J_{-1}(x)] \sin \theta + [J_2(x) - J_{-2}(x)] \sin 2\theta + \cdots\}$$

$$= \{J_0(x) + 2J_2(x) \cos 2\theta + \cdots\} + i\{2J_1(x) \sin \theta + 2J_3(x) \sin 3\theta + \cdots\}$$

where we have used Problem 6.5(a). Equating real and imaginary parts gives the required results.

6.15. Prove $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta \quad n = 0, 1, 2, \dots$

Multiply the first and second results of Problem 6.14 by $\cos n\theta$ and $\sin n\theta$ respectively and integrate from 0 to π using

$$\int_0^\pi \cos m\theta \cos n\theta d\theta = \begin{cases} 0 & m \neq n \\ \pi/2 & m = n \end{cases}, \quad \int_0^\pi \sin m\theta \sin n\theta d\theta = \begin{cases} 0 & m \neq n \\ \pi/2 & m = n \end{cases}$$

Then if n is even or zero, we have

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) \cos n\theta d\theta, \quad 0 = \frac{1}{\pi} \int_0^\pi \sin(x \sin \theta) \sin n\theta d\theta$$

or on adding,

$$J_n(x) = \frac{1}{\pi} \int_0^\pi [\cos(x \sin \theta) \cos n\theta + \sin(x \sin \theta) \sin n\theta] d\theta = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$$

Similarly, if n is odd,

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \sin(x \sin \theta) \sin n\theta d\theta, \quad 0 = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) \cos n\theta d\theta$$

and by adding,

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$$

Thus we have the required result whether n is even or odd, i.e. $n = 0, 1, 2, \dots$

6.16. Prove the result of Problem 6.8(b) for integer values of n by using the generating function.

Differentiating both sides of the generating function with respect to t , we have, omitting the limits $-\infty$ to ∞ for n ,

$$e^{\frac{x}{2}(t - \frac{1}{t})} \frac{x}{2} \left(1 + \frac{1}{t^2}\right) = \sum n J_n(x) t^{n-1}$$

or

$$\frac{x}{2} \left(1 + \frac{1}{t^2}\right) \sum J_n(x) t^n = \sum n J_n(x) t^{n-1}$$

i.e.

$$\sum \frac{x}{2} \left(1 + \frac{1}{t^2}\right) J_n(x) t^n = \sum n J_n(x) t^{n-1}$$

This can be written as

$$\sum \frac{x}{2} J_n(x) t^n + \sum \frac{x}{2} J_n(x) t^{n-2} = \sum n J_n(x) t^{n-1}$$

or

$$\sum \frac{x}{2} J_n(x) t^n + \sum \frac{x}{2} J_{n+2}(x) t^n = \sum (n+1) J_{n+1}(x) t^n$$

i.e.

$$\sum \left[\frac{x}{2} J_n(x) + \frac{x}{2} J_{n+2}(x) \right] t^n = \sum (n+1) J_{n+1}(x) t^n$$

Since coefficients of t^n must be equal, we have

$$\frac{x}{2} J_n(x) + \frac{x}{2} J_{n+2}(x) = (n+1) J_{n+1}(x)$$

from which the required result is obtained on replacing n by $n-1$.

BESSEL FUNCTIONS OF THE SECOND KIND

6.17. (a) Show that if n is not an integer, the general solution of Bessel's equation is

$$y = EJ_n(x) + F \left[\frac{J_n(x) \cos n\pi - J_{-n}(x)}{\sin n\pi} \right]$$

where E and F are arbitrary constants.

(b) Explain how to use part (a) to obtain the general solution of Bessel's equation in case n is an integer.

(a) Since J_{-n} and J_n are linearly independent, the general solution of Bessel's equation can be written

$$y = c_1 J_n(x) + c_2 J_{-n}(x)$$

and the required result follows on replacing the arbitrary constants c_1, c_2 by E, F , where

$$c_1 = E + \frac{F \cos n\pi}{\sin n\pi}, \quad c_2 = \frac{-F}{\sin n\pi}$$

Note that we define the Bessel function of the second kind if n is not an integer by

$$Y_n(x) = \frac{J_n(x) \cos n\pi - J_{-n}(x)}{\sin n\pi}$$

(b) The expression

$$\frac{J_n(x) \cos n\pi - J_{-n}(x)}{\sin n\pi}$$

becomes an "indeterminate" of the form $0/0$ for the case when n is an integer. This is because for an integer n we have $\cos n\pi = (-1)^n$ and $J_{-n}(x) = (-1)^n J_n(x)$ [see Problem 6.5]. This "indeterminate form" can be evaluated by using L'Hospital's rule, i.e.

$$\lim_{p \rightarrow n} \left[\frac{J_p(x) \cos p\pi - J_{-p}(x)}{\sin p\pi} \right] = \lim_{p \rightarrow n} \frac{\frac{\partial}{\partial p} [J_p(x) \cos p\pi - J_{-p}(x)]}{\frac{\partial}{\partial p} [\sin p\pi]}$$

This motivates the definition (11) on page 98.

6.18. Use Problem 6.17 to obtain the general solution of Bessel's equation for $n = 0$.

In this case we must evaluate

$$\lim_{p \rightarrow 0} \left[\frac{J_p(x) \cos p\pi - J_{-p}(x)}{\sin p\pi} \right] \quad (1)$$

Using L'Hospital's rule (differentiating the numerator and denominator with respect to p), we find for the limit in (1)

$$\lim_{p \rightarrow 0} \left[\frac{(\partial J_p / \partial p) \cos p\pi - (\partial J_{-p} / \partial p)}{\pi \cos p\pi} \right] = \frac{1}{\pi} \left[\frac{\partial J_p}{\partial p} - \frac{\partial J_{-p}}{\partial p} \right]_{p=0}$$

where the notation indicates that we are to take the partial derivatives of $J_p(x)$ and $J_{-p}(x)$ with respect to p and then put $p = 0$. Since $\partial J_{-p} / \partial (-p) = -\partial J_{-p} / \partial p$, the required limit is also equal to

$$\frac{2}{\pi} \frac{\partial J_p}{\partial p} \Big|_{p=0}$$

To obtain $\partial J_p / \partial p$ we differentiate the series

$$J_p(x) = \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{p+2r}}{r! \Gamma(p+r+1)}$$

with respect to p and obtain

$$\frac{\partial J_p}{\partial p} = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!} \frac{\partial}{\partial p} \left\{ \frac{(x/2)^{p+2r}}{\Gamma(p+r+1)} \right\} \quad (2)$$

Now if we let $\frac{(x/2)^{p+2r}}{\Gamma(p+r+1)} = G$, then $\ln G = (p+2r) \ln(x/2) - \ln \Gamma(p+r+1)$ so that differentiation with respect to p gives

$$\frac{1}{G} \frac{\partial G}{\partial p} = \ln(x/2) - \frac{\Gamma'(p+r+1)}{\Gamma(p+r+1)}$$

Then for $p = 0$, we have

$$\frac{\partial G}{\partial p} \Big|_{p=0} = \frac{(x/2)^{2r}}{\Gamma(r+1)} \left[\ln(x/2) - \frac{\Gamma'(r+1)}{\Gamma(r+1)} \right] \quad (3)$$

Using (2) and (3), we have

$$\begin{aligned} \frac{2}{\pi} \frac{\partial J_p}{\partial p} \Big|_{p=0} &= \frac{2}{\pi} \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{2r}}{r! \Gamma(r+1)} \left[\ln(x/2) - \frac{\Gamma'(r+1)}{\Gamma(r+1)} \right] \\ &= \frac{2}{\pi} \{ \ln(x/2) + \gamma \} J_0(x) + \frac{2}{\pi} \left[\frac{x^2}{2^2} - \frac{x^4}{2^2 4^2} (1 + \frac{1}{2}) + \frac{x^6}{2^2 4^2 6^2} (1 + \frac{1}{2} + \frac{1}{3}) - \dots \right] \end{aligned}$$

where the last series is obtained on using the result 6. on page 69. This last series is the series for $Y_0(x)$. We can in a similar manner obtain the series (12), page 98, for $Y_n(x)$ where n is an integer. The general solution if n is an integer is then given by $y = c_1 J_n(x) + c_2 Y_n(x)$.

FUNCTIONS RELATED TO BESSEL FUNCTIONS

6.19. Prove that the recurrence formula for the modified Bessel function of the first kind $I_n(x)$ is

$$I_{n+1}(x) = I_{n-1}(x) - \frac{2n}{x} I_n(x)$$

From Problem 6.8(b) we have

$$J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x) \quad (1)$$

Replace x by ix to obtain

$$J_{n+1}(ix) = -\frac{2in}{x} J_n(ix) - J_{n-1}(ix) \quad (2)$$

Now by definition $I_n(x) = i^{-n} J_n(ix)$ or $J_n(ix) = i^n I_n(x)$, so that (2) becomes

$$i^{n+1} I_{n+1}(x) = -\frac{2in}{x} i^n I_n(x) - i^{n-1} I_{n-1}(x)$$

Dividing by i^{n+1} then gives the required result.

6.20. If n is not an integer, show that

$$(a) \quad H_n^{(1)}(x) = \frac{J_{-n}(x) - e^{-in\pi} J_n(x)}{i \sin n\pi}, \quad (b) \quad H_n^{(2)}(x) = \frac{e^{in\pi} J_n(x) - J_{-n}(x)}{i \sin n\pi}$$

(a) By definition of $H_n^{(1)}(x)$ and $Y_n(x)$ (see pages 99 and 98 respectively) we have

$$\begin{aligned} H_n^{(1)}(x) &= J_n(x) + iY_n(x) = J_n(x) + i \left[\frac{J_n(x) \cos n\pi - J_{-n}(x)}{\sin n\pi} \right] \\ &= \frac{J_n(x) \sin n\pi + iJ_n(x) \cos n\pi - iJ_{-n}(x)}{\sin n\pi} \\ &= i \left[\frac{J_n(x) (\cos n\pi - i \sin n\pi) - J_{-n}(x)}{\sin n\pi} \right] \\ &= i \left[\frac{J_n(x) e^{-in\pi} - J_{-n}(x)}{\sin n\pi} \right] = \frac{J_{-n}(x) - e^{-in\pi} J_n(x)}{i \sin n\pi} \end{aligned}$$

(b) Since $H_n^{(2)}(x) = J_n(x) - iY_n(x)$, we find on replacing i by $-i$ in the result of part (a),

$$H_n^{(2)}(x) = \frac{J_{-n}(x) - e^{in\pi} J_n(x)}{-i \sin n\pi} = \frac{e^{in\pi} J_n(x) - J_{-n}(x)}{i \sin n\pi}$$

6.21. Show that (a) $\text{Ber}(x) = 1 - \frac{x^4}{2^2 4^2} + \frac{x^8}{2^2 4^2 6^2 8^2} - \dots$

$$(b) \quad \text{Bei}(x) = \frac{x^2}{2^2} - \frac{x^6}{2^2 4^2 6^2} + \frac{x^{10}}{2^2 4^2 6^2 8^2 10^2} - \dots$$

We have

$$\begin{aligned}
 J_0(i^{3/2}x) &= 1 - \frac{(i^{3/2}x)^2}{2^2} + \frac{(i^{3/2}x)^4}{2^2 4^2} - \frac{(i^{3/2}x)^6}{2^2 4^2 6^2} + \frac{(i^{3/2}x)^8}{2^2 4^2 6^2 8^2} - \dots \\
 &= 1 - \frac{i^3 x^2}{2^2} + \frac{i^6 x^4}{2^2 4^2} - \frac{i^9 x^6}{2^2 4^2 6^2} + \frac{i^{12} x^8}{2^2 4^2 6^2 8^2} - \dots \\
 &= 1 + \frac{ix^2}{2^2} - \frac{x^4}{2^2 4^2} - \frac{ix^6}{2^2 4^2 6^2} + \frac{x^8}{2^2 4^2 6^2 8^2} - \dots \\
 &= \left(1 - \frac{x^4}{2^2 4^2} + \frac{x^8}{2^2 4^2 6^2 8^2} - \dots\right) + i\left(\frac{x^2}{2^2} - \frac{x^6}{2^2 4^2 6^2} + \dots\right)
 \end{aligned}$$

and the required result follows on noting that $J_0(i^{3/2}x) = \text{Ber}(x) + i \text{Bei}(x)$ and equating real and imaginary parts. Note that the subscript zero has been omitted from $\text{Ber}_0(x)$ and $\text{Bei}_0(x)$.

EQUATIONS TRANSFORMABLE INTO BESSEL'S EQUATION

6.22. Find the general solution of the equation $xy'' + y' + ay = 0$.

The equation can be written as $x^2 y'' + xy' + axy = 0$ and is a special case of equation (26), page 101, where $k = 0$, $\alpha = \sqrt{a}$, $r = 1/2$, $\beta = 0$. Then the solution as given by (27), page 101, is

$$y = c_1 J_0(2\sqrt{ax}) + c_2 Y_0(2\sqrt{ax})$$

ORTHOGONALITY OF BESSEL FUNCTIONS

6.23. Prove that $\int_0^1 x J_n(\lambda x) J_n(\mu x) dx = \frac{\mu J_n(\lambda) J_n'(\mu) - \lambda J_n(\mu) J_n'(\lambda)}{\lambda^2 - \mu^2}$ if $\lambda \neq \mu$.

From (3) and (4), page 97, we see that $y_1 = J_n(\lambda x)$ and $y_2 = J_n(\mu x)$ are solutions of the equations

$$x^2 y_1'' + x y_1' + (\lambda^2 x^2 - n^2) y_1 = 0, \quad x^2 y_2'' + x y_2' + (\mu^2 x^2 - n^2) y_2 = 0$$

Multiplying the first equation by y_2 , the second by y_1 and subtracting, we find

$$x^2 [y_2 y_1'' - y_1 y_2''] + x [y_2 y_1' - y_1 y_2'] = (\mu^2 - \lambda^2) x^2 y_1 y_2$$

which on division by x can be written as

$$x \frac{d}{dx} [y_2 y_1' - y_1 y_2'] + [y_2 y_1' - y_1 y_2'] = (\mu^2 - \lambda^2) x y_1 y_2$$

or

$$\frac{d}{dx} \{x [y_2 y_1' - y_1 y_2']\} = (\mu^2 - \lambda^2) x y_1 y_2$$

Then by integrating and omitting the constant of integration,

$$(\mu^2 - \lambda^2) \int x y_1 y_2 dx = x [y_2 y_1' - y_1 y_2']$$

or, using $y_1 = J_n(\lambda x)$, $y_2 = J_n(\mu x)$ and dividing by $\mu^2 - \lambda^2 \neq 0$,

$$\int x J_n(\lambda x) J_n(\mu x) dx = \frac{x [\lambda J_n(\mu x) J_n'(\lambda x) - \mu J_n(\lambda x) J_n'(\mu x)]}{\mu^2 - \lambda^2}$$

Thus

$$\int_0^1 x J_n(\lambda x) J_n(\mu x) dx = \frac{\lambda J_n(\mu) J_n'(\lambda) - \mu J_n(\lambda) J_n'(\mu)}{\mu^2 - \lambda^2}$$

which is equivalent to the required result.

6.24. Prove that $\int_0^1 x J_n^2(\lambda x) dx = \frac{1}{2} \left[J_n'^2(\lambda) + \left(1 - \frac{n^2}{\lambda^2}\right) J_n^2(\lambda) \right]$.

Let $\mu \rightarrow \lambda$ in the result of Problem 6.23. Then, using L'Hospital's rule, we find

$$\begin{aligned}
 \int_0^1 x J_n^2(\lambda x) dx &= \lim_{\mu \rightarrow \lambda} \frac{\lambda J_n'(\mu) J_n'(\lambda) - J_n(\lambda) J_n'(\mu) - \mu J_n(\lambda) J_n'(\mu)}{2\mu} \\
 &= \frac{\lambda J_n'^2(\lambda) - J_n(\lambda) J_n''(\lambda) - \lambda J_n(\lambda) J_n''(\lambda)}{2\lambda}
 \end{aligned}$$

But since $\lambda^2 J_n''(\lambda) + \lambda J_n'(\lambda) + (\lambda^2 - n^2) J_n(\lambda) = 0$, we find on solving for $J_n''(\lambda)$ and substituting,

$$\int_0^1 x J_n^2(\lambda x) dx = \frac{1}{2} \left[J_n^2(\lambda) + \left(1 - \frac{n^2}{\lambda^2} \right) J_n^2(\lambda) \right]$$

6.25. Prove that if λ and μ are any two different roots of the equation $RJ_n(x) + SxJ_n'(x) = 0$, where R and S are constants, then

$$\int_0^1 x J_n(\lambda x) J_n(\mu x) dx = 0$$

i.e. $\sqrt{x} J_n(\lambda x)$ and $\sqrt{x} J_n(\mu x)$ are orthogonal in $(0, 1)$.

Since λ and μ are roots of $RJ_n(x) + SxJ_n'(x) = 0$, we have

$$RJ_n(\lambda) + S\lambda J_n'(\lambda) = 0, \quad RJ_n(\mu) + S\mu J_n'(\mu) = 0 \quad (1)$$

Then since R and S are not both zero we find from (1),

$$\mu J_n(\lambda) J_n'(\mu) - \lambda J_n(\mu) J_n'(\lambda) = 0$$

and so from Problem 6.23 we have the required result

$$\int_0^1 x J_n(\lambda x) J_n(\mu x) dx = 0$$

SERIES OF BESSEL FUNCTIONS OF THE FIRST KIND

6.26. If $f(x) = \sum_{p=1}^{\infty} A_p J_n(\lambda_p x)$, $0 < x < 1$, where λ_p , $p = 1, 2, 3, \dots$, are the positive roots of $J_n(x) = 0$, show that

$$A_p = \frac{2}{J_{n+1}^2(\lambda_p)} \int_0^1 x J_n(\lambda_p x) f(x) dx$$

Multiply the series for $f(x)$ by $x J_n(\lambda_k x)$ and integrate term by term from 0 to 1. Then

$$\begin{aligned} \int_0^1 x J_n(\lambda_k x) f(x) dx &= \sum_{p=1}^{\infty} A_p \int_0^1 x J_n(\lambda_k x) J_n(\lambda_p x) dx \\ &= A_k \int_0^1 x J_n^2(\lambda_k x) dx \\ &= \frac{1}{2} A_k J_{n+1}^2(\lambda_k) \end{aligned}$$

where we have used Problems 6.24 and 6.25 together with the fact that $J_n(\lambda_k) = 0$. It follows that

$$A_k = \frac{2}{J_{n+1}^2(\lambda_k)} \int_0^1 x J_n(\lambda_k x) f(x) dx$$

To obtain the required result from this, we note that from the recurrence formula 3, page 99, which is equivalent to the formula 6 on that page, we have

$$\lambda_k J_n'(\lambda_k) = n J_n(\lambda_k) - \lambda_k J_{n+1}(\lambda_k)$$

or since $J_n(\lambda_k) = 0$,

$$J_n'(\lambda_k) = -J_{n+1}(\lambda_k)$$

6.27. Expand $f(x) = 1$ in a series of the form

$$\sum_{p=1}^{\infty} A_p J_0(\lambda_p x)$$

for $0 < x < 1$, if λ_p , $p = 1, 2, 3, \dots$, are the positive roots of $J_0(x) = 0$.

From Problem 6.26 we have

$$\begin{aligned} A_p &= \frac{2}{J_1^2(\lambda_p)} \int_0^1 x J_0(\lambda_p x) dx = \frac{2}{\lambda_p^2 J_1^2(\lambda_p)} \int_0^{\lambda_p} v J_0(v) dv \\ &= \frac{2}{\lambda_p^2 J_1^2(\lambda_p)} v J_1(v) \Big|_0^{\lambda_p} = \frac{2}{\lambda_p J_1(\lambda_p)} \end{aligned}$$

where we have made the substitution $v = \lambda_p x$ in the integral and used the result of Problem 6.10(a) with $n = 1$.

Thus we have the required series

$$f(x) = 1 = \sum_{p=1}^{\infty} \frac{2}{\lambda_p J_1(\lambda_p)} J_0(\lambda_p x)$$

which can be written

$$\frac{J_0(\lambda_1 x)}{\lambda_1 J_1(\lambda_1)} + \frac{J_0(\lambda_2 x)}{\lambda_2 J_1(\lambda_2)} + \dots = \frac{1}{2}$$

SOLUTIONS USING BESSEL FUNCTIONS OF THE FIRST KIND

6.28. A circular plate of unit radius (see Fig. 6-7) has its plane faces insulated. If the initial temperature is $F(\rho)$ and if the rim is kept at temperature zero, find the temperature of the plate at any time.

Since the temperature is independent of ϕ , the boundary value problem for determining $u(\rho, t)$ is

$$\frac{\partial u}{\partial t} = \kappa \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} \right) \quad (1)$$

$$u(1, t) = 0, \quad u(\rho, 0) = F(\rho), \quad |u(\rho, t)| < M$$

Let $u = P(\rho) T(t) = PT$ in equation (1). Then

$$PT' = \kappa \left(P''T + \frac{1}{\rho} P'T \right)$$

or dividing by κPT ,

$$\frac{T'}{\kappa T} = \frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} = -\lambda^2$$

from which

$$T' + \kappa \lambda^2 T = 0, \quad P'' + \frac{1}{\rho} P' + \lambda^2 P = 0$$

These have general solutions

$$T = c_1 e^{-\kappa \lambda^2 t}, \quad P = A_1 J_0(\lambda \rho) + B_1 Y_0(\lambda \rho)$$

Since $u = PT$ is bounded at $\rho = 0$, $B_1 = 0$. Then

$$u(\rho, t) = A e^{-\kappa \lambda^2 t} J_0(\lambda \rho)$$

where $A = A_1 c_1$.

From the first boundary condition,

$$u(1, t) = A e^{-\kappa \lambda^2 t} J_0(\lambda) = 0$$

from which $J_0(\lambda) = 0$ and $\lambda = \lambda_1, \lambda_2, \dots$ are the positive roots.

Thus a solution is

$$u(\rho, t) = A e^{-\kappa \lambda_m^2 t} J_0(\lambda_m \rho) \quad m = 1, 2, 3, \dots$$

By superposition, a solution is

$$u(\rho, t) = \sum_{m=1}^{\infty} A_m e^{-\kappa \lambda_m^2 t} J_0(\lambda_m \rho)$$

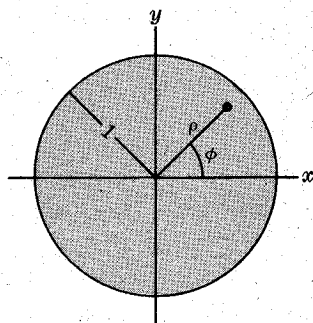


Fig. 6-7

From the second boundary condition,

$$u(\rho, 0) = F(\rho) = \sum_{m=1}^{\infty} A_m J_0(\lambda_m \rho)$$

Then from Problem 6.26 with $n = 0$ we have

$$A_m = \frac{2}{J_1^2(\lambda_m)} \int_0^1 \rho F(\rho) J_0(\lambda_m \rho) d\rho$$

and so

$$u(\rho, t) = \sum_{m=1}^{\infty} \left\{ \left[\frac{2}{J_1^2(\lambda_m)} \int_0^1 \rho F(\rho) J_0(\lambda_m \rho) d\rho \right] e^{-\kappa \lambda_m^2 t} J_0(\lambda_m \rho) \right\} \quad (2)$$

which can be established as the required solution.

Note that this solution also gives the temperature of an infinitely long solid cylinder whose convex surface is kept at temperature zero and whose initial temperature is $F(\rho)$.

- 6.29. A solid conducting cylinder of unit height and radius and with diffusivity κ is initially at temperature $f(\rho, z)$ (see Fig. 6-8). The entire surface is suddenly lowered to temperature zero and kept at this temperature. Find the temperature at any point of the cylinder at any subsequent time.

Since there is no ϕ -dependence, as is evident from symmetry, the heat conduction equation is

$$\frac{\partial u}{\partial t} = \kappa \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{\partial^2 u}{\partial z^2} \right) \quad (1)$$

where $u = u(\rho, z, t)$. The boundary conditions are given by

$$u(\rho, z, 0) = f(\rho, z), \quad u(\rho, 0, t) = 0, \quad u(\rho, 1, t) = 0, \quad u(1, z, t) = 0, \quad |u(\rho, z, t)| < M \quad (2)$$

where $0 \leq \rho < 1$, $0 < z < 1$, $t > 0$.

To solve this boundary value problem let $U = PZT = P(\rho)Z(z)T(t)$ in (1) to obtain

$$PZT' = \kappa \left(P''ZT + \frac{1}{\rho} P'ZT + PZ''T \right)$$

Then dividing by κPZT we have

$$\frac{T'}{\kappa T} = \frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{Z''}{Z}$$

Since the left side depends only on t while the right side depends only on ρ and z , each side must be a constant, say $-\lambda^2$. Thus

$$T' + \kappa \lambda^2 T = 0$$

$$\frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{Z''}{Z} = -\lambda^2 \quad (3)$$

The last equation can be written as

$$\frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} = -\lambda^2 - \frac{Z''}{Z}$$

from which we see that each side must be a constant, say $-\mu^2$. From this we obtain the two equations

$$\rho P'' + P' + \mu^2 \rho P = 0 \quad (4)$$

$$Z'' - \mu^2 Z = 0 \quad (5)$$

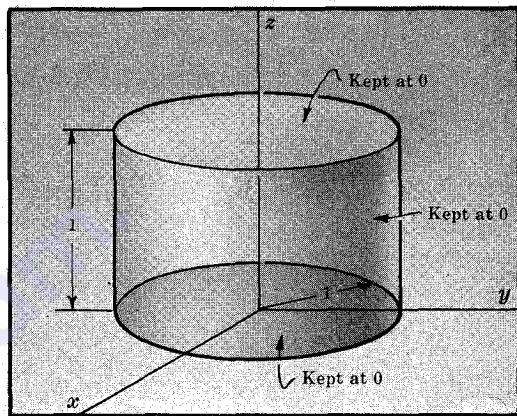


Fig. 6-8

where we have written

$$\nu^2 = \mu^2 - \lambda^2 \quad (6)$$

The solutions of (3), (4) and (5) are given by

$$T = c_1 e^{-\kappa \lambda^2 t}, \quad P = c_2 J_0(\mu \rho) + c_3 Y_0(\mu \rho), \quad Z = c_4 e^{\nu z} + c_5 e^{-\nu z}$$

Thus a solution to (1) is given by the product of these, i.e.

$$u(\rho, z, t) = [c_1 e^{-\kappa \lambda^2 t}] [c_2 J_0(\mu \rho) + c_3 Y_0(\mu \rho)] [c_4 e^{\nu z} + c_5 e^{-\nu z}]$$

Now from the boundedness condition at $\rho = 0$ we must have $c_3 = 0$. Thus the solution becomes

$$u(\rho, z, t) = e^{-\kappa \lambda^2 t} J_0(\mu \rho) [A e^{\nu z} + B e^{-\nu z}] \quad (7)$$

From the second boundary condition in (2) we see that

$$u(\rho, 0, t) = e^{-\kappa \lambda^2 t} J_0(\mu \rho) (A + B) = 0$$

so that we must have $A + B = 0$ or $B = -A$. Then (7) becomes

$$u(\rho, z, t) = A e^{-\kappa \lambda^2 t} J_0(\mu \rho) [e^{\nu z} - e^{-\nu z}]$$

From the third condition we have

$$u(\rho, 1, t) = A e^{-\kappa \lambda^2 t} J_0(\mu \rho) [e^{\nu} - e^{-\nu}] = 0$$

which can be satisfied only if $e^{\nu} - e^{-\nu} = 0$ or

$$e^{2\nu} = 1 = e^{2k\pi i} \quad k = 0, 1, 2, \dots$$

It follows that we must have $2\nu = 2k\pi i$ or

$$\nu = k\pi i \quad k = 0, 1, 2, \dots \quad (8)$$

Using this in (7), it becomes

$$u(\rho, z, t) = C e^{-\kappa \lambda^2 t} J_0(\mu \rho) \sin k\pi z$$

where C is a new constant.

From the fourth condition in (2) we obtain

$$u(1, z, t) = C e^{-\kappa \lambda^2 t} J_0(\mu) \sin k\pi z = 0$$

which can be satisfied only if $J_0(\mu) = 0$ so that

$$\mu = r_1, r_2, \dots \quad (9)$$

where r_m ($m = 1, 2, \dots$) is the m th positive root of $J_0(x) = 0$. Now from (6), (8) and (9) it follows that

$$\lambda^2 = \mu^2 - \nu^2 = r_m^2 + k^2 \pi^2$$

so that a solution satisfying all conditions in (2) but the first is given by

$$u(\rho, z, t) = C e^{-\kappa(r_m^2 + k^2 \pi^2)t} J_0(r_m \rho) \sin k\pi z \quad (10)$$

where $k = 1, 2, 3, \dots$, $m = 1, 2, 3, \dots$. Replacing C by C_{km} and summing over k and m we obtain by the superposition principle the solution

$$u(\rho, z, t) = \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} C_{km} e^{-\kappa(r_m^2 + k^2 \pi^2)t} J_0(r_m \rho) \sin k\pi z \quad (11)$$

The first condition in (2) now leads to

$$f(\rho, z) = \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} C_{km} J_0(r_m \rho) \sin k\pi z$$

This can be written as

$$f(\rho, z) = \sum_{k=1}^{\infty} \left\{ \sum_{m=1}^{\infty} C_{km} J_0(r_m \rho) \right\} \sin k\pi z = \sum_{k=1}^{\infty} b_k \sin k\pi z$$

where

$$b_k = \sum_{m=1}^{\infty} C_{km} J_0(r_m \rho) \quad (12)$$

It follows from this that b_k are the Fourier coefficients obtained when $f(\rho, z)$ is expanded into a Fourier sine series in z [we think of ρ as kept constant in this case]. Thus by the methods of Chapter 2 we have

$$b_k = \frac{2}{1} \int_0^1 f(\rho, z) \sin k\pi z \, dz \quad (13)$$

We now must find C_{km} from the expansion (12). Since b_k is a function of ρ , this is simply the expansion of b_k into a Bessel series as in Problem 6.26, and we find

$$C_{km} = \frac{2}{J_1^2(r_m)} \int_0^1 \rho b_k J_0(r_m \rho) \, d\rho \quad (14)$$

This becomes on using (13)

$$C_{km} = \frac{4u}{J_1^2(r_m)} \int_0^1 \int_0^1 \rho f(\rho, z) J_0(r_m \rho) \sin k\pi z \, d\rho \, dz \quad (15)$$

The required solution is thus given by (11) with the coefficients (15).

6.30. Work Problem 6.29 if $f(\rho, z) = u_0$, a constant.

In this case we find from (15) of Problem 6.29

$$\begin{aligned} C_{km} &= \frac{4u_0}{J_1^2(r_m)} \int_0^1 \int_0^1 \rho J_0(r_m \rho) \sin k\pi z \, d\rho \, dz \\ &= \frac{4u_0}{J_1^2(r_m)} \left\{ \int_0^1 \rho J_0(r_m \rho) \, d\rho \right\} \left\{ \int_0^1 \sin k\pi z \, dz \right\} \\ &= \frac{4u_0}{J_1^2(r_m)} \left\{ \frac{J_1(r_m)}{r_m} \right\} \left\{ \frac{1 - \cos k\pi}{k\pi} \right\} \\ &= \frac{4u_0(1 - \cos k\pi)}{k\pi r_m J_1(r_m)} \end{aligned}$$

on using the same procedure as in Problem 6.27. The required solution is thus

$$u(\rho, z, t) = \frac{4u_0}{\pi} \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} \frac{1 - \cos k\pi}{k r_m J_1(r_m)} e^{-\kappa(r_m^2 + k^2 \pi^2)t} J_0(r_m \rho) \sin k\pi z$$

6.31. A drum consists of a stretched circular membrane of unit radius whose rim, represented by the circle of Fig. 6-7, is fixed. If the membrane is struck so that its initial displacement is $F(\rho, \phi)$ and is then released, find the displacement at any time.

The boundary value problem for the displacement $z(\rho, \phi, t)$ from the equilibrium or rest position (the xy -plane) is

$$\frac{\partial^2 z}{\partial t^2} = a^2 \left(\frac{\partial^2 z}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial z}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 z}{\partial \phi^2} \right)$$

$$z(1, \phi, t) = 0, \quad z(\rho, \phi, 0) = 0, \quad z_t(\rho, \phi, 0) = 0, \quad z(\rho, \phi, 0) = F(\rho, \phi)$$

Let $z = P(\rho) \Phi(\phi) T(t) = P\Phi T$. Then

$$P\Phi T'' = a^2 \left(P''\Phi T + \frac{1}{\rho} P'\Phi T + \frac{1}{\rho^2} P\Phi'' T \right)$$

Dividing by $a^2 P\Phi T$,

$$\frac{T''}{a^2 T} = \frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{1}{\rho^2} \frac{\Phi''}{\Phi} = -\lambda^2$$

and so

$$T'' + \lambda^2 a^2 T = 0 \quad (1)$$

$$\frac{P''}{P} + \frac{1}{\rho} \frac{P'}{P} + \frac{1}{\rho^2} \frac{\Phi''}{\Phi} = -\lambda^2 \quad (2)$$

Multiplying (2) by ρ^2 , the variables can be separated to yield

$$\frac{\rho^2 P''}{P} + \frac{\rho P'}{P} + \lambda^2 \rho^2 = -\frac{\Phi''}{\Phi} = \mu^2$$

so that

$$\Phi'' + \mu^2 \Phi = 0 \quad (3)$$

$$\rho^2 P'' + \rho P' + (\lambda^2 \rho^2 - \mu^2) P = 0 \quad (4)$$

General solutions of (1), (3) and (4) are

$$T = A_1 \cos \lambda a t + B_1 \sin \lambda a t \quad (5)$$

$$\Phi = A_2 \cos \mu \phi + B_2 \sin \mu \phi \quad (6)$$

$$P = A_3 J_\mu(\lambda \rho) + B_3 Y_\mu(\lambda \rho) \quad (7)$$

A solution $z(\rho, \phi, t)$ is given by the product of these.

Since z must have period 2π in the variable ϕ , we must have $\mu = m$ where $m = 0, 1, 2, 3, \dots$ from (6).

Also, since z is bounded at $\rho = 0$ we must take $B_3 = 0$.

Furthermore, to satisfy $z_t(\rho, \phi, 0) = 0$ we must choose $B_1 = 0$.

Then a solution is

$$u(\rho, \phi, t) = J_m(\lambda \rho) \cos \lambda a t (A \cos m\phi + B \sin m\phi)$$

Since $z(1, \phi, t) = 0$, $J_m(\lambda) = 0$ so that $\lambda = \lambda_{mk}$, $k = 1, 2, 3, \dots$, are the positive roots.

By superposition (summing over both m and k),

$$\begin{aligned} z(\rho, \phi, t) &= \sum_{m=0}^{\infty} \sum_{k=1}^{\infty} J_m(\lambda_{mk} \rho) \cos(\lambda_{mk} a t) (A_{mk} \cos m\phi + B_{mk} \sin m\phi) \\ &= \sum_{m=0}^{\infty} \left\{ \left[\sum_{k=1}^{\infty} A_{mk} J_m(\lambda_{mk} \rho) \right] \cos m\phi \right. \\ &\quad \left. + \left[\sum_{k=1}^{\infty} B_{mk} J_m(\lambda_{mk} \rho) \right] \sin m\phi \right\} \cos \lambda_{mk} a t \end{aligned} \quad (8)$$

Putting $t = 0$, we have

$$z(\rho, \phi, 0) = F(\rho, \phi) = \sum_{m=0}^{\infty} \{C_m \cos m\phi + D_m \sin m\phi\} \quad (9)$$

where

$$\begin{aligned} C_m &= \sum_{k=1}^{\infty} A_{mk} J_m(\lambda_{mk} \rho) \\ D_m &= \sum_{k=1}^{\infty} B_{mk} J_m(\lambda_{mk} \rho) \end{aligned} \quad (10)$$

But (9) is simply a Fourier series and we can determine C_m and D_m by the usual methods. We find

$$\begin{aligned} C_m &= \begin{cases} \frac{1}{\pi} \int_0^{2\pi} F(\rho, \phi) \cos m\phi \, d\phi & m = 1, 2, 3, \dots \\ \frac{1}{2\pi} \int_0^{2\pi} F(\rho, \phi) \, d\phi & m = 0 \end{cases} \\ D_m &= \frac{1}{\pi} \int_0^{2\pi} F(\rho, \phi) \sin m\phi \, d\phi \quad m = 0, 1, 2, 3, \dots \end{aligned}$$

From (10), using the results of Bessel series expansions, we have

$$\begin{aligned}
 A_{mk} &= \frac{2}{[J_{m+1}(\lambda_{mk})]^2} \int_0^1 \rho J_m(\lambda_{mk}\rho) C_m d\rho \\
 &= \begin{cases} \frac{2}{\pi [J_{m+1}(\lambda_{mk})]^2} \int_0^1 \int_0^{2\pi} \rho F(\rho, \phi) J_m(\lambda_{mk}\rho) \cos m\phi d\rho d\phi & \text{if } m = 1, 2, 3, \dots \\ \frac{1}{\pi [J_1(\lambda_{0k})]^2} \int_0^1 \int_0^{2\pi} \rho F(\rho, \phi) J_0(\lambda_{0k}\rho) d\rho d\phi & \text{if } m = 0 \end{cases} \\
 B_{mk} &= \frac{2}{[J_{m+1}(\lambda_{mk})]^2} \int_0^1 \rho J_m(\lambda_{mk}\rho) D_m d\rho \\
 &= \frac{2}{\pi [J_{m+1}(\lambda_{mk})]^2} \int_0^1 \int_0^{2\pi} \rho F(\rho, \phi) J_m(\lambda_{mk}\rho) \sin m\phi d\rho d\phi \quad \text{if } m = 0, 1, 2, \dots
 \end{aligned}$$

Using these values of A_{mk} and B_{mk} in (8) yields the required solution.

Note that the various modes of vibration of the drum are obtained by specifying particular values of m and k . The frequencies of vibration are then given by

$$f_{mk} = \frac{\lambda_{mk}}{2\pi} a$$

Because these are not integer multiples of the lowest frequency, we would expect noise rather than a musical tone.

SERIES USING BESSEL FUNCTIONS OF THE SECOND KIND

6.32. Let $u_0(\lambda m \rho) = Y_0(\lambda m a) J_0(\lambda m \rho) - J_0(\lambda m a) Y_0(\lambda m \rho)$ where λ_m , $m = 1, 2, 3, \dots$, are the positive roots of $Y_0(\lambda a) J_0(\lambda b) - J_0(\lambda a) Y_0(\lambda b) = 0$. Show that

$$\int_a^b \rho u_0(\lambda_m \rho) u_0(\lambda_n \rho) d\rho = 0 \quad m \neq n$$

The functions $P_m = u_0(\lambda_m \rho)$ and $P_n = u_0(\lambda_n \rho)$ satisfy the equations

$$\rho P_m'' + P_m' + \lambda_m^2 \rho P_m = 0 \quad (1)$$

$$\rho P_n'' + P_n' + \lambda_n^2 \rho P_n = 0 \quad (2)$$

Multiplying (1) by P_n , (2) by P_m , and subtracting, we find

$$\rho(P_n P_m'' - P_m P_n'') + P_n P_m' - P_m P_n' = (\lambda_n^2 - \lambda_m^2) \rho P_m P_n$$

which can be written

$$\rho \frac{d}{d\rho} (P_n P_m' - P_m P_n') + P_n P_m' - P_m P_n' = (\lambda_n^2 - \lambda_m^2) \rho P_m P_n$$

or

$$\frac{d}{d\rho} [\rho (P_n P_m' - P_m P_n')] = (\lambda_n^2 - \lambda_m^2) \rho P_m P_n$$

Then by integrating both sides from a to b we have

$$\begin{aligned}
 (\lambda_n^2 - \lambda_m^2) \int_a^b \rho P_m P_n d\rho &= \left. \rho (P_n P_m' - P_m P_n') \right|_a^b \\
 &= \left. \rho [\lambda_m u_0(\lambda_n \rho) u_0'(\lambda_m \rho) - \lambda_n u_0(\lambda_m \rho) u_0'(\lambda_n \rho)] \right|_a^b \\
 &= 0
 \end{aligned}$$

on using the facts $u_0(\lambda_m a) = 0$, $u_0(\lambda_n a) = 0$, $u_0(\lambda_m b) = 0$, $u_0(\lambda_n b) = 0$. Then since $\lambda_m \neq \lambda_n$ we have

$$\int_a^b \rho P_m P_n d\rho = \int_a^b \rho u_0(\lambda_m \rho) u_0(\lambda_n \rho) d\rho = 0$$

- 6.33.** Show how to expand a function $F(\rho)$ into a series of the form $\sum_{m=1}^{\infty} A_m u_0(\lambda_m \rho)$ where the functions $u_0(\lambda_m \rho)$ are given in Problem 6.32.

Suppose that

$$F(\rho) = \sum_{m=1}^{\infty} A_m u_0(\lambda_m \rho) \quad (1)$$

Then on multiplying both sides by $\rho u_0(\lambda_n \rho)$ and integrating from a to b we find

$$\begin{aligned} \int_a^b \rho F(\rho) u_0(\lambda_n \rho) d\rho &= \sum_{m=1}^{\infty} A_m \int_a^b \rho u_0(\lambda_m \rho) u_0(\lambda_n \rho) d\rho \\ &= A_n \int_a^b \rho [u_0(\lambda_n \rho)]^2 d\rho \end{aligned}$$

on making use of Problem 6.32.

$$\text{Thus} \quad A_n = \frac{\int_a^b \rho F(\rho) u_0(\lambda_n \rho) d\rho}{\int_a^b \rho [u_0(\lambda_n \rho)]^2 d\rho} \quad (2)$$

Although these coefficients have been obtained formally, we can show that when these coefficients are used in the right side of (1) it does converge to $F(\rho)$ at points of continuity, assuming that $F(\rho)$ and $F'(\rho)$ are piecewise continuous, while at points of discontinuity it converges to $\frac{1}{2}[F(\rho+0) + F(\rho-0)]$.

- 6.34.** A very long hollow cylinder of inner radius a and outer radius b (whose cross section is indicated in Fig. 6-9) is made of conducting material of diffusivity κ . If the inner and outer surfaces are kept at temperature zero while the initial temperature is a given function $f(\rho)$, where ρ is the distance from the axis, find the temperature at any point at any later time t .

Since symmetry shows that there is no ϕ - or z -dependence, the boundary value problem which we must solve for $u = u(\rho, t)$ is

$$\frac{\partial u}{\partial t} = \kappa \left(\frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} \right) \quad (1)$$

$$u(a, t) = 0, \quad u(b, t) = 0, \quad u(\rho, 0) = f(\rho), \quad |u(\rho, t)| < M \quad (2)$$

By separation of variables we have as in Problem 6.28

$$u(\rho, t) = e^{-\kappa \lambda^2 t} [a_1 J_0(\lambda \rho) + b_1 Y_0(\lambda \rho)] \quad (3)$$

From $u(a, t) = 0$ and $u(b, t) = 0$ we find

$$a_1 J_0(\lambda a) + b_1 Y_0(\lambda a) = 0, \quad a_1 J_0(\lambda b) + b_1 Y_0(\lambda b) = 0 \quad (4)$$

These equations lead to the equation

$$Y_0(\lambda a) J_0(\lambda b) - J_0(\lambda a) Y_0(\lambda b) = 0 \quad (5)$$

for determining λ . The equation (5) has infinitely many positive roots $\lambda_1, \lambda_2, \dots$

From the first equation in (4) we find

$$b_1 = -\frac{a_1 J_0(\lambda a)}{Y_0(\lambda a)}$$

so that (3) can be written

$$u(\rho, t) = A e^{-\kappa \lambda^2 t} [Y_0(\lambda a) J_0(\lambda \rho) - J_0(\lambda a) Y_0(\lambda \rho)] \quad (6)$$

where A is a constant.

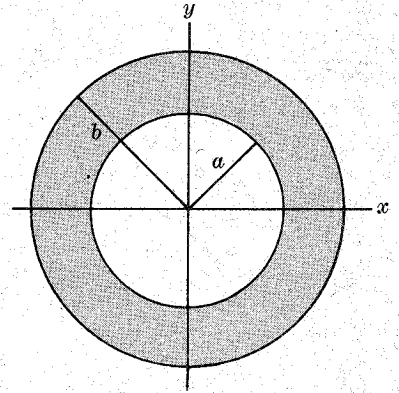


Fig. 6-9

Using the fact that for $\lambda = \lambda_m$ (6) is a solution, together with the principle of superposition, we obtain the solution

$$u(\rho, t) = \sum_{m=1}^{\infty} A_m e^{-\kappa \lambda_m^2 t} u_0(\lambda_m \rho) \quad (7)$$

where

$$u_0(\lambda_m \rho) = Y_0(\lambda_m a) J_0(\lambda_m \rho) - J_0(\lambda_m a) Y_0(\lambda_m \rho) \quad (8)$$

From the condition $u(\rho, 0) = f(\rho)$ we now obtain from (7)

$$f(\rho) = \sum_{m=1}^{\infty} A_m u_0(\lambda_m \rho) \quad (9)$$

Then

$$A_m = \frac{\int_a^b \rho f(\rho) u_0(\lambda_m \rho) d\rho}{\int_a^b \rho [u_0(\lambda_m \rho)]^2 d\rho} \quad (10)$$

Substitution of these coefficients into (7) gives the required solution.

- 6.35.** A simple pendulum initially has a length of l_0 and makes an angle θ_0 with the vertical. It is then released from this position. If the length l of the pendulum increases with time t according to $l = l_0 + \epsilon t$ where ϵ is a constant, find the position of the pendulum at any time assuming the oscillations to be small.

Let m be the mass of the bob and θ the angle which the pendulum makes with the vertical at any time t . The weight mg can be resolved into two components, one tangential to the path and given by $mg \sin \theta$ and the other perpendicular to it and given by $mg \cos \theta$, as shown in Fig. 6-10. From mechanics we know that

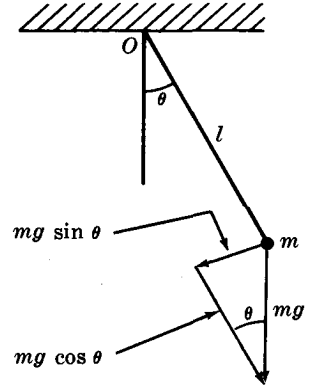


Fig. 6-10

$$\text{Torque about } O = \frac{d}{dt} (\text{Angular momentum about } O)$$

$$\text{or} \quad (-mg \sin \theta) l = \frac{d}{dt} (ml^2 \dot{\theta}) \quad (1)$$

where $\dot{\theta} = d\theta/dt$. This equation can be written as

$$l \ddot{\theta} + 2 \dot{l} \dot{\theta} + g \sin \theta = 0$$

or since $l = l_0 + \epsilon t$,

$$(l_0 + \epsilon t) \ddot{\theta} + 2\epsilon \dot{\theta} + g \sin \theta = 0$$

Letting $x = l_0 + \epsilon t$ in this equation it becomes

$$x \frac{d^2 \theta}{dx^2} + 2 \frac{d\theta}{dx} + \frac{g}{\epsilon^2} \theta = 0 \quad (2)$$

Multiplying by x and comparing with equations (26) and (27), page 101, we find that the solution is

$$\theta = \frac{1}{\sqrt{l_0 + \epsilon t}} \left[A J_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) + B Y_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) \right] \quad (3)$$

Since $\theta = \theta_0$ at $t = 0$ we have

$$\theta_0 = \frac{1}{\sqrt{l_0}} \left[A J_1 \left(\frac{2\sqrt{g l_0}}{\epsilon} \right) + B Y_1 \left(\frac{2\sqrt{g l_0}}{\epsilon} \right) \right] \quad (4)$$

To satisfy $\dot{\theta} = 0$ at $t=0$ we must first obtain $\dot{\theta} = d\theta/dt$. We find

$$\begin{aligned}\dot{\theta} = \frac{d\theta}{dt} &= -\frac{\epsilon}{2(l_0 + \epsilon t)^{3/2}} \left[AJ_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) + BY_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) \right] \\ &\quad + \frac{\sqrt{g}}{l_0 + \epsilon t} \left[AJ'_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) + BY'_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) \right]\end{aligned}$$

Now since $\dot{\theta} = 0$ for $t = 0$ we find

$$\begin{aligned}0 &= -\frac{\epsilon}{2l_0^{3/2}} \left[AJ_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) + BY_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) \right] \\ &\quad + \frac{\sqrt{g}}{l_0} \left[AJ'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) + BY'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) \right]\end{aligned}$$

or using (4)

$$AJ'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) + BY'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) = \frac{\epsilon \theta_0}{2\sqrt{g}} \quad (5)$$

Solving for A and B from (4) and (5) we find

$$\begin{aligned}A &= \frac{\sqrt{l_0} Y'_1 - (\epsilon/2\sqrt{g}) Y_1}{J_1 Y'_1 - Y_1 J'_1} \theta_0 \\ B &= \frac{(\epsilon/2\sqrt{g}) J_1 - \sqrt{l_0} J'_1}{J_1 Y'_1 - Y_1 J'_1} \theta_0\end{aligned} \quad (6)$$

where the argument $2\sqrt{gl_0}/\epsilon$ in J_1, J'_1, Y_1, Y'_1 has been omitted.

Now from Problem 6.58 with $n = 1$ we know that

$$J_1(x) Y'_1(x) - Y_1(x) J'_1(x) = \frac{2}{\pi x}$$

so that

$$J_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) Y'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) - Y_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) J'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) = \frac{\epsilon}{\pi \sqrt{gl_0}}$$

Thus (6) becomes

$$\begin{aligned}A &= \frac{\pi \sqrt{g} l_0 \theta_0}{\epsilon} Y'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) - \frac{\pi \sqrt{l_0} \theta_0}{2} Y_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) \\ B &= \frac{\pi \sqrt{l_0} \theta_0}{2} J_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) - \frac{\pi \sqrt{g} l_0 \theta_0}{\epsilon} J'_1 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right)\end{aligned} \quad (7)$$

Now from formula 3, page 99, with $n = 1$ and the corresponding formula involving Y_n for $n = 1$, we have from (7)

$$\begin{aligned}A &= -\frac{\pi \sqrt{l_0} \theta_0}{2} Y_2 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) \\ B &= \frac{\pi \sqrt{l_0} \theta_0}{2} J_2 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right)\end{aligned} \quad (8)$$

Using these in (3) we thus find

$$\theta = \frac{\pi \sqrt{l_0} \theta_0}{2\sqrt{l_0 + \epsilon t}} \left[J_2 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) Y_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) - Y_2 \left(\frac{2\sqrt{gl_0}}{\epsilon} \right) J_1 \left(\frac{2\sqrt{g}}{\epsilon} \sqrt{l_0 + \epsilon t} \right) \right] \quad (9)$$

Supplementary Problems

BESSEL FUNCTIONS OF THE FIRST KIND

6.36. (a) Show that $J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 4} + \frac{x^5}{2^2 4^2 6} - \frac{x^7}{2^2 4^2 6^2 8} + \cdots$ and verify that the interval of convergence is $-\infty < x < \infty$.

(b) Show that $J_0'(x) = -J_1(x)$.

(c) Show that $\frac{d}{dx}[xJ_1(x)] = xJ_0(x)$.

6.37. Evaluate (a) $J_{5/2}(x)$ and (b) $J_{-5/2}(x)$ in terms of sines and cosines.

6.38. Find $J_3(x)$ in terms of $J_0(x)$ and $J_1(x)$.

6.39. Prove that (a) $J_n''(x) = \frac{1}{4}[J_{n-2}(x) - 2J_n(x) + J_{n+2}(x)]$

$$(b) \quad J_n'''(x) = \frac{1}{8}[J_{n-3}(x) - 3J_{n-1}(x) + 3J_{n+1}(x) - J_{n+3}(x)]$$

and generalize these results.

6.40. Evaluate (a) $\int x^3 J_2(x) dx$, (b) $\int_0^1 x^3 J_0(x) dx$, (c) $\int x^2 J_0(x) dx$.

6.41. Evaluate (a) $\int J_1(\sqrt[3]{x}) dx$, (b) $\int \frac{J_2(x)}{x^2} dx$.

6.42. Evaluate $\int J_0(x) \sin x dx$.

6.43. Verify directly the result $J_n'(x)J_{-n}(x) - J_{-n}'(x)J_n(x) = \frac{2 \sin n\pi}{\pi x}$ for (a) $n = \frac{1}{2}$ and (b) $n = \frac{3}{2}$.

GENERATING FUNCTION AND MISCELLANEOUS RESULTS

6.44. Use the generating function to prove that $J_n'(x) = \frac{1}{2}[J_{n-1}(x) + J_{n+1}(x)]$ for the case where n is an integer.

6.45. Use the generating function to work Problem 6.39 for the case where n is an integer.

6.46. Show that (a) $1 = J_0(x) + 2J_2(x) + 2J_4(x) + \cdots$

$$(b) \quad J_1(x) - J_3(x) + J_5(x) - J_7(x) + \cdots = \frac{1}{2} \sin x$$

6.47. Show that $\frac{x}{4}J_1(x) = J_2(x) - 2J_4(x) + 3J_6(x) - \cdots$.

6.48. Show that $J_0(x) = \frac{2}{\pi} \int_0^{\pi/2} \cos(x \sin \theta) d\theta$.

6.49. Show that (a) $\int_0^{\pi/2} J_1(x \cos \theta) d\theta = \frac{1 - \cos x}{x}$

$$(b) \quad \int_0^{\pi/2} J_0(x \sin \theta) \cos \theta \sin \theta d\theta = \frac{J_1(x)}{x}.$$

6.50. Show that $\int_0^x J_0(t) dt = 2 \sum_{k=0}^{\infty} J_{2k+1}(x).$

6.51. Show that (a) $\int_0^{\infty} e^{-ax} J_0(bx) dx = \frac{1}{\sqrt{a^2 + b^2}}$

(b) $\int_0^{\infty} e^{-ax} J_n(bx) dx = \frac{(\sqrt{a^2 + b^2} - a)^n}{\sqrt{a^2 + b^2}}, \quad n > -1$

6.52. Show that $\int_0^{\infty} J_0(x) dx = 1.$

6.53. Prove that $|J_n(x)| \leq 1$ for all integers n . Is the result true if n is not an integer?

BESSEL FUNCTIONS OF THE SECOND KIND

6.54. Show that (a) $Y_{n+1}(x) = \frac{2n}{x} Y_n(x) - Y_{n-1}(x),$ (b) $Y'_n(x) = \frac{1}{2} [Y_{n-1}(x) - Y_{n+1}(x)].$

6.55. Explain why the recurrence formulas for $J_n(x)$ on page 99 hold if $J_n(x)$ is replaced by $Y_n(x)$.

6.56. Prove that $Y'_0(x) = -Y_1(x).$

6.57. Evaluate (a) $Y_{1/2}(x),$ (b) $Y_{-1/2}(x),$ (c) $Y_{3/2}(x),$ (d) $Y_{-3/2}(x).$

6.58. Prove that $J_n(x) Y'_n(x) - J'_n(x) Y_n(x) = \frac{2}{\pi x}.$

6.59. Evaluate (a) $\int x^3 Y_2(x) dx,$ (b) $\int Y_3(x) dx,$ (c) $\int \frac{Y_3(x)}{x^3} dx.$

6.60. Prove the result (11), page 98.

FUNCTIONS RELATED TO BESSEL FUNCTIONS

6.61. Show that $I_0(x) = 1 + \frac{x^2}{2^2} + \frac{x^4}{2^2 4^2} + \frac{x^6}{2^2 4^2 6^2} + \cdots.$

6.62. Show that (a) $I'_n(x) = \frac{1}{2} \{I_{n-1}(x) + I_{n+1}(x)\},$ (b) $x I'_n(x) = x I_{n-1}(x) - n I_n(x).$

6.63. Show that $e^{\frac{x}{2}(t + \frac{1}{t})} = \sum_{n=-\infty}^{\infty} I_n(x) t^n$ is the generating function for $I_n(x).$

6.64. Show that $I_0(x) = \frac{2}{\pi} \int_0^{\pi/2} \cosh(x \sin \theta) d\theta.$

6.65. Show that (a) $\sinh x = 2[I_1(x) + I_3(x) + \cdots]$
(b) $\cosh x = I_0(x) + 2[I_2(x) + I_4(x) + \cdots].$

6.66. Show that (a) $I_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\cosh x - \frac{\sinh x}{x} \right),$ (b) $I_{-3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\sinh x - \frac{\cosh x}{x} \right).$

6.67. (a) Show that $K_{n+1}(x) = K_{n-1}(x) + \frac{2n}{x} K_n(x).$ (b) Explain why the functions $K_n(x)$ satisfy the same recurrence formulas as $I_n(x).$

6.68. Give asymptotic formulas for (a) $H_n^{(1)}(x)$, (b) $H_n^{(2)}(x)$.

6.69. Show that (a) $\text{Ber}_n(x) = \sum_{k=0}^{\infty} \frac{(x/2)^{2k+n}}{k! \Gamma(n+k+1)} \cos\left(\frac{3n+2k}{4}\pi\right)$.

$$(b) \quad \text{Bei}_n(x) = \sum_{k=0}^{\infty} \frac{(x/2)^{2k+n}}{k! \Gamma(n+k+1)} \sin\left(\frac{3n+2k}{4}\pi\right).$$

6.70. Show that

$$\text{Ker}(x) = -\{\ln(x/2) + \gamma\} \text{Ber}(x) + \frac{\pi}{4} \text{Bei}(x) + 1 - \frac{(x/2)^4}{2!^2} (1 + \frac{1}{2}) + \frac{(x/2)^8}{4!^2} (1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{4}) - \dots$$

EQUATIONS TRANSFORMABLE INTO BESSEL'S EQUATION

6.71. Prove that (27), page 101, is a solution of (26).

6.72. Solve $4xy'' + 4y' + y = 0$.

6.73. Solve (a) $xy'' + 2y' + xy = 0$, (b) $y''' + x^2y = 0$.

6.74. Solve $y'' + e^{2x}y = 0$. [Hint. Let $e^x = u$].

6.75. (a) Show by direct substitution that $y = J_0(2\sqrt{x})$ is a solution of $xy'' + y' + y = 0$ and (b) write the general solution.

6.76. (a) Show by direct substitution that $y = \sqrt{x} J_{1/3}(\frac{2}{3}x^{3/2})$ is a solution of $y'' + xy = 0$ and (b) write the general solution.

6.77. (a) Show that Bessel's equation $x^2y'' + xy' + (x^2 - n^2)y = 0$ can be transformed into

$$\frac{d^2u}{dx^2} + \left(1 - \frac{n^2 - 1/4}{x^2}\right)u = 0$$

where $y = u/\sqrt{x}$. (b) Discuss the case where $n = \pm 1/2$.

(b) Discuss the case where x is large and explain the connection with the asymptotic formulas on page 101.

6.78. Solve $x^2y'' - xy' + x^2y = 0$.

6.79. Show that the equation (26) on page 101 has the solution (28) if $\alpha = 0$. [Hint. Let $y = x^p$ and choose p appropriately, or make the transformation $x = e^t$.]

ORTHOGONAL SERIES OF BESSEL FUNCTIONS

6.80. Is the result of Problem 6.27, page 113, valid for $-1 \leq x \leq 1$? Justify your answer.

6.81. Show that $\int x J_n^2(\lambda x) dx = \frac{x^2}{2} [J_n^2(\lambda x) + J_{n+1}^2(\lambda x)] - \frac{nx}{\lambda} J_n(\lambda x) J_{n+1}(\lambda x) + c$

6.82. Prove the results (34) and (35), page 102.

6.83. Show that $\frac{1-x^2}{8} = \sum_{p=1}^{\infty} \frac{J_0(\lambda_p x)}{\lambda_p^3 J_1(\lambda_p)}$ $-1 < x < 1$

where λ_p are the positive roots of $J_0(\lambda) = 0$.

6.84. Show that $x = 2 \sum_{p=1}^{\infty} \frac{J_1(\lambda_p x)}{\lambda J_2(\lambda_p)}$ $-1 < x < 1$

where λ_p are the positive roots of $J_1(\lambda) = 0$.

6.85. Show that
$$x^3 = \sum_{p=1}^{\infty} \frac{2(8 - \lambda_p^2) J_1(\lambda_p x)}{\lambda_p^3 J_1'(\lambda_p)} \quad -1 < x < 1$$

where λ_p are the positive roots of $J_1(\lambda) = 0$.

6.86. Show that
$$x^2 = \sum_{p=1}^{\infty} \frac{2(\lambda_p^2 - 4) J_0(\lambda_p x)}{\lambda_p^3 J_1(\lambda_p)} \quad -1 < x < 1$$

where λ_p are the positive roots of $J_0(\lambda) = 0$.

6.87. Show that
$$\frac{J_0(\alpha x)}{2J_0(\alpha)} = \sum_{p=1}^{\infty} \frac{\lambda_p J_0(\lambda_p x)}{(\lambda_p^2 - \alpha^2) J_1(\lambda_p)} \quad -1 < x < 1$$

where λ_p are the positive roots of $J_0(\lambda) = 0$.

6.88. If $f(x) = \sum_{p=1}^{\infty} A_p J_0(\lambda_p x)$ where $J_0(\lambda_p) = 0$, $p = 1, 2, 3, \dots$, show that

$$\int_0^1 x[f(x)]^2 dx = \sum_{p=1}^{\infty} A_p^2 J_1^2(\lambda_p)$$

Compare with Parseval's identity for Fourier series.

6.89. Use Problems 6.84 and 6.88 to show that

$$\sum_{p=1}^{\infty} \frac{1}{\lambda_p^2} = \frac{1}{4}$$

where λ_p are the positive roots of $J_0(\lambda) = 0$.

6.90. Derive the results (a) (35) on page 102, (b) (36) on page 102, and (c) (37) on page 102.

SOLUTIONS USING BESSEL FUNCTIONS

6.91. The temperature of a long solid circular cylinder of unit radius is initially zero. At $t = 0$ the surface is given a constant temperature u_0 which is then maintained. Show that the temperature of the cylinder is given by

$$u(\rho, t) = u_0 \left\{ 1 - 2 \sum_{n=1}^{\infty} \frac{J_0(\lambda_n \rho)}{\lambda_n J_1(\lambda_n)} e^{-\kappa \lambda_n^2 t} \right\}$$

where λ_n , $n = 1, 2, 3, \dots$, are the positive roots of $J_0(\lambda) = 0$ and κ is the diffusivity.

6.92. Show that if $F(\rho) = u_0(1 - \rho^2)$, then the temperature of the plate of Problem 6.28 is given by

$$u(\rho, t) = 4\kappa u_0 \sum_{n=1}^{\infty} \frac{J_0(\lambda_n \rho) J_2(\lambda_n)}{\lambda_n^2 J_1^2(\lambda_n)} e^{-\kappa \lambda_n^2 t}$$

6.93. A cylinder $0 < \rho < a$, $0 < z < l$ has the end $z = 0$ at temperature $f(\rho)$ while the other surfaces are kept at temperature zero. Show that the steady-state temperature at any point is given by

$$u(\rho, z) = \frac{2}{a^2} \sum_{n=1}^{\infty} \frac{J_0(\lambda_n \rho) \sinh \lambda_n(l - z)}{J_1^2(\lambda_n a) \sinh \lambda_n l} \int_0^a \rho f(\rho) J_0(\lambda_n \rho) d\rho$$

where $J_0(\lambda_n a) = 0$, $n = 1, 2, 3, \dots$.

6.94. A circular membrane of unit radius lies in the xy -plane with its center at the origin. Its edge $\rho = 1$ is fixed in the xy -plane and it is set into vibration by displacing it an amount $f(\rho)$ and then releasing it. Show that the displacement is given by

$$z(\rho, t) = \sum_{n=1}^{\infty} \frac{J_0(\lambda_n \rho) \cos \lambda_n t}{2J_1^2(\lambda_n)} \int_0^1 \rho f(\rho) J_0(\lambda_n \rho) d\rho$$

where λ_n are the roots of $J_0(\lambda) = 0$.

- 6.95. (a) Solve the boundary value problem

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2}$$

where $0 < \rho < 1$, $0 < \phi < 2\pi$, $t > 0$, u is bounded, and

$$u(1, \phi, t) = 0, \quad u(\rho, \phi, 0) = \rho \cos 3\phi, \quad u_t(\rho, \phi, 0) = 0$$

- (b) Give a physical interpretation to the solution.

- 6.96. Solve and interpret the boundary value problem

$$\frac{\partial}{\partial x} \left(x \frac{\partial y}{\partial x} \right) = \frac{\partial^2 y}{\partial t^2}$$

given that $y(x, 0) = f(x)$, $y_t(x, 0) = 0$, $y(1, t) = 0$ and $y(x, t)$ is bounded for $0 \leq x \leq 1$, $t > 0$.

- 6.97. (a) Work Problem 6.93 if the end
- $z = 0$
- is kept at temperature
- $f(\rho, \phi)$
- . (b) Determine the temperature in the special case where
- $f(\rho, \phi) = \rho^2 \cos \phi$
- .

- 6.98. (a) Work Problem 6.93 if there is radiation obeying Newton's law of cooling at the end
- $z = 0$
- .

- 6.99. A chain of constant mass per unit length is suspended vertically from one end
- O
- as indicated in Fig. 6-11. If the chain is displaced slightly at time
- $t = 0$
- so that its shape is given by
- $f(x)$
- ,
- $0 < x < L$
- , and then released, show that the displacement of any point
- x
- at time
- t
- is given by

$$y(x, t) = \sum_{n=1}^{\infty} A_n J_0 \left(2\lambda_n \sqrt{\frac{L-x}{g}} \right) \cos \lambda_n t$$

where λ_n are the roots of $J_0(2\lambda\sqrt{L/g}) = 0$ and

$$A_n = \frac{2}{J_1^2(\lambda_n)} \int_0^1 v J_0(\lambda_n v) f(L - \frac{1}{2}gv^2) dv$$

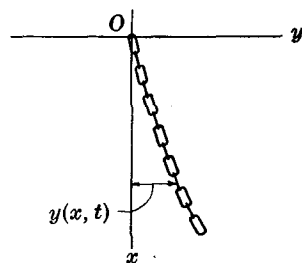


Fig. 6-11

- 6.100. Determine the frequencies of the normal modes for the vibrating chain of Problem 6.99 and indicate whether you would expect music or noise from the vibrations.

- 6.101. A solid circular cylinder
- $0 < \rho < a$
- ,
- $0 < z < L$
- has its bases kept at temperature zero and the convex surface at constant temperature
- u_0
- . Show that the steady-state temperature at any point of the cylinder is

$$u(\rho, z) = \frac{4u_0}{\pi} \sum_{n=1}^{\infty} \frac{I_0[(2n-1)\pi\rho/L] \sin[(2n-1)\pi z/L]}{(2n-1)I_0[(2n-1)\pi a/L]}$$

where I_0 is the modified Bessel function of order zero.

- 6.102. Suppose that the chain in Problem 6.99, which is initially at rest, is given an initial velocity distribution defined by
- $h(x)$
- ,
- $0 < x < L$
- . Show that the displacement of any point
- x
- of the string at any time
- t
- is given by

$$y(x, t) = \sum_{n=1}^{\infty} B_n J_0 \left(2\lambda_n \sqrt{\frac{L-x}{g}} \right) \sin \lambda_n t$$

where λ_n are the roots of $J_0(2\lambda\sqrt{L/g}) = 0$ and

$$B_n = \frac{2}{\lambda_n J_1^2(\lambda_n)} \int_0^1 v J_0(\lambda_n v) h(L - \frac{1}{2}gv^2) dv$$

- 6.103. Work Problem 6.99 if the chain is given both an initial shape
- $f(x)$
- and initial velocity distribution
- $h(x)$
- .

- 6.104. The surface $\rho = 1$ of an infinite cylinder is kept at temperature $f(z)$. Show that the steady-state temperature everywhere in the cylinder is given by

$$u(\rho, z) = \frac{1}{\pi} \int_{\lambda=0}^{\infty} \int_{v=-\infty}^{\infty} \frac{f(v) \cos \lambda(v-z) I_0(\lambda \rho)}{I_0(\lambda)} d\lambda dv$$

- 6.105. A string stretched between $x = 0$ and $x = L$ has a variable density given by $\sigma = \sigma_0 + \epsilon x$ where σ_0 and ϵ are constants. The string is given an initial shape $f(x)$ and then released.

(a) Show that if the tension τ is constant the boundary value problem is given by

$$\tau \frac{\partial^2 y}{\partial x^2} = (\sigma_0 + \epsilon x) \frac{\partial^2 y}{\partial t^2} \quad 0 < x < L, \quad t > 0$$

$$y(0, t) = 0, \quad y(L, t) = 0, \quad y(x, 0) = f(x), \quad y_t(x, 0) = 0, \quad |y(x, t)| < M$$

(b) Show that the frequencies of the normal modes of vibration are given by $f_n = \omega_n/2\pi$ where the ω_n ($n = 1, 2, 3, \dots$) are the positive roots of the equation.

$$J_{1/3}(\alpha\omega) J_{-1/3}(\beta\omega) = J_{1/3}(\beta\omega) J_{-1/3}(\alpha\omega)$$

in which
$$\alpha = \frac{2\sigma_0}{3\epsilon} \sqrt{\frac{\sigma_0}{\tau}}, \quad \beta = \frac{2(\sigma_0 + \epsilon L)}{3\epsilon} \sqrt{\frac{\sigma_0 + \epsilon L}{\tau}}$$

MISCELLANEOUS PROBLEMS

- 6.106. A particle moves along the positive x -axis with a force of repulsion per unit mass equal to a constant α^2 times the instantaneous distance from the origin. If the mass m increases with time according to $m = m_0 + \epsilon t$, where m_0 and ϵ are constants, and if initially the particle is located at the origin and traveling with speed v_0 , show that the position x at any time $t > 0$ is given by

$$x = \frac{m_0 v_0}{\epsilon} \left\{ K_0 \left(\frac{\alpha m_0}{\epsilon} \right) I_0 \left(\frac{\alpha m_0}{\epsilon} + \alpha t \right) - I_0 \left(\frac{\alpha m_0}{\epsilon} \right) K_0 \left(\frac{\alpha m_0}{\epsilon} + \alpha t \right) \right\}$$

- 6.107. Show that if $m \neq n$

$$\int \frac{J_m(\lambda x) J_n(\lambda x)}{x} dx = \frac{\lambda x}{m^2 - n^2} \{ J'_m(\lambda x) J_n(\lambda x) - J_m(\lambda x) J'_n(\lambda x) \} + c$$

- 6.108. Deduce the integral $\int \frac{J_m^2(\lambda x)}{x} dx$ by using a limiting procedure in the result of Problem 6.107.

- 6.109. Show that
$$\int_0^\infty \frac{J_n(x)}{x^{n-1}} dx = \frac{1}{2^{n-1} \Gamma(n)} \quad n > 0$$

- 6.110. Explain how the Sturm-Liouville theory of Chapter 3 can be used to arrive at various results involving Bessel functions obtained in this chapter.

- 6.111. A cylinder of unit height and radius (see Fig. 6-8, page 115) has its top surface kept at temperature u_0 and the other surfaces at temperature zero. Show that the steady-state temperature at any point is given by

$$u(\rho, z) = 2u_0 \sum_{n=1}^{\infty} \frac{(\sinh \lambda_n z) J_0(\lambda_n \rho)}{(\lambda_n \sinh \lambda_n) J_1(\lambda_n)}$$

where λ_n are the positive roots of $J_0(\lambda) = 0$.

- 6.112. Work Problem 6.29 if the base $z = 1$ is insulated.

- 6.113. Work Problem 6.29 if the convex surface is insulated.
- 6.114. Work Problem 6.29 if the bases $z = 0$ and $z = 1$ are kept at constant temperatures u_1 and u_2 respectively. [Hint. Let $u(\rho, z, t) = v(\rho, z, t) + w(\rho, z)$ and choose $w(\rho, z)$ appropriately, noting that physically it represents the steady-state solution.]
- 6.115. Show how Problem 6.29 can be solved if the radius of the cylinder is a while the height is h .
- 6.116. Work Problem 6.29 if the initial temperature is $f(\rho, \phi, z)$.

- 6.117. A membrane has the form of the region bounded by two concentric circles of radii a and b as shown in Fig. 6-12.

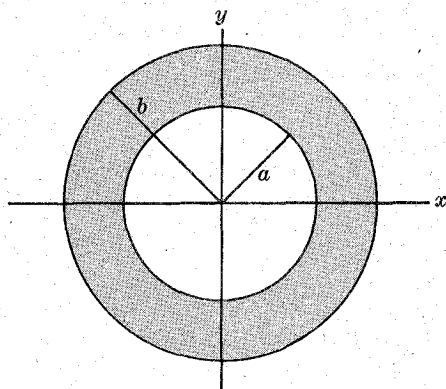


Fig. 6-12

- (a) Show that the frequencies of the various modes of vibration are given by

$$f_{mn} = \frac{\lambda_{mn}}{2\pi} \sqrt{\frac{\tau}{\mu}}$$

where τ is the tension per unit length, μ is the mass per unit area, and λ_{mn} are roots of the equation

$$J_m(\lambda_m a) Y_m(\lambda b) - J_m(\lambda b) Y_m(\lambda a) = 0$$

- (b) Find the displacement at any time of any point of the membrane if the membrane is given an initial shape and then released.
- 6.118. A metal conducting pipe of diffusivity κ has inner radius a , outer radius b and height h . A coordinate system is chosen so that one of the bases lies in the xy -plane and the axis of the pipe is chosen to be the z -axis. If the initial temperature of the pipe is $f(\rho, z)$, $a < \rho < b$, $0 < z < h$, while the surface is kept at temperature zero, find the temperature at any point at any time.
- 6.119. Work Problem 6.118 if the initial temperature is $f(\rho, \phi, z)$.
- 6.120. Work Problem 6.118 if (a) the bases are insulated, (b) the convex surfaces are insulated, (c) the entire surface is insulated.

Chapter 7

Legendre Functions and Applications

LEGENDRE'S DIFFERENTIAL EQUATION

Legendre functions arise as solutions of the differential equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad (1)$$

which is called *Legendre's differential equation*. The general solution of (1) in the case where $n = 0, 1, 2, 3, \dots$ is given by

$$y = c_1 P_n(x) + c_2 Q_n(x) \quad (2)$$

where $P_n(x)$ are polynomials called *Legendre polynomials* and $Q_n(x)$ are called *Legendre functions of the second kind*. The $Q_n(x)$ are unbounded at $x = \pm 1$.

The differential equation (1) is obtained, for example, from Laplace's equation $\nabla^2 u = 0$ expressed in spherical coordinates (r, θ, ϕ) , when it is assumed that u is independent of ϕ . See Problem 7.1.

LEGENDRE POLYNOMIALS

The Legendre polynomials are defined by

$$P_n(x) = \frac{(2n-1)(2n-3)\cdots 1}{n!} \left\{ x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} x^{n-4} - \dots \right\} \quad (3)$$

Note that $P_n(x)$ is a polynomial of degree n . The first few Legendre polynomials are as follows:

$$P_0(x) = 1$$

$$P_3(x) = \frac{1}{2}(5x^3 - 3x)$$

$$P_1(x) = x$$

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

$$P_2(x) = \frac{1}{2}(3x^2 - 1)$$

$$P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x)$$

In all cases $P_n(1) = 1$, $P_n(-1) = (-1)^n$.

The Legendre polynomials can also be expressed by *Rodrigue's formula*:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n \quad (4)$$

GENERATING FUNCTION FOR LEGENDRE POLYNOMIALS

The function

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n \quad (5)$$

is called the *generating function* for Legendre polynomials and is useful in obtaining their properties.

RECURRENCE FORMULAS

$$1. \quad P_{n+1}(x) = \frac{2n+1}{n+1} x P_n(x) - \frac{n}{n+1} P_{n-1}(x)$$

$$2. \quad P'_{n+1}(x) - P'_{n-1}(x) = (2n+2)P_n(x)$$

LEGENDRE FUNCTIONS OF THE SECOND KIND

If $|x| < 1$, the Legendre functions of the second kind are given by the following, according as n is even or odd respectively:

$$Q_n(x) = \frac{(-1)^{n/2} 2^n [(n/2)!]^2}{n!} \left\{ x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-1)(n-3)(n+2)(n+4)}{5!} x^5 - \dots \right\} \quad (6)$$

$$Q_n(x) = \frac{(-1)^{(n+1)/2} 2^{n-1} \{[(n-1)/2]!\}^2}{1 \cdot 3 \cdot 5 \cdots n} \left\{ 1 - \frac{n(n+1)}{2!} x^2 + \frac{n(n-2)(n+1)(n+3)}{4!} x^4 - \dots \right\} \quad (7)$$

For $n > 1$, the leading coefficients are taken so that the recurrence formulas for $P_n(x)$ above apply also $Q_n(x)$.

ORTHOGONALITY OF LEGENDRE POLYNOMIALS

The following results are fundamental:

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0 \quad \text{if } m \neq n \quad (8)$$

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1} \quad (9)$$

The first shows that any two different Legendre polynomials are orthogonal in the interval $-1 < x < 1$.

SERIES OF LEGENDRE POLYNOMIALS

If $f(x)$ and $f'(x)$ are piecewise continuous then at every point of continuity of $f(x)$ in the interval $-1 < x < 1$ there will exist a Legendre series expansion having the form

$$f(x) = A_0 P_0(x) + A_1 P_1(x) + A_2 P_2(x) + \dots = \sum_{k=0}^{\infty} A_k P_k(x) \quad (10)$$

where

$$A_k = \frac{2k+1}{2} \int_{-1}^1 f(x) P_k(x) dx \quad (11)$$

At any point of discontinuity the series on the right in (10) converges to $\frac{1}{2}[f(x+0) + f(x-0)]$, which can be used to replace the left side of (10).

ASSOCIATED LEGENDRE FUNCTIONS

The differential equation

$$(1-x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1-x^2} \right] y = 0 \quad (12)$$

is called *Legendre's associated differential equation*. If $m=0$ this reduces to Legendre's equation (1). Solutions to (12) are called *associated Legendre functions*. We consider the case where m and n are non-negative integers. In this case the general solution of (12) is given by

$$y = c_1 P_n^m(x) + c_2 Q_n^m(x) \quad (13)$$

where $P_n^m(x)$ and $Q_n^m(x)$ are called *associated Legendre functions of the first and second kinds* respectively. They are given in terms of the ordinary Legendre functions by

$$P_n^m(x) = (1-x^2)^{m/2} \frac{d^m}{dx^m} P_n(x) \quad (14)$$

$$Q_n^m(x) = (1-x^2)^{m/2} \frac{d^m}{dx^m} Q_n(x) \quad (15)$$

Note that if $m > n$, $P_n^m(x) = 0$. The functions $Q_n^m(x)$ are unbounded for $x = \pm 1$.

The differential equation (12) is obtained from Laplace's equation $\nabla^2 u = 0$ expressed in spherical coordinates (r, θ, ϕ) . See Problem 7.21.

ORTHOGONALITY OF ASSOCIATED LEGENDRE FUNCTIONS

As in the case of Legendre polynomials, the Legendre functions $P_n^m(x)$ are orthogonal in $-1 < x < 1$, i.e.

$$\int_{-1}^1 P_n^m(x) P_k^m(x) dx = 0 \quad n \neq k \quad (16)$$

We also have

$$\int_{-1}^1 [P_n^m(x)]^2 dx = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!} \quad (17)$$

Using these, we can expand a function $f(x)$ in a series of the form

$$f(x) = \sum_{k=0}^{\infty} A_k P_k^m(x) \quad (18)$$

SOLUTIONS TO BOUNDARY VALUE PROBLEMS USING LEGENDRE FUNCTIONS

Various boundary value problems can be solved by use of Legendre functions. See Problems 7.18–7.20 and 7.28–7.30.

Solved Problems

LEGENDRE'S DIFFERENTIAL EQUATION

7.1. By letting $u = R\Theta$, where R depends only on r and Θ depends only on θ , in Laplace's equation $\nabla^2 u = 0$ expressed in spherical coordinates, show that R and Θ satisfy the equations

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} + \lambda^2 R = 0 \qquad \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \lambda^2 (\sin \theta) \Theta = 0$$

Laplace's equation in spherical coordinates is given by

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2} = 0 \quad (1)$$

See (4), page 5. If u is independent of ϕ , then the equation can be written

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) = 0 \quad (2)$$

Letting $u = R\Theta$ in this equation, where it is supposed that R depends only on r while Θ depends only on θ , we have

$$\frac{\Theta}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{R}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = 0$$

Multiplying by r^2 , dividing by $R\Theta$ and rearranging, we find

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right)$$

Since one side depends only on r while the other depends only on θ , it follows that each side must be a constant, say $-\lambda^2$. Then we have

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\lambda^2 \quad (3)$$

and

$$\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = \lambda^2 \quad (4)$$

which can be rewritten respectively as

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} + \lambda^2 R = 0 \quad (5)$$

and

$$\frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \lambda^2 (\sin \theta) \Theta = 0 \quad (6)$$

as required.

7.2. Show that the solution for the R -equation in Problem 7.1 can be written as

$$R = Ar^n + \frac{B}{r^{n+1}}$$

where $\lambda^2 = -n(n+1)$.

The R -equation of Problem 7.1 is

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} + \lambda^2 R = 0$$

This is an Euler or Cauchy equation and can be solved by letting $R = r^p$ and determining p . Alternatively, comparison with (26) and (28), page 101, for the case where $x = r$, $y = R$, $k = \frac{1}{2}$, $\alpha = 0$, $\beta = \lambda$ shows that the general solution is

$$R = r^{-1/2}[Ar^{\sqrt{1/4-\lambda^2}} + Br^{-\sqrt{1/4-\lambda^2}}]$$

or

$$R = Ar^{-1/2+\sqrt{1/4-\lambda^2}} + Br^{-1/2-\sqrt{1/4-\lambda^2}} \quad (1)$$

This solution can be simplified if we write

$$-\frac{1}{2} + \sqrt{\frac{1}{4} - \lambda^2} = n \quad (2)$$

so that

$$-\frac{1}{2} - \sqrt{\frac{1}{4} - \lambda^2} = -n - 1 \quad (3)$$

In such case (1) becomes

$$R = Ar^n + \frac{B}{r^{n+1}} \quad (4)$$

Multiplying equations (2) and (3) together leads to

$$\lambda^2 = -n(n+1) \quad (5)$$

- 7.3. Show that the Θ -equation (6) of Problem 7.1 becomes Legendre's differential equation (1), page 130, on making the transformation $\xi = \cos \theta$.

Using the value $\lambda^2 = -n(n+1)$ from (5) of Problem 7.2 in the Θ -equation (6) of Problem 7.1, it becomes

$$\frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + n(n+1)(\sin \theta)\Theta = 0 \quad (1)$$

We now let $\xi = \cos \theta$ in this equation. Then

$$\frac{d\Theta}{d\theta} = \frac{d\Theta}{d\xi} \frac{d\xi}{d\theta} = -\sin \theta \frac{d\Theta}{d\xi}$$

Thus
$$\sin \theta \frac{d\Theta}{d\theta} = -\sin^2 \theta \frac{d\Theta}{d\xi} = (\xi^2 - 1) \frac{d\Theta}{d\xi}$$

since $\sin^2 \theta = 1 - \cos^2 \theta = 1 - \xi^2$. It follows that

$$\begin{aligned} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) &= \frac{d}{d\theta} \left[(\xi^2 - 1) \frac{d\Theta}{d\xi} \right] \\ &= \frac{d}{d\xi} \left[(\xi^2 - 1) \frac{d\Theta}{d\xi} \right] \frac{d\xi}{d\theta} = \frac{d}{d\xi} \left[(1 - \xi^2) \frac{d\Theta}{d\xi} \right] \sin \theta \end{aligned} \quad (2)$$

Using this in (1) and canceling the factor $\sin \theta$, we obtain

$$\frac{d}{d\xi} \left[(1 - \xi^2) \frac{d\Theta}{d\xi} \right] + n(n+1)\Theta = 0 \quad (3)$$

Replacing Θ by y and ξ by x , and carrying out the indicated differentiation, yields the required Legendre equation

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0 \quad (4)$$

- 7.4. Use the method of Frobenius to find series solutions of Legendre's differential equation $(1 - x^2)y'' - 2xy' + n(n+1)y = 0$.

Assuming a solution of the form $y = \sum c_k x^{k+\beta}$ where the summation index k goes from $-\infty$ to ∞ and $c_k = 0$ for $k < 0$, we have

$$\begin{aligned}
 n(n+1)y &= \sum n(n+1)c_k x^{k+\beta} \\
 -2xy' &= \sum -2(k+\beta)c_k x^{k+\beta} \\
 (1-x^2)y'' &= \sum (k+\beta)(k+\beta-1)c_k x^{k+\beta-2} - \sum (k+\beta)(k+\beta-1)c_k x^{k+\beta} \\
 &= \sum (k+\beta+2)(k+\beta+1)c_{k+2} x^{k+\beta} - \sum (k+\beta)(k+\beta-1)c_k x^{k+\beta}
 \end{aligned}$$

Then by addition,

$$\sum [(k+\beta+2)(k+\beta+1)c_{k+2} - (k+\beta)(k+\beta-1)c_k - 2(k+\beta)c_k + n(n+1)c_k]x^{k+\beta} = 0$$

and since the coefficient of $x^{k+\beta}$ must be zero, we find

$$(k+\beta+2)(k+\beta+1)c_{k+2} + [n(n+1) - (k+\beta)(k+\beta+1)]c_k = 0 \quad (1)$$

Letting $k = -2$ we obtain, since $c_{-2} = 0$, the indicial equation $\beta(\beta-1)c_0 = 0$ or, assuming $c_0 \neq 0$, $\beta = 0$ or 1 .

Case 1: $\beta = 0$.

In this case (1) becomes

$$(k+2)(k+1)c_{k+2} + [n(n+1) - k(k+1)]c_k = 0 \quad (2)$$

Putting $k = -1, 0, 1, 2, 3, \dots$ in succession, we find that c_1 is arbitrary while

$$c_2 = -\frac{n(n+1)}{2!}c_0, \quad c_3 = \frac{1 \cdot 2 - n(n+1)}{3!}c_1, \quad c_4 = \frac{[2 \cdot 3 - n(n+1)]}{4!}c_2, \quad \dots$$

and so we obtain

$$\begin{aligned}
 y &= c_0 \left[1 - \frac{n(n+1)}{2!}x^2 + \frac{n(n-2)(n+1)(n+3)}{4!}x^4 - \dots \right] \\
 &\quad + c_1 \left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-1)(n-3)(n+2)(n+4)}{5!}x^5 - \dots \right] \quad (3)
 \end{aligned}$$

Since we have a solution with two arbitrary constants, we need not consider Case 2: $\beta = 1$.

For an even integer $n \geq 0$, the first of the above series terminates and gives a polynomial solution. For an odd integer $n > 0$, the second series terminates and gives a polynomial solution. Thus for any integer $n \geq 0$ the equation has polynomial solutions. If $n = 0, 1, 2, 3$, for example, we obtain from (3) the polynomials

$$c_0, \quad c_1 x, \quad c_0(1-3x^2), \quad c_1 \left(\frac{3x-5x^3}{2} \right)$$

which are, apart from a multiplicative constant, the Legendre polynomials $P_n(x)$. This multiplicative constant is chosen so that $P_n(1) = 1$.

The series solution in (3) which does not terminate can be shown to diverge for $x = \pm 1$. This second solution, which is unbounded for $x = \pm 1$ or equivalently for $\theta = 0, \pi$, is called a *Legendre function of the second kind* and is denoted by $Q_n(x)$. It follows that the general solution of Legendre's differential equation can be written as

$$y = c_1 P_n(x) + c_2 Q_n(x)$$

In case n is not an integer both series solutions are unbounded for $x = \pm 1$.

7.5. Show that a solution of Laplace's equation $\nabla^2 u = 0$ which is independent of ϕ is given by

$$u = \left(A_1 r^n + \frac{B_1}{r^{n+1}} \right) [A_2 P_n(\xi) + B_2 Q_n(\xi)]$$

where $\xi = \cos \theta$.

This result follows at once from Problems 7.1 through 7.4 since $u = R\theta$ where

$$R = A_1 r^n + \frac{B_1}{r^{n+1}}$$

and the general solution of the θ -equation (Legendre's equation) is written in terms of two linearly independent solutions $P_n(\xi)$ and $Q_n(\xi)$ as

$$\theta = A_2 P_n(\xi) + B_2 Q_n(\xi)$$

The functions $P_n(\xi)$ and $Q_n(\xi)$ are the Legendre functions of the first and second kinds respectively.

LEGENDRE POLYNOMIALS

7.6. Derive formula (3), page 130, for the Legendre polynomials.

From (2) of Problem 7.4 we see that if $k = n$ then $c_{n+2} = 0$ and thus $c_{n+4} = 0$, $c_{n+6} = 0$, ... Then letting $k = n-2, n-4, \dots$ we find from (2) of Problem 7.4,

$$c_{n-2} = -\frac{n(n-1)}{2(2n-1)} c_n, \quad c_{n-4} = -\frac{(n-2)(n-3)}{4(2n-3)} c_{n-2} = \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} c_n, \quad \dots$$

This leads to the polynomial solutions

$$y = c_n \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} x^{n-4} - \dots \right]$$

The Legendre polynomials $P_n(x)$ are defined by choosing

$$c_n = \frac{(2n-1)(2n-3)\cdots 3 \cdot 1}{n!}$$

This choice is made in order that $P_n(1) = 1$.

7.7. Derive Rodrigue's formula $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$.

By Problem 7.6 the Legendre polynomials are given by

$$P_n(x) = \frac{(2n-1)(2n-3)\cdots 3 \cdot 1}{n!} \left\{ x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} x^{n-4} - \dots \right\}$$

Now integrating this n times from 0 to x , we obtain

$$\frac{(2n-1)(2n-3)\cdots 3 \cdot 1}{(2n)!} \left\{ x^{2n} - nx^{2n-2} + \frac{n(n-1)}{2!} x^{2n-4} - \dots \right\}$$

which can be written

$$\frac{(2n-1)(2n-3)\cdots 3 \cdot 1}{(2n)(2n-1)(2n-2)\cdots 2 \cdot 1} (x^2 - 1)^n \quad \text{or} \quad \frac{1}{2^n n!} (x^2 - 1)^n$$

which proves that

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

GENERATING FUNCTION

7.8. Prove that $\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n$.

Using the binomial theorem

$$(1+v)^p = 1 + pv + \frac{p(p-1)}{2!} v^2 + \frac{p(p-1)(p-2)}{3!} v^3 + \dots$$

we have

$$\begin{aligned}\frac{1}{\sqrt{1-2xt+t^2}} &= [1-t(2x-t)]^{-1/2} \\ &= 1 + \frac{1}{2}t(2x-t) + \frac{1 \cdot 3}{2 \cdot 4}t^2(2x-t)^2 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}t^3(2x-t)^3 + \dots\end{aligned}$$

and the coefficient of t^n in this expansion is

$$\begin{aligned}\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} (2x)^n - \frac{1 \cdot 3 \cdot 5 \cdots (2n-3)}{2 \cdot 4 \cdot 6 \cdots (2n-2)} \cdot \frac{(n-1)}{1!} (2x)^{n-2} \\ + \frac{1 \cdot 3 \cdot 5 \cdots 2n-5}{2 \cdot 4 \cdot 6 \cdots 2n-4} \cdot \frac{(n-2)(n-3)}{2!} (2x)^{n-4} - \dots\end{aligned}$$

which can be written as

$$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} \left\{ x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4(2n-1)(2n-3)} x^{n-4} - \dots \right\}$$

i.e. $P_n(x)$. The required result thus follows.

RECURRENCE FORMULAS FOR LEGENDRE POLYNOMIALS

7.9. Prove that $P_{n+1}(x) = \frac{2n+1}{n+1} xP_n(x) - \frac{n}{n+1} P_{n-1}(x)$.

From the generating function of Problem 7.8 we have

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n \quad (1)$$

Differentiating with respect to t ,

$$\frac{x-t}{(1-2xt+t^2)^{3/2}} = \sum_{n=0}^{\infty} nP_n(x)t^{n-1}$$

Multiplying by $1-2xt+t^2$,

$$\frac{x-t}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} (1-2xt+t^2)nP_n(x)t^{n-1} \quad (2)$$

Now the left side of (2) can be written in terms of (1) and we have

$$\sum_{n=0}^{\infty} (x-t)P_n(x)t^n = \sum_{n=0}^{\infty} (1-2xt+t^2)nP_n(x)t^{n-1}$$

i.e.

$$\sum_{n=0}^{\infty} xP_n(x)t^n - \sum_{n=0}^{\infty} P_n(x)t^{n+1} = \sum_{n=0}^{\infty} nP_n(x)t^{n-1} - \sum_{n=0}^{\infty} 2nxP_n(x)t^n + \sum_{n=0}^{\infty} nP_n(x)t^{n+1}$$

Equating the coefficients of t^n on each side, we find

$$xP_n(x) - P_{n-1}(x) = (n+1)P_{n+1}(x) - 2nxP_n(x) + (n-1)P_{n-1}(x)$$

which yields the required result.

7.10. Given that $P_0(x) = 1$, $P_1(x) = x$, find (a) $P_2(x)$ and (b) $P_3(x)$.

Using the recurrence formula of Problem 7.9, we have on letting $n = 1$,

$$P_2(x) = \frac{3}{2}xP_1(x) - \frac{1}{2}P_0(x) = \frac{3}{2}x^2 - \frac{1}{2} = \frac{1}{2}(3x^2 - 1)$$

Similarly letting $n = 2$,

$$P_3(x) = \frac{5}{3}xP_2(x) - \frac{2}{3}P_1(x) = \frac{5}{3}x\left(\frac{3x^2-1}{2}\right) - \frac{2}{3}x = \frac{1}{2}(5x^3 - 3x)$$

LEGENDRE FUNCTIONS OF THE SECOND KIND

7.11. Obtain the results (6) and (7), page 131, for the Legendre functions of the second kind in the case where n is a non-negative integer.

The Legendre functions of the second kind are the series solutions of Legendre's equation which do not terminate. From (3) of Problem 7.4 we see that if n is even the series which does not terminate is

$$x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-1)(n-3)(n+2)(n+4)}{5!}x^5 - \dots$$

while if n is odd the series which does not terminate is

$$1 - \frac{n(n+1)}{2!}x^2 + \frac{n(n-2)(n+1)(n+3)}{4!}x^4 - \dots$$

These series solutions, apart from multiplicative constants, provide definitions for Legendre functions of the second kind and are given by (6) and (7) on page 131. The multiplicative constants are chosen so that the Legendre functions of the second kind will satisfy the same recurrence formulas (page 131) as the Legendre polynomials.

7.12. Obtain the Legendre functions of the second kind (a) $Q_0(x)$, (b) $Q_1(x)$, and (c) $Q_2(x)$.

(a) From (6), page 131, we have if $n = 0$,

$$\begin{aligned} Q_0(x) &= x + \frac{2}{3!}x^3 + \frac{1 \cdot 3 \cdot 2 \cdot 4}{5!}x^5 + \frac{1 \cdot 3 \cdot 5 \cdot 2 \cdot 4 \cdot 6}{7!}x^7 + \dots \\ &= x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \dots = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) \end{aligned}$$

where we have used the expansion $\ln(1+u) = u - u^2/2 + u^3/3 - u^4/4 + \dots$.

(b) From (7), page 131, we have if $n = 1$,

$$\begin{aligned} Q_1(x) &= - \left\{ 1 - \frac{(1)(2)}{2!}x^2 + \frac{(1)(-1)(2)(4)}{4!}x^4 - \frac{(1)(-1)(-3)(2)(4)(6)}{6!}x^6 + \dots \right\} \\ &= x \left\{ x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \right\} - 1 = \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1 \end{aligned}$$

(c) The recurrence formulas for $Q_n(x)$ are identical with those of $P_n(x)$. Then from Problem 7.9,

$$Q_{n+1}(x) = \frac{2n+1}{n+1}xQ_n(x) - \frac{n}{n+1}Q_{n-1}(x)$$

Putting $n = 1$, we have on using parts (a) and (b),

$$Q_2(x) = \frac{3}{2}xQ_1(x) - \frac{1}{2}Q_0(x) = \left(\frac{3x^2-1}{4} \right) \ln \left(\frac{1+x}{1-x} \right) - \frac{3x}{2}$$

ORTHOGONALITY OF LEGENDRE POLYNOMIALS

7.13. Prove that $\int_{-1}^1 P_m(x)P_n(x)dx = 0$ if $m \neq n$.

Since $P_m(x)$, $P_n(x)$ satisfy Legendre's equation,

$$(1-x^2)P_m'' - 2xP_m' + m(m+1)P_m = 0$$

$$(1-x^2)P_n'' - 2xP_n' + n(n+1)P_n = 0$$

Then multiplying the first equation by P_n , the second equation by P_m and subtracting, we find

$$(1-x^2)[P_nP_m'' - P_mP_n''] - 2x[P_nP_m' - P_mP_n'] = [n(n+1) - m(m+1)]P_mP_n$$

which can be written

$$(1-x^2) \frac{d}{dx} [P_n P'_m - P_m P'_n] - 2x[P_n P'_m - P_m P'_n] = [n(n+1) - m(m+1)] P_m P_n$$

or
$$\frac{d}{dx} \{(1-x^2)[P_n P'_m - P_m P'_n]\} = [n(n+1) - m(m+1)] P_m P_n$$

Thus by integrating we have

$$[n(n+1) - m(m+1)] \int_{-1}^1 P_m(x) P_n(x) dx = (1-x^2)[P_n P'_m - P_m P'_n] \Big|_{-1}^1 = 0$$

Then since $m \neq n$,

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0$$

7.14. Prove that
$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}.$$

From the generating function

$$\frac{1}{\sqrt{1-2tx+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$$

we have on squaring both sides,

$$\frac{1}{1-2tx+t^2} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} P_m(x) P_n(x) t^{m+n}$$

Then by integrating from -1 to 1 we have

$$\int_{-1}^1 \frac{dx}{1-2tx+t^2} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \left\{ \int_{-1}^1 P_m(x) P_n(x) dx \right\} t^{m+n}.$$

Using the result of Problem 7.13 on the right side and performing the integration on the left side,

$$-\frac{1}{2t} \ln(1-2tx+t^2) \Big|_{-1}^1 = \sum_{n=0}^{\infty} \left\{ \int_{-1}^1 [P_n(x)]^2 dx \right\} t^{2n}$$

or
$$\frac{1}{t} \ln \left(\frac{1+t}{1-t} \right) = \sum_{n=0}^{\infty} \left\{ \int_{-1}^1 [P_n(x)]^2 dx \right\} t^{2n}$$

i.e.
$$\sum_{n=0}^{\infty} \frac{2t^{2n}}{2n+1} = \sum_{n=0}^{\infty} \left\{ \int_{-1}^1 [P_n(x)]^2 dx \right\} t^{2n}$$

Equating coefficients of t^{2n} , it follows that

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$$

SERIES OF LEGENDRE POLYNOMIALS

7.15. If $f(x) = \sum_{k=0}^{\infty} A_k P_k(x)$, $-1 < x < 1$, show that

$$A_k = \frac{2k+1}{2} \int_{-1}^1 P_k(x) f(x) dx$$

Multiplying the given series by $P_m(x)$ and integrating from -1 to 1 , we have on using Problems 7.13 and 7.14,

$$\begin{aligned}\int_{-1}^1 P_m(x) f(x) dx &= \sum_{k=0}^{\infty} A_k \int_{-1}^1 P_m(x) P_k(x) dx \\ &= A_m \int_{-1}^1 [P_m(x)]^2 dx = \frac{2A_m}{2m+1}\end{aligned}$$

Then as required,

$$A_m = \frac{2m+1}{2} \int_{-1}^1 P_m(x) f(x) dx$$

7.16. Expand the function $f(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & -1 < x < 0 \end{cases}$ in a series of the form $\sum_{k=0}^{\infty} A_k P_k(x)$.

By Problem 7.15

$$\begin{aligned}A_k &= \frac{2k+1}{2} \int_{-1}^1 P_k(x) f(x) dx = \frac{2k+1}{2} \int_{-1}^0 P_k(x)(0) dx + \frac{2k+1}{2} \int_0^1 P_k(x)(1) dx \\ &= \frac{2k+1}{2} \int_0^1 P_k(x) dx\end{aligned}$$

$$\text{Then } A_0 = \frac{1}{2} \int_0^1 P_0(x) dx = \frac{1}{2} \int_0^1 (1) dx = \frac{1}{2}$$

$$A_1 = \frac{3}{2} \int_0^1 P_1(x) dx = \frac{3}{2} \int_0^1 x dx = \frac{3}{4}$$

$$A_2 = \frac{5}{2} \int_0^1 P_2(x) dx = \frac{5}{2} \int_0^1 \frac{3x^2-1}{2} dx = 0$$

$$A_3 = \frac{7}{2} \int_0^1 P_3(x) dx = \frac{7}{2} \int_0^1 \frac{5x^3-3x}{2} dx = -\frac{7}{16}$$

$$A_4 = \frac{9}{2} \int_0^1 P_4(x) dx = \frac{9}{2} \int_0^1 \frac{35x^4-30x^2+3}{8} dx = 0$$

$$A_5 = \frac{11}{2} \int_0^1 P_5(x) dx = \frac{11}{2} \int_0^1 \frac{63x^5-70x^3+15x}{8} dx = \frac{11}{32}$$

etc. Thus

$$f(x) = \frac{1}{2}P_0(x) + \frac{3}{4}P_1(x) - \frac{7}{16}P_3(x) + \frac{11}{32}P_5(x) - \dots$$

The general term for the coefficients in this series can be obtained by using the recurrence formula 2 on page 131 and the results of Problem 7.34. We find

$$A_n = \frac{2n+1}{2} \int_0^1 P_n(x) dx = \frac{1}{2} \int_0^1 [P'_{n+1}(x) - P'_{n-1}(x)] dx = \frac{1}{2} [P_{n-1}(0) - P_{n+1}(0)]$$

For n even $A_n = 0$, while for n odd we can use Problem 7.34(c).

7.17. Expand $f(x) = x^2$ in a series of the form $\sum_{k=0}^{\infty} A_k P_k(x)$.

Method 1.

We must find A_k , $k = 0, 1, 2, 3, \dots$, such that

$$\begin{aligned}x^2 &= A_0 P_0(x) + A_1 P_1(x) + A_2 P_2(x) + A_3 P_3(x) + \dots \\ &= A_0(1) + A_1(x) + A_2\left(\frac{3x^2-1}{2}\right) + A_3\left(\frac{5x^3-3x}{2}\right) + \dots\end{aligned}$$

Since the left side is a polynomial of degree 2 we must have $A_3 = 0$, $A_4 = 0$, $A_5 = 0$, Thus

$$x^2 = A_0 - \frac{A_2}{2} + A_1x + \frac{3}{2}A_2x^2$$

from which

$$A_0 - \frac{A_2}{2} = 0, \quad A_1 = 0, \quad \frac{3}{2}A_2 = 1$$

Then

$$A_0 = \frac{1}{3}, \quad A_1 = 0, \quad A_2 = \frac{2}{3}$$

i.e.

$$x^2 = \frac{1}{3}P_0(x) + \frac{2}{3}P_2(x)$$

Method 2.

Using the method of Problem 7.15 we see that if

$$x^2 = \sum_{k=0}^{\infty} A_k P_k(x)$$

then

$$A_k = \frac{2k+1}{2} \int_{-1}^1 x^2 P_k(x) dx$$

Putting $k = 0, 1, 2, \dots$, we find as before $A_0 = \frac{1}{3}$, $A_1 = 0$, $A_2 = \frac{2}{3}$, $A_3 = 0$, $A_4 = 0$, ... so that

$$x^2 = \frac{1}{3}P_0(x) + \frac{2}{3}P_2(x)$$

In general when we expand a polynomial in a series of Legendre polynomials, the series, which terminates, can most easily be found by using Method 1.

SOLUTIONS USING LEGENDRE FUNCTIONS

7.18. Find the potential v (a) interior to and (b) exterior to a hollow sphere of unit radius if half of its surface is charged to potential v_0 and the other half to potential zero.

Choose the sphere in the position shown in Fig. 7-1. Then v is independent of ϕ and we can use the results of Problem 7.5. A solution is

$$v(r, \theta) = \left(A_1 r^n + \frac{B_1}{r^{n+1}} \right) [A_2 P_n(\xi) + B_2 Q_n(\xi)]$$

where $\xi = \cos \theta$. Since v must be bounded at $\theta = 0$ and π , i.e. $\xi = \pm 1$, we must choose $B_2 = 0$. Then

$$v(r, \theta) = \left(A r^n + \frac{B}{r^{n+1}} \right) P_n(\xi) \quad (1)$$

The boundary conditions are

$$v(1, \theta) = \begin{cases} v_0 & \text{if } 0 < \theta < \frac{\pi}{2} \quad \text{i.e.} \quad 0 < \xi < 1 \\ 0 & \text{if } \frac{\pi}{2} < \theta < \pi \quad \text{i.e.} \quad -1 < \xi < 0 \end{cases}$$

and v is bounded.

(a) **Interior Potential**, $0 \leq r < 1$.

Since v is bounded at $r = 0$, choose $B = 0$ in (1). Then a solution is

$$A r^n P_n(\xi) = A r^n P_n(\cos \theta)$$

By superposition,

$$v(r, \theta) = \sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta) = \sum_{n=0}^{\infty} A_n r^n P_n(\xi)$$

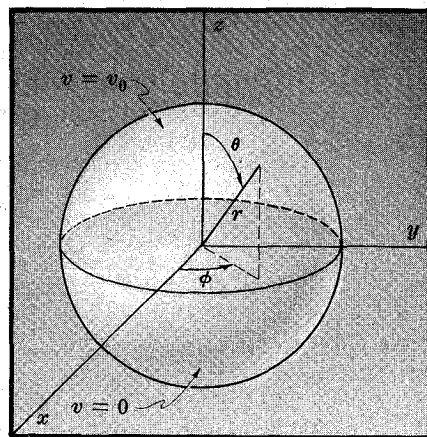


Fig. 7-1

When $r = 1$,

$$v(1, \theta) = \sum_{n=0}^{\infty} A_n P_n(\xi)$$

Then as in Problem 7.15,

$$A_n = \frac{2n+1}{2} \int_{-1}^1 v(1, \theta) P_n(\xi) d\xi = \left(\frac{2n+1}{2} \right) v_0 \int_0^1 P_n(\xi) d\xi$$

from which

$$A_0 = \frac{1}{2} v_0, \quad A_1 = \frac{3}{4} v_0, \quad A_2 = 0, \quad A_3 = -\frac{7}{16} v_0, \quad A_4 = 0, \quad A_5 = \frac{11}{32} v_0, \quad \dots$$

$$\text{Thus} \quad v(r, \theta) = \frac{v_0}{2} \left[1 + \frac{3}{2} r P_1(\cos \theta) - \frac{7}{8} r^3 P_3(\cos \theta) + \frac{11}{16} r^5 P_5(\cos \theta) + \dots \right] \quad (2)$$

(b) **Exterior Potential**, $1 < r < \infty$.

Since v is bounded as $r \rightarrow \infty$, choose $A = 0$ in (1). Then a solution is

$$\frac{B}{r^{n+1}} P_n(\xi) = \frac{B}{r^{n+1}} P_n(\cos \theta)$$

By superposition,

$$v(r, \theta) = \sum_{n=0}^{\infty} \frac{B_n}{r^{n+1}} P_n(\xi)$$

When $r = 1$,

$$v(1, \theta) = \sum_{n=0}^{\infty} B_n P_n(\xi)$$

Then $B_n = A_n$ of part (a) and so

$$v(r, \theta) = \frac{v_0}{2r} \left[1 + \frac{3}{2r} P_1(\cos \theta) - \frac{7}{8r^3} P_3(\cos \theta) + \frac{11}{16r^5} P_5(\cos \theta) + \dots \right] \quad (3)$$

7.19. A uniform hemisphere (see Fig. 7-2) has its convex surface kept at temperature u_0 while its base is kept at temperature zero. Find the steady-state temperature inside.

The boundary value problem in this case is

$$\nabla^2 u = 0$$

where

$$u = u_0 \quad \text{on the convex surface}$$

$$u = 0 \quad \text{on the base}$$

The solution can be obtained from the results of Problem 7.18. To see this we note that the present problem is equivalent to the problem of solving Laplace's equation inside a sphere of which the top half surface is kept at temperature u_0 and the bottom half surface is kept at temperature $-u_0$. By symmetry, the plane of separation will then automatically be at temperature zero as required in this problem.

We can then obtain the required solution by first subtracting $v_0/2$ from the solution in Problem 7.18 and then replacing $v_0/2$ by u_0 . The result is

$$u(r, \theta) = u_0 \left[\frac{3}{2} r P_1(\cos \theta) - \frac{7}{8} r^3 P_3(\cos \theta) + \frac{11}{16} r^5 P_5(\cos \theta) + \dots \right]$$

The problem can also, of course, be solved directly without use of the results in Problem 7.18.

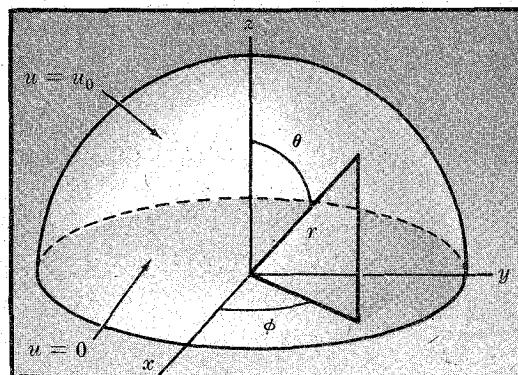


Fig. 7-2

7.20. (a) Find the gravitational potential at any point on the axis of a thin uniform ring of radius a . (b) Find the potential of the ring in part (a) at any point in space.

- (a) Choose the ring to be in the xy -plane so that the axis is the z -axis as indicated in Fig. 7-3. Then the potential at any point P on the z -axis is seen to be the mass of the ring divided by the distance $\sqrt{a^2 + z^2}$ from any point Q on the ring to the point P . Letting σ denote the mass per unit length of the ring it follows that the potential at P is

$$v_P = \frac{2\pi a \sigma}{\sqrt{a^2 + z^2}} \quad (1)$$

- (b) In this case we must solve Laplace's equation $\nabla^2 v = 0$ where v reduces to v_P for points P on the z -axis. Now we know that because of the manner in which the ring has been located that v is independent of ϕ . We thus have as a solution to Laplace's equation

$$v = \left(A_1 r^n + \frac{B_1}{r^{n+1}} \right) [A_2 P_n(\xi) + B_2 Q_n(\xi)]$$

where $\xi = \cos \theta$. Since v must be bounded at $\theta = 0$ and π , i.e. $\xi = \pm 1$, we must choose $B_2 = 0$. Then

$$v = \left(A r^n + \frac{B}{r^{n+1}} \right) P_n(\xi) \quad (2)$$

There are two cases to be considered, corresponding to the regions $0 \leq r < a$ and $r > a$.

Case 1: $0 \leq r < a$.

In this case we must choose $B = 0$ in (2) since otherwise the solution is unbounded at $r = 0$. Then $v = A r^n P_n(\xi)$. By superposition we are led to consider the solution

$$v = \sum_{n=0}^{\infty} A_n r^n P_n(\xi) \quad (3)$$

Now when $\theta = 0$, i.e. $\xi = 1$, this must reduce to the potential on the z -axis, in which case $r = z$. Then we must have

$$\frac{2\pi a \sigma}{\sqrt{a^2 + z^2}} = \sum_{n=0}^{\infty} A_n z^n \quad (4)$$

In order to obtain A_n we must expand the left side as a power series in z . We use the binomial theorem to obtain

$$\begin{aligned} \frac{2\pi a \sigma}{\sqrt{a^2 + z^2}} &= 2\pi \sigma \left(1 + \frac{z^2}{a^2} \right)^{-1/2} \\ &= 2\pi \sigma \left[1 - \frac{1}{2} \left(\frac{z}{a} \right)^2 + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{z}{a} \right)^4 - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \left(\frac{z}{a} \right)^6 + \dots \right] \end{aligned} \quad (5)$$

Comparison of (4) and (5) leads to

$$A_0 = 2\pi \sigma, \quad A_1 = 0, \quad A_2 = -\frac{2\pi \sigma}{2a^2}, \quad A_3 = 0, \quad A_4 = \frac{2\pi \sigma \cdot 1 \cdot 3}{a^4 \cdot 2 \cdot 4}, \quad \dots$$

Using these in (3) we then find

$$v = 2\pi \sigma \left[P_0(\cos \theta) - \frac{1}{2} \left(\frac{r}{a} \right)^2 P_2(\cos \theta) + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{r}{a} \right)^4 P_4(\cos \theta) - \dots \right] \quad (6)$$

where $0 \leq r < a$.

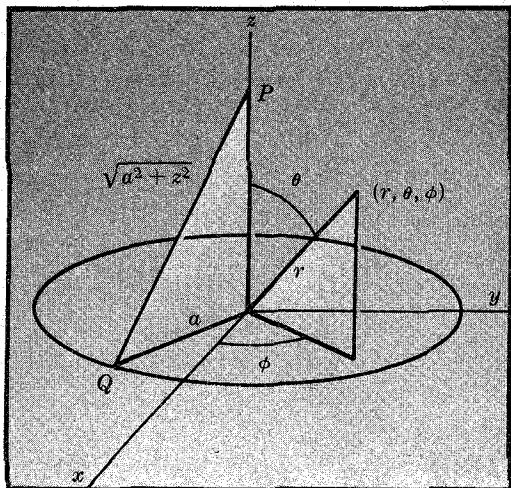


Fig. 7-3

Case 2: $r > a$.

In this case we must choose $A = 0$ in (2) since otherwise the solution becomes unbounded as $r \rightarrow \infty$. Then $v = BP_n(\xi)/r^{n+1}$ and by superposition we are led to consider the solution

$$v = \sum_{n=0}^{\infty} \frac{B_n}{r^{n+1}} P_n(\xi) \quad (7)$$

As in Case 1, this must reduce to the potential on the z -axis for $\theta = 0$ and $r = z$, i.e.

$$\frac{2\pi a\sigma}{\sqrt{a^2 + z^2}} = \sum_{n=0}^{\infty} \frac{B_n}{z^{n+1}} \quad (8)$$

Thus, to find B_n we must expand the left side in inverse powers of z . Again we use the binomial theorem to obtain

$$\begin{aligned} \frac{2\pi a\sigma}{\sqrt{a^2 + z^2}} &= \frac{2\pi a\sigma}{z} \left(1 + \frac{a^2}{z^2}\right)^{-1/2} \\ &= \frac{2\pi a\sigma}{z} \left[1 - \frac{1}{2} \left(\frac{a}{z}\right)^2 + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{a}{z}\right)^4 - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \left(\frac{a}{z}\right)^6 + \dots\right] \end{aligned} \quad (9)$$

Comparison of (8) and (9) leads to

$$B_0 = 2\pi a\sigma, \quad B_1 = 0, \quad B_2 = -2\pi a\sigma \left(\frac{1}{2}a^2\right), \quad B_3 = 0, \quad B_4 = 2\pi a\sigma \left(\frac{1 \cdot 3}{2 \cdot 4}a^4\right), \quad \dots$$

Using these in (7) we then find

$$v = \frac{2\pi a\sigma}{r} \left[P_0(\cos \theta) - \frac{1}{2} \left(\frac{a}{r}\right)^2 P_2(\cos \theta) + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{a}{r}\right)^4 P_4(\cos \theta) - \dots \right] \quad (10)$$

where $r > a$.

ASSOCIATED LEGENDRE FUNCTIONS

7.21. Show how Legendre's associated differential equation (12), page 132, is obtained from Laplace's equation $\nabla^2 u = 0$ expressed in spherical coordinates (r, θ, ϕ) .

In this case we must modify the results obtained in Problem 7.1 by including the ϕ -dependence. Then letting $u = R\Theta\Phi$ in (1) of Problem 7.1 we obtain

$$\frac{\Theta\Phi}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{R\Phi}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{R\Theta}{r^2 \sin^2 \theta} \frac{d^2\Phi}{d\phi^2} = 0 \quad (1)$$

Multiplying by r^2 , dividing by $R\Theta\Phi$ and rearranging, we find

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{1}{\Phi \sin^2 \theta} \frac{d^2\Phi}{d\phi^2}$$

Since one side depends only on r , while the other depends only on θ and ϕ , it follows that each side must be a constant, say $-\lambda^2$. Then we have

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\lambda^2 \quad (2)$$

and

$$\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi \sin^2 \theta} \frac{d^2\Phi}{d\phi^2} = \lambda^2 \quad (3)$$

The equation (2) is identical with (2) in Problem 7.1, so that we have as solution according to Problem 7.2

$$R = A_1 r^n + \frac{B_1}{r^{n+1}} \quad (4)$$

where we use $\lambda^2 = -n(n+1)$.

If now we multiply equation (3) by $\sin^2 \theta$ and rearrange, it can be written as

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - n(n+1) \sin^2 \theta.$$

Since one side depends only on ϕ while the other side depends only on θ each side must be a constant, say $-m^2$. Then we have

$$\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + [n(n+1) \sin^2 \theta - m^2] \Theta = 0 \quad (5)$$

$$\frac{d^2 \Phi}{d\phi^2} + m^2 \Phi = 0 \quad (6)$$

If we now make the transformation $\xi = \cos \theta$ in equation (5) we find as in Problem 7.3 that it can be written as

$$(1-\xi^2) \frac{d}{d\xi} \left[(1-\xi^2) \frac{d\Theta}{d\xi} \right] + [n(n+1)(1-\xi^2) - m^2] \Theta = 0$$

Dividing by $1-\xi^2$ the equation becomes

$$(1-\xi^2) \frac{d^2 \Theta}{d\xi^2} - 2\xi \frac{d\Theta}{d\xi} + \left[n(n+1) - \frac{m^2}{1-\xi^2} \right] \Theta = 0 \quad (7)$$

which is Legendre's associated differential equation (12) on page 132 if we replace θ by y and ξ by x .

The general solution of (7) is shown in Problem 7.22 to be

$$\Theta = A_2 P_n^m(\xi) + B_2 Q_n^m(\xi) \quad (8)$$

where $\xi = \cos \theta$ and

$$P_n^m(\xi) = (1-\xi^2)^{m/2} \frac{d^m}{d\xi^m} P_n(\xi) \quad (9)$$

$$Q_n^m(\xi) = (1-\xi^2)^{m/2} \frac{d^m}{d\xi^m} Q_n(\xi) \quad (10)$$

We call $P_n^m(\xi)$ and $Q_n^m(\xi)$ *associated Legendre functions of the first and second kinds* respectively.

The general solution of (6) is

$$\Phi = A_3 \cos m\phi + B_3 \sin m\phi \quad (11)$$

If the function $u(r, \theta, \phi)$ is to be periodic of period 2π in ϕ , we must have m equal to an integer, which we take as positive. For the case $m=0$ the solution $u(r, \theta, \phi)$ is independent of ϕ and reduces to that given in Problem 7.5.

7.22. (a) Show that if m is a positive integer and u_n is any solution of Legendre's differential equation, then $d^m u_n / dx^m$ is a solution of Legendre's associated differential equation.

(b) Obtain the general solution of Legendre's associated equation.

(a) If Legendre's differential equation has the solution u_n then we must have

$$(1-x^2)u_n'' - 2xu_n' + n(n+1)u_n = 0$$

By differentiating this equation m times and letting $v_n^m = d^m u_n / dx^m$ we obtain

$$(1-x^2) \frac{d^2 v_n^m}{dx^2} - 2(m+1)x \frac{dv_n^m}{dx} + [n(n+1) - m(m+1)]v_n^m = 0$$

In this equation we now let $v_n^m = (1-x^2)^p y$. Then it becomes

$$(1-x^2)^2 y'' - [2(m+1)x(1-x^2) + 4px(1-x^2)]y' + \{4(m+1)px^2 + (4p^2 - 2p)x^2 - 2p + [n(n+1) - m(m+1)](1-x^2)\}y = 0$$

If we now choose $p = -m/2$, this equation becomes after dividing by $1 - x^2$

$$(1 - x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] y = 0 \quad (1)$$

which is Legendre's associated differential equation. Since $v_n^m = (1 - x^2)^{-m/2}y$, it follows that $y = (1 - x^2)^{m/2}v_n^m$, or

$$y = (1 - x^2)^{m/2} \frac{d^m u_n}{dx^m} \quad (2)$$

is a solution of (1).

(b) Since the general solution of Legendre's equation is $c_1 P_n(x) + c_2 Q_n(x)$, we can show that the general solution of Legendre's associated differential equation is

$$y = c_1 P_n^m(x) + c_2 Q_n^m(x) \quad (3)$$

$$\text{where} \quad P_n^m(x) = (1 - x^2)^{m/2} \frac{d^m P_n}{dx^m}, \quad Q_n^m(x) = (1 - x^2)^{m/2} \frac{d^m Q_n}{dx^m} \quad (4)$$

7.23. Obtain the associated Legendre functions (a) $P_2^1(x)$, (b) $P_3^2(x)$, (c) $P_2^3(x)$, (d) $Q_2^1(x)$.

$$(a) \quad P_2^1(x) = (1 - x^2)^{1/2} \frac{d}{dx} P_2(x) = (1 - x^2)^{1/2} \frac{d}{dx} \left(\frac{3x^2 - 1}{2} \right) = 3x(1 - x^2)^{1/2}$$

$$(b) \quad P_3^2(x) = (1 - x^2)^{2/2} \frac{d^2}{dx^2} P_3(x) = (1 - x^2) \frac{d^2}{dx^2} \left(\frac{5x^3 - 3x}{2} \right) = 15x - 15x^3$$

$$(c) \quad P_2^3(x) = (1 - x^2)^{3/2} \frac{d^3}{dx^3} P_2(x) = 0. \quad \text{Note that in general } P_n^m(x) = 0 \text{ if } m > n.$$

(d) Using Problem 7.12(c) we find

$$\begin{aligned} Q_2^1(x) &= (1 - x^2)^{1/2} \frac{d}{dx} Q_2(x) = (1 - x^2)^{1/2} \frac{d}{dx} \left\{ \frac{3x^2 - 1}{4} \ln \left(\frac{1+x}{1-x} \right) - \frac{3x}{2} \right\} \\ &= (1 - x^2)^{1/2} \left[\frac{3x}{2} \ln \left(\frac{1+x}{1-x} \right) + \frac{3x^2 - 2}{1 - x^2} \right] \end{aligned}$$

7.24. Verify that $P_3^2(x)$ is a solution of Legendre's associated equation (12), page 132, for $m = 2$, $n = 3$.

By Problem 7.23, $P_3^2(x) = 15x - 15x^3$. Substituting this in the equation

$$(1 - x^2)y'' - 2xy' + \left[3 \cdot 4 - \frac{4}{1 - x^2} \right] y = 0$$

we find after simplifying,

$$(1 - x^2)(-90x) - 2x(15 - 45x^2) + \left[12 - \frac{4}{1 - x^2} \right] [15x - 15x^3] = 0$$

and so $P_3^2(x)$ is a solution.

7.25. Verify the result (16), page 132, for the functions $P_2^1(x)$ and $P_3^1(x)$.

We have from Problem 7.23(a), $P_2^1(x) = 3x(1 - x^2)^{1/2}$. Also,

$$P_3^1(x) = (1 - x^2)^{3/2} \frac{d}{dx} P_3(x) = (1 - x^2)^{3/2} \frac{d}{dx} \left(\frac{5x^3 - 3x}{2} \right) = \frac{15x^2}{2} (1 - x^2)^{3/2}$$

$$\text{Then} \quad \int_{-1}^1 P_2^1(x) P_3^1(x) dx = \int_{-1}^1 \frac{45x^3}{2} (1 - x^2)^2 dx = 0$$

7.26. Verify the result (17), page 132, for the function $P_2^1(x)$.

Since $P_2^1(x) = 3x(1-x^2)^{1/2}$,

$$\int_{-1}^1 [P_2^1(x)]^2 dx = 9 \int_{-1}^1 x^2(1-x^2) dx = 9 \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_{-1}^1 = \frac{36}{15} = \frac{12}{5}$$

Now according to (17), page 132, the required result should be

$$\frac{2}{2(2)+1} \frac{(2+1)!}{(2-1)!} = \frac{2}{5} \cdot \frac{3!}{1!} = \frac{12}{5}$$

so that the verification is achieved.

7.27. Expand $v_0(1-x^2)$ in a series of the form $\sum_{k=0}^{\infty} A_k P_k^m(x)$ where v_0 is a constant and $m=2$.

We must find A_k , $k=0, 1, 2, \dots$, so that

$$v_0(1-x^2) = A_0 P_0^2(x) + A_1 P_1^2(x) + A_2 P_2^2(x) + \dots \quad (1)$$

Method 1.

$$\text{Since} \quad P_k^2(x) = (1-x^2) \frac{d^2}{dx^2} P_k(x)$$

we have

$$P_0^2(x) = 0, \quad P_1^2(x) = 0, \quad P_2^2(x) = (1-x^2) \frac{d^2}{dx^2} \left(\frac{3x^2-1}{2} \right) = 3(1-x^2),$$

$$P_3^2(x) = (1-x^2) \frac{d^2}{dx^2} \left(\frac{5x^3-3x}{2} \right) = 15x(1-x^2), \quad \dots$$

Then (1) becomes

$$v_0(1-x^2) = 3A_2(1-x^2) + 15A_3x(1-x^2) + \dots$$

By comparing coefficients on each side we see that this can be satisfied if $3A_2 = v_0$, $15A_3 = 0$ and $A_k = 0$ for $k > 3$. Thus we have

$$v_0(1-x^2) = \frac{v_0}{3} P_2^2(x) \quad (2)$$

so that the required expansion consists of only one term.

Method 2.

If $f(x) = \sum_{k=0}^{\infty} A_k P_k^m(x)$, then on multiplying by $P_n^m(x)$ and integrating from -1 to 1 we obtain

$$\int_{-1}^1 f(x) P_n^m(x) dx = \sum_{k=0}^{\infty} A_k \int_{-1}^1 P_n^m(x) P_k^m(x) dx$$

Using (16) and (17), page 132, we see that the right side reduces to the single term

$$\frac{2}{2n+1} \frac{(n+m)!}{(n-m)!} A_n$$

so that

$$A_n = \frac{(2n+1)(n-m)!}{2(n+m)!} \int_{-1}^1 f(x) P_n^m(x) dx$$

If $f(x) = v_0(1-x^2)$ and $m=2$, then

$$A_n = \frac{(2n+1)(n-2)!}{2(n+2)!} \int_{-1}^1 v_0(1-x^2) P_n^2(x) dx$$

Using this we can show that $A_3 = v_0/3$, $A_4 = 0$, $A_5 = 0$, $\dots$ and so we obtain the result (2) as in Method 1.

7.28. Show that a solution to Laplace's equation $\nabla^2 v = 0$ in spherical coordinates is given by

$$v = \left(A_1 r^n + \frac{B_1}{r^{n+1}} \right) [A_2 P_n^m(\cos \theta) + B_2 Q_n^m(\cos \theta)] [A_3 \cos m\phi + B_3 \sin m\phi]$$

This follows at once from Problems 7.21 and 7.22 since $u = R\Theta\Phi$ where

$$R = A_1 r^n + \frac{B_1}{r^{n+1}}$$

$$\Theta = A_2 P_n^m(\cos \theta) + B_2 Q_n^m(\cos \theta)$$

$$\Phi = A_3 \cos m\phi + B_3 \sin m\phi$$

7.29. Suppose that the surface of the sphere of Problem 7.18 is kept at potential $v_0 \sin^2 \theta \cos 2\phi$. Determine the potential (a) inside and (b) outside the surface.

(a) **Interior Potential**, $0 \leq r < 1$.

Since v is bounded at $r = 0$ we must choose $B_1 = 0$ in the solution as given in Problem 7.28. Also since v is bounded at $\theta = 0$ and π , we must choose $B_2 = 0$. Then a bounded solution is given by

$$v(r, \theta, \phi) = r^n P_n^m(\cos \theta) (A \cos m\phi + B \sin m\phi)$$

Since m and n can be any non-negative integers we can replace A by A_{mn} , B by B_{mn} and then, using the superposition principle, sum over m and n to obtain the solution

$$v(r, \theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} r^n P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (1)$$

Now the boundary potential is given by

$$v(1, \theta, \phi) = v_0 \sin^2 \theta \cos 2\phi \quad (2)$$

By comparison of (2) with

$$v(1, \theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (3)$$

obtained from (1) with $r = 1$ it is seen that we must have $B_{mn} = 0$ for all m and $A_{mn} = 0$ for $m \neq 2$. Hence, (3) becomes

$$v(1, \theta, \phi) = \sum_{n=0}^{\infty} A_{2n} P_n^2(\cos \theta) \cos 2\phi$$

Comparison with (2) then shows that we must have

$$v_0 \sin^2 \theta = \sum_{n=0}^{\infty} A_{2n} P_n^2(\cos \theta)$$

or using $\cos \theta = \xi$

$$\begin{aligned} v_0(1 - \xi^2) &= \sum_{n=0}^{\infty} A_{2n} P_n^2(\xi) \\ &= A_{20} P_0^2(\xi) + A_{21} P_1^2(\xi) + A_{22} P_2^2(\xi) + \cdots \end{aligned} \quad (4)$$

We have already obtained this expansion in Problem 7.27, from which we see that $A_{22} = v_0/3$, while all other coefficients are zero. It thus follows from (1) that

$$v(r, \theta, \phi) = \frac{v_0}{3} r^2 P_2^2(\cos \theta) \cos 2\phi = v_0 r^2 \sin^2 \theta \cos 2\phi \quad (5)$$

(b) **Exterior Potential**, $r > 1$.

Since v must be bounded as $r \rightarrow \infty$ in this case and is also bounded at $\theta = 0$ and π , we choose $A_1 = 0$, $B_2 = 0$ in the solution of Problem 7.28. Thus a solution is

$$v(r, \theta, \phi) = \frac{P_n^m(\cos \theta)}{r^{n+1}} (A \cos m\phi + B \sin m\phi)$$

or by superposition

$$v(r, \theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{P_n^m(\cos \theta)}{r^{n+1}} (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (6)$$

Using the fact that $v(1, \theta, \phi) = v_0 \sin^2 \theta \cos 2\phi$ we again find $m = 2$, $B_{mn} = 0$ which leads to equation (4) of part (a). As before we then find $A_{22} = v_0/3$, while all other coefficients are zero, leading to the required solution

$$\begin{aligned} v(r, \theta, \phi) &= \frac{v_0}{3r^3} P_2^2(\cos \theta) \cos 2\phi \\ &= \frac{v_0}{r^3} \sin^2 \theta \cos 2\phi \end{aligned} \quad (7)$$

It is easy to check that the above are the required solutions by direct substitution.

7.30. Solve Problem 7.18 if the surface potential is $f(\theta, \phi)$.

As in Problem 7.29 we are led to the following solutions inside and outside the sphere:

Inside the sphere, $0 \leq r < 1$

$$v(r, \theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} r^n P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (1)$$

Outside the sphere, $r > 1$

$$v(r, \theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{P_n^m(\cos \theta)}{r^{n+1}} (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (2)$$

For the case $r = 1$ both of these lead to

$$f(\theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} P_n^m(\cos \theta) (A_{mn} \cos m\phi + B_{mn} \sin m\phi)$$

This is equivalent to the expansion

$$F(\xi, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} P_n^m(\xi) (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (3)$$

where $\xi = \cos \theta$. Let us write this as

$$F(\xi, \phi) = \sum_{n=0}^{\infty} C_n P_n^m(\xi) \quad (4)$$

where

$$C_n = \sum_{m=0}^{\infty} (A_{mn} \cos m\phi + B_{mn} \sin m\phi) \quad (5)$$

As in Method 2 of Problem 7.27 we find from (4)

$$C_n = \frac{(2n+1)(n-m)!}{2(n+m)!} \int_{-1}^1 F(\xi, \phi) P_n^m(\xi) d\xi \quad (6)$$

We also see from (5) that A_{mn} and B_{mn} are simply the Fourier coefficients obtained by expansion of C_n (which is a function of ϕ) in a Fourier series. Using the methods of Fourier series it follows that

$$\begin{aligned} A_{0n} &= \frac{1}{2\pi} \int_0^{2\pi} C_n d\phi \\ A_{mn} &= \frac{1}{\pi} \int_0^{2\pi} C_n \cos m\phi d\phi \quad m = 1, 2, 3, \dots \\ B_{mn} &= \frac{1}{\pi} \int_0^{2\pi} C_n \sin m\phi d\phi \quad m = 1, 2, 3, \dots \end{aligned}$$

Combining these results we see that

$$A_{0n} = \frac{(2n+1)(n-m)!}{4\pi(n+m)!} \int_{-1}^1 \int_0^{2\pi} F(\xi, \phi) P_n^m(\xi) d\xi d\phi$$

while for $m = 1, 2, 3, \dots$

$$A_{mn} = \frac{(2n+1)(n-m)!}{2\pi(n+m)!} \int_{-1}^1 \int_0^{2\pi} F(\xi, \phi) P_n^m(\xi) \cos m\phi \, d\xi \, d\phi$$

$$B_{mn} = \frac{(2n+1)(n-m)!}{2\pi(n+m)!} \int_{-1}^1 \int_0^{2\pi} F(\xi, \phi) P_n^m(\xi) \sin m\phi \, d\xi \, d\phi$$

Using these results in (1) and (2) we obtain the required solutions.

Supplementary Problems

LEGENDRE POLYNOMIALS

7.31. Use Rodrigue's formula (4), page 130, to verify the formulas for $P_0(x)$, $P_1(x)$, $\dots$, $P_5(x)$, on page 130.

7.32. Obtain the formulas for $P_4(x)$ and $P_5(x)$ using a recurrence formula.

7.33. Evaluate (a) $\int_0^1 x P_5(x) \, dx$, (b) $\int_{-1}^1 [P_2(x)]^2 \, dx$, (c) $\int_{-1}^1 P_2(x) P_4(x) \, dx$.

7.34. Show that (a) $P_n(1) = 1$ (c) $P_{2n-1}(0) = 0$
 (b) $P_n(-1) = (-1)^n$ (d) $P_{2n}(0) = (-1)^n \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$
 for $n = 1, 2, 3, \dots$

7.35. Use the generating function to prove that $P'_{n+1}(x) - P'_{n-1}(x) = (2n+1)P_n(x)$.

7.36. Prove that (a) $P'_{n+1}(x) - xP'_n(x) = (n+1)P_n(x)$, (b) $xP'_n(x) - P'_{n-1}(x) = nP_n(x)$.

7.37. Show that $\sum_{n=0}^{\infty} P_n(\cos \theta) = \frac{1}{2} \csc \frac{\theta}{2}$.

7.38. Show that (a) $P_2(\cos \theta) = \frac{1}{4}(1 + 3 \cos 2\theta)$, (b) $P_3(\cos \theta) = \frac{1}{8}(3 \cos \theta + 5 \cos 3\theta)$.

7.39. Show that $P_7(x) = \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x)$.

7.40. Show from the generating function that (a) $P_n(1) = 1$, (b) $P_n(-1) = (-1)^n$.

7.41. Show that $\sum_{k=1}^{\infty} \frac{x^k P_{k-1}(x)}{k} = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, $-1 < x < 1$.

LEGENDRE FUNCTIONS OF THE SECOND KIND

7.42. Prove that the series (6) and (7) on page 131 which are nonterminating are convergent for $-1 < x < 1$ but divergent for $x = \pm 1$.

7.43. Find $Q_3(x)$.

7.44. Write the general solution of $(1-x^2)y'' - 2xy' + 2y = 0$.

SERIES OF LEGENDRE POLYNOMIALS

7.45. Expand $x^4 - 3x^2 + x$ in a series of the form $\sum_{k=0}^{\infty} A_k P_k(x)$.

7.46. Expand $f(x) = \begin{cases} 2x + 1 & 0 < x \leq 1 \\ 0 & -1 \leq x < 0 \end{cases}$ in a series of the form $\sum_{k=0}^{\infty} A_k P_k(x)$, writing the first four nonzero terms.

7.47. If $f(x) = \sum_{k=0}^{\infty} A_k P_k(x)$, obtain Parseval's identity

$$\int_{-1}^1 [f(x)]^2 dx = \sum_{k=0}^{\infty} \frac{A_k^2}{2k+1}$$

and illustrate by using the function of Problem 7.45.

SOLUTIONS USING LEGENDRE FUNCTIONS

7.48. Find the potential v (a) interior and (b) exterior to a hollow sphere of unit radius with center at the origin if the surface is charged to potential $v_0(1 + 3 \cos \theta)$ where v_0 is constant.

7.49. Solve Problem 7.48 if the surface potential is $v_0 \sin^2 \theta$.

7.50. Find the steady-state temperature within the region bounded by two concentric spheres of radii a and $2a$ if the temperatures of the outer and inner spheres are u_0 and 0 respectively.

7.51. Find the gravitational potential at any point outside a solid uniform sphere of radius a of mass m .

7.52. Is there a solution to Problem 7.51, if the point is inside the sphere? Explain.

7.53. Interpret Problem 7.48 as a temperature problem.

7.54. Show that the potential due to a uniform spherical shell of inner radius a and outer radius b is given by

$$v = \begin{cases} 2\pi\sigma(b^2 - a^2) & r < a \\ 2\pi\sigma(3b^2r - 2a^3 - r^3)/3r & a < r < b \\ 4\pi\sigma(b^2 - a^2)/3r & r > b \end{cases}$$

7.55. A solid uniform circular disc of radius a and mass M is located in the xy -plane with center at the origin. Show that the gravitational potential at any point of the plane is given by

$$v = \frac{2M}{a} \left[1 - \frac{r}{a} P_1(\cos \theta) + \frac{1}{2} \left(\frac{r}{a} \right)^2 P_2(\cos \theta) - \frac{1}{2 \cdot 4} \left(\frac{r}{a} \right)^4 P_4(\cos \theta) + \frac{1 \cdot 3}{2 \cdot 4 \cdot 6} \left(\frac{r}{a} \right)^6 P_6(\cos \theta) - \dots \right]$$

if $r < a$ and

$$v = \frac{M}{r} \left[1 - \frac{1}{4} \left(\frac{a}{r} \right)^2 P_2(\cos \theta) + \frac{1 \cdot 3}{4 \cdot 6} \left(\frac{a}{r} \right)^4 P_4(\cos \theta) - \frac{1 \cdot 3 \cdot 5}{4 \cdot 6 \cdot 8} \left(\frac{a}{r} \right)^6 P_6(\cos \theta) + \dots \right]$$

if $r > a$.

ASSOCIATED LEGENDRE FUNCTIONS

7.56. Find (a) $P_2^2(x)$, (b) $P_4^2(x)$, (c) $P_4^3(x)$.

7.57. Find (a) $Q_1^1(x)$, (b) $Q_1^2(x)$.

- 7.58. Verify that the expressions for $P_2^1(x)$ and $Q_2^1(x)$ are solutions of the corresponding differential equation and thus write the general solution.
- 7.59. Verify formulas (16) and (17), page 132, for the case where (a) $m = 1$, $n = 1$, $l = 2$, (b) $m = 1$, $n = 1$, $l = 1$.
- 7.60. Obtain a generating function for $P_n^m(x)$.
- 7.61. Use the generating function to obtain results (16) and (17) on page 132.
- 7.62. Show how to expand $f(x)$ in a series of the form $\sum_{k=0}^{\infty} A_k P_k^m(x)$ and illustrate by using the cases (a) $f(x) = x^2$, $m = 2$ and (b) $f(x) = x(1-x)$, $m = 1$. Verify the corresponding Parseval's identity in each case.
- 7.63. Work Problem 7.18 if the potential on the surface is $v_0 \sin^3 \theta \cos \theta \cos 3\phi$.

MISCELLANEOUS PROBLEMS

- 7.64. Show that
$$P_n(x) = \frac{1}{2^n} \sum_{k=0}^{[n/2]} \frac{(-1)^k (2n-2k)!}{k! (n-k)! (n-2k)!} x^{n-2k}$$
 where $[n/2]$ is the largest integer $\leq n/2$.

- 7.65. Show that

$$P_n(x) = \frac{1}{\pi} \int_0^\pi (x + \sqrt{x^2 - 1} \cos u)^n du$$

Use the result to find $P_2(x)$ and $P_3(x)$.

- 7.66. Show that

$$\int_{-1}^1 (1-x^2) P_m'(x) P_n'(x) dx = \begin{cases} 0 & m \neq n \\ \frac{2n(n+1)}{2n+1} & m = n \end{cases}$$

- 7.67. Show that

$$\int_{-1}^1 P_n(x) \ln(1-x) dx = \begin{cases} -2/n(n+1) & n \neq 0 \\ 2(\ln 2 - 1) & n = 0 \end{cases}$$

- 7.68. (a) Show that $\int_{-1}^1 x^m P_n(x) dx = 0$ if $m < n$ or if $m - n$ is an odd positive integer.

- (b) Show that

$$\int_{-1}^1 x^{n+2p} P_n(x) dx = \frac{(n+2p)! \Gamma(p + \frac{1}{2})}{2^n (2p)! \Gamma(p + n + \frac{3}{2})}$$

for any non-negative integers n and p .

- 7.69. Show that a solution of the wave equation

$$\nabla^2 V = \frac{1}{c^2} \frac{\partial^2 V}{\partial t^2}$$

depending on r , θ , and t , but not on ϕ , is given by

$$V = [A_1 J_{n+1/2}(\omega r/c) + B_1 J_{-n-1/2}(\omega r/c)] [A_2 P_n(\cos \theta) + B_2 Q_n(\cos \theta)] [A_3 \cos \omega t + B_3 \sin \omega t]$$

- 7.70. Work Problem 7.69 if there is also ϕ -dependence.
- 7.71. A heat-conducting region is bounded by two concentric spheres of radii a and b ($a < b$) which have their surfaces maintained at constant temperatures u_1 and u_2 respectively. Find the steady-state temperature at any point of the region.
- 7.72. Interpret Problem 7.18 as a temperature problem.
- 7.73. Obtain a solution similar to that given in Problem 7.69 for the heat conduction equation

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u$$

where u depends on r, θ , and t but not on ϕ .



Hermite, Laguerre and Other Orthogonal Polynomials

HERMITE'S DIFFERENTIAL EQUATION. HERMITE POLYNOMIALS

An important equation which arises in problems of physics is called *Hermite's differential equation*; it is given by

$$y'' - 2xy' + 2ny = 0 \quad (1)$$

where $n = 0, 1, 2, 3, \dots$

The equation (1) has polynomial solutions called *Hermite polynomials* given by *Rodrigue's formula*

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}) \quad (2)$$

The first few Hermite polynomials are

$$H_0(x) = 1, \quad H_1(x) = 2x, \quad H_2(x) = 4x^2 - 2, \quad H_3(x) = 8x^3 - 12x \quad (3)$$

Note that $H_n(x)$ is a polynomial of degree n .

GENERATING FUNCTION FOR HERMITE POLYNOMIALS

The generating function for Hermite polynomials is given by

$$e^{2tx - t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n \quad (4)$$

This result is useful in obtaining many properties of $H_n(x)$.

RECURRENCE FORMULAS FOR HERMITE POLYNOMIALS

We can show (see Problems 8.2 and 8.20) that the Hermite polynomials satisfy the recurrence formulas

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x) \quad (5)$$

$$H'_n(x) = 2nH_{n-1}(x) \quad (6)$$

Starting with $H_0(x) = 1$, $H_1(x) = 2x$, we can use (5) to obtain higher-degree Hermite polynomials.

ORTHOGONALITY OF HERMITE POLYNOMIALS

We can show (see Problem 8.4) that

$$\int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx = 0 \quad m \neq n \quad (7)$$

so that the Hermite polynomials are mutually orthogonal with respect to the weight or density function e^{-x^2} .

In the case where $m = n$ we can show (see Problem 8.4) that the left side of (7) becomes

$$\int_{-\infty}^{\infty} e^{-x^2} H_n^2(x) dx = 2^n n! \sqrt{\pi} \quad (8)$$

From this we can normalize the Hermite polynomials so as to obtain an orthonormal set.

SERIES OF HERMITE POLYNOMIALS

Using the orthogonality of the Hermite polynomials it is possible to expand a function in a series having the form

$$f(x) = A_0 H_0(x) + A_1 H_1(x) + A_2 H_2(x) + \dots \quad (9)$$

where

$$A_n = \frac{1}{2^n n! \sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2} f(x) H_n(x) dx \quad (10)$$

See Problem 8.6.

In general such series expansions are possible when $f(x)$ and $f'(x)$ are piecewise continuous.

LAGUERRE'S DIFFERENTIAL EQUATION. LAGUERRE POLYNOMIALS

Another differential equation of importance in physics is *Laguerre's differential equation* given by

$$xy'' + (1-x)y' + ny = 0 \quad (11)$$

where $n = 0, 1, 2, 3, \dots$

This equation has polynomial solutions called *Laguerre polynomials* given by

$$L_n(x) = e^x \frac{d^n}{dx^n} (x^n e^{-x}) \quad (12)$$

which is also referred to as *Rodrigue's formula* for the Laguerre polynomials.

The first few Laguerre polynomials are

$$L_0(x) = 1, \quad L_1(x) = 1 - x, \quad L_2(x) = x^2 - 4x + 2, \quad L_3(x) = 6 - 18x + 9x^2 - x^3 \quad (13)$$

Note that $L_n(x)$ is a polynomial of degree n .

SOME IMPORTANT PROPERTIES OF LAGUERRE POLYNOMIALS

In the following we list some properties of the Laguerre polynomials.

1. Generating function.

$$\frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^n \quad (14)$$

2. Recurrence formulas.

$$L_{n+1}(x) = (2n+1-x)L_n(x) - n^2 L_{n-1}(x) \quad (15)$$

$$L'_n(x) - nL'_{n-1}(x) + nL_{n-1}(x) = 0 \quad (16)$$

$$xL'_n(x) = nL_n(x) - n^2 L_{n-1}(x) \quad (17)$$

3. Orthogonality.

$$\int_0^{\infty} e^{-x} L_m(x) L_n(x) dx = \begin{cases} 0 & \text{if } m \neq n \\ (n!)^2 & \text{if } m = n \end{cases} \quad (18)$$

4. Series expansions.

$$\text{If} \quad f(x) = A_0 L_0(x) + A_1 L_1(x) + A_2 L_2(x) + \dots \quad (19)$$

$$\text{then} \quad A_n = \frac{1}{(n!)^2} \int_0^{\infty} e^{-x} f(x) L_n(x) dx \quad (20)$$

MISCELLANEOUS ORTHOGONAL POLYNOMIALS AND THEIR PROPERTIES

There are many other examples of orthogonal polynomials. Some of the more important ones, together with their properties, are given in the following list.

1. Associated Laguerre polynomials $L_n^m(x)$.

These are polynomials defined by

$$L_n^m(x) = \frac{d^m}{dx^m} L_n(x) \quad (21)$$

and satisfying the equation

$$xy'' + (m+1-x)y' + (n-m)y = 0 \quad (22)$$

If $m > n$ then $L_n^m(x) = 0$.

We have

$$\int_0^{\infty} x^m e^{-x} L_n^m(x) L_p^m(x) dx = 0 \quad p \neq n \quad (23)$$

$$\int_0^{\infty} x^m e^{-x} \{L_n^m(x)\}^2 dx = \frac{(n!)^3}{(n-m)!} \quad (24)$$

2. Chebyshev polynomials $T_n(x)$.

These are polynomials defined by

$$T_n(x) = \cos(n \cos^{-1} x) = x^n - \binom{n}{2} x^{n-2} (1-x^2) + \binom{n}{4} x^{n-4} (1-x^2)^2 - \dots \quad (25)$$

and satisfying the differential equation

$$(1-x^2)y'' - xy' + n^2y = 0 \quad (26)$$

where $n = 0, 1, 2, \dots$

A recurrence formula for $T_n(x)$ is given by

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad (27)$$

and the generating function is

$$\frac{1-tx}{1-2tx+t^2} = \sum_{n=0}^{\infty} T_n(x)t^n \quad (28)$$

We also have

$$\int_{-1}^1 \frac{T_m(x) T_n(x)}{\sqrt{1-x^2}} dx = 0 \quad m \neq n \quad (29)$$

$$\int_{-1}^1 \frac{\{T_n(x)\}^2}{\sqrt{1-x^2}} dx = \begin{cases} \pi & n=0 \\ \pi/2 & n=1, 2, \dots \end{cases} \quad (30)$$

Solved Problems

HERMITE POLYNOMIALS

8.1. Use the generating function for the Hermite polynomials to find (a) $H_0(x)$, (b) $H_1(x)$, (c) $H_2(x)$, (d) $H_3(x)$.

We have

$$e^{2tx-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)t^n}{n!} = H_0(x) + H_1(x)t + \frac{H_2(x)}{2!}t^2 + \frac{H_3(x)}{3!}t^3 + \dots$$

$$\begin{aligned} \text{Now } e^{2tx-t^2} &= 1 + (2tx-t^2) + \frac{(2tx-t^2)^2}{2!} + \frac{(2tx-t^2)^3}{3!} + \dots \\ &= 1 + (2x)t + (2x^2-1)t^2 + \left(\frac{4x^3-6x}{3}\right)t^3 + \dots \end{aligned}$$

Comparing the two series, we have

$$H_0(x) = 1, \quad H_1(x) = 2x, \quad H_2(x) = 4x^2 - 2, \quad H_3(x) = 8x^3 - 12x$$

8.2. Prove that $H'_n(x) = 2nH_{n-1}(x)$.

$$\text{Differentiating } e^{2tx-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n \text{ with respect to } x,$$

$$2te^{2tx-t^2} = \sum_{n=0}^{\infty} \frac{H'_n(x)}{n!} t^n$$

$$\text{or } \sum_{n=0}^{\infty} \frac{2H_n(x)}{n!} t^{n+1} = \sum_{n=0}^{\infty} \frac{H'_n(x)}{n!} t^n$$

Equating coefficients of t^n on both sides,

$$\frac{2H_{n-1}(x)}{(n-1)!} = \frac{H'_n(x)}{n!} \quad \text{or} \quad H'_n(x) = 2nH_{n-1}(x)$$

8.3. Prove that $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2})$.

$$\text{We have } e^{2tx-t^2} = e^{x^2-(t-x)^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n$$

$$\text{Then } \left. \frac{\partial^n}{\partial t^n} (e^{2tx-t^2}) \right|_{t=0} = H_n(x)$$

$$\begin{aligned} \text{But } \left. \frac{\partial^n}{\partial t^n} (e^{2tx-t^2}) \right|_{t=0} &= e^{x^2} \left. \frac{\partial^n}{\partial t^n} [e^{-(t-x)^2}] \right|_{t=0} \\ &= e^{x^2} \left. \frac{\partial^n}{\partial (-x)^n} [e^{-(t-x)^2}] \right|_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}) \end{aligned}$$

8.4. Prove that $\int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx = \begin{cases} 0 & m \neq n \\ 2^n n! \sqrt{\pi} & m = n \end{cases}$.

$$\text{We have } e^{2tx-t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)t^n}{n!}, \quad e^{2sx-s^2} = \sum_{m=0}^{\infty} \frac{H_m(x)s^m}{m!}$$

Multiplying these,

$$e^{2tx-t^2+2sx-s^2} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{H_m(x) H_n(x) s^m t^n}{m! n!}$$

Multiplying by e^{-x^2} and integrating from $-\infty$ to ∞ ,

$$\int_{-\infty}^{\infty} e^{-[(x+s+t)^2 - 2st]} dx = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{g^m t^n}{m! n!} \int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx$$

Now the left side is equal to

$$e^{2st} \int_{-\infty}^{\infty} e^{-(x+s+t)^2} dx = e^{2st} \int_{-\infty}^{\infty} e^{-u^2} du = e^{2st} \sqrt{\pi} = \sqrt{\pi} \sum_{m=0}^{\infty} \frac{2^m g^m t^m}{m!}$$

By equating coefficients the required result follows.

The result

$$\int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x) dx = 0 \quad m \neq n$$

can also be proved by using a method similar to that of Problem 7.13, page 138 (see Problem 8.24).

8.5. Show that the Hermite polynomials satisfy the differential equation

$$y'' - 2xy' + 2ny = 0$$

From (5) and (6), page 154, we have on eliminating $H_{n-1}(x)$:

$$H_{n+1}(x) = 2xH_n(x) - H'_n(x) \quad (1)$$

Differentiating both sides we have

$$H'_{n+1}(x) = 2xH'_n(x) + 2H_n(x) - H''_n(x) \quad (2)$$

But from (6), page 154, we have on replacing n by $n+1$:

$$H'_{n+1}(x) = 2(n+1)H_n(x) \quad (3)$$

Using (3) in (2) we then find on simplifying:

$$H''_n(x) - 2xH'_n(x) + 2nH_n(x) = 0$$

which is the required result.

We can also proceed as in Problem 8.25.

8.6. (a) If $f(x) = \sum_{k=0}^{\infty} A_k H_k(x)$ show that $A_k = \frac{1}{2^k k! \sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2} f(x) H_k(x) dx$.

(b) Expand x^3 in a series of Hermite polynomials.

(a) If $f(x) = \sum_{k=0}^{\infty} A_k H_k(x)$ then on multiplying both sides by $e^{-x^2} H_n(x)$ and integrating term by term from $-\infty$ to ∞ (assuming this to be possible) we arrive at

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) H_n(x) dx = \sum_{k=0}^{\infty} A_k \int_{-\infty}^{\infty} e^{-x^2} H_k(x) H_n(x) dx \quad (1)$$

But from Problem 8.4

$$\int_{-\infty}^{\infty} e^{-x^2} H_k(x) H_n(x) dx = \begin{cases} 0 & k \neq n \\ 2^n n! \sqrt{\pi} & k = n \end{cases}$$

Thus (1) becomes

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) H_n(x) dx = A_n 2^n n! \sqrt{\pi}$$

or

$$A_n = \frac{1}{2^n n! \sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2} f(x) H_n(x) dx \quad (2)$$

which yields the required result on replacing n by k .

- (b) We must find coefficients A_k , $k = 1, 2, 3, \dots$, such that

$$x^3 = \sum_{k=0}^{\infty} A_k H_k(x) \quad (3)$$

Method 1.

The expansion (3) can be written

$$x^3 = A_0 H_0(x) + A_1 H_1(x) + A_2 H_2(x) + A_3 H_3(x) + \dots \quad (4)$$

$$\text{or} \quad x^3 = A_0(1) + A_1(2x) + A_2(4x^2 - 2) + A_3(8x^3 - 12x) + \dots \quad (5)$$

Since $H_k(x)$ is a polynomial of degree k we see that we must have $A_4 = 0$, $A_5 = 0$, $A_6 = 0$, $\dots$; otherwise the left side of (5) is a polynomial of degree 3 while the right side would be a polynomial of degree greater than 3. Thus we have from (5)

$$x^3 = (A_0 - 2A_2) + (2A_1 - 12A_3)x + 4A_2x^2 + 8A_3x^3$$

Then equating coefficients of like powers of x on both sides we find

$$8A_3 = 1, \quad 4A_2 = 0, \quad 2A_1 - 12A_3 = 0, \quad A_0 - 2A_2 = 0$$

from which

$$A_0 = 0, \quad A_1 = \frac{3}{4}, \quad A_2 = 0, \quad A_3 = \frac{1}{8}$$

Thus (2) becomes

$$x^3 = \frac{3}{4}H_1(x) + \frac{1}{8}H_3(x)$$

which is the required expansion.

Check.

$$\frac{3}{4}H_1(x) + \frac{1}{8}H_3(x) = \frac{3}{4}(2x) + \frac{1}{8}(8x^3 - 12x) = x^3$$

Method 2.

The coefficients A_k in (1) are given by

$$A_k = \frac{1}{2^k k! \sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2} x^3 H_k(x) dx$$

as obtained in part (a) with $f(x) = x^3$.

Putting $k = 0, 1, 2, 3, 4, \dots$ and integrating we then find

$$A_0 = 0, \quad A_1 = \frac{3}{4}, \quad A_2 = 0, \quad A_3 = \frac{1}{8}, \quad A_4 = 0, \quad A_5 = 0, \quad \dots$$

and we are led to the same result as in Method 1.

In general, for expansion of polynomials the first of the above methods will be easier and faster.

- 8.7. (a) Write Parseval's identity corresponding to the series expansion $f(x) = \sum_{k=0}^{\infty} A_k H_k(x)$.

- (b) Verify the result of part (a) for the case where $f(x) = x^3$.

- (a) We can obtain Parseval's identity formally by first squaring both sides of $f(x) = \sum_{k=0}^{\infty} A_k H_k(x)$ to obtain

$$\{f(x)\}^2 = \sum_{k=0}^{\infty} \sum_{p=0}^{\infty} A_k A_p H_k(x) H_p(x)$$

Then multiplying by e^{-x^2} and integrating from $-\infty$ to ∞ we find

$$\int_{-\infty}^{\infty} e^{-x^2} \{f(x)\}^2 dx = \sum_{k=0}^{\infty} \sum_{p=0}^{\infty} A_k A_p \int_{-\infty}^{\infty} e^{-x^2} H_k(x) H_p(x) dx$$

Making use of the results of Problem 8.4 this can be written as

$$\int_{-\infty}^{\infty} e^{-x^2} \{f(x)\}^2 dx = \sqrt{\pi} \sum_{k=0}^{\infty} 2^k k! A_k^2$$

which is Parseval's identity for the Hermite polynomials.

- (b) From Problem 8.6 it follows that if $f(x) = x^3$ then $A_0 = 0$, $A_1 = \frac{3}{4}$, $A_2 = 0$, $A_3 = \frac{1}{8}$, $A_4 = 0$, $A_5 = 0$, ... Thus Parseval's identity becomes

$$\int_{-\infty}^{\infty} e^{-x^2} \{x^3\}^2 dx = \sqrt{\pi} [2(1!)(\frac{3}{4})^2 + 2^3(3!)(\frac{1}{8})^2]$$

Now the right side reduces to $15\sqrt{\pi}/8$. The left side is

$$\begin{aligned} \int_{-\infty}^{\infty} x^6 e^{-x^2} dx &= 2 \int_0^{\infty} x^6 e^{-x^2} dx = \int_0^{\infty} u^{5/2} e^{-u} du \\ &= \Gamma(\frac{7}{2}) = (\frac{5}{2})(\frac{3}{2})(\frac{1}{2})\sqrt{\pi} \\ &= \frac{15}{8}\sqrt{\pi} \end{aligned}$$

where we have made the transformation $x = \sqrt{u}$. Thus Parseval's identity is verified.

LAGUERRE POLYNOMIALS

- 8.8. Determine the Laguerre polynomials (a) $L_0(x)$, (b) $L_1(x)$, (c) $L_2(x)$, (d) $L_3(x)$.

We have $L_n(x) = e^x \frac{d^n}{dx^n} (x^n e^{-x})$. Then

$$\begin{aligned} (a) \quad L_0(x) &= 1 \\ (b) \quad L_1(x) &= e^x \frac{d}{dx} (x e^{-x}) = 1 - x \\ (c) \quad L_2(x) &= e^x \frac{d^2}{dx^2} (x^2 e^{-x}) = 2 - 4x + x^2 \\ (d) \quad L_3(x) &= e^x \frac{d^3}{dx^3} (x^3 e^{-x}) = 6 - 18x + 9x^2 - x^3 \end{aligned}$$

- 8.9. Prove that the Laguerre polynomials $L_n(x)$ are orthogonal in $(0, \infty)$ with respect to the weight function e^{-x} .

From Laguerre's differential equation we have for any two Laguerre polynomials $L_m(x)$ and $L_n(x)$,

$$\begin{aligned} xL_m'' + (1-x)L_m' + mL_m &= 0 \\ xL_n'' + (1-x)L_n' + nL_n &= 0 \end{aligned}$$

Multiplying these equations by L_n and L_m respectively and subtracting, we find

$$x[L_n L_m'' - L_m L_n''] + (1-x)[L_n L_m' - L_m L_n'] = (n-m)L_m L_n$$

$$\text{or} \quad \frac{d}{dx} [L_n L_m' - L_m L_n'] + \frac{1-x}{x} [L_n L_m' - L_m L_n'] = \frac{(n-m)L_m L_n}{x}$$

Multiplying by the integrating factor

$$e^{\int (1-x)/x dx} = e^{\ln x - x} = x e^{-x}$$

this can be written as

$$\frac{d}{dx} \{x e^{-x} [L_n L_m' - L_m L_n']\} = (n-m) e^{-x} L_m L_n$$

so that by integrating from 0 to ∞ ,

$$(n-m) \int_0^{\infty} e^{-x} L_m(x) L_n(x) dx = x e^{-x} [L_n L'_m - L_m L'_n] \Big|_0^{\infty} = 0.$$

Thus if $m \neq n$,

$$\int_0^{\infty} e^{-x} L_m(x) L_n(x) dx = 0$$

which proves the required result.

8.10. Prove that $L_{n+1}(x) = (2n+1-x)L_n(x) - n^2 L_{n-1}(x)$.

The generating function for the Laguerre polynomials is

$$\frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^n \quad (1)$$

Differentiating both sides with respect to t yields

$$\frac{e^{-xt/(1-t)}}{(1-t)^2} - \frac{x e^{-xt/(1-t)}}{(1-t)^3} = \sum_{n=0}^{\infty} \frac{n L_n(x)}{n!} t^{n-1} \quad (2)$$

Multiplying both sides by $(1-t)^2$ and using (1) on the left side we find

$$\sum_{n=0}^{\infty} (1-t) \frac{L_n(x)}{n!} t^n - \sum_{n=0}^{\infty} \frac{x L_n(x)}{n!} t^n = \sum_{n=0}^{\infty} (1-t)^2 \frac{n L_n(x)}{n!} t^{n-1}$$

which can be written as

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^n - \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^{n+1} - \sum_{n=0}^{\infty} \frac{x L_n(x)}{n!} t^n \\ = \sum_{n=0}^{\infty} \frac{n L_n(x)}{n!} t^{n-1} - \sum_{n=0}^{\infty} \frac{2n L_n(x)}{n!} t^n + \sum_{n=0}^{\infty} \frac{n L_n(x)}{n!} t^{n+1} \end{aligned}$$

If we now equate coefficients of t^n on both sides of this equation we find

$$\frac{L_n(x)}{n!} - \frac{L_{n-1}(x)}{(n-1)!} - \frac{x L_n(x)}{n!} = \frac{(n+1)L_{n+1}(x)}{(n+1)!} - \frac{2n L_n(x)}{n!} + \frac{(n-1)L_{n-1}(x)}{(n-1)!}$$

Multiplying by $n!$ and simplifying we then obtain, as required,

$$L_{n+1}(x) = (2n+1-x)L_n(x) - n^2 L_{n-1}(x)$$

8.11. Expand $x^3 + x^2 - 3x + 2$ in a series of Laguerre polynomials, i.e. $\sum_{k=0}^{\infty} A_k L_k(x)$.

We shall use a method similar to Method 1 of Problem 8.6(b). Since we must expand a polynomial of degree 3 we need only take terms up to $L_3(x)$. Thus

$$x^3 + x^2 - 3x + 2 = A_0 L_0(x) + A_1 L_1(x) + A_2 L_2(x) + A_3 L_3(x)$$

Using the results of Problem 8.8 this can be written

$$x^3 + x^2 - 3x + 2 = (A_0 + A_1 + 2A_2 + 6A_3) - (A_1 + 4A_2 + 18A_3)x + (A_2 + 9A_3)x^2 - A_3 x^3$$

Then, equating like powers of x on both sides we have

$$A_0 + A_1 + 2A_2 + 6A_3 = 2, \quad A_1 + 4A_2 + 18A_3 = 3, \quad A_2 + 9A_3 = 1, \quad -A_3 = 1$$

Solving these we find

$$A_0 = 7, \quad A_1 = -19, \quad A_2 = 10, \quad A_3 = -1$$

Then the required expansion is

$$x^3 + x^2 - 3x + 2 = 7L_0(x) - 19L_1(x) + 10L_2(x) - L_3(x)$$

We can also work the problem by using (19) and (20), page 156.

MISCELLANEOUS ORTHOGONAL POLYNOMIALS

8.12. Obtain the associated Laguerre polynomials (a) $L_2^1(x)$, (b) $L_2^2(x)$, (c) $L_3^2(x)$, (d) $L_3^4(x)$.

$$(a) \quad L_2^1(x) = \frac{d}{dx} L_2(x) = \frac{d}{dx} (2 - 4x + x^2) = 2x - 4$$

$$(b) \quad L_2^2(x) = \frac{d^2}{dx^2} L_2(x) = \frac{d^2}{dx^2} (2 - 4x + x^2) = 2$$

$$(c) \quad L_3^2(x) = \frac{d^2}{dx^2} L_3(x) = \frac{d^2}{dx^2} (6 - 18x + 9x^2 - x^3) = 18 - 6x$$

$$(d) \quad L_3^4(x) = \frac{d^4}{dx^4} L_3(x) = 0. \quad \text{In general } L_n^m(x) = 0 \text{ if } m > n.$$

8.13. Verify the result (24), page 156, for $m = 1$, $n = 2$.

We must show that

$$\int_0^\infty x e^{-x} \{L_2^1(x)\}^2 dx = \frac{(2!)^3}{1!} = 8$$

Now since $L_2^1(x) = 2x - 4$ by Problem 8.12(a) we have

$$\begin{aligned} \int_0^\infty x e^{-x} (2x - 4)^2 dx &= 4 \int_0^\infty x^3 e^{-x} dx - 16 \int_0^\infty x^2 e^{-x} dx + 16 \int_0^\infty x e^{-x} dx \\ &= 4 \Gamma(4) - 16 \Gamma(3) + 16 \Gamma(2) \\ &= 4(3!) - 16(2!) + 16(1!) \\ &= 8 \end{aligned}$$

so that the result is verified.

8.14. Verify the result (23), page 156, with $m = 2$, $n = 2$, $p = 3$.

We must show that

$$\int_0^\infty x^2 e^{-x} L_2^2(x) L_3^2(x) dx = 0$$

Since $L_2^2(x) = 2$, $L_3^2(x) = 18 - 6x$ by Problem 8.12(a) and (b) respectively the integral is

$$\begin{aligned} \int_0^\infty x^2 e^{-x} (2)(18 - 6x) dx &= 36 \int_0^\infty x^2 e^{-x} dx - 12 \int_0^\infty x^3 e^{-x} dx \\ &= 36 \Gamma(3) - 12 \Gamma(4) \\ &= 36(2!) - 12(3!) = 0 \end{aligned}$$

as required.

8.15. Verify that $L_3^2(x)$ satisfies the differential equation (22), page 156, in the special case $m = 2$, $n = 3$.

From Problem 8.12(c) we have $L_3^2(x) = 18 - 6x$. The differential equation (22), page 156, with $m = 2$, $n = 3$ is

$$xy'' + (3 - x)y' + y = 0$$

Substituting $y = 18 - 6x$ in this equation we have

$$x(0) + (3 - x)(-6) + 18 - 6x = 0$$

which is an identity. Thus $L_3^2(x)$ satisfies the differential equation.

8.16. Show that the Chebyshev polynomial $T_n(x)$ is given by

$$T_n(x) = x^n - \binom{n}{2}x^{n-2}(1-x^2) + \binom{n}{4}x^{n-4}(1-x^2)^2 - \binom{n}{6}x^{n-6}(1-x^2)^3 + \dots$$

We have by definition

$$T_n(x) = \cos(n \cos^{-1} x)$$

Let $u = \cos^{-1} x$ so that $x = \cos u$. Then $T_n(x) = \cos nu$. Now by De Moivre's theorem

$$(\cos u + i \sin u)^n = \cos nu + i \sin nu$$

Thus $\cos nu$ is the real part of $(\cos u + i \sin u)^n$. But this expansion is, by the binomial theorem,

$$(\cos u)^n + \binom{n}{1}(\cos u)^{n-1}(i \sin u) + \binom{n}{2}(\cos u)^{n-2}(i \sin u)^2 + \binom{n}{3}(\cos u)^{n-3}(i \sin u)^3 + \dots$$

and the real part of this is given by

$$\cos^n u - \binom{n}{2}\cos^{n-2}u \sin^2 u + \binom{n}{4}\cos^{n-4}u \sin^4 u - \dots$$

Then since $\cos u = x$ and $\sin^2 u = 1 - x^2$, this becomes

$$x^n - \binom{n}{2}x^{n-2}(1-x^2) + \binom{n}{4}x^{n-4}(1-x^2)^2 - \dots$$

8.17. Find (a) $T_2(x)$ and (b) $T_3(x)$.

Using Problem 8.16 we find for $n = 2$ and $n = 3$ respectively:

$$(a) \quad T_2(x) = x^2 - \binom{2}{2}x^0(1-x^2) = x^2 - (1-x^2) = 2x^2 - 1$$

$$(b) \quad T_3(x) = x^3 - \binom{3}{2}x^1(1-x^2) = x^3 - 3x(1-x^2) = 4x^3 - 3x$$

Another method.

Since $T_0(x) = \cos 0 = 1$, $T_1(x) = \cos(\cos^{-1} x) = x$ we have from the recurrence formula (27), page 156, on putting $n = 1$ and $n = 2$ respectively,

$$T_2(x) = 2xT_1(x) - T_0(x) = 2x^2 - 1$$

$$T_3(x) = 2xT_2(x) - T_1(x) = 2x(2x^2 - 1) - x = 4x^3 - 3x$$

8.18. Verify that $T_n(x) = \cos(n \cos^{-1} x)$ satisfies the differential equation

$$(1-x^2)y'' - xy' + n^2y = 0$$

for the case $n = 3$.

From Problem 8.17(b), $T_3(x) = 4x^3 - 3x$ and the differential equation for $n = 3$ is

$$(1-x^2)y'' - xy' + 9y = 0$$

Then if $y = 4x^3 - 3x$ the left side becomes

$$(1-x^2)(24x) - x(12x^2 - 3) + 9(4x^3 - 3x) = 0$$

so that the differential equation reduces to an identity.

Supplementary Problems

HERMITE POLYNOMIALS

- 8.19. Use Rodrigue's formula (2), page 154, to obtain the Hermite polynomials $H_0(x)$, $H_1(x)$, $H_2(x)$, $H_3(x)$.
- 8.20. Use the generating function to obtain the recurrence formula (5) on page 154, and obtain $H_2(x)$, $H_3(x)$ given that $H_0(x) = 1$, $H_1(x) = 2x$.
- 8.21. Show directly that (a) $\int_{-\infty}^{\infty} e^{-x^2} H_2(x) H_3(x) dx = 0$, (b) $\int_{-\infty}^{\infty} e^{-x^2} [H_2(x)]^2 dx = 8\sqrt{\pi}$.
- 8.22. Evaluate $\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) dx$.
- 8.23. Show that $H_{2n}(0) = \frac{(-1)^n (2n)!}{n!}$.
- 8.24. Prove the result (7), page 154, by using a method similar to that in Problem 7.13, pages 138 and 139.
- 8.25. Work Problem 8.5, page 158, by using (a) Rodrigue's formula, (b) the method of Frobenius.
- 8.26. (a) Expand $f(x) = x^3 - 3x^2 + 2x$ in a series of the form $\sum_{k=0}^{\infty} A_k H_k(x)$. (b) Verify Parseval's identity for the function in part (a).
- 8.27. Find the general solution of Hermite's differential equation for the cases (a) $n = 0$ and (b) $n = 1$.

LAGUERRE POLYNOMIALS

- 8.28. Find $L_4(x)$ and show that it satisfies Laguerre's equation (11), page 155, for $n = 4$.
- 8.29. Use the generating function to obtain the recurrence formula (16) on page 155.
- 8.30. Use formula (15) to determine $L_2(x)$, $L_3(x)$ and $L_4(x)$ if we define $L_n(x) = 0$ when $n = -1$ and $L_n(x) = 1$ when $n = 0$.
- 8.31. Show that $nL_{n-1}(x) = nL'_{n-1}(x) - L'_n(x)$.
- 8.32. Prove that $\int_0^{\infty} e^{-x} \{L_n(x)\}^2 dx = (n!)^2$.
- 8.33. Prove the results (19) and (20), page 156.
- 8.34. Expand $f(x) = x^3 - 3x^2 + 2x$ in a series of the form $\sum_{k=0}^{\infty} A_k L_k(x)$.
- 8.35. Illustrate Parseval's identity for Problem 8.34.
- 8.36. Find the general solution of Laguerre's differential equation for $n = 0$.
- 8.37. Obtain Laguerre's differential equation (11), page 155, from the generating function (14), page 155.

MISCELLANEOUS ORTHOGONAL POLYNOMIALS

- 8.38. Find (a) $L_4^2(x)$, (b) $L_5^3(x)$.
- 8.39. Verify the results (23) and (24), page 156, for $m = 2$, $n = 3$.
- 8.40. Verify that $L_4^2(x)$ satisfies the differential equation (22), page 156, in the special case $m = 2$, $n = 4$.

8.41. Evaluate $\int_0^{\infty} x^2 e^{-x} L_4^2(x) dx$.

8.42. Show that a generating function for the associated Laguerre polynomials is given by

$$\frac{(-t)^m e^{-xt/(1-t)}}{(1-t)^{m+1}} = \sum_{k=m}^{\infty} \frac{L_k^m(x)}{k!} t^k$$

8.43. Solve Chebyshev's differential equation (26), page 156, for the case where $n = 0$.

8.44. Find (a) $T_4(x)$ and (b) $T_5(x)$.

8.45. Expand $f(x) = x^3 + x^2 - 4x + 2$ in a series of Chebyshev polynomials $\sum_{k=0}^{\infty} A_k T_k(x)$.

8.46. (a) Write Parseval's identity corresponding to the expansion of $f(x)$ in a series of Chebyshev polynomials and (b) verify the identity by using the function of Problem 8.45.

8.47. Prove the recurrence formula (27), page 156.

8.48. Prove the results (29) and (30) on page 156.

MISCELLANEOUS PROBLEMS

8.49. (a) Find the general solution of Hermite's differential equation. (b) Write the general solution for the cases where $n = 1$ and $n = 2$. [Hint: Let $y = vH_n(x)$ and determine v so that Hermite's equation is satisfied.]

8.50. In quantum mechanics the Schrodinger equation for a harmonic oscillator is given by

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}(E - \frac{1}{2}\kappa x^2)\psi = 0$$

where E, m, h, κ are constants. Show that solutions of this equation are given by

$$\psi = C_n H_n(x/a) e^{-x^2/4a^2}$$

where $n = 0, 1, 2, 3, \dots$ and

$$a = \sqrt{\frac{4}{16\pi^2\kappa m}} \quad E = (n + \frac{1}{2}) \frac{h}{2\pi} \sqrt{\frac{\kappa}{m}}$$

The differential equation is a Sturm-Liouville differential equation whose eigenvalues and eigenfunctions are given by E and ψ respectively.

8.51. (a) Find the general solution of Laguerre's differential equation. (b) Write the general solution for the cases $n = 1$ and $n = 2$. [Hint: Let $y = vL_n(x)$. See also Problem 8.49.]

8.52. Prove the results (18) on page 156 by using the generating function.

8.53. (a) Show that Laguerre's associated differential equation (22), page 156, is obtained by differentiating Laguerre's equation (11) m times with respect to x , and thus (b) show that a solution is $d^m L_n/dx^m$.

8.54. Prove the results (23) and (24) on page 156.

8.55. (a) Find the general solution of Chebyshev's differential equation. (b) Write the general solution for the cases $n = 1$ and $n = 2$. [Hint: Let $y = vT_n(x)$.]

8.56. Discuss the theory of (a) Hermite polynomials, (b) Laguerre polynomials, (c) associated Laguerre polynomials, and (d) Chebyshev polynomials from the viewpoint of Sturm-Liouville theory.

8.57. Discuss the relationship between the expansion of a function in Fourier series and in Chebyshev polynomials.



Appendix A

Uniqueness of Solutions

A proof establishing the uniqueness of solutions to boundary value problems can often be accomplished by assuming the existence of two solutions and then arriving at a contradiction. We illustrate the procedure by an example involving heat conduction.

Consider a finite closed region $\mathcal{R}$ having surface S . Suppose that the initial temperature inside $\mathcal{R}$ and the surface temperature are specified. Then the boundary value problem for the temperature $u(x, y, z, t)$ at any point (x, y, z) at time t is given by

$$\frac{\partial u}{\partial t} = \kappa \nabla^2 u \quad \text{inside } \mathcal{R} \quad (1)$$

$$u(x, y, z, 0) = f(x, y, z) \quad \text{at points } (x, y, z) \text{ of } \mathcal{R} \quad (2)$$

$$u(x, y, z, t) = g(x, y, z, t) \quad \text{at points } (x, y, z) \text{ of } S \quad (3)$$

We shall assume that all functions are at least differentiable at points of $\mathcal{R}$ and S .

Assume the existence of two different solutions, say u_1 and u_2 , of the above boundary value problem. Then letting $U = u_1 - u_2$ we find that U satisfies the boundary value problem

$$\frac{\partial U}{\partial t} = \kappa \nabla^2 U \quad \text{inside } \mathcal{R} \quad (4)$$

$$U(x, y, z, 0) = 0 \quad \text{at points of } \mathcal{R} \quad (5)$$

$$U(x, y, z, t) = 0 \quad \text{at points of } S \quad (6)$$

Let us now consider

$$W(t) = \frac{1}{2} \iiint_{\mathcal{R}} [U(x, y, z, t)]^2 dx dy dz \quad (7)$$

where the integration is performed over the region $\mathcal{R}$. Using (5) we see that

$$W(0) = 0 \quad (8)$$

Also from (7) we have

$$\frac{dW}{dt} = \iiint_{\mathcal{R}} U \frac{\partial U}{\partial t} dx dy dz = \kappa \iiint_{\mathcal{R}} U \nabla^2 U dx dy dz \quad (9)$$

where we have used (4).

We now make use of *Green's theorem* to show that

$$\iiint_{\mathcal{R}} U \nabla^2 U dx dy dz = \iint_S U \frac{\partial U}{\partial n} dS - \iiint_{\mathcal{R}} \left[\left(\frac{\partial U}{\partial x} \right)^2 + \left(\frac{\partial U}{\partial y} \right)^2 + \left(\frac{\partial U}{\partial z} \right)^2 \right] dx dy dz \quad (10)$$

where n is a unit outward-drawn normal to S . Since $U = 0$ on S the first integral on the right of (10) is zero and we have

$$\iiint_{\mathcal{R}} U \nabla^2 U \, dx \, dy \, dz = - \iiint_{\mathcal{R}} \left[\left(\frac{\partial U}{\partial x} \right)^2 + \left(\frac{\partial U}{\partial y} \right)^2 + \left(\frac{\partial U}{\partial z} \right)^2 \right] dx \, dy \, dz \quad (11)$$

Thus we have from (9),

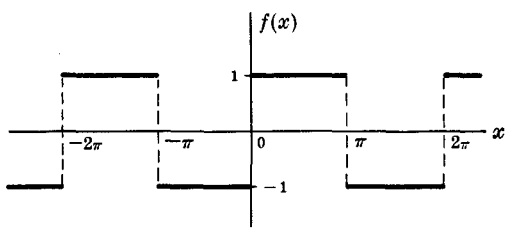
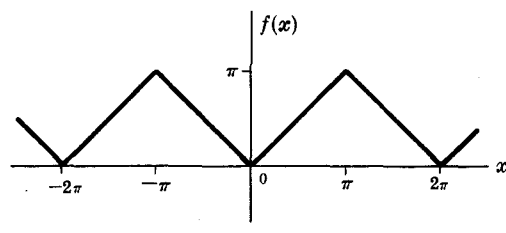
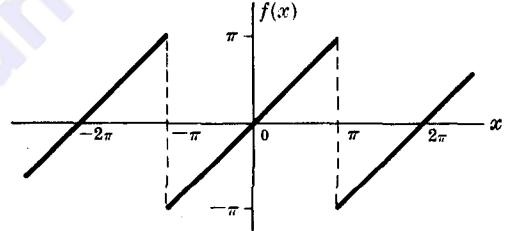
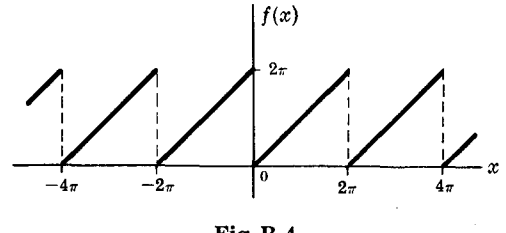
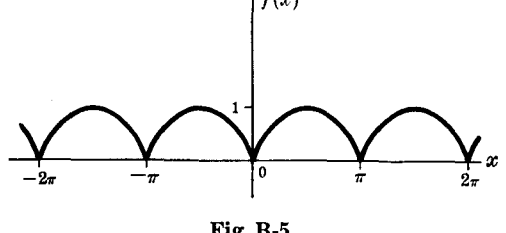
$$\frac{dW}{dt} = -\kappa \iiint_{\mathcal{R}} \left[\left(\frac{\partial U}{\partial x} \right)^2 + \left(\frac{\partial U}{\partial y} \right)^2 + \left(\frac{\partial U}{\partial z} \right)^2 \right] dx \, dy \, dz \quad (12)$$

It follows from this that $dW/dt \leq 0$, i.e. W is a nonincreasing function of t , and, in view of (8), that $W(t) \leq 0$. But from (7) we see that $W(t) \geq 0$. Thus it follows that $W(t) = 0$ identically.

Now if $U(x, y, z, t)$ is not zero at a point of $\mathcal{R}$ it follows by its continuity that there will be a neighborhood of the point in which it is not zero. Then the integral in (7) would have to be greater than zero, i.e. $W(t) > 0$. This contradiction with $W(t) = 0$ shows that $U(x, y, z, t)$ must be identically zero, which shows that $u_1 = u_2$ and the solution is unique.



Special Fourier Series

B-1 $f(x) = \begin{cases} 1 & 0 < x < \pi \\ -1 & -\pi < x < 0 \end{cases}$ $\frac{4}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots \right)$	 Fig. B-1
B-2 $f(x) = x = \begin{cases} x & 0 < x < \pi \\ -x & -\pi < x < 0 \end{cases}$ $\frac{\pi}{2} - \frac{4}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right)$	 Fig. B-2
B-3 $f(x) = x, -\pi < x < \pi$ $2 \left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots \right)$	 Fig. B-3
B-4 $f(x) = x, 0 < x < 2\pi$ $\pi - 2 \left(\frac{\sin x}{1} + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \dots \right)$	 Fig. B-4
B-5 $f(x) = \sin x , -\pi < x < \pi$ $\frac{2}{\pi} - \frac{4}{\pi} \left(\frac{\cos 2x}{1 \cdot 3} + \frac{\cos 4x}{3 \cdot 5} + \frac{\cos 6x}{5 \cdot 7} + \dots \right)$	 Fig. B-5

B-6

$$f(x) = \begin{cases} \sin x & 0 < x < \pi \\ 0 & \pi < x < 2\pi \end{cases}$$

$$\frac{1}{\pi} + \frac{1}{2} \sin x - \frac{2}{\pi} \left(\frac{\cos 2x}{1 \cdot 3} + \frac{\cos 4x}{3 \cdot 5} + \frac{\cos 6x}{5 \cdot 7} + \dots \right)$$

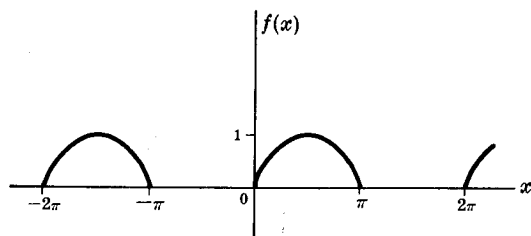


Fig. B-6

B-7

$$f(x) = \begin{cases} \cos x & 0 < x < \pi \\ -\cos x & -\pi < x < 0 \end{cases}$$

$$\frac{8}{\pi} \left(\frac{\sin 2x}{1 \cdot 3} + \frac{2 \sin 4x}{3 \cdot 5} + \frac{3 \sin 6x}{5 \cdot 7} + \dots \right)$$

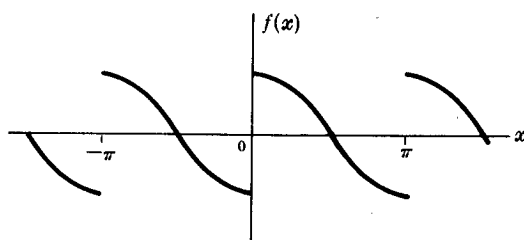


Fig. B-7

B-8

$$f(x) = x^2, \quad -\pi < x < \pi$$

$$\frac{\pi^2}{3} - 4 \left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \dots \right)$$

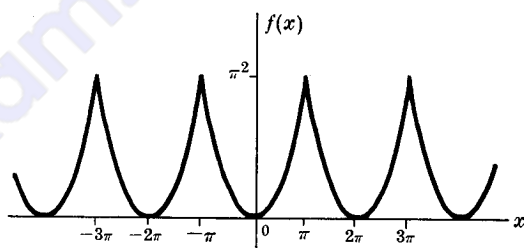


Fig. B-8

B-9

$$f(x) = x(\pi - x), \quad 0 < x < \pi$$

$$\frac{\pi^2}{6} - \left(\frac{\cos 2x}{1^2} + \frac{\cos 4x}{2^2} + \frac{\cos 6x}{3^2} + \dots \right)$$

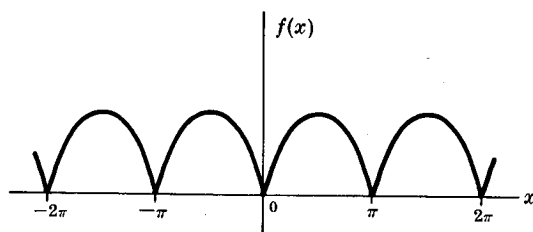


Fig. B-9

B-10

$$f(x) = x(\pi - x)(\pi + x), \quad -\pi < x < \pi$$

$$12 \left(\frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \dots \right)$$

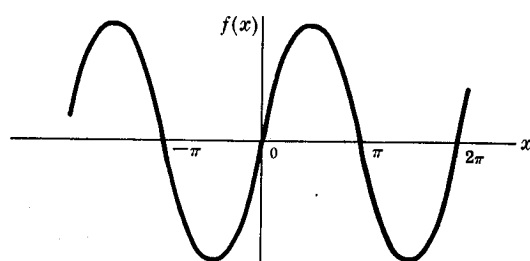
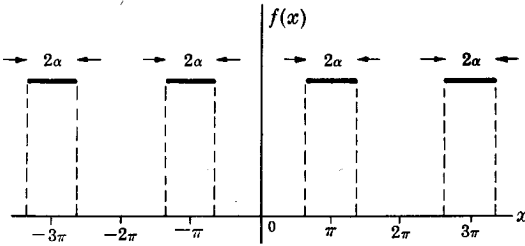
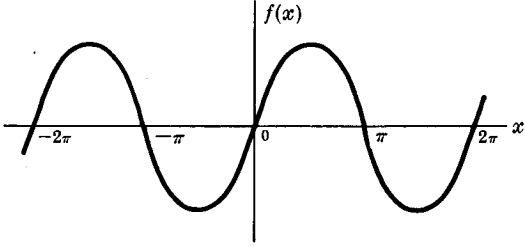


Fig. B-10

B-11 $f(x) = \begin{cases} 0 & 0 < x < \pi - \alpha \\ 1 & \pi - \alpha < x < \pi + \alpha \\ 0 & \pi + \alpha < x < 2\pi \end{cases}$	 Fig. B-11
$\frac{\alpha}{\pi} - \frac{2}{\pi} \left(\frac{\sin \alpha \cos x}{1} - \frac{\sin 2\alpha \cos 2x}{2} + \frac{\sin 3\alpha \cos 3x}{3} - \dots \right)$	 Fig. B-12
B-12 $f(x) = \begin{cases} x(\pi - x) & 0 < x < \pi \\ -x(\pi - x) & -\pi < x < 0 \end{cases}$	
$\frac{8}{\pi} \left(\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \dots \right)$	
B-13 $f(x) = \sin \mu x, \quad -\pi < x < \pi, \quad \mu \neq \text{integer}$ $\frac{2 \sin \mu \pi}{\pi} \left(\frac{\sin x}{1^2 - \mu^2} - \frac{2 \sin 2x}{2^2 - \mu^2} + \frac{3 \sin 3x}{3^2 - \mu^2} - \dots \right)$	
B-14 $f(x) = \cos \mu x, \quad -\pi < x < \pi, \quad \mu \neq \text{integer}$ $\frac{2\mu \sin \mu \pi}{\pi} \left(\frac{1}{2\mu^2} + \frac{\cos x}{1^2 - \mu^2} - \frac{\cos 2x}{2^2 - \mu^2} + \frac{\cos 3x}{3^2 - \mu^2} - \dots \right)$	
B-15 $f(x) = \tan^{-1} [(a \sin x)/(1 - a \cos x)], \quad -\pi < x < \pi, \quad a < 1$ $a \sin x + \frac{a^2}{2} \sin 2x + \frac{a^3}{3} \sin 3x + \dots$	
B-16 $f(x) = \ln (1 - 2a \cos x + a^2), \quad -\pi < x < \pi, \quad a < 1$ $-2 \left(a \cos x + \frac{a^2}{2} \cos 2x + \frac{a^3}{3} \cos 3x + \dots \right)$	
B-17 $f(x) = \frac{1}{2} \tan^{-1} [(2a \sin x)/(1 - a^2)], \quad -\pi < x < \pi, \quad a < 1$ $a \sin x + \frac{a^3}{3} \sin 3x + \frac{a^5}{5} \sin 5x + \dots$	
B-18 $f(x) = \frac{1}{2} \tan^{-1} [(2a \cos x)/(1 - a^2)], \quad -\pi < x < \pi, \quad a < 1$ $a \cos x - \frac{a^3}{3} \cos 3x + \frac{a^5}{5} \cos 5x - \dots$	

B-19 $f(x) = e^{\mu x}, -\pi < x < \pi$

$$\frac{2 \sinh \mu \pi}{\pi} \left(\frac{1}{2\mu} + \sum_{n=1}^{\infty} \frac{(-1)^n (\mu \cos nx - n \sin nx)}{\mu^2 + n^2} \right)$$

B-20 $f(x) = \sinh \mu x, -\pi < x < \pi$

$$\frac{2 \sinh \mu \pi}{\pi} \left(\frac{\sin x}{1^2 + \mu^2} - \frac{2 \sin 2x}{2^2 + \mu^2} + \frac{3 \sin 3x}{3^2 + \mu^2} - \dots \right)$$

B-21 $f(x) = \cosh \mu x, -\pi < x < \pi$

$$\frac{2\mu \sinh \mu \pi}{\pi} \left(\frac{1}{2\mu^2} - \frac{\cos x}{1^2 + \mu^2} + \frac{\cos 2x}{2^2 + \mu^2} - \frac{\cos 3x}{3^2 + \mu^2} + \dots \right)$$

B-22 $f(x) = \ln |\sin \frac{1}{2}x|, -\pi < x < \pi$

$$-\left(\ln 2 + \frac{\cos x}{1} + \frac{\cos 2x}{2} + \frac{\cos 3x}{3} + \dots \right)$$

B-23 $f(x) = \ln |\cos \frac{1}{2}x|, -\pi < x < \pi$

$$-\left(\ln 2 - \frac{\cos x}{1} + \frac{\cos 2x}{2} - \frac{\cos 3x}{3} + \dots \right)$$

B-24 $f(x) = \frac{1}{6}\pi^2 - \frac{1}{2}\pi x + \frac{1}{4}x^2, 0 \leq x \leq 2\pi$

$$\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \dots$$

B-25 $f(x) = \frac{1}{12}x(x-\pi)(x-2\pi), 0 \leq x \leq 2\pi$

$$\frac{\sin x}{1^3} + \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} + \dots$$

B-26 $f(x) = \frac{1}{90}\pi^4 - \frac{1}{12}\pi^2 x^2 + \frac{1}{12}\pi x^3 - \frac{1}{48}x^4, 0 \leq x \leq 2\pi$

$$\frac{\cos x}{1^4} + \frac{\cos 2x}{2^4} + \frac{\cos 3x}{3^4} + \dots$$

Appendix C

Special Fourier Transforms

SPECIAL FOURIER TRANSFORM PAIRS

	$f(x)$	$F(\alpha)$
C-1	$\begin{cases} 1 & x < b \\ 0 & x > b \end{cases}$	$\frac{2 \sin b\alpha}{\alpha}$
C-2	$\frac{1}{x^2 + b^2}$	$\frac{\pi e^{-b\alpha}}{b}$
C-3	$\frac{x}{x^2 + b^2}$	$-\frac{\pi i\alpha}{b} e^{-b\alpha}$
C-4	$f^{(n)}(x)$	$i^n \alpha^n F(\alpha)$
C-5	$x^n f(x)$	$i^n \frac{d^n F}{d\alpha^n}$
C-6	$f(bx)e^{itz}$	$\frac{1}{b} F\left(\frac{\alpha - t}{b}\right)$

SPECIAL FOURIER COSINE TRANSFORMS

	$f(x)$	$F_C(\alpha)$
C-7	$\begin{cases} 1 & 0 < x < b \\ 0 & x > b \end{cases}$	$\frac{\sin b\alpha}{\alpha}$
C-8	$\frac{1}{x^2 + b^2}$	$\frac{\pi e^{-b\alpha}}{2b}$
C-9	e^{-bx}	$\frac{b}{\alpha^2 + b^2}$
C-10	$x^{n-1} e^{-bx}$	$\frac{\Gamma(n) \cos(n \tan^{-1} \alpha/b)}{(\alpha^2 + b^2)^{n/2}}$
C-11	e^{-bx^2}	$\frac{1}{2} \sqrt{\frac{\pi}{b}} e^{-\alpha^2/4b}$
C-12	$x^{-1/2}$	$\sqrt{\frac{\pi}{2\alpha}}$
C-13	x^{-n}	$\frac{\pi \alpha^{n-1} \sec(n\pi/2)}{2\Gamma(n)}, \quad 0 < n < 1$
C-14	$\ln\left(\frac{x^2 + b^2}{x^2 + c^2}\right)$	$\frac{e^{-c\alpha} - e^{-b\alpha}}{\pi\alpha}$
C-15	$\frac{\sin bx}{x}$	$\begin{cases} \pi/2 & \alpha < b \\ \pi/4 & \alpha = b \\ 0 & \alpha > b \end{cases}$
C-16	$\sin bx^2$	$\sqrt{\frac{\pi}{8b}} \left(\cos \frac{\alpha^2}{4b} - \sin \frac{\alpha^2}{4b} \right)$
C-17	$\cos bx^2$	$\sqrt{\frac{\pi}{8b}} \left(\cos \frac{\alpha^2}{4b} + \sin \frac{\alpha^2}{4b} \right)$
C-18	$\operatorname{sech} bx$	$\frac{\pi}{2b} \operatorname{sech} \frac{\pi\alpha}{2b}$
C-19	$\frac{\cosh(\sqrt{\pi} x/2)}{\cosh(\sqrt{\pi} x)}$	$\sqrt{\frac{\pi}{2}} \frac{\cosh(\sqrt{\pi} \alpha/2)}{\cosh(\sqrt{\pi} \alpha)}$
C-20	$\frac{e^{-b\sqrt{x}}}{\sqrt{x}}$	$\sqrt{\frac{\pi}{2\alpha}} \{ \cos(2b\sqrt{\alpha}) - \sin(2b\sqrt{\alpha}) \}$

SPECIAL FOURIER SINE TRANSFORMS

	$f(x)$	$F_S(\alpha)$
C-21	$\begin{cases} 1 & 0 < x < b \\ 0 & x > b \end{cases}$	$\frac{1 - \cos b\alpha}{\alpha}$
C-22	x^{-1}	$\frac{\pi}{2}$
C-23	$\frac{x}{x^2 + b^2}$	$\frac{\pi}{2} e^{-b\alpha}$
C-24	e^{-bx}	$\frac{\alpha}{\alpha^2 + b^2}$
C-25	$x^{n-1} e^{-bx}$	$\frac{\Gamma(n) \sin(n \tan^{-1} \alpha/b)}{(\alpha^2 + b^2)^{n/2}}$
C-26	xe^{-bx^2}	$\frac{\sqrt{\pi}}{4b^{3/2}} \alpha e^{-\alpha^2/4b}$
C-27	$x^{-1/2}$	$\sqrt{\frac{\pi}{2\alpha}}$
C-28	x^{-n}	$\frac{\pi \alpha^{n-1} \csc(n\pi/2)}{2 \Gamma(n)} \quad 0 < n < 2$
C-29	$\frac{\sin bx}{x}$	$\frac{1}{2} \ln \left(\frac{\alpha + b}{\alpha - b} \right)$
C-30	$\frac{\sin bx}{x^2}$	$\begin{cases} \pi\alpha/2 & \alpha < b \\ \pi b/2 & \alpha > b \end{cases}$
C-31	$\frac{\cos bx}{x}$	$\begin{cases} 0 & \alpha < b \\ \pi/4 & \alpha = b \\ \pi/2 & \alpha > b \end{cases}$
C-32	$\tan^{-1}(x/b)$	$\frac{\pi}{2\alpha} e^{-b\alpha}$
C-33	$\csc bx$	$\frac{\pi}{2b} \tanh \frac{\pi\alpha}{2b}$
C-34	$\frac{1}{e^{2x} - 1}$	$\frac{\pi}{4} \coth \left(\frac{\pi\alpha}{2} \right) - \frac{1}{2\alpha}$

Appendix D

Tables of Values for $J_0(x)$ and $J_1(x)$

$J_0(x)$

x	0	1	2	3	4	5	6	7	8	9
0.	1.0000	.9975	.9900	.9776	.9604	.9385	.9120	.8812	.8463	.8075
1.	.7652	.7196	.6711	.6201	.5669	.5118	.4554	.3980	.3400	.2818
2.	.2239	.1666	.1104	.0555	.0025	-.0484	-.0968	-.1424	-.1850	-.2243
3.	-.2601	-.2921	-.3202	-.3443	-.3643	-.3801	-.3918	-.3992	-.4026	-.4018
4.	-.3971	-.3887	-.3766	-.3610	-.3423	-.3205	-.2961	-.2693	-.2404	-.2097
5.	-.1776	-.1443	-.1103	-.0758	-.0412	-.0068	.0270	.0599	.0917	.1220
6.	.1506	.1773	.2017	.2238	.2433	.2601	.2740	.2851	.2931	.2981
7.	.3001	.2991	.2951	.2882	.2786	.2663	.2516	.2346	.2154	.1944
8.	.1717	.1475	.1222	.0960	.0692	.0419	.0146	-.0125	-.0392	-.0653
9.	-.0903	-.1142	-.1367	-.1577	-.1768	-.1939	-.2090	-.2218	-.2323	-.2403

$J_1(x)$

x	0	1	2	3	4	5	6	7	8	9
0.	.0000	.0499	.0995	.1483	.1960	.2423	.2867	.3290	.3688	.4059
1.	.4401	.4709	.4983	.5220	.5419	.5579	.5699	.5778	.5815	.5812
2.	.5767	.5683	.5560	.5399	.5202	.4971	.4708	.4416	.4097	.3754
3.	.3391	.3009	.2613	.2207	.1792	.1374	.0955	.0538	.0128	-.0272
4.	-.0660	-.1033	-.1386	-.1719	-.2028	-.2311	-.2566	-.2791	-.2985	-.3147
5.	-.3276	-.3371	-.3432	-.3460	-.3453	-.3414	-.3343	-.3241	-.3110	-.2951
6.	-.2767	-.2559	-.2329	-.2081	-.1816	-.1538	-.1250	-.0953	-.0652	-.0349
7.	-.0047	.0252	.0543	.0826	.1096	.1352	.1592	.1813	.2014	.2192
8.	.2346	.2476	.2580	.2657	.2708	.2731	.2728	.2697	.2641	.2559
9.	.2453	.2324	.2174	.2004	.1816	.1613	.1395	.1166	.0928	.0684

Appendix E

Zeros of Bessel Functions

The following table lists the first few positive roots of $J_n(x) = 0$ and $J'_n(x) = 0$. Note that for all cases listed successive large roots differ approximately by $\pi = 3.14159\dots$

	$n = 0$	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$	$n = 6$
$J_n(x) = 0$	2.4048	3.8317	5.1356	6.3802	7.5883	8.7715	9.9361
	5.5201	7.0156	8.4172	9.7610	11.0647	12.3386	13.5893
	8.6537	10.1735	11.6198	13.0152	14.3725	15.7002	17.0038
	11.7915	13.3237	14.7960	16.2235	17.6160	18.9801	20.3208
	14.9309	16.4706	17.9598	19.4094	20.8269	22.2178	23.5861
	18.0711	19.6159	21.1170	22.5827	24.0190	25.4303	26.8202
$J'_n(x) = 0$	0.0000	1.8412	3.0542	4.2012	5.3176	6.4156	7.5013
	3.8317	5.3314	6.7061	8.0152	9.2824	10.5199	11.7349
	7.0156	8.5363	9.9695	11.3459	12.6819	13.9872	15.2682
	10.1735	11.7060	13.1704	14.5859	15.9641	17.3128	18.6374
	13.3237	14.8636	16.3475	17.7888	19.1960	20.5755	21.9317
	16.4706	18.0155	19.5129	20.9725	22.4010	23.8036	25.1839



Answers to Supplementary Problems

CHAPTER 1

1.27. $u(0, t) = T_1, \quad u(L, t) = T_2, \quad u(x, 0) = f(x), \quad |u(x, t)| < M$

1.28. (a) $u_x(0, t) = \frac{\partial u}{\partial x}\bigg|_{x=0} = 0, \quad u_x(L, t) = \frac{\partial u}{\partial x}\bigg|_{x=L} = 0, \quad u(x, 0) = f(x), \quad |u(x, t)| < M$

(b) $u_x(0, t) = \frac{\partial u}{\partial x}\bigg|_{x=0} = B(u_1 - u_0), \quad u_x(L, t) = \frac{\partial u}{\partial x}\bigg|_{x=L} = B(u_2 - u_0),$

$u(x, 0) = f(x), \quad |u(x, t)| < M$

where $u_1 = u(0, t), \quad u_2 = u(L, t)$

1.31. $\frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}, \quad y(x, 0) = \begin{cases} 2hx/L & 0 \leq x \leq L/2 \\ 2h(L-x)/L & L/2 \leq x \leq L \end{cases},$

$y(0, t) = 0, \quad y(L, t) = 0, \quad y_t(x, 0) = 0, \quad |y(x, t)| < M$

- 1.32. (a) linear, dep. var. u , ind. var. x, y , order 2 (d) linear, dep. var. y , ind. var. x, t , order 2
(b) linear, dep. var. T , ind. var. x, y, z , order 4 (e) nonlinear, dep. var. z , ind. var. r, s , order 1
(c) nonlinear, dep. var. ϕ , ind. var. x, y , order 3

- 1.33. (a) hyperbolic (b) hyperbolic (c) elliptic (d) parabolic
(e) elliptic if $x^2 + y^2 < 1$, hyperbolic if $x^2 + y^2 > 1$, parabolic if $x^2 + y^2 = 1$
(f) elliptic if $M < 1$, hyperbolic if $M > 1$, parabolic if $M = 1$

1.35. (b) $x(2x + y - 2)^2$

1.36. $3 \frac{\partial^2 u}{\partial x^2} - 5 \frac{\partial^2 u}{\partial x \partial y} - 2 \frac{\partial^2 u}{\partial y^2} = 0$

1.37. (a) $2 \frac{\partial z}{\partial x} + 3 \frac{\partial z}{\partial y} = 2z$ (b) $2 \frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} = 0$

1.38. (a) $xz = F(x) + G(y)$ (b) $xz = x^6 + x^2 + 6y^4 - 68$

1.39. (a) $u = F(x + y) + G(x - y)$ (d) $z = F(3x + y) + G(y - x)$
(b) $u = e^{3x}F(y - 2x)$ (e) $z = F(x + y) + xG(x + y)$
(c) $u = F(x + iy) + G(x - iy)$

1.40. (a) $u = F(y - 2x) + \frac{x^2}{2}$ (c) $u = F(y) + xG(y) + x^2H(y) + I(y - 2x) + \frac{x^4}{6}$

(b) $y = F(x - t) + G(x + t) - t^4$ (d) $z = F(x + y) + G(2x + y) - \frac{x}{2} \sin y + \frac{3}{4} \cos y$

1.41. $u = F(x + iy) + G(x - iy) + xH(x + iy) + xJ(x - iy) + (x^2 + y^2)^{2/4}$

1.43. (a) $u = 4e^{(3y-2x)/2}$ (e) $u = 8e^{-2x-6t}$

(b) $u = 3e^{-5x-3y} + 2e^{-3x-2y}$ (f) $u = 10e^{-x-3t} - 6e^{-4x-6t}$

(c) $u = 2e^{-36t} \sin 3x - 4e^{-100t} \sin 5x$ (g) $u = 6e^{-\pi^2 t/4} \sin(\pi x/2) + 3e^{-\pi^2 t} \sin \pi x$

(d) $u = 8e^{-9\pi^2 t/16} \cos \frac{3\pi x}{4} - 6e^{-81\pi^2 t/16} \cos \frac{9\pi x}{4}$

1.44. (a) $y = \frac{5}{2\pi} \sin \pi x \sin 2\pi t$ (b) $y = \frac{3}{4\pi} \sin 2\pi x \sin 4\pi t - \frac{1}{5\pi} \sin 5\pi x \sin 10\pi t$

1.45. $u = e^{-2t}(2e^{-\pi^2 t} \sin \pi x - e^{-16\pi^2 t} \sin 4\pi x)$

CHAPTER 2

$$2.34. \quad (a) \quad \frac{16}{\pi} \sum_{n=1}^{\infty} \frac{(1 - \cos n\pi)}{n} \sin \frac{n\pi x}{2} \quad (c) \quad 20 - \frac{40}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi x}{5}$$

$$(b) \quad 2 - \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{(1 - \cos n\pi)}{n^2} \cos \frac{n\pi x}{4} \quad (d) \quad \frac{3}{2} + \sum_{n=1}^{\infty} \left\{ \frac{6(\cos n\pi - 1)}{n^2 \pi^2} \cos \frac{n\pi x}{3} - \frac{6 \cos n\pi}{n\pi} \sin \frac{n\pi x}{3} \right\}$$

$$2.35. \quad (a) \quad x = 0, \pm 2, \pm 4, \dots; 0 \quad (c) \quad x = 0, \pm 10, \pm 20, \dots; 20$$

$$(b) \quad \text{no discontinuities} \quad (d) \quad x = \pm 3, \pm 9, \pm 15, \dots; 3$$

$$2.36. \quad \frac{16}{\pi} \left\{ \cos \frac{\pi x}{4} + \frac{1}{3^2} \cos \frac{3\pi x}{4} + \frac{1}{5^2} \cos \frac{5\pi x}{4} + \dots \right\}$$

$$2.37. \quad (a) \quad \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{n \sin 2n\pi}{4n^2 - 1} \quad (b) \quad f(0) = f(\pi) = 0$$

2.38. Same answer as 2.37.

$$2.39. \quad (a) \quad \frac{32}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi}{2} \sin \frac{n\pi x}{8} \quad (b) \quad \frac{16}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{2 \cos n\pi/2 - \cos n\pi - 1}{n^2} \right) \cos \frac{n\pi x}{8}$$

$$2.42. \quad u(x, t) = \psi(x) + \frac{2}{L} \sum_{n=1}^{\infty} \left\{ \int_0^{\infty} [f(u) - \psi(u)] \sin \frac{n\pi u}{L} du \right\} e^{-\kappa n^2 \pi^2 t / L^2} \sin \frac{n\pi x}{L}$$

$$\text{where} \quad \psi(x) = \frac{\beta(1 - e^{-\beta L})}{\kappa L} - \frac{\beta}{\kappa} (1 - e^{-\gamma x})$$

$$2.50. \quad (a) \quad u(x, t) = -\frac{200}{\pi} \sum_{m=1}^{\infty} \frac{e^{-m^2 \pi^2 t / 8} \cos m\pi}{m} \sin \frac{m\pi x}{4}$$

$$2.52. \quad 150 - 5x$$

$$2.53. \quad u(\rho, \phi) = 120 + 60\rho^2 \cos 2\phi$$

$$2.55. \quad y(x, t) = \frac{.96}{\pi^3} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} \sin \frac{(2n-1)\pi x}{2} \cos \frac{(2n-1)\pi ct}{2}$$

$$2.57. \quad u(x, y) = \sum_{k=1}^{\infty} \left[\frac{2}{a \sinh(k\pi)} \int_0^a f_1(x) \sin \frac{k\pi x}{a} dx \right] \sin \frac{k\pi x}{a} \sinh \frac{k\pi y}{a} \\ + \sum_{l=1}^{\infty} \left[\frac{2}{a \sinh(l\pi)} \int_0^a f_2(x) \sin \frac{l\pi x}{a} dx \right] \sin \frac{l\pi x}{a} \sinh \frac{l\pi}{a} (a - y) \\ + \sum_{m=0}^{\infty} \left[\frac{2}{a \sinh(m\pi)} \int_0^a g_1(y) \sin \frac{m\pi y}{a} dy \right] \sin \frac{m\pi y}{a} \sinh \frac{m\pi x}{a} \\ + \sum_{n=0}^{\infty} \left[\frac{2}{a \sinh(n\pi)} \int_0^a g_2(y) \sin \frac{n\pi y}{a} dy \right] \sin \frac{n\pi y}{a} \sinh \frac{n\pi x}{a}$$

$$2.59. \quad y(x, t) = \sum_{k=1}^{\infty} \left[\frac{2}{k\pi a} \int_0^L g(u) \sin \frac{k\pi u}{L} \sin \frac{k\pi at}{L} du + \frac{2}{L} \int_0^L f(u) \sin \frac{k\pi u}{L} \cos \frac{k\pi at}{L} du \right] \sin \frac{k\pi x}{L}$$

$$\text{or} \quad \frac{1}{2} [f(x+at) + f(x-at)] + \frac{1}{2a} \int_{x-at}^{x+at} g(u) du$$

$$2.61. \quad z(x, y, t) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} [A_{mn} \cos \lambda_{mn} at + B_{mn} \sin \lambda_{mn} at] \sin \frac{m\pi x}{b} \sin \frac{n\pi y}{c}$$

where

$$A_{mn} = \frac{4}{bc} \int_0^b \int_0^c f(x, y) \sin \frac{m\pi x}{b} \sin \frac{n\pi y}{c} dx dy,$$

$$B_{mn} = \frac{4}{abc\lambda_{mn}} \int_0^b \int_0^c g(x, y) \sin \frac{m\pi x}{b} \sin \frac{n\pi y}{c} dx dy,$$

and

$$\lambda_{mn} = \pi \sqrt{\frac{m^2}{b^2} + \frac{n^2}{c^2}}$$

$$2.63. \quad u(x, t) = \psi(x) - \frac{2}{L} \sum_{n=1}^{\infty} e^{-(\alpha^2 + n^2\pi^2/L^2)t} \left[\int_0^L \psi(u) \sin \frac{n\pi u}{L} du \right] \sin \frac{n\pi x}{L}$$

where

$$\psi(x) = \frac{(u_1 e^{\alpha L} - u_2) e^{-\alpha x} - (u_1 e^{-\alpha L} - u_2) e^{\alpha x}}{e^{\alpha L} - e^{-\alpha L}}$$

$$2.64. \quad u(x, t) = \psi(x) + \frac{2}{L} \sum_{n=1}^{\infty} e^{-(\alpha^2 + n^2\pi^2/L^2)t} \left[\int_0^L (f(u) - \psi(u)) \sin \frac{n\pi u}{L} du \right] \sin \frac{n\pi x}{L}$$

where

$$\psi(x) = \frac{(u_1 e^{\alpha L} - u_2) e^{-\alpha x} - (u_1 e^{-\alpha L} - u_2) e^{\alpha x}}{e^{\alpha L} - e^{-\alpha L}}$$

$$2.65. \quad y(x, t) = \frac{2}{L} \sum_{n=1}^{\infty} \left[\int_0^L f(x) \sin \frac{n\pi x}{L} dx \right] \sin \frac{n\pi x}{L} \cos \frac{n^2\pi^2 b t}{L^2}$$

$$2.66. \quad y(x, t) = \frac{2}{L} \sum_{n=1}^{\infty} \left[\left\{ \int_0^L f(u) \sin \frac{n\pi u}{L} du \right\} \sin \frac{n\pi x}{L} \cos \frac{n^2\pi^2 b t}{L^2} \right. \\ \left. + \left\{ \frac{L^2}{n^2\pi^2 b} \int_0^L g(u) \sin \frac{n\pi u}{L} du \right\} \sin \frac{n^2\pi^2 b t}{L^2} \sin \frac{n\pi x}{L} \right]$$

$$2.69. \quad u(x, t) = \psi(x) + \frac{2}{L} \sum_{n=1}^{\infty} \left\{ \int_0^L [f(u) - \psi(u)] \sin \frac{n\pi u}{L} du \right\} e^{-(\kappa n^2\pi^2/L^2)t} \sin \frac{n\pi x}{L}$$

where

$$\psi(x) = \frac{B(1 - e^{-\gamma L})x}{\gamma^2 \kappa L} + \frac{B}{\kappa \gamma^2} (1 - e^{-\gamma x})$$

$$2.70. \quad \text{Same as 2.69 but with } \psi(x) = \frac{u_0}{\kappa \alpha^2} \left(\sin \alpha x - \frac{\kappa}{L} \sin \alpha L \right)$$

$$2.71. \quad y(x, t) = \psi(x) + \frac{2}{L} \sum_{n=1}^{\infty} \left\{ \int_0^L (f(u) - \psi(u)) \sin \frac{n\pi u}{L} du \right\} \sin \frac{n\pi x}{L} \cos \frac{n\pi a t}{L} \quad \text{where } \psi(x) = \frac{gx}{2} (x - L)$$

$$2.72. \quad u(x, y, z) = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} B_{mn} \sin m\pi x \sin n\pi y \sinh \sqrt{m^2 + n^2} \pi z$$

where

$$B_{mn} = \frac{4}{\sinh \sqrt{m^2 + n^2} \pi} \int_0^1 \int_0^1 f(x, y) \sin m\pi x \sin n\pi y dx dy$$

$$2.76. \quad u(r, \theta) = \frac{2}{\beta} \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n \left\{ \int_0^{\beta} f(\phi) \sin \frac{n\pi \phi}{\beta} d\phi \right\} \sin \frac{n\pi \theta}{\beta}$$

CHAPTER 3

3.15. (a) $a_0 = 1$, $a_1 = -\sqrt{3}$, $a_2 = 2\sqrt{3}$, $a_3 = \sqrt{5}$, $a_4 = -6\sqrt{5}$, $a_5 = 6\sqrt{5}$

(b) 1 , $\sqrt{3}(2x-1)$, $\sqrt{5}(6x^2-6x+1)$

3.17. (b) 1 , $1-x$, $1-2x+\frac{1}{2}x^2$

3.19. (b) $\sqrt{\frac{1}{\pi}} \cos(0 \cos^{-1} x) = \sqrt{\frac{1}{\pi}}$, $\sqrt{\frac{2}{\pi}} \cos(n \cos^{-1} x)$ ($n = 1, 2, \dots$)

3.20. $f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x)$ where $c_n = \int_a^b w(x) f(x) \phi_n(x) dx$

3.21. $f(x) = \frac{c_0}{2} + \sum_{n=1}^{\infty} c_n \phi_n(x)$

where $c_n = \sqrt{\frac{2}{\pi}} \int_{-1}^1 (1-x^2)^{-1/2} f(x) \cos(n \cos^{-1} x) dx = \sqrt{\frac{2}{\pi}} \int_{-\pi/2}^{\pi/2} f(\cos u) \cos nu du$

3.24. (a) $2, -1, \frac{2}{3}$ (b) $\sqrt{(18\pi^3-49)/18\pi}$

3.30. $\sqrt{2}$, $\sqrt{\frac{2}{3}}x$, $\sqrt{\frac{2}{5}}\left(\frac{3x^2-1}{2}\right)$, i.e. the normalized Legendre polynomials

3.31. $1-x$, $\frac{1}{2}(2-4x+x^2)$, $\frac{1}{6}(6-18x+9x^2-x^3)$

3.33. (b) eigenvalues $m\pi$, eigenfunctions $B_m \sin m\pi x$, where $m = 1, 2, \dots$

(c) $\sqrt{2} \sin m\pi x$, $m = 1, 2, \dots$

3.34. (a) $\frac{(2m-1)\pi}{2}$, $B_m \cos \frac{(2m-1)\pi x}{2}$, $\sqrt{2} \cos \frac{(2m-1)\pi x}{2}$, $m = 1, 2, \dots$

(b) $m\pi$, $B_m \sin m\pi x$, $\sqrt{2} \sin m\pi x$, $m = 1, 2, \dots$

3.40. $u(x, t) = \frac{2}{L} \sum_{n=1}^{\infty} \left\{ \int_0^L f(u) \cos \frac{(2n-1)\pi u}{2L} du \right\} \cos \frac{(2n-1)\pi x}{2L} e^{-\kappa(2n-1)^2 \pi^2 t / 4L^2}$

(b) Heat conduction in an infinite strip of width L , with one side at 0° , the other side insulated, and initial temperature distribution given by $f(x)$.

3.41. (a) $y(x, t) = \frac{2}{L} \sum_{n=1}^{\infty} \left[\int_0^L f(u) \sin \frac{(2n-1)\pi u}{2L} \pi u du \right] \sin \frac{(2n-1)\pi x}{2L} \cos \frac{(2n-1)\pi at}{2L}$

(b) Vibrating string with end $x = 0$ fixed, end $x = L$ free, initial shape $f(x)$, initial speed zero

CHAPTER 4

4.26. (a) 30, (b) $16/105$, (c) $\frac{8}{3}\pi^{3/2}$

4.37. (a) $1/105$, (b) $4/15$, (c) $2\pi/\sqrt{3}$

4.27. (a) 24, (b) $\frac{80}{243}$, (c) $\frac{\sqrt{2\pi}}{16}$

4.38. (a) $1/60$, (b) $\pi/2$, (c) 3π

4.28. (a) $\frac{1}{3} \Gamma(\frac{1}{3})$, (b) $\frac{3\sqrt{\pi}}{2}$, (c) $\frac{\Gamma(4/5)}{5\sqrt[5]{16}}$

4.39. (a) 12π , (b) π

4.31. (a) 24, (b) $-3/128$, (c) $\frac{1}{8} \Gamma(\frac{1}{8})$

4.41. (a) $3\pi/256$, (b) $5\pi/8$

4.32. (a) $(16\sqrt{\pi})/105$, (b) $-3 \Gamma(2/3)$

4.42. (a) $16/15$, (b) $8/105$

CHAPTER 5

5.24. (a) $\frac{\sin \alpha \epsilon}{\alpha \epsilon}$ (b) 1

5.30. (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{4}$

5.25. (a) $\frac{4}{\alpha^3}(\alpha \cos \alpha - \sin \alpha)$ (b) $\frac{3\pi}{16}$

5.37. $y(u) = \left(\frac{4}{\pi}\right)^{1/4} e^{-2u^2}$

5.26. (a) $\frac{1 - \cos \alpha}{\alpha}$ (b) $\frac{\sin \alpha}{\alpha}$

5.44. $u(x, y) = \frac{2}{\pi} \tan^{-1} \frac{x}{y}$

5.27. (a) $\frac{\alpha}{1 + \alpha^2}$

5.45. $u(x, y) = \frac{u_0}{2} + \frac{u_0}{\pi} \tan^{-1} \frac{x}{y}$

5.28. $y(x) = (2 + 2 \cos x - 4 \cos 2x)/\pi x$

5.46. $u(x, y) = \frac{1}{\pi} \left[\tan^{-1} \left(\frac{1+x}{y} \right) + \tan^{-1} \left(\frac{1-x}{y} \right) \right]$

CHAPTER 6

6.37. (a) $\sqrt{\frac{2}{\pi x}} \left[\frac{(3-x^2) \sin x - 3x \cos x}{x^2} \right]$

(b) $\sqrt{\frac{2}{\pi x}} \left[\frac{3x \sin x + (3-x^2) \cos x}{x^2} \right]$

6.38. $\left(\frac{8-x^2}{x^2} \right) J_1(x) - \frac{4}{x} J_0(x)$

6.40. (a) $x^3 J_3(x) + c$ (b) $2J_0(1) - 3J_1(1)$ (c) $x^2 J_1(x) + x J_0(x) - \int J_0(x) dx$

6.41. (a) $6\sqrt[3]{x} J_1(\sqrt[3]{x}) - 3\sqrt[3]{x^2} J_0(\sqrt[3]{x}) + c$ (b) $-\frac{J_2(x)}{3x} - \frac{J_1(x)}{3} + \frac{1}{3} \int J_0(x) dx$

6.42. $x J_0(x) \sin x - x J_1(x) \cos x + c$

6.57. (a) $-\sqrt{\frac{2}{\pi x}} \cos x$ (b) $\sqrt{\frac{2}{\pi x}} \sin x$ (c) $-\sqrt{\frac{2}{\pi x}} \left(\sin x + \frac{\cos x}{x} \right)$ (d) $\sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$

6.59. (a) $x^3 Y_3(x) + c$ (b) $-Y_2(x) - 2Y_1(x)/x + c$

(c) $-\frac{1}{15} Y_1(x) - \frac{1}{15x} Y_2(x) - \frac{1}{5x^2} Y_3(x) + \frac{1}{15} \int Y_0(x) dx$

6.68. (a) $\sqrt{\frac{2}{\pi x}} e^{i(x - \pi/4 - n\pi/2)}$ (b) $\sqrt{\frac{2}{\pi x}} e^{-i(x - \pi/4 - n\pi/2)}$

6.72. $y = A J_0(\sqrt{x}) + B Y_0(\sqrt{x})$

6.73. (a) $y = \frac{A \sin x + B \cos x}{x}$ (b) $y = \sqrt{x} [A J_{1/4}(\frac{1}{2}x^2) + B J_{-1/4}(\frac{1}{2}x^2)]$

6.74. $y = A J_0(e^x) + B Y_0(e^x)$

6.75. (b) $y = A J_0(2\sqrt{x}) + B Y_0(2\sqrt{x})$

6.76. (b) $y = A\sqrt{x} J_{1/3}(\frac{2}{3}x^{3/2}) + B\sqrt{x} J_{-1/3}(\frac{2}{3}x^{3/2})$

6.78. $y = Ax J_1(x) + Bx Y_1(x)$

6.95. $u(\rho, \phi, t) = \sum_{n=1}^{\infty} A_n J_3(\lambda_n \rho) \cos 3\phi \cos \lambda_n t$ where λ_n are the positive roots of $J_3(\lambda) = 0$ and

$$A_n = \frac{2[(\lambda_n^2 - 8)J_0(\lambda_n) - 6\lambda_n J_1(\lambda_n) + 8]}{\lambda_n^3 J_4^2(\lambda_n)}$$

6.96. $y(x, t) = \sum_{n=1}^{\infty} \frac{J_0(\lambda_n \sqrt{x})}{J_1^2(\lambda_n)} \cos(\frac{1}{2}\lambda_n t) \int_0^1 f(x) J_0(\lambda_n \sqrt{x}) dx$ where $J_0(\lambda_n) = 0$, $n = 1, 2, \dots$

6.97. (a) $u(\rho, \phi, z) = \frac{2}{a^2 \pi} \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} \frac{J_n(\lambda_k \rho) \sinh \lambda_k(\rho - z)}{J_{n+1}^2(\lambda_k a) \sinh \lambda_k a} (A_{n,k} \sin n\phi + B_{n,k} \cos n\phi)$

where

$$A_{n,k} = \int_0^a \int_0^{2\pi} \rho f(\rho, \phi) J_n(\lambda_k \rho) \sin n\phi \, d\rho \, d\phi$$

$$B_{n,k} = \int_0^a \int_0^{2\pi} \rho f(\rho, \phi) J_n(\lambda_k \rho) \cos n\phi \, d\rho \, d\phi$$

and $J_n(\lambda_k a) = 0$

(b) $u(\rho, \phi, z) = \frac{2}{a^2} \sum_{k=0}^{\infty} \frac{J_1(\lambda_k \rho) \sinh \lambda_k(\rho - z)}{J_2^2(\lambda_k a) \sinh \lambda_k a} B_k \cos \phi$

where

$$B_k = \frac{1}{\lambda_k^3} \left[(3 - \lambda_k^2) J_0(\lambda_k a) - 3 \int_0^a J_0(\lambda_k \rho) \, d\rho \right]$$

and $J_1(\lambda_k a) = 0$

6.113. $u(\rho, z, t) = \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} c_{km} e^{-\kappa(r_m^2 + k^2 \pi^2)t} J_0(r_m \rho) \sin k\pi z$

where

$$c_{km} = \frac{4}{J_0^2(r_m)} \int_0^1 \int_0^1 \rho J_0(r_m \rho) f(\rho, z) \sin k\pi z \, d\rho \, dz$$

and $J_1(r_m) = 0$

6.117. $z(\rho, \phi, t) = \sum_{m=0}^{\infty} \frac{1}{2} A_{m0} u_0(\lambda_{m0} \rho) \cos \lambda_{m0} \phi \cos \lambda_{m0} t$
 $+ \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} (A_{mn} \cos n\phi + B_{mn} \sin n\phi) u_n(\lambda_{mn} \rho) \cos \lambda_{mn} t$

where $u_n(\lambda_{mn} \rho) = J_n(\lambda_{mn} \rho) Y_n(\lambda_{mn} a) - J_n(\lambda_{mn} a) Y_n(\lambda_{mn} \rho)$,

$$A_{mn} = \frac{1}{\pi L} \int_0^{2\pi} \int_a^b \rho f(\rho, \phi) u_n(\lambda_{mn} \rho) \cos n\phi \, d\rho \, d\phi,$$

$$B_{mn} = \frac{1}{\pi L} \int_0^{2\pi} \int_a^b \rho f(\rho, \phi) u_n(\lambda_{mn} \rho) \sin n\phi \, d\rho \, d\phi,$$

$$L = \int_a^b \rho [u_n(\lambda_{mn} \rho)]^2 \, d\rho$$

and $c = \sqrt{\tau/\mu}$

CHAPTER 7

$$7.32. \quad P_4(x) = \frac{1}{8}(3 - 30x^2 + 35x^4)$$

$$P_5(x) = \frac{1}{8}(15x - 70x^3 + 63x^5)$$

$$7.33. \quad (a) 0 \quad (b) 2/5 \quad (c) 0$$

$$7.43. \quad a_3(x) = \frac{x}{4}(5x^2 - 3) \ln\left(\frac{1+x}{1-x}\right) - \frac{5x^2}{2} + \frac{2}{3}$$

$$7.44. \quad y = Ax + B\left[1 + \frac{x}{2} \ln\left(\frac{1+x}{1-x}\right)\right]$$

$$7.45. \quad -\frac{4}{5}P_0(x) + P_1(x) - \frac{10}{7}P_2(x) + \frac{8}{35}P_4(x)$$

$$7.46. \quad P_0(x) + \frac{7}{4}P_1(x) + \frac{5}{8}P_2(x) - \frac{7}{16}P_3(x) + \dots$$

$$7.48. \quad (a) \quad v = v_0 + 3v_0r \cos \theta$$

$$(b) \quad v = \frac{v_0}{r} + \frac{3v_0}{r^2} \cos \theta$$

$$7.49. \quad (a) \quad v = \frac{2v_0}{3}[1 - r^2P_2(\cos \theta)]$$

$$(b) \quad v = \frac{2v_0}{3}\left[\frac{1}{r} - \frac{P_2(\cos \theta)}{r^3}\right]$$

$$7.50. \quad 2u_0\left(1 - \frac{a}{r}\right)$$

$$7.51. \quad m/r \text{ where } r > a \text{ is the distance from the center of the sphere}$$

$$7.56. \quad (a) \quad 3(1 - x^2)$$

$$(b) \quad -\frac{5}{2}(3 - 24x^2 - 21x^4)$$

$$(c) \quad -105x(1 - x^2)^{3/2}$$

$$7.57. \quad (a) \quad -\sqrt{1-x^2}\left[\frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) + \frac{x}{1-x^2}\right]$$

$$(b) \quad \frac{x^4 - x^2 + 2}{1 - x^2}$$

$$7.63. \quad \text{Inside, } v = v_0 r^4 \sin^3 \theta \cos \theta \cos 3\phi$$

$$\text{Outside, } v = \frac{v_0}{r^5} \sin^3 \theta \cos \theta \cos 3\phi$$

$$7.72. \quad u = \frac{bu_2 - au_1}{b-a} + \frac{ab(u_1 - u_2)}{(b-a)r}$$

$$7.73. \quad u = [A_1 J_{n+1/2}(\lambda r) + B_1 J_{-n-1/2}(\lambda r)] \cdot [A_2 P_n(\cos \theta) + B_2 Q_n(\cos \theta)] e^{-\kappa \lambda^2 t}$$

CHAPTER 8

$$8.17. \quad 1, 2x, 4x^2 - 2, 8x^3 - 12x$$

$$8.20. \quad 4x^2 - 2, 8x^3 - 12x$$

$$8.22. \quad \frac{1}{2}\sqrt{\pi} \text{ if } n = 0, \quad 2\sqrt{\pi} \text{ if } n = 2, \\ 0 \text{ otherwise}$$

$$8.26. \quad (a) \quad -\frac{3}{2}H_0(x) + \frac{7}{4}H_1(x) - \frac{3}{4}H_2(x) + \frac{1}{8}H_3(x)$$

$$8.27. \quad (a) \quad y = c_1 + c_2 \int e^{x^2} dx$$

$$(b) \quad y = c_1 x + c_2 x \int \frac{e^{x^2}}{x^2} dx$$

$$8.28. \quad L_4(x) = 24 - 96x + 72x^2 - 16x^3 + x^4$$

$$8.30. \quad L_2(x) = 2 - 4x + x^2$$

$$L_3(x) = 6 - 18x + 9x^2 - x^3$$

$$8.34. \quad 2L_0(x) - 8L_1(x) + 6L_2(x) - L_3(x)$$

$$8.36. \quad (a) \quad y = c_1 + c_2 \int \frac{e^x}{x} dx$$

$$(b) \quad y = c_1(1-x) + c_2(1-x) \int \frac{e^x}{x(1-x)^2} dx$$

$$8.38. \quad L_4^2(x) = 144 - 96x + 12x^2$$

$$L_5^3(x) = -1296 + 600x - 60x^2$$

$$8.41. \quad 180$$

$$8.43. \quad y = A \sin^{-1} x + B$$

$$8.44. \quad (a) \quad T_4(x) = 8x^4 - 8x^2 + 1$$

$$(b) \quad T_5(x) = 16x^5 - 20x^3 + 2x$$

$$8.45. \quad \frac{1}{4}[T_3(x) + 2T_2(x) - 13T_1(x) + 10T_0(x)]$$

$$8.49. \quad (a) \quad y = AH_n(x) + BH_n(x) \int \frac{e^{x^2} dx}{[H_n(x)]^2}$$

$$8.51. \quad (a) \quad y = AL_n(x) + BL_n(x) \int \frac{e^x dx}{x[L_n(x)]^2}$$

$$8.55. \quad (a) \quad y = AT_n(x) + BT_n(x) \int \frac{dx}{\sqrt{1-x^2} [T_n(x)]^2}$$



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